Learn Being Forward

COURSES · CAMPUS PARTNERSHIPS · PLACEMENTS

STUDY NOTES · BASIC TO ADVANCED

Non-Verbal Reasoning
See the Pattern, Find the Rule

Complete notes for the Learn Being Forward Non-Verbal Reasoning syllabus: every question type explained with figures, rules and shortcut tricks, followed by basic, intermediate and advanced questions with fully worked solutions.

22	3	31	548
modules	levels	minimum days	solved questions

How to use these notes

These notes follow the **Learn Being Forward Non-Verbal Reasoning syllabus** exactly: the same 22 modules, in the same order, with every topic and subtopic explained step by step. Each subtopic number (for example **10.2.1**) matches the syllabus, so you can always find where you are.

Level	Modules	Minimum	Outcome
1. Beginner	1 to 7	8 days	See and describe changes in figures, and solve figure series, analogies, classification, mirror images and water images
2. Intermediate	8 to 15	9 days	Find embedded figures, complete patterns, solve paper folding and cutting, count figures, crack figure matrices and form figures from pieces
3. Advanced	16 to 22	14 days	Master painted cubes, dice, 3D nets and views, abstract rule-based puzzles, visual Venn diagrams and diagrammatic operator tests, and perform in company-specific assessments
Total	1 to 22	31 days	About 6 weeks at 6 study days per week

1 · Understand
Read the method and study every figure until you can name the rule in your own words.

2 · Learn the checklist
Use the element checklist (number, shape, position, rotation, shading, size) on every question.

3 · Solve with a timer
Attempt the basic, intermediate and advanced questions before looking at any solution.

4 · Check and log
Compare with the solutions, check folding and dice questions with real paper or a die, and log every mistake.

Every topic follows the same pattern: the idea in plain words, worked figure examples, the standard method and **shortcut tricks** for exam speed. Each module then gives about 25 practice questions in three levels, **basic**, **intermediate** and **advanced**, in the styles used by campus tests (TCS, Infosys, Wipro, Accenture, Cognizant, Capgemini, AMCAT) and SHL, Aon and bank exams. All figures were drawn by a program from the rule of each question, so the correct answer figure is generated from the rule and the wrong options each break exactly one part of it; every question is fully solved.

HOW TO READ THE BOXES

Gold-bordered boxes hold **shortcut tricks**, worked examples and notes, red boxes warn about common traps, and navy boxes hold the key rule to remember. Problem figures are shown on the left and answer figures, labelled (a) to (d), on the right or below.

Course plan at a glance

Module	Topic	Level	Min days
01	Introduction to non-verbal reasoning	Beginner	1
02	Visual perception basics	Beginner	1
03	Figure series	Beginner	2
04	Figure analogy	Beginner	1

Module	Topic	Level	Min days
05	Figure classification (odd one out)	Beginner	1
06	Mirror images	Beginner	1
07	Water images	Beginner	1
08	Embedded figures	Intermediate	1
09	Completion of figures and pattern completion	Intermediate	1
10	Paper folding	Intermediate	1
11	Paper cutting and punching	Intermediate	1
12	Counting figures	Intermediate	2
13	Figure matrix	Intermediate	1
14	Grouping of figures	Intermediate	1
15	Dot situation, figure formation and figure analysis	Intermediate	1
16	Cubes and cuboids	Advanced	2
17	Dice	Advanced	2
18	3D spatial visualisation	Advanced	2
19	Abstract and rule-based reasoning	Advanced	2
20	Venn diagrams and visual relationships	Advanced	1
21	Diagrammatic and flowchart reasoning	Advanced	2
22	Exam strategy, mock tests and company-specific preparation	Advanced	3

Table of contents

LEVEL 1 · BEGINNER LEVEL · MODULES 1 TO 7

Module 01 · Introduction to non-verbal reasoning 8
- 1.1 What is non-verbal reasoning? 8
- 1.2 What it measures 8
- 1.3 Why do we need it? 8
- 1.4 Where is it tested? 8
- 1.5 How to train visual reasoning 8
- 1.6 Diagnostic test 8

Module 02 · Visual perception basics 19
- 2.1 Elements of a figure 19
- 2.2 Types of change 19
- 2.3 Symmetry 19
- 2.4 The element checklist 19
- 2.5 Practice questions 19
- 2.6 Solutions 19

Module 03 · Figure series 30
- 3.1 Single-rule series 30
- 3.2 Multi-rule series 30
- 3.3 Alternate and cyclic series 30
- 3.4 Types of questions 30
- 3.5 Method 30
- 3.6 Practice questions 30
- 3.7 Solutions 30

Module 04 · Figure analogy 44
- 4.1 Concept 44
- 4.2 Common relationships 44
- 4.3 Method 44
- 4.4 Practice questions 44
- 4.5 Solutions 44

Module 05 · Figure classification (odd one out) 58
- 5.1 Common grouping rules 58
- 5.2 Method 58
- 5.3 Practice questions 58
- 5.4 Solutions 58

Module 06 · Mirror images 67
- 6.1 Concepts 67
- 6.2 Mirror images of letters 67
- 6.3 Numbers and clocks 67
- 6.4 Complex figures 67
- 6.5 Method 67
- 6.6 Practice questions 67
- 6.7 Solutions 67

Module 07 · Water images 77
7.1 Concepts 77
7.2 Letters, numbers and words 77
7.3 Figures and clocks 77
7.4 Combined mirror and water images 77
7.5 Practice questions 77
7.6 Solutions 77

LEVEL 2 · INTERMEDIATE LEVEL · MODULES 8 TO 15

Module 08 · Embedded figures 88
8.1 Concepts 88
8.2 Method 88
8.3 Reverse questions 88
8.4 Practice questions 88
8.5 Solutions 88

Module 09 · Completion of figures and pattern completion 100
9.1 Completion of incomplete figures 100
9.2 Pattern completion 100
9.3 Method 100
9.4 Practice questions 100
9.5 Solutions 100

Module 10 · Paper folding 115
10.1 Concepts 115
10.2 Question types 115
10.3 Method 115
10.4 Practice questions 115
10.5 Solutions 115

Module 11 · Paper cutting and punching 125
11.1 Concepts 125
11.2 Question types 125
11.3 Method 125
11.4 Practice questions 125
11.5 Solutions 125

Module 12 · Counting figures 135
12.1 Counting triangles 135
12.2 Counting squares and rectangles 135
12.3 Counting straight lines 135
12.4 Counting other shapes 135
12.5 Method 135
12.6 Practice questions 135
12.7 Solutions 135

Module 13 · Figure matrix 147
13.1 Matrix rules 147
13.2 Common rule types 147
13.3 Raven's Progressive Matrices style questions 147
13.4 Method 147

13.5 Practice questions 147
13.6 Solutions 147

Module 14 · Grouping of figures 167

14.1 Concepts 167
14.2 Method 167
14.3 Practice questions 167
14.4 Solutions 167

Module 15 · Dot situation, figure formation and figure analysis 183

15.1 Dot situation 183
15.2 Figure formation 183
15.3 Construction of squares and triangles 183
15.4 Shape overlay and transparency 183
15.5 Practice questions 183
15.6 Solutions 183

LEVEL 3 · ADVANCED LEVEL · MODULES 16 TO 22

Module 16 · Cubes and cuboids 198

16.1 Basics 198
16.2 Painted cube formulas 198
16.3 Variations 198
16.4 Cube counting in stacks 198
16.5 Practice questions 198
16.6 Solutions 198

Module 17 · Dice 208

17.1 Basics 208
17.2 Finding opposite faces 208
17.3 Unfolded dice (nets) 208
17.4 Other variations 208
17.5 Practice questions 208
17.6 Solutions 208

Module 18 · 3D spatial visualisation 219

18.1 Nets of 3D shapes 219
18.2 Orthographic views 219
18.3 3D rotation 219
18.4 Cross-sections 219
18.5 Isometric drawing (overview) 219
18.6 Practice questions 219
18.7 Solutions 219

Module 19 · Abstract and rule-based reasoning 229

19.1 Abstract reasoning formats 229
19.2 Rule types 229
19.3 Method 229
19.4 Practice questions 229
19.5 Solutions 229

Module 20 · Venn diagrams and visual relationships 243

20.1 Choosing the right diagram 243

20.2 Reading Venn diagrams 243
20.3 Mixed-shape diagrams 243
20.4 Practice questions 243
20.5 Solutions 243

Module 21 · Diagrammatic and flowchart reasoning 253

21.1 Operators and transformations 253
21.2 Flowchart reasoning 253
21.3 Sequences of operations 253
21.4 Switch and switch-challenge games 253
21.5 Practice questions 253
21.6 Solutions 253

Module 22 · Exam strategy, mock tests and company-specific preparation 267

22.1 Strategy 267
22.2 Company-specific patterns 267
22.3 Mock tests and analysis 267
22.4 Revision plan 267
22.5 Mixed mock test 267
22.6 Solutions 267

LEVEL 1

Beginner level

See and describe changes in figures, and solve figure series, analogies, classification, mirror images and water images

7 MODULES · 8 DAYS MINIMUM

Module 01 · Introduction to non-verbal reasoning	1 day
Module 02 · Visual perception basics	1 day
Module 03 · Figure series	2 days
Module 04 · Figure analogy	1 day
Module 05 · Figure classification (odd one out)	1 day
Module 06 · Mirror images	1 day
Module 07 · Water images	1 day

MODULE 01 · BEGINNER

Introduction to non-verbal reasoning

Minimum duration **1 day** | Level **Beginner**

This module explains what non-verbal reasoning is, what it measures, why companies test it, where it is tested and how to train visual thinking, and ends with a 25-question diagnostic test covering the main question types.

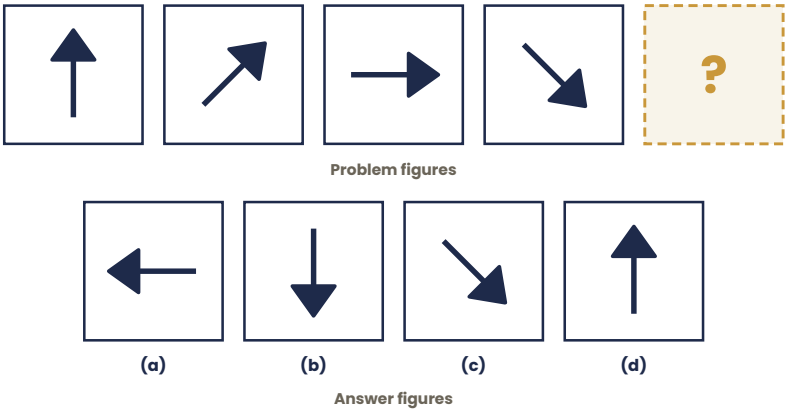
IN THIS MODULE

1.1 What is non-verbal reasoning?
1.2 What it measures
1.3 Why do we need it?
1.4 Where is it tested?
1.5 How to train visual reasoning
1.6 Diagnostic test

1.1 What is non-verbal reasoning?

1.1.1 Definition

Non-verbal reasoning means solving problems using figures, shapes and patterns instead of words or numbers. Each question hides a rule (a rotation, a movement, a count, a reflection) inside a set of figures, and you must find the rule and apply it.



A TYPICAL NON-VERBAL QUESTION: THE ARROW TURNS 45° CLOCKWISE EACH TIME

1.1.2 Other names

The same skill is called abstract reasoning, diagrammatic reasoning, figural reasoning or visual reasoning, depending on the test provider.

1.1.3 How it differs from logical and verbal reasoning

Area	Material	Example
Verbal reasoning	words and sentences	analogies of words, reading comprehension

Area	Material	Example
Logical reasoning	words, letters and numbers with rules	coding, seating, syllogisms
Non-verbal reasoning	figures and shapes only	figure series, mirror images, cubes

1.2 What it measures

1.2.1 Pattern recognition

Seeing what repeats or changes regularly across a set of figures.

1.2.2 Spatial visualisation

Turning, flipping, folding and unfolding shapes in your mind: mirror images, paper folding, cubes and dice.

1.2.3 Rule identification and fluid intelligence

Finding a rule you have never seen before, using only the evidence in front of you. Psychologists call this fluid intelligence; it is largely independent of language and schooling.

1.2.4 Attention to detail

Many wrong options differ from the right one in a single small detail (one line, one shading, a mirrored element). Careful checking wins marks.

1.3 Why do we need it?

1.3.1 Language-independent measure

Because it uses no words, non-verbal reasoning lets companies compare candidates from different language and educational backgrounds fairly.

1.3.2 Link to technical work

Spotting patterns, tracking several changing parts at once and testing a rule against evidence are the same habits used in programming, debugging, design, engineering and data work.

1.4 Where is it tested?

1.4.1 Campus tests

Capgemini, Accenture, Cognizant, TCS, Infosys and Wipro include figure-based questions in their reasoning sections.

1.4.2 Assessment providers

AMCAT, SHL, Aon (cut-e), Mettl and HirePro all offer abstract or diagrammatic reasoning tests used by many employers.

1.4.3 Game-based assessments

Some companies use short games (for example matching or rotating shapes under time pressure) that test the same skills.

1.4.4 Government, defence and banking exams

Defence entrance tests, SSC, railway and some banking exams include figure series, mirror images, paper folding and counting figures.

1.5 How to train visual reasoning

1.5.1 Break figures into elements

Never look at a figure as a whole. List its elements (each shape, line and dot) and track each one separately from figure to figure.

1.5.2 Use a checklist, not guesswork



THE ELEMENT CHECKLIST USED THROUGHOUT THIS COURSE

1.5.3 Build speed

> **TIP**
> Spend 10 minutes a day on timed sets of one question type. For folding and dice questions, check your answer with real paper or a real die until you can see it in your head.

1.6 Diagnostic test

Attempt the 25 questions below in 25 minutes. Record your score by type (series, analogy, classification, mirror and water images, counting, folding, cubes and dice); later modules cover each type in depth.

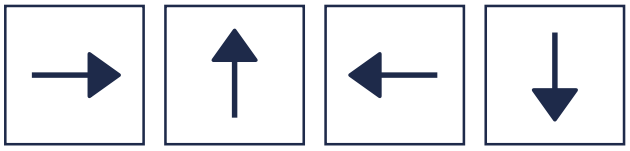
1.6.1 Diagnostic questions

1.6.1.1 Basic level

Q1 Which figure comes next in the series?



Problem figures

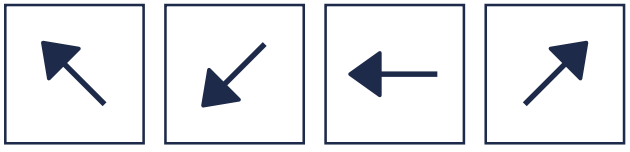


Answer figures

Q2 Which figure comes next in the series?



Problem figures



Answer figures

Q3 Which figure comes next in the series?

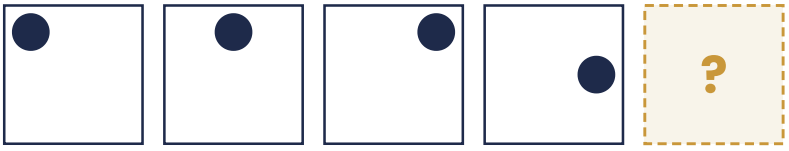


Problem figures

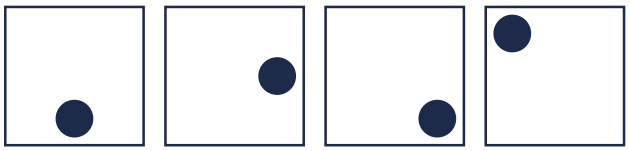


Answer figures

Q4 Which figure comes next in the series?



Problem figures



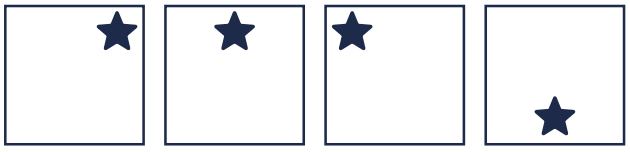
(a) (b) (c) (d)

Answer figures

Q5 Which figure comes next in the series?



Problem figures



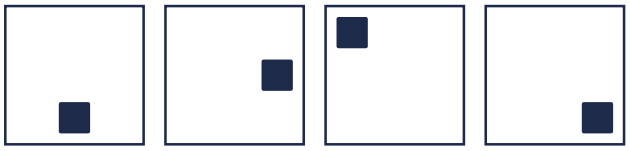
(a) (b) (c) (d)

Answer figures

Q6 Which figure comes next in the series?



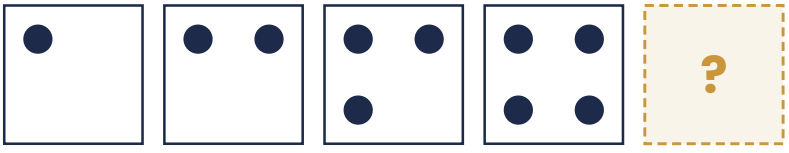
Problem figures



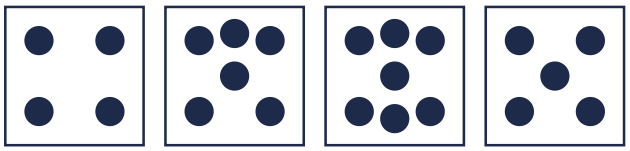
(a) (b) (c) (d)

Answer figures

Q7 Which figure comes next in the series?



Problem figures



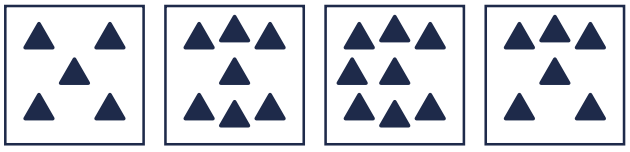
(a) (b) (c) (d)

Answer figures

Q8 Which figure comes next in the series?



Problem figures

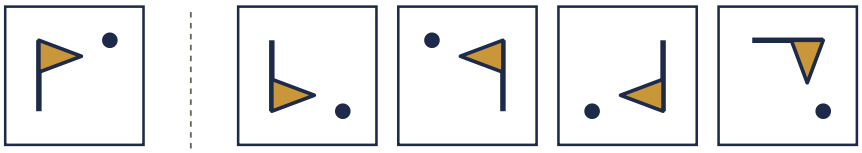


(a) (b) (c) (d)

Answer figures

1.6.1.2 Intermediate level

Q9 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?

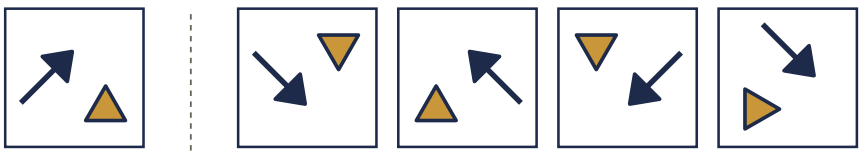


Problem figure

(a) (b) (c) (d)

Answer figures

Q10 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?

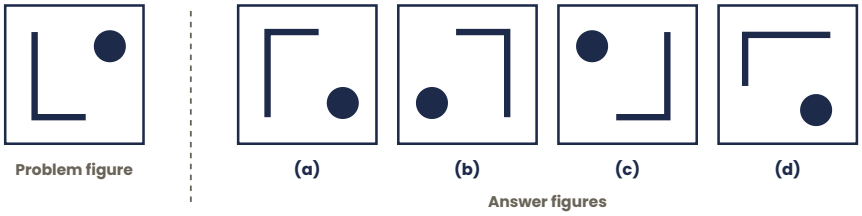


Problem figure

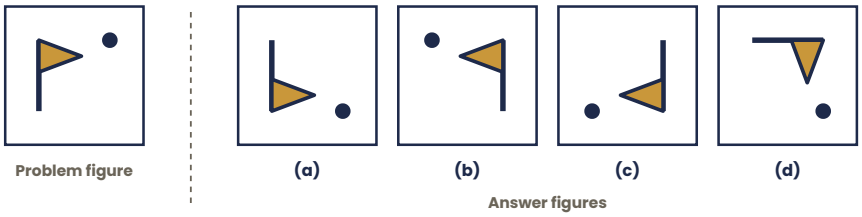
(a) (b) (c) (d)

Answer figures

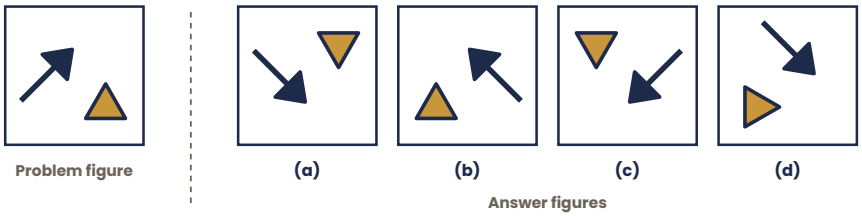
Q11 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?



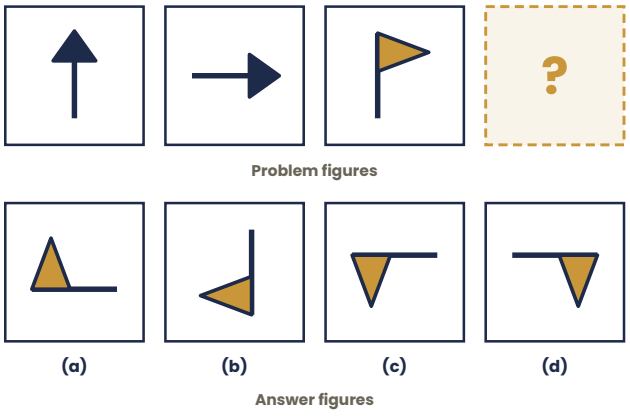
Q12 Which answer figure is the water image of the problem figure?



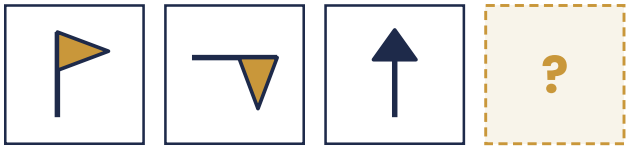
Q13 Which answer figure is the water image of the problem figure?



Q14 Figure 1 is related to figure 2 in the same way as figure 3 is related to which answer figure?



Q15 Figure 1 is related to figure 2 in the same way as figure 3 is related to which answer figure?



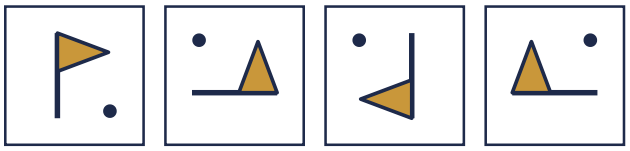
Problem figures



(a) (b) (c) (d)

Answer figures

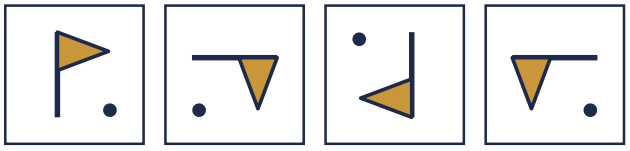
Q16 Which figure is different from the others?



(a) (b) (c) (d)

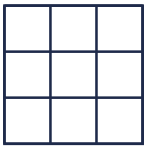
1.6.1.3 Advanced level

Q17 Which figure is different from the others?



(a) (b) (c) (d)

Q18 How many squares are there in a 3 × 3 grid of equal squares? Options: (a) 9, (b) 10, (c) 14, (d) 16.



Q19 A sheet of paper is folded in half once and a single hole is punched through both layers. How many holes are there when it is unfolded? Options: (a) 1, (b) 2, (c) 3, (d) 4.

Q20 A sheet is folded in half twice and one hole is punched. How many holes appear when it is unfolded? Options: (a) 2, (b) 3, (c) 4, (d) 8.

Q21 On a standard die, opposite faces add to 7. Which number is opposite 2? Options: (a) 3, (b) 4, (c) 5, (d) 6.

Q22 A cube painted on all faces is cut into 27 equal small cubes. How many small cubes have no paint? Options: (a) 0, (b) 1, (c) 6, (d) 8.

Q23 A clock shows 3:00. What time does its mirror image show? Options: (a) 3:00, (b) 6:00, (c) 9:00, (d) 12:00.

1.6.2 Solutions

Q1 **Answer: (b)**
The arrow turns 90° clockwise at every step, so the next figure is turned another 90° clockwise: answer **(b)**.

Q2 **Answer: (b)**
The arrow turns 45° anticlockwise at every step, so the next figure is turned another 45° anticlockwise: answer **(b)**.

Q3 **Answer: (c)**
The flag turns 90° clockwise at every step, so the next figure is turned another 90° clockwise: answer **(c)**.

Q4 **Answer: (c)**
The circle moves 1 place clockwise around the frame each time; continuing the movement gives **(c)**.

Q5 **Answer: (b)**
The star moves 2 places clockwise around the frame each time; continuing the movement gives **(b)**.

Q6 **Answer: (d)**
The square moves 1 place anticlockwise around the frame each time; continuing the movement gives **(d)**.

Q7 **Answer: (d)**
The number of circles increases by 1 each time: 1, 2, 3, 4, so the next figure has 5: answer **(d)**.

Q8 **Answer: (d)**
The number of triangles increases by 1 each time: 2, 3, 4, 5, so the next figure has 6: answer **(d)**.

Q9 **Answer: (b)**
A vertical mirror swaps left and right but keeps top and bottom: answer **(b)**.

Q10 **Answer: (b)**

A vertical mirror swaps left and right but keeps top and bottom: answer **(b)**.

Q11 **Answer: (c)**

A vertical mirror swaps left and right but keeps top and bottom: answer **(c)**.

Q12 **Answer: (a)**

A water image swaps top and bottom but keeps left and right: answer **(a)**.

Q13 **Answer: (a)**

A water image swaps top and bottom but keeps left and right: answer **(a)**.

Q14 **Answer: (d)**

Figure 2 is figure 1 turned 90° clockwise. Turning figure 3 the same way gives **(d)**.

Q15 **Answer: (a)**

Figure 2 is figure 1 turned 90° clockwise. Turning figure 3 the same way gives **(a)**.

Q16 **Answer: (b)**

Three figures are the same figure turned through different angles; figure **(b)** is a mirror image, which no rotation can produce.

Q17 **Answer: (d)**

Three figures are the same figure turned through different angles; figure **(d)** is a mirror image, which no rotation can produce.

Q18 **Answer: (c)**

Squares of size 1: 9, size 2: 4, size 3: 1. Total 9 + 4 + 1 = 14. Answer **(c) 14**.

Q19 **Answer: (b)**

Each fold doubles the number of layers, so one punch makes 2 holes: **(b)**.

Q20 **Answer: (c)**

Two folds make 4 layers: **(c) 4** holes.

Q21 **Answer: (c)**

$7 - 2 =$ **(c) 5**.

Q22 **Answer: (b)**

Only the centre cube is unpainted: $(3 - 2)^3 =$ **(b) 1**.

Q23 **Answer: (c)**

Mirror time = 11:60 - 3:00 = **(c) 9:00**.

Non-verbal reasoning is rule-finding with figures. Break every figure into elements, check number, shape, position, rotation, shading and size in that order, and confirm the rule on every figure before choosing an answer.

MODULE 02 · BEGINNER

Visual perception basics

Minimum duration **1 day** | Level **Beginner**

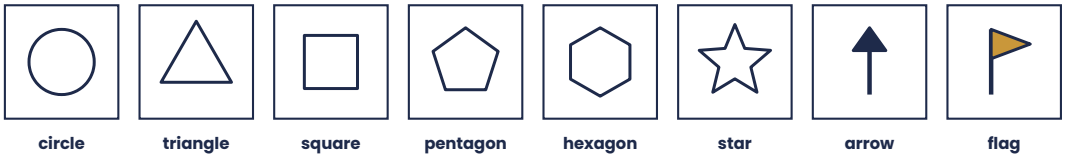
Every non-verbal question is built from a few basic changes to shapes. This module teaches you to see them: the elements of a figure (shape, size, shading, number, position), the types of change (rotation, reflection, movement, addition and deletion, shading, size, interchange), symmetry, and the element checklist used for every later question type, followed by 25 solved questions.

IN THIS MODULE

2.1 Elements of a figure
2.2 Types of change
2.3 Symmetry
2.4 The element checklist
2.5 Practice questions
2.6 Solutions

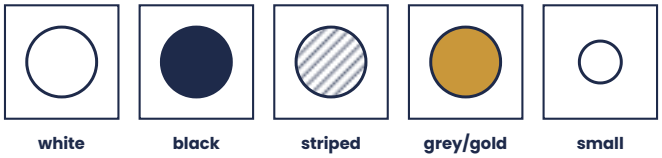
2.1 Elements of a figure

2.1.1 Shapes



THE BASIC SHAPES USED IN FIGURE QUESTIONS

2.1.2 Size, colour and shading



SHADING AND SIZE

2.1.3 Number of elements, sides and corners

Count elements (how many shapes), sides (a pentagon has 5) and corners. Many series simply add or remove one element or one side at each step.

2.1.4 Position inside a frame

Positions are usually the four corners, the four side-middles and the centre. Movements go clockwise or anticlockwise around these nine places.

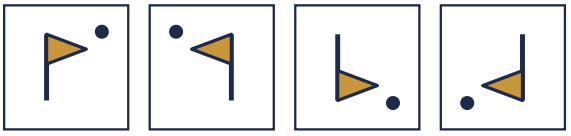
2.2 Types of change

2.2.1 Rotation



ROTATION KEEPS THE FIGURE THE SAME BUT TURNS IT

2.2.2 Reflection vs rotation

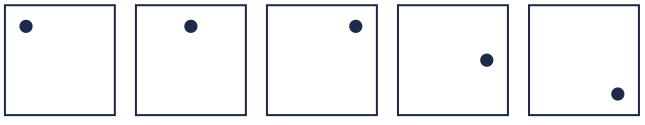


A FLIP REVERSES THE FIGURE; A TURN DOES NOT

WATCH OUT

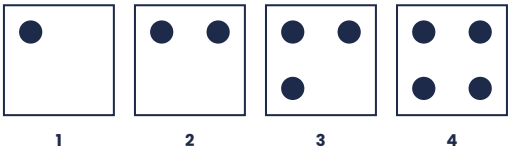
A reflection can never be produced by a rotation of an asymmetric figure. This is the most common trap in analogy and odd-one-out questions.

2.2.3 Movement



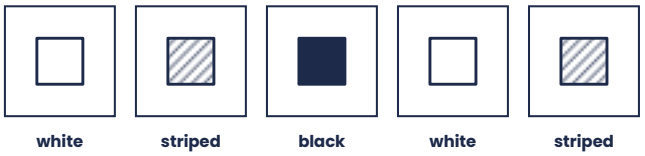
A DOT MOVING ONE PLACE CLOCKWISE EACH TIME

2.2.4 Addition and deletion



ONE ELEMENT ADDED AT EACH STEP

2.2.5 Change of shading

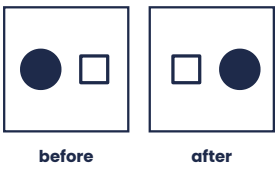


A REPEATING SHADING CYCLE

2.2.6 Size change

Elements may grow or shrink step by step, or swap sizes with one another.

2.2.7 Interchange of elements



TWO ELEMENTS SWAP PLACES

2.3 Symmetry

2.3.1 Lines of symmetry

Shape	Lines of symmetry
Equilateral triangle	3
Square	4
Rectangle	2
Regular pentagon	5
Regular hexagon	6
Circle	infinitely many
Parallelogram (non-rectangle)	0

2.3.2 Rotational symmetry

A figure has rotational symmetry of order n if it looks the same n times in one full turn: a square has order 4, an equilateral triangle order 3, the letter S order 2.

2.4 The element checklist

2.4.1 Number → shape → position → rotation → shading → size

For every pair of figures, go through the checklist in order and write down each change. The cheaper checks (counting, shape) come first; rotation and reflection, which need more care, come later.

> **SHORTCUT TRICK**
> Use a small asymmetric detail (a dot, a flag, an arrowhead) to decide between rotation and reflection: in a rotation the detail stays on the same side relative to the rest of the figure.

2.5 Practice questions

2.5.1 Basic level

Q1 Which change turns figure 1 into figure 2?

(a) 90° anticlockwise rotation
(b) 180° rotation
(c) Top–bottom reflection
(d) 90° clockwise rotation

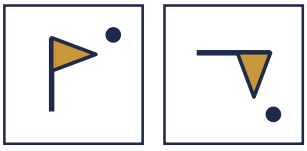


figure 1 figure 2

Q2 Which change turns figure 1 into figure 2?

(a) 180° rotation
(b) Left–right reflection
(c) Top–bottom reflection
(d) 90° anticlockwise rotation

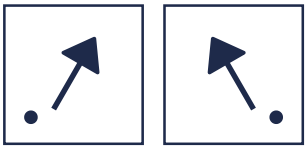


figure 1 figure 2

Q3 Which change turns figure 1 into figure 2?

(a) Left–right reflection
(b) Top–bottom reflection
(c) 90° anticlockwise rotation
(d) 45° clockwise rotation

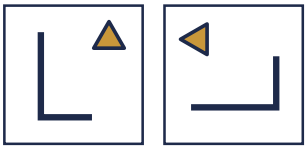


figure 1 figure 2

Q4 Which change turns figure 1 into figure 2?

(a) 180° rotation
(b) Top–bottom reflection
(c) 90° clockwise rotation
(d) 45° clockwise rotation

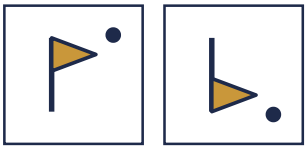


figure 1 figure 2

Q5 Which change turns figure 1 into figure 2?

(a) Top–bottom reflection
(b) 180° rotation
(c) Left–right reflection
(d) 90° clockwise rotation

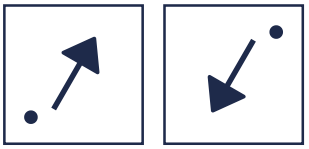


figure 1 figure 2

Q6 Which change turns figure 1 into figure 2?

(a) 45° clockwise rotation
(b) Top–bottom reflection
(c) 90° anticlockwise rotation
(d) 90° clockwise rotation



figure 1 figure 2

Q7 Which change turns figure 1 into figure 2?

(a) Left–right reflection
(b) 180° rotation
(c) 45° clockwise rotation
(d) Top–bottom reflection

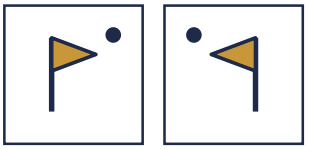


figure 1 figure 2

Q8 Which change turns figure 1 into figure 2?

(a) 90° clockwise rotation
(b) Top–bottom reflection
(c) Left–right reflection
(d) 90° anticlockwise rotation

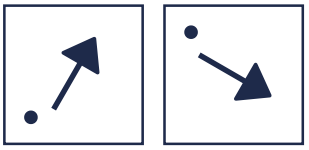


figure 1 figure 2

2.5.2 Intermediate level

Q9 Which change turns figure 1 into figure 2?

(a) Left–right reflection
(b) Top–bottom reflection
(c) 180° rotation
(d) 90° clockwise rotation



figure 1 figure 2

Q10 Which change turns figure 1 into figure 2?

(a) 180° rotation
(b) 90° anticlockwise rotation
(c) Top–bottom reflection
(d) 90° clockwise rotation

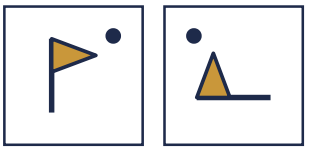


figure 1 figure 2

Q11 Which change turns figure 1 into figure 2?

(a) 90° clockwise rotation
(b) 180° rotation
(c) 45° clockwise rotation
(d) Left–right reflection

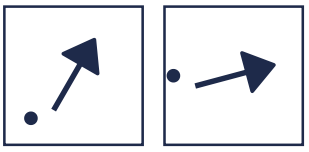


figure 1 figure 2

Q12 Which change turns figure 1 into figure 2?

(a) Top–bottom reflection
(b) 45° clockwise rotation
(c) 90° anticlockwise rotation
(d) 180° rotation

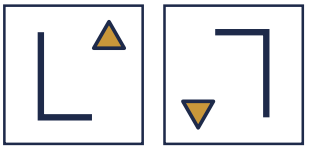
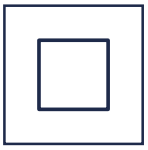


figure 1 figure 2

Q13 How many lines of symmetry does this regular figure have?

(a) 3 **(b)** 5 **(c)** 4 **(d)** 8



Q14 How many lines of symmetry does this regular figure have?

(a) 2 **(b)** 4 **(c)** 6 **(d)** 3



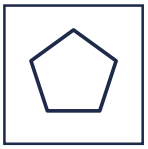
Q15 How many lines of symmetry does this regular figure have?

(a) 5 **(b)** 6 **(c)** 7 **(d)** 12



Q16 How many lines of symmetry does this regular figure have?

(a) 4 **(b)** 6 **(c)** 10 **(d)** 5



Q17 How many lines of symmetry does this regular figure have?

(a) 4 **(b)** 6 **(c)** 10 **(d)** 5



2.5.3 Advanced level

Q18 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 1 **(b)** 0 **(c)** 2 **(d)** 3

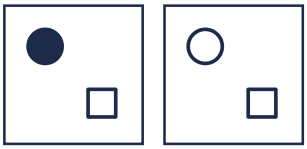


figure 1 figure 2

Q19 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 0 **(b)** 2 **(c)** 1 **(d)** 3

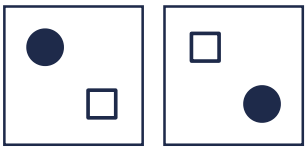


figure 1 figure 2

Q20 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 3 **(b)** 0 **(c)** 1 **(d)** 2

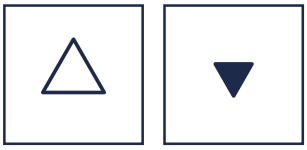


figure 1 figure 2

Q21 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 0 **(b)** 2 **(c)** 1 **(d)** 3

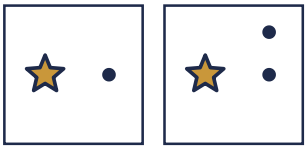


figure 1 figure 2

Q22 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 3 **(b)** 0 **(c)** 1 **(d)** 2

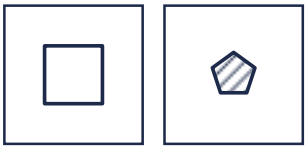


figure 1 figure 2

Q23 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 0 **(b)** 2 **(c)** 3 **(d)** 1



figure 1 figure 2

Q24 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

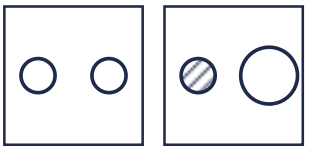


figure 1 figure 2

Q25 How many separate changes (shape, size, shading, position, rotation, number) turn figure 1 into figure 2?

(a) 0 **(b)** 2 **(c)** 1 **(d)** 3

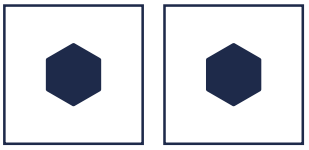


figure 1 figure 2

2.6 Solutions

Q1 **Answer: (d) 90° clockwise rotation**

Track the small detail: the result is a **90° clockwise rotation**.

Q2 **Answer: (b) Left–right reflection**

Track the small detail: the result is a **left–right reflection**.

Q3 **Answer: (c) 90° anticlockwise rotation**

Track the small detail: the result is a **90° anticlockwise rotation**.

Q4 **Answer: (b) Top–bottom reflection**

Track the small detail: the result is a **top–bottom reflection**.

Q5 **Answer: (b) 180° rotation**

Track the small detail: the result is a **180° rotation**.

Q6 **Answer: (a) 45° clockwise rotation**

Track the small detail: the result is a **45° clockwise rotation**.

Q7 **Answer: (a) Left–right reflection**

Track the small detail: the result is a **left–right reflection**.

Q8 **Answer: (a) 90° clockwise rotation**

Track the small detail: the result is a **90° clockwise rotation**.

Q9 **Answer: (b) Top–bottom reflection**

Track the small detail: the result is a **top–bottom reflection**.

Q10 **Answer: (b) 90° anticlockwise rotation**

Track the small detail: the result is a **90° anticlockwise rotation**.

Q11 **Answer: (c) 45° clockwise rotation**

Track the small detail: the result is a **45° clockwise rotation**.

Q12 **Answer: (d) 180° rotation**

Track the small detail: the result is a **180° rotation**.

Q13 **Answer: (c) 4**

A regular square with this outline has **4** lines of symmetry.

Q14 **Answer: (d) 3**

A regular triangle with this outline has **3** lines of symmetry.

Q15 **Answer: (b) 6**

A regular hexagon with this outline has **6** lines of symmetry.

Q16 **Answer: (d) 5**

A regular pentagon with this outline has **5** lines of symmetry.

Q17 **Answer: (d) 5**

A regular five-pointed star has **5** lines of symmetry, one through each point.

Q18 **Answer: (a) 1**

Using the checklist: only the circle's shading changes, so **1** change.

Q19 **Answer: (c) 1**

Using the checklist: the two elements interchange positions (one change), so **1** change.

Q20 **Answer: (a) 3**

Using the checklist: size, rotation and shading all change, so **3** changes.

Q21 **Answer: (c) 1**

Using the checklist: one dot is added, so **1** change.

Q22 **Answer: (a) 3**

Using the checklist: shape, size and shading change, so **3** changes.

Q23 **Answer: (d) 1**

Using the checklist: only the rotation changes, so **1** change.

Q24 **Answer: (c) 2**

Using the checklist: one circle is shaded and the other grows, so **2** changes.

Q25 **Answer: (c) 1**

Using the checklist: a white dot is added in the centre, so **1** change.

Every figure question is built from a few changes: rotation, reflection, movement, addition or deletion, shading, size and interchange. Check number, shape, position, rotation, shading and size in that order, and use an asymmetric detail to tell a rotation from a reflection.

MODULE 03 · BEGINNER

Figure series

Minimum duration **2 days** | Level **Beginner**

Figure series questions show a sequence of figures and ask for the next one, a missing one, or the one that breaks the rule. This module covers single-rule series (rotation, movement, addition or removal, shading), multi-rule series with elements moving at different speeds, alternate and cyclic series, the three question types, and the one-element-at-a-time method, followed by 25 solved questions whose figures were generated from their rules.

IN THIS MODULE

3.1 Single-rule series
3.2 Multi-rule series
3.3 Alternate and cyclic series
3.4 Types of questions
3.5 Method
3.6 Practice questions
3.7 Solutions

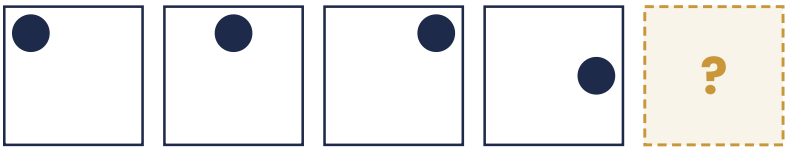
3.1 Single-rule series

3.1.1 Rotation of one element



THE ARROW TURNS 45° CLOCKWISE EACH TIME

3.1.2 Movement around the frame



THE CIRCLE MOVES ONE PLACE CLOCKWISE AROUND THE FRAME

3.1.3 Addition or removal of elements



ONE TRIANGLE IS ADDED EACH TIME

3.1.4 Changing shading



SHADING CYCLES WHITE → STRIPED → BLACK

3.2 Multi-rule series

3.2.1 Two or more elements changing by different rules



THE ARROW TURNS AND THE DOT MOVES, EACH BY ITS OWN RULE

3.2.2 Different speeds



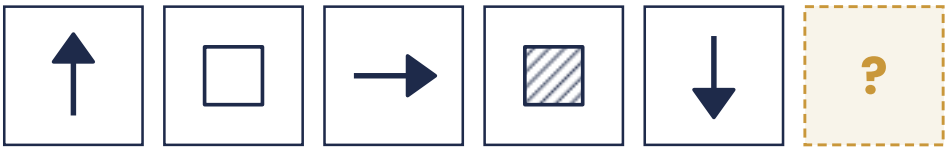
THE STAR MOVES ONE STEP AND THE CIRCLE TWO STEPS AT A TIME

TIP

When two elements move, write their positions as numbers (1 to 8 clockwise from top-left). Their step sizes become obvious: 1, 2, 3, 4 and 5, 7, 1, 3.

3.3 Alternate and cyclic series

3.3.1 Every other figure



ODD AND EVEN POSITIONS FOLLOW DIFFERENT RULES

3.3.2 Repeating cycles

Some series repeat a short cycle of 2, 3 or 4 figures. Number the figures and find the cycle length; the next figure equals the one a full cycle earlier.

3.4 Types of questions

3.4.1 Next figure

The usual type: find the rule from the given figures and continue it one step.

3.4.2 Missing figure in the middle

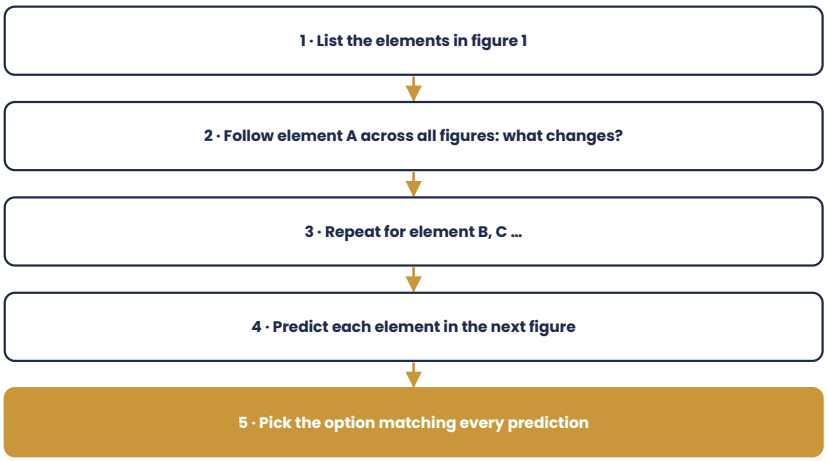
The rule must fit the figures on both sides of the gap. Check the figure before and after the gap.

3.4.3 Wrong figure in a series

Exactly one figure breaks the rule. Find the rule from the majority, then compare each figure with the regenerated series.

3.5 Method

3.5.1 Track one element at a time



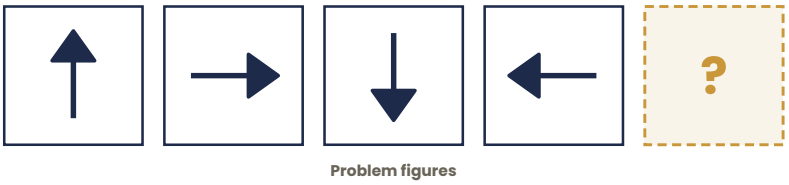
THE ONE-ELEMENT-AT-A-TIME METHOD

WATCH OUT
Do not choose an option because it "looks right". Eliminate options that fail any single rule; usually only one survives.

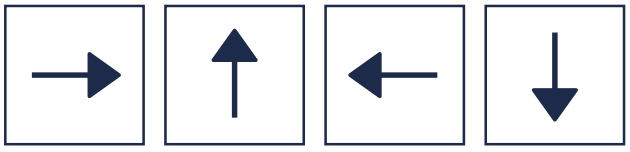
3.6 Practice questions

3.6.1 Basic level

Q1 Which figure comes next?

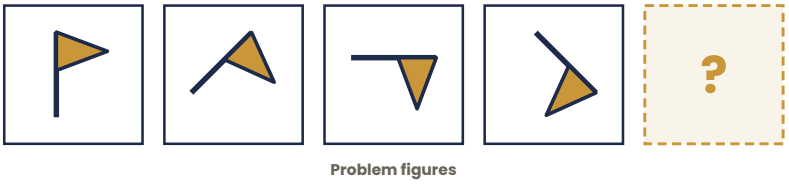


Problem figures

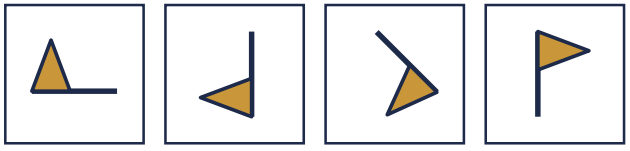


Answer figures

Q2 Which figure comes next?

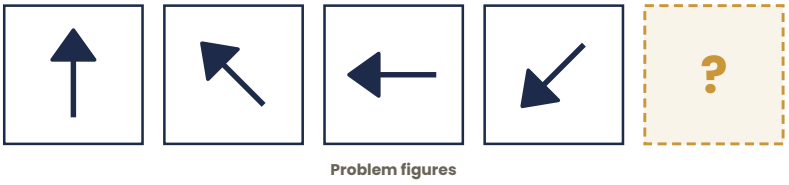


Problem figures

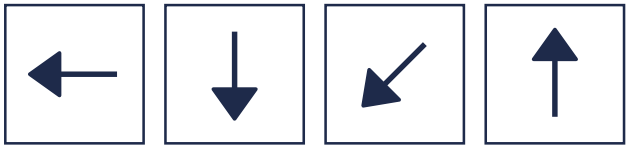


Answer figures

Q3 Which figure comes next?



Problem figures

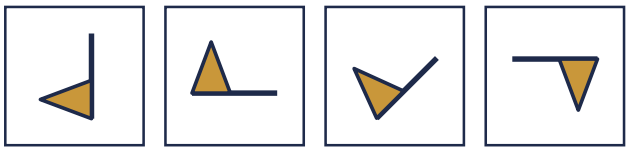


Answer figures

Q4 Which figure comes next?



Problem figures



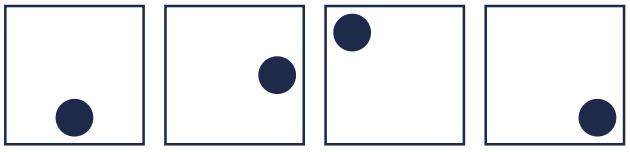
(a) (b) (c) (d)

Answer figures

Q5 Which figure comes next?



Problem figures



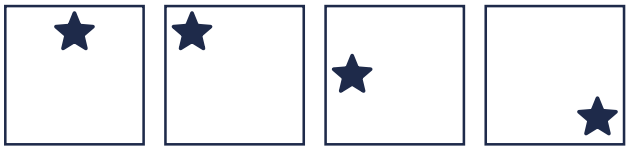
(a) (b) (c) (d)

Answer figures

Q6 Which figure comes next?



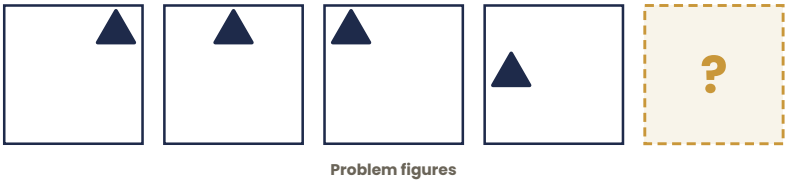
Problem figures



(a) (b) (c) (d)

Answer figures

Q7 Which figure comes next?

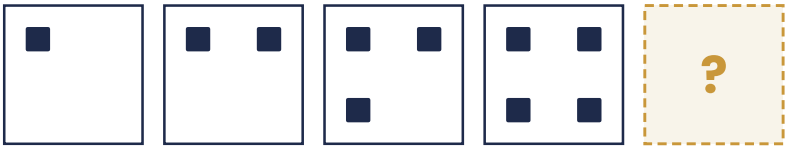


Problem figures

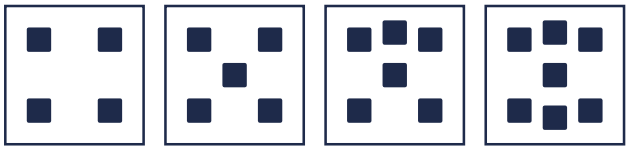


Answer figures

Q8 Which figure comes next?



Problem figures



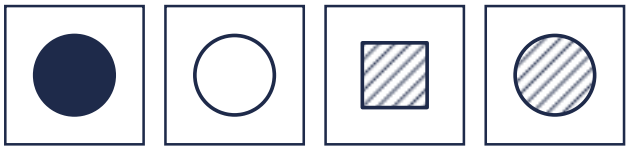
Answer figures

3.6.2 Intermediate level

Q9 Which figure comes next?

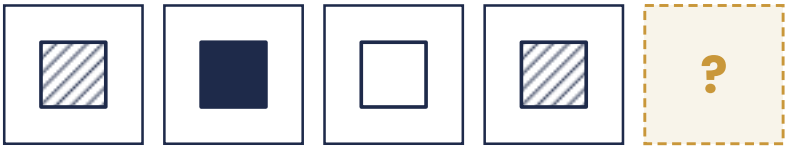


Problem figures



Answer figures

Q10 Which figure comes next?



Problem figures



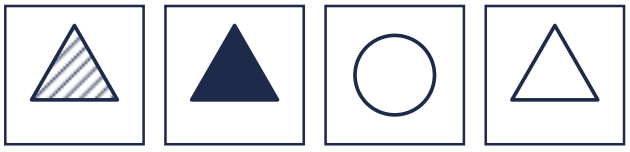
(a) (b) (c) (d)

Answer figures

Q11 Which figure comes next?



Problem figures



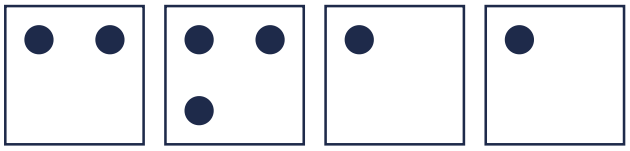
(a) (b) (c) (d)

Answer figures

Q12 Which figure comes next?



Problem figures



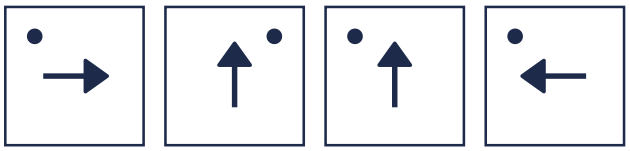
(a) (b) (c) (d)

Answer figures

Q13 Which figure comes next?



Problem figures



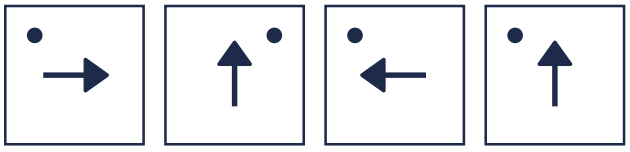
(a) (b) (c) (d)

Answer figures

Q14 Which figure comes next?



Problem figures



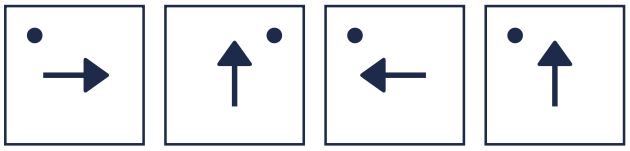
(a) (b) (c) (d)

Answer figures

Q15 Which figure comes next?



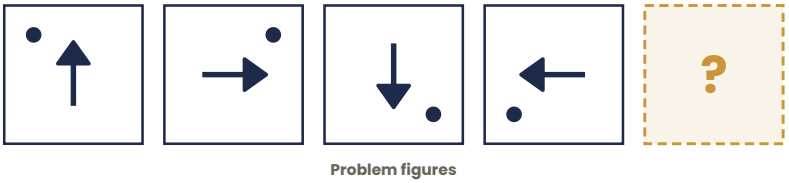
Problem figures



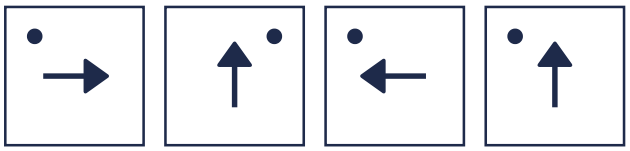
(a) (b) (c) (d)

Answer figures

Q16 Which figure comes next?



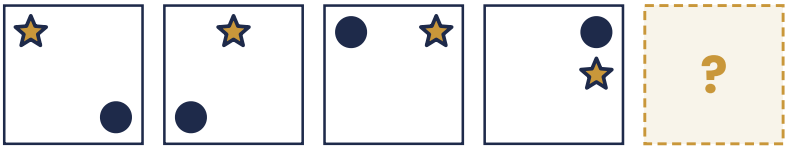
Problem figures



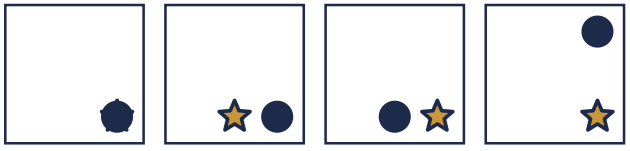
Answer figures

3.6.3 Advanced level

Q17 Which figure comes next?

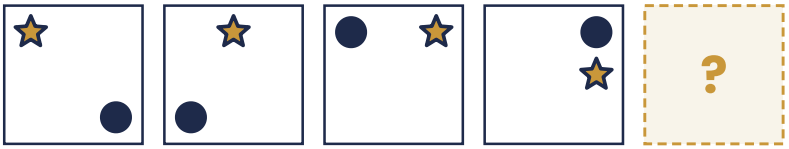


Problem figures

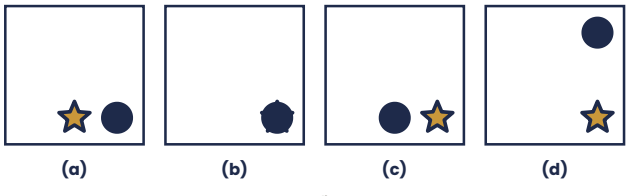


Answer figures

Q18 Which figure comes next?

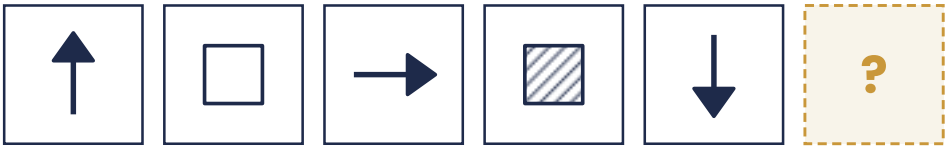


Problem figures

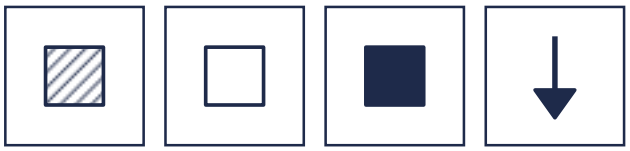


Answer figures

Q19 Which figure comes next?

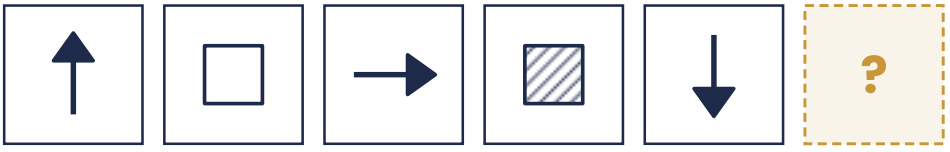


Problem figures

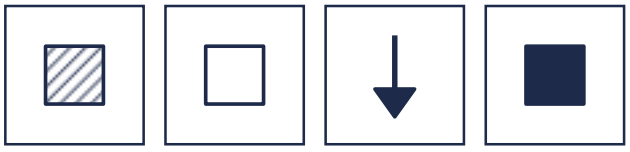


Answer figures

Q20 Which figure comes next?



Problem figures

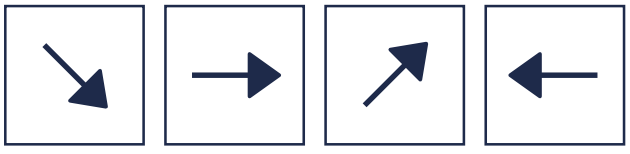


Answer figures

Q21 Which figure fits in the place of the question mark?



Problem figures



Answer figures

Q22 Which figure fits in the place of the question mark?



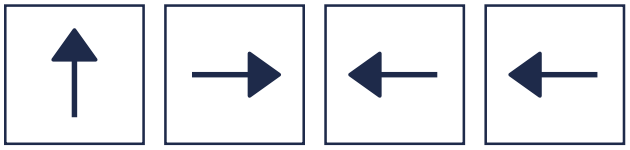
Problem figures



(a) (b) (c) (d)

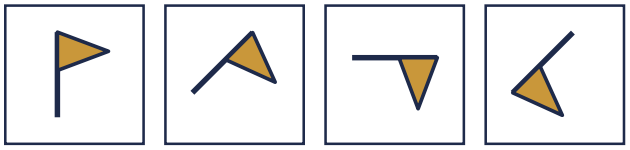
Answer figures

Q23 Which figure breaks the rule of the series?



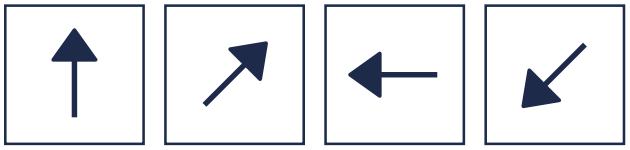
(a) (b) (c) (d)

Q24 Which figure breaks the rule of the series?



(a) (b) (c) (d)

Q25 Which figure breaks the rule of the series?



(a) (b) (c) (d)

3.7 Solutions

Q1 **Answer: (b)**

The arrow turns 90° clockwise at every step, so the next figure is turned another 90° clockwise: answer **(b)**.

Q2 **Answer: (b)**

The flag turns 45° clockwise at every step, so the next figure is turned another 45° clockwise: answer **(b)**.

Q3 **Answer: (b)**

The arrow turns 45° anticlockwise at every step, so the next figure is turned another 45° anticlockwise: answer **(b)**.

Q4 **Answer: (d)**

The flag turns 90° anticlockwise at every step, so the next figure is turned another 90° anticlockwise: answer **(d)**.

Q5 **Answer: (d)**

The circle moves 1 place clockwise around the frame each time; continuing the movement gives **(d)**.

Q6 **Answer: (b)**

The star moves 2 places clockwise around the frame each time; continuing the movement gives **(b)**.

Q7 **Answer: (d)**

The triangle moves 1 place anticlockwise around the frame each time; continuing the movement gives **(d)**.

Q8 **Answer: (b)**

The number of squares increases by 1 each time: 1, 2, 3, 4, so the next figure has 5: answer **(b)**.

Q9 **Answer: (d)**

The shading repeats in the cycle white, striped, black. The next figure is striped: answer **(d)**.

Q10 **Answer: (c)**

The shading repeats in the cycle white, striped, black. The next figure is black: answer **(c)**.

Q11 **Answer: (d)**

The shading repeats in the cycle white, striped, black. The next figure is white: answer **(d)**.

Q12 **Answer: (a)**

The number of circles decreases by 1 each time: 6, 5, 4, 3, so the next figure has 2: answer **(a)**.

Q13 **Answer: (c)**

Two rules work together: the arrow turns 90° clockwise and the dot moves one corner clockwise each time. Only **(c)** continues both rules.

Q14 **Answer: (d)**

Two rules work together: the arrow turns 90° clockwise and the dot moves one corner clockwise each time. Only **(d)** continues both rules.

Q15 **Answer: (d)**

Two rules work together: the arrow turns 90° clockwise and the dot moves one corner clockwise each time. Only **(d)** continues both rules.

Q16 **Answer: (d)**

Two rules work together: the arrow turns 90° clockwise and the dot moves one corner clockwise each time. Only **(d)** continues both rules.

Q17 **Answer: (a)**

The star moves one place clockwise and the black circle two places clockwise each time. Continuing both movements gives **(a)**.

Q18 **Answer: (b)**

The star moves one place clockwise and the black circle two places clockwise each time. Continuing both movements gives **(b)**.

Q19 **Answer: (c)**

Figures 1, 3, 5 follow one rule (the arrow turns 90° clockwise) and figures 2, 4 another (the square goes white → striped → black). The next figure belongs to the second series: **(c)**.

Q20 **Answer: (d)**

Figures 1, 3, 5 follow one rule (the arrow turns 90° clockwise) and figures 2, 4 another (the square goes white → striped → black). The next figure belongs to the second series: **(d)**.

Q21 **Answer: (b)**

The figure turns 45° clockwise each step; the missing third figure is turned 90° from the first: **(b)**.

Q22 **Answer: (a)**

The figure turns 90° clockwise each step; the missing third figure is turned 180° from the first: **(a)**.

Q23 **Answer: (c)**

The figure should turn 90° clockwise each time; figure **(c)** does not follow this (it is turned the wrong amount).

Q24 **Answer: (d)**

The figure should turn 45° clockwise each time; figure **(d)** does not follow this (it is flipped).

Q25 **Answer: (b)**

The figure should turn 45° anticlockwise each time; figure **(b)** does not follow this (it is turned the wrong amount).

Track one element at a time: rotation angle and direction, position around the frame, number of elements, shading cycle. Allow different speeds for different elements and separate rules for odd and even positions. Eliminate any option that breaks a single rule.

MODULE 04 · BEGINNER

Figure analogy

Minimum duration **1 day** | Level **Beginner**

Figure analogies apply the change seen in one pair of figures to another figure: figure 1 is to figure 2 as figure 3 is to ?. This module covers the common relationships (rotation, reflection, adding or removing elements or sides, shading swaps, inner and outer shapes interchanging, combinations of two changes) and the list-apply-eliminate method, followed by 25 solved questions generated from their rules.

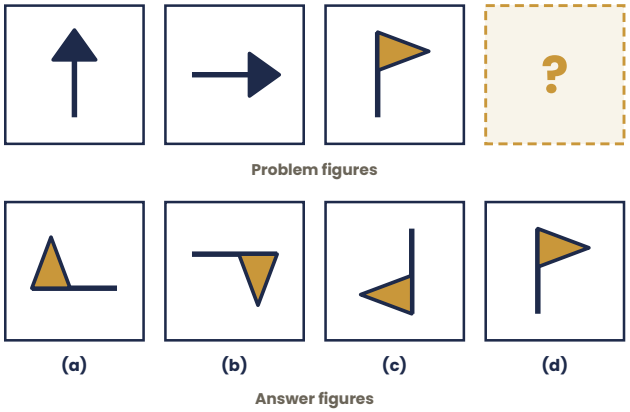
IN THIS MODULE

4.1 Concept
4.2 Common relationships
4.3 Method
4.4 Practice questions
4.5 Solutions

4.1 Concept

4.1.1 Figure 1 : figure 2 :: figure 3 : ?

Find every change that turns figure 1 into figure 2, then apply exactly the same changes to figure 3. The answer must show **all** the changes and **nothing else**.



A ROTATION ANALOGY

4.2 Common relationships

4.2.1 Rotation by a fixed angle

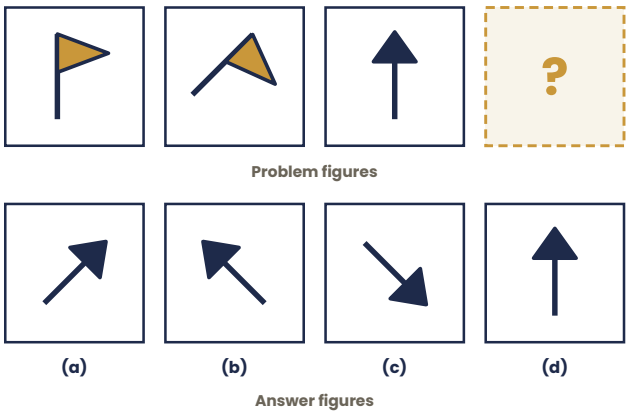


FIGURE 2 IS FIGURE 1 TURNED 45° CLOCKWISE

4.2.2 Reflection

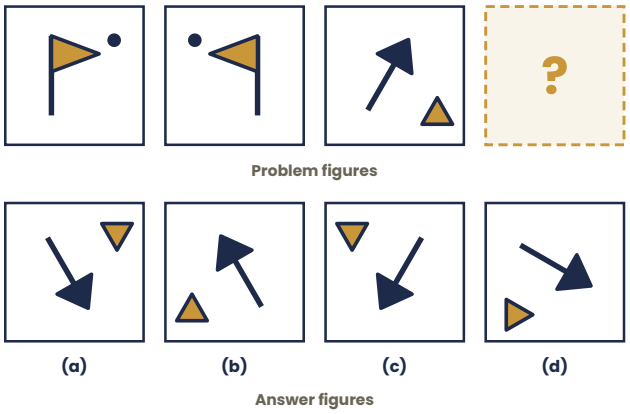
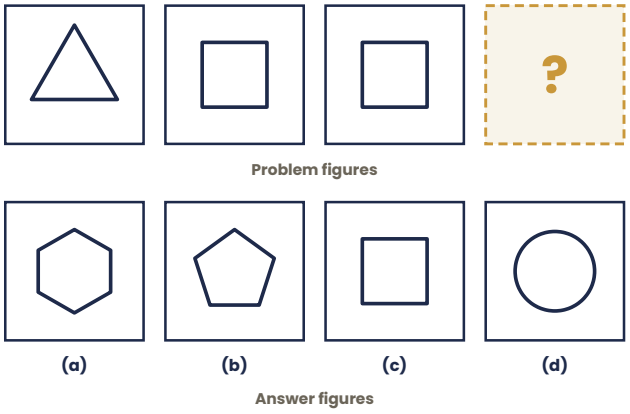


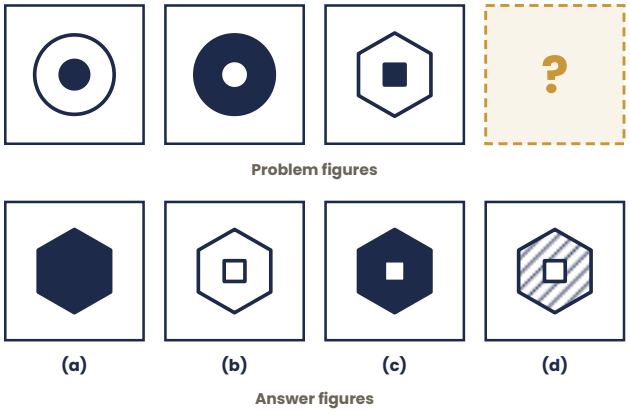
FIGURE 2 IS THE LEFT–RIGHT MIRROR IMAGE OF FIGURE 1

4.2.3 Increase or decrease in elements or sides



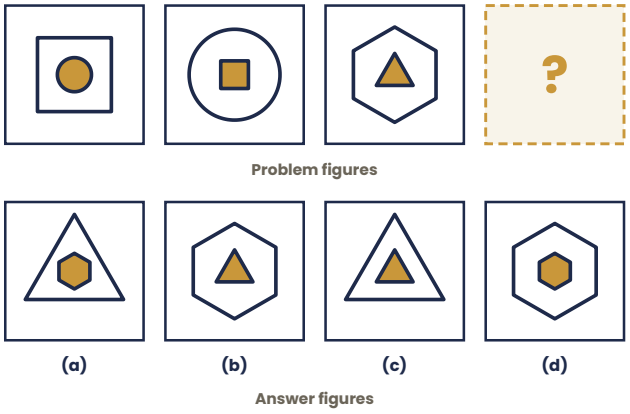
FROM FIGURE 1 TO FIGURE 2 THE NUMBER OF SIDES INCREASES BY ONE

4.2.4 Shading swaps



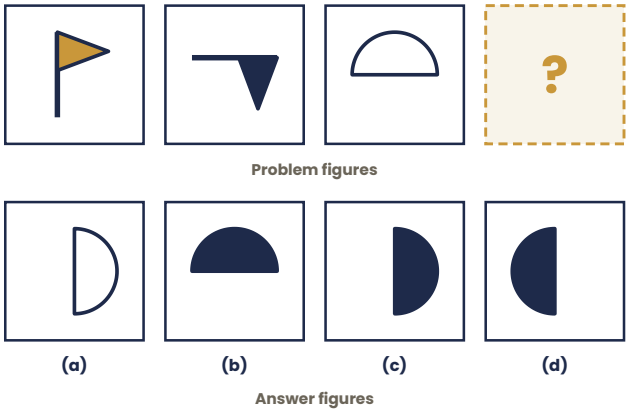
THE OUTER AND INNER SHADINGS SWAP (WHITE ↔ BLACK)

4.2.5 Interchange of inner and outer shapes



THE INNER AND OUTER SHAPES INTERCHANGE

4.2.6 Combination of two changes



TWO CHANGES HAPPEN TOGETHER: A 90° CLOCKWISE TURN AND WHITE BECOMING BLACK

4.3 Method

4.3.1 Write down every change from figure 1 to figure 2

Use the element checklist: number, shape, position, rotation, shading, size. Write each change in two or three words (“turns 90° cw”, “inner black”).

4.3.2 Apply all changes to figure 3

Predict the answer before looking at the options. If figure 3 has a different shape, apply the change to that shape, not to the shape of figure 1.

4.3.3 Eliminate options that fail one change

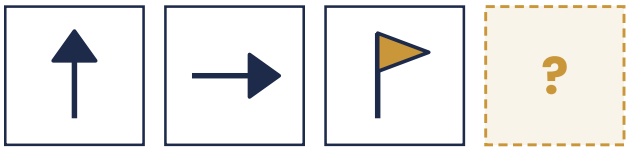
SHORTCUT TRICK

Wrong options are usually built by applying only one of two changes, or by applying a reflection instead of a rotation. Check each change separately and cross out options as soon as one fails.

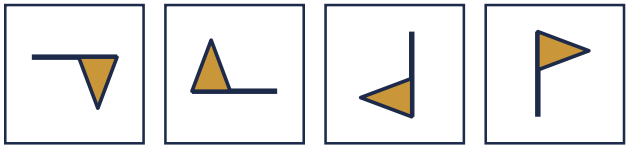
4.4 Practice questions

4.4.1 Basic level

Q1 Figure 1 is to figure 2 as figure 3 is to which answer figure?



Problem figures



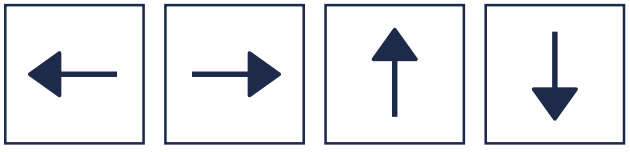
(a) (b) (c) (d)

Answer figures

Q2 Figure 1 is to figure 2 as figure 3 is to which answer figure?



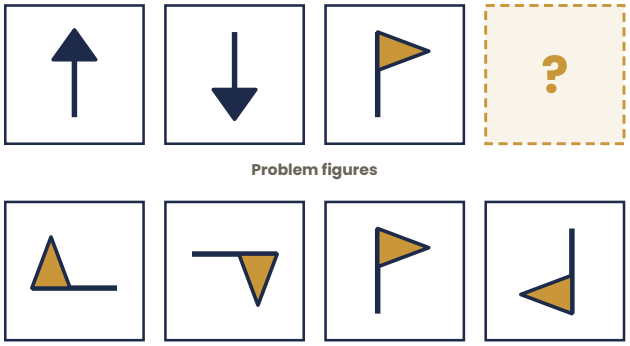
Problem figures



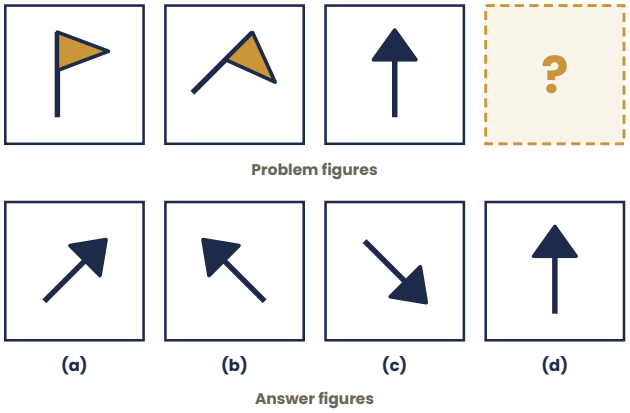
(a) (b) (c) (d)

Answer figures

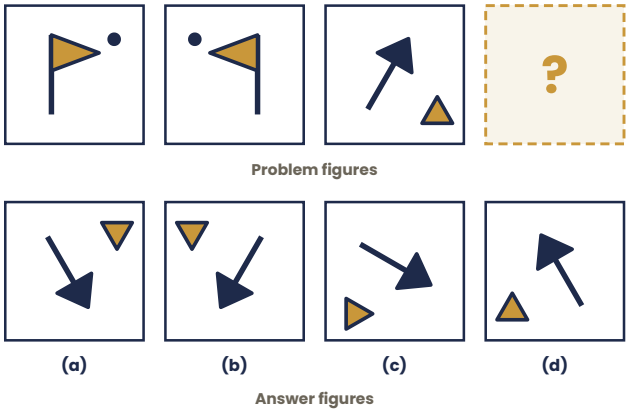
Q3 Figure 1 is to figure 2 as figure 3 is to which answer figure?



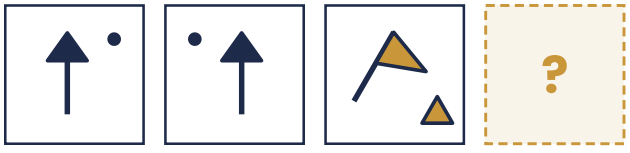
Q4 Figure 1 is to figure 2 as figure 3 is to which answer figure?



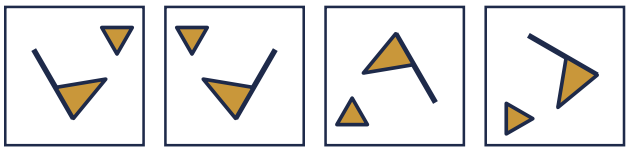
Q5 Figure 1 is to figure 2 as figure 3 is to which answer figure?



Q6 Figure 1 is to figure 2 as figure 3 is to which answer figure?



Problem figures



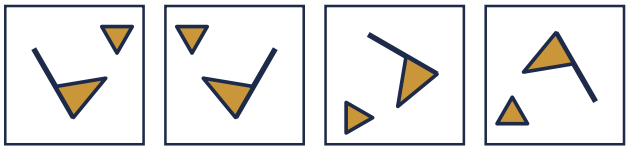
(a) (b) (c) (d)

Answer figures

Q7 Figure 1 is to figure 2 as figure 3 is to which answer figure?



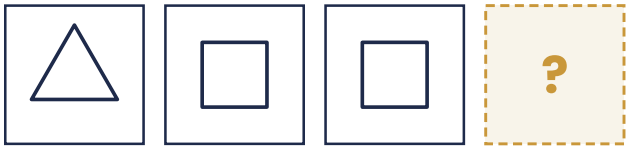
Problem figures



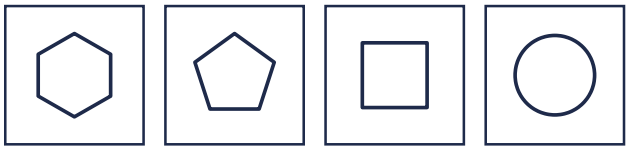
(a) (b) (c) (d)

Answer figures

Q8 Figure 1 is to figure 2 as figure 3 is to which answer figure?



Problem figures

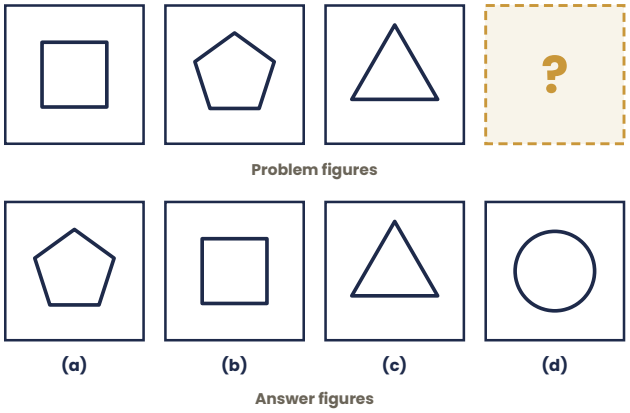


(a) (b) (c) (d)

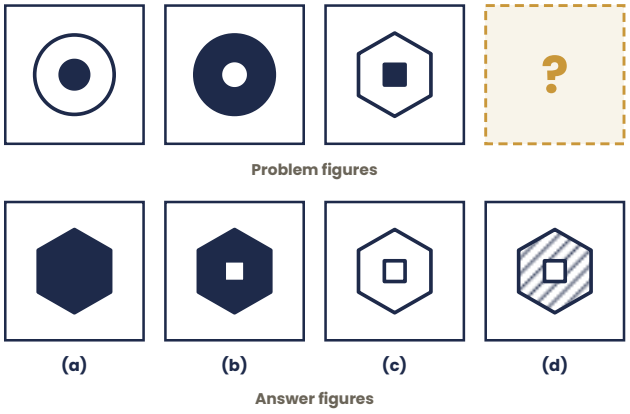
Answer figures

4.4.2 Intermediate level

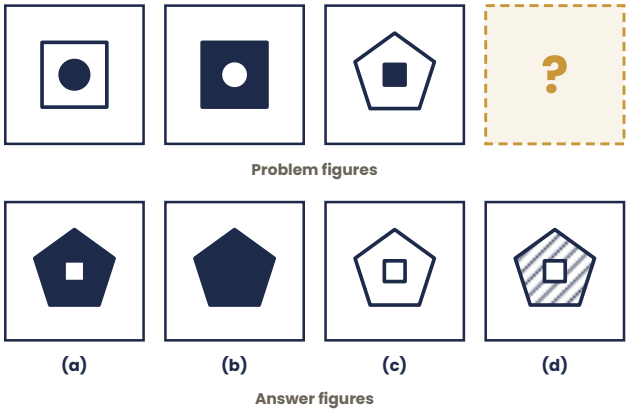
Q9 Figure 1 is to figure 2 as figure 3 is to which answer figure?



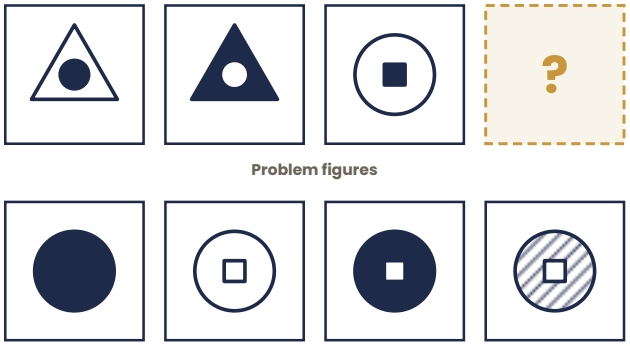
Q10 Figure 1 is to figure 2 as figure 3 is to which answer figure?



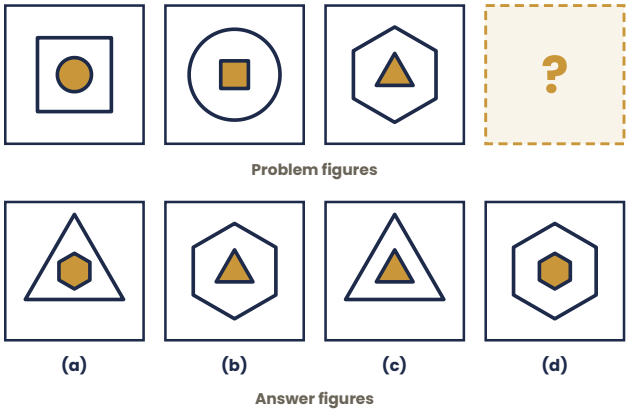
Q11 Figure 1 is to figure 2 as figure 3 is to which answer figure?



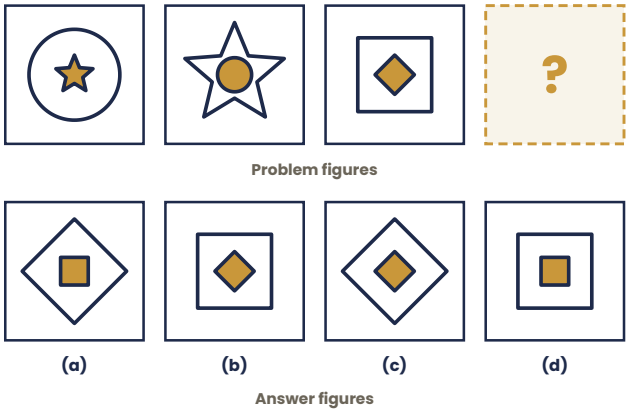
Q12 Figure 1 is to figure 2 as figure 3 is to which answer figure?



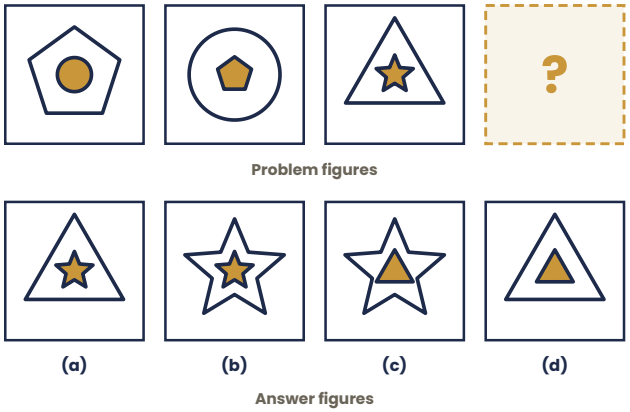
Q13 Figure 1 is to figure 2 as figure 3 is to which answer figure?



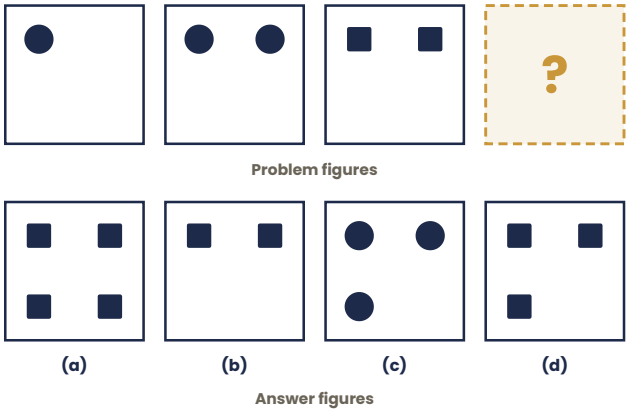
Q14 Figure 1 is to figure 2 as figure 3 is to which answer figure?



Q15 Figure 1 is to figure 2 as figure 3 is to which answer figure?

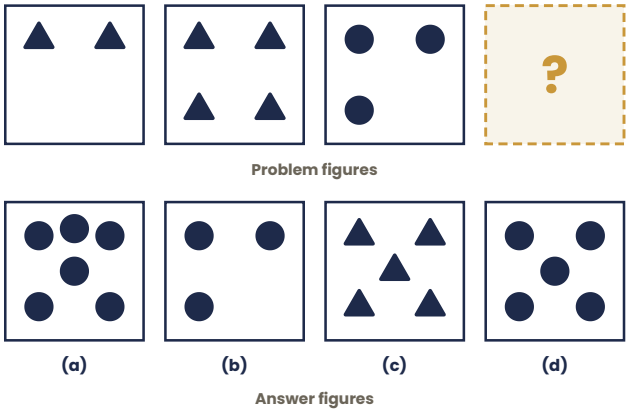


Q16 Figure 1 is to figure 2 as figure 3 is to which answer figure?

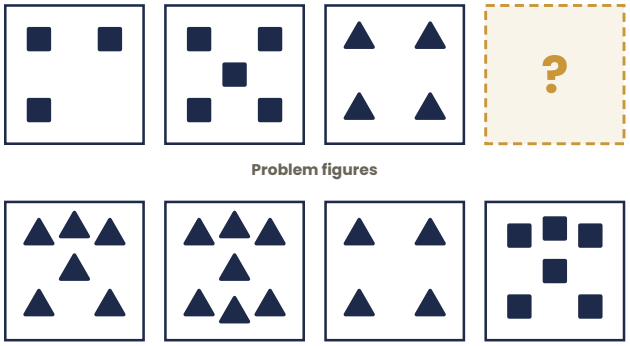


4.4.3 Advanced level

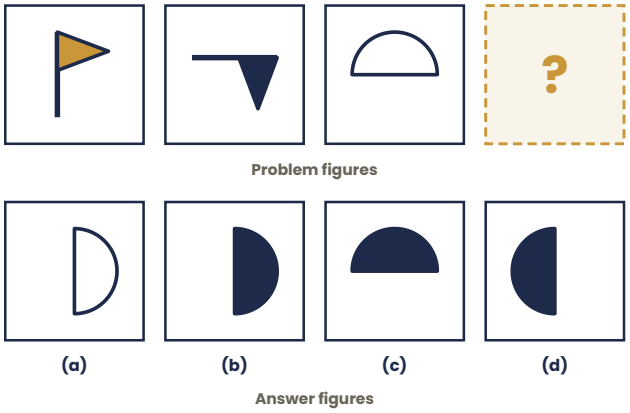
Q17 Figure 1 is to figure 2 as figure 3 is to which answer figure?



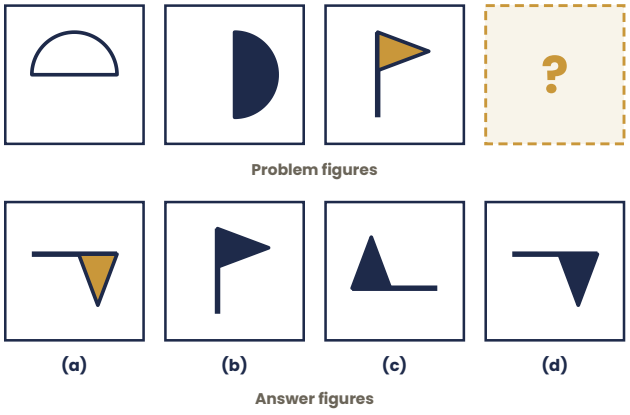
Q18 Figure 1 is to figure 2 as figure 3 is to which answer figure?



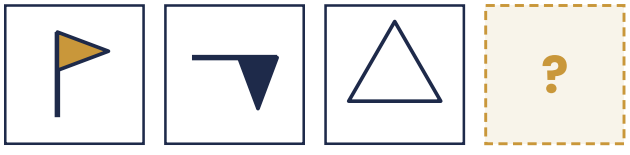
Q19 Figure 1 is to figure 2 as figure 3 is to which answer figure?



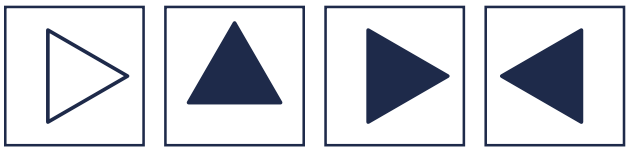
Q20 Figure 1 is to figure 2 as figure 3 is to which answer figure?



Q21 Figure 1 is to figure 2 as figure 3 is to which answer figure?



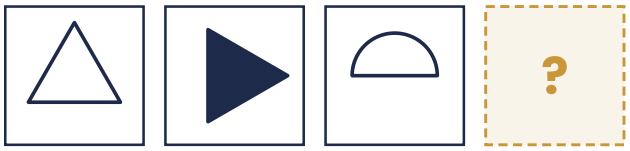
Problem figures



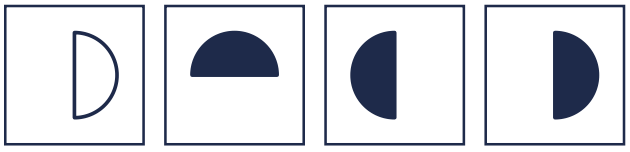
(a) (b) (c) (d)

Answer figures

Q22 Figure 1 is to figure 2 as figure 3 is to which answer figure?



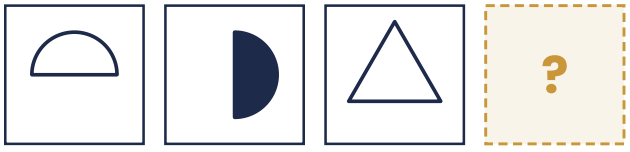
Problem figures



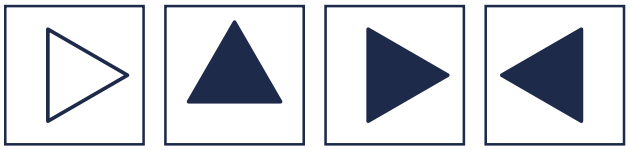
(a) (b) (c) (d)

Answer figures

Q23 Figure 1 is to figure 2 as figure 3 is to which answer figure?



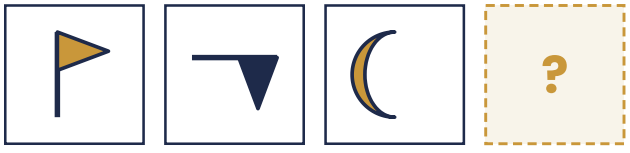
Problem figures



(a) (b) (c) (d)

Answer figures

Q24 Figure 1 is to figure 2 as figure 3 is to which answer figure?



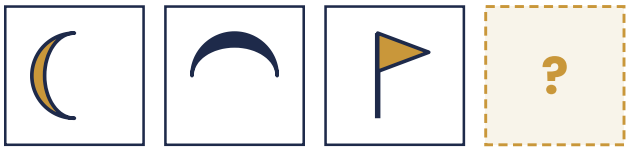
Problem figures



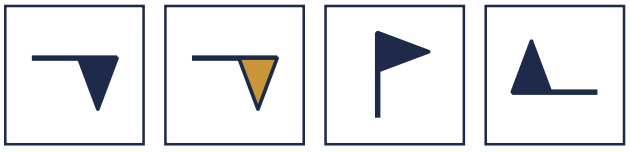
(a) (b) (c) (d)

Answer figures

Q25 Figure 1 is to figure 2 as figure 3 is to which answer figure?



Problem figures



(a) (b) (c) (d)

Answer figures

4.5 Solutions

Q1 **Answer: (a)**

Figure 2 is figure 1 turned 90° clockwise. Turning figure 3 the same way gives **(a)**.

Q2 **Answer: (a)**

Figure 2 is figure 1 turned 90° anticlockwise. Turning figure 3 the same way gives **(a)**.

Q3 **Answer: (d)**

Figure 2 is figure 1 turned 180° clockwise. Turning figure 3 the same way gives **(d)**.

Q4 **Answer: (a)**

Figure 2 is figure 1 turned 45° clockwise. Turning figure 3 the same way gives **(a)**.

Q5 **Answer: (d)**

Figure 2 is the left–right mirror image of figure 1. The mirror image of figure 3 is **(d)**.

Q6 **Answer: (c)**

Figure 2 is the left–right mirror image of figure 1. The mirror image of figure 3 is **(c)**.

Q7 **Answer: (d)**

Figure 2 is the left–right mirror image of figure 1. The mirror image of figure 3 is **(d)**.

Q8 **Answer: (b)**

From figure 1 to figure 2 the number of sides increases by one. Doing the same to figure 3 gives **(b)**.

Q9 **Answer: (b)**

From figure 1 to figure 2 the number of sides increases by one. Doing the same to figure 3 gives **(b)**.

Q10 **Answer: (b)**

The outer and inner shadings swap (white ↔ black). Swapping them in figure 3 gives **(b)**.

Q11 **Answer: (a)**

The outer and inner shadings swap (white ↔ black). Swapping them in figure 3 gives **(a)**.

Q12 **Answer: (c)**

The outer and inner shadings swap (white ↔ black). Swapping them in figure 3 gives **(c)**.

Q13 **Answer: (a)**

The inner and outer shapes interchange. Interchanging them in figure 3 gives **(a)**.

Q14 **Answer: (a)**

The inner and outer shapes interchange. Interchanging them in figure 3 gives **(a)**.

Q15 **Answer: (c)**

The inner and outer shapes interchange. Interchanging them in figure 3 gives **(c)**.

Q16 **Answer: (d)**

Figure 2 has 1 more element than figure 1, of the same kind. Adding 1 to figure 3 (keeping its shape) gives **(d)**.

Q17 **Answer: (d)**

Figure 2 has 2 more elements than figure 1, of the same kind. Adding 2 to figure 3 (keeping its shape) gives **(d)**.

Q18 **Answer: (a)**

Figure 2 has 2 more elements than figure 1, of the same kind. Adding 2 to figure 3 (keeping its shape) gives **(a)**.

Q19 **Answer: (b)**

Two changes happen together: a 90° clockwise turn and white becoming black. Only **(b)** shows both changes applied to figure 3.

Q20 **Answer: (d)**

Two changes happen together: a 90° clockwise turn and white becoming black. Only **(d)** shows both changes applied to figure 3.

Q21 **Answer: (c)**

Two changes happen together: a 90° clockwise turn and white becoming black. Only **(c)** shows both changes applied to figure 3.

Q22 **Answer: (d)**

Two changes happen together: a 90° clockwise turn and white becoming black. Only **(d)** shows both changes applied to figure 3.

Q23 **Answer: (c)**

Two changes happen together: a 90° clockwise turn and white becoming black. Only **(c)** shows both changes applied to figure 3.

Q24 **Answer: (d)**

Two changes happen together: a 90° clockwise turn and white becoming black. Only **(d)** shows both changes applied to figure 3.

Q25 **Answer: (a)**

Two changes happen together: a 90° clockwise turn and white becoming black. Only **(a)** shows both changes applied to figure 3.

List every change from figure 1 to figure 2 with the checklist, apply all of them (and nothing else) to figure 3, and predict the answer before reading the options. Watch for options that apply only one of two changes or use a reflection in place of a rotation.

MODULE 05 · BEGINNER

Figure classification (odd one out)

Minimum duration **1 day** | Level **Beginner**

Classification questions ask you to find the figure that does not follow the common rule. This module covers the usual grouping rules (same number of sides, lines or elements; same rotation or direction; symmetry; open vs closed; position of a small element; rotated copies vs a mirrored one) and the compare-then-check method with its rotation-vs-reflection trap, followed by 25 solved questions generated from their rules.

IN THIS MODULE

5.1 Common grouping rules
5.2 Method
5.3 Practice questions
5.4 Solutions

5.1 Common grouping rules

5.1.1 Same number of sides, lines or elements



FOUR FIGURES HAVE 4 SIDES; FIGURE (C) DOES NOT.

5.1.2 Same rotation or direction



IN FOUR FIGURES THE ARROW POINTS TO THE UPPER RIGHT; IN FIGURE (D) IT POINTS TO THE UPPER LEFT.

5.1.3 Symmetry

Symmetric figures (with at least one line of symmetry) may be grouped against one asymmetric figure. Imagine folding each figure down the middle: the odd one does not match itself.

5.1.4 Open vs closed figures



FOUR FIGURES ARE CLOSED SHAPES; FIGURE (E) IS OPEN.

5.1.5 Position of a small element



IN FOUR FIGURES THE DOT IS INSIDE THE LARGE SHAPE; IN FIGURE (B) IT IS OUTSIDE.

5.1.6 Rotated copies vs a mirrored one



FOUR FIGURES ARE THE SAME FIGURE TURNED THROUGH DIFFERENT ANGLES. FIGURE (C) IS ITS MIRROR IMAGE, WHICH NO TURN CAN PRODUCE.

5.2 Method

5.2.1 Compare the first figures to find the rule

Take three or four figures and ask what they share, using the checklist. Then check the remaining figure(s). Exactly one figure must fail the rule.

5.2.2 Rotation vs reflection trap

WATCH OUT
When all figures look like the same shape in different orientations, the odd one is usually the mirror image. Track one asymmetric detail (a dot, a flag) clockwise around each figure to detect it.

SHORTCUT TRICK
If you find two figures that fail your rule, your rule is wrong; if none fails, it is too weak. Sharpen it until exactly one fails.

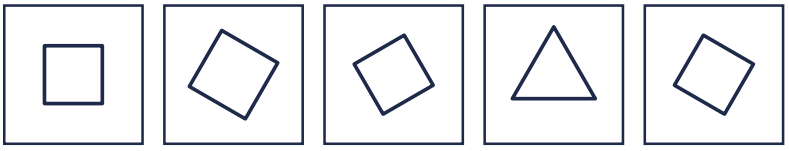
5.3 Practice questions

5.3.1 Basic level

Q1 Which figure is different from the others?



Q2 Which figure is different from the others?



Q3 Which figure is different from the others?



Q4 Which figure is different from the others?



Q5 Which figure is different from the others?



Q6 Which figure is different from the others?



Q7 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q8 Which figure is different from the others?



(a) (b) (c) (d) (e)

5.3.2 Intermediate level

Q9 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q10 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q11 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q12 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q13 Which figure is different from the others?



Q14 Which figure is different from the others?



Q15 Which figure is different from the others?



Q16 Which figure is different from the others?



5.3.3 Advanced level

Q17 Which figure is different from the others?



Q18 Which figure is different from the others?



Q19 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q20 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q21 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q22 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q23 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q24 Which figure is different from the others?



(a) (b) (c) (d) (e)

Q25 Which figure is different from the others?



(a) (b) (c) (d) (e)

5.4 Solutions

Q1 **Answer: (b)**
Four figures have 3 sides; figure **(b)** does not.

Q2 **Answer: (d)**
Four figures have 4 sides; figure **(d)** does not.

Q3 **Answer: (a)**
Four figures have 5 sides; figure **(a)** does not.

Q4 **Answer: (e)**
Four figures have 6 sides; figure **(e)** does not.

Q5 **Answer: (c)**
Four figures contain 2 elements; figure **(c)** contains 3.

Q6 **Answer: (a)**
Four figures contain 3 elements; figure **(a)** contains 4.

Q7 **Answer: (e)**
Four figures contain 4 elements; figure **(e)** contains 5.

Q8 **Answer: (a)**
In four figures the arrow points to the upper right; in figure **(a)** it points to the upper left.

Q9 **Answer: (c)**
In four figures the arrow points to the upper right; in figure **(c)** it points to the upper left.

Q10 **Answer: (e)**
In four figures the arrow points to the upper right; in figure **(e)** it points to the upper left.

Q11 **Answer: (d)**
In four figures the dot is inside the large shape; in figure **(d)** it is outside.

Q12 **Answer: (a)**

In four figures the dot is inside the large shape; in figure **(a)** it is outside.

Q13 **Answer: (c)**

In four figures the dot is inside the large shape; in figure **(c)** it is outside.

Q14 **Answer: (c)**

In four figures the circle is black and the square white; in figure **(c)** the colours are reversed.

Q15 **Answer: (e)**

In four figures the circle is black and the square white; in figure **(e)** the colours are reversed.

Q16 **Answer: (a)**

Four figures are closed shapes; figure **(a)** is open.

Q17 **Answer: (b)**

Four figures are the same figure turned through different angles. Figure **(b)** is its mirror image, which no turn can produce.

Q18 **Answer: (d)**

Four figures are the same figure turned through different angles. Figure **(d)** is its mirror image, which no turn can produce.

Q19 **Answer: (a)**

Four figures are the same figure turned through different angles. Figure **(a)** is its mirror image, which no turn can produce.

Q20 **Answer: (c)**

Four figures are the same figure turned through different angles. Figure **(c)** is its mirror image, which no turn can produce.

Q21 **Answer: (b)**

Four figures are closed shapes; figure **(b)** is open.

Q22 **Answer: (d)**

Four figures are closed shapes; figure **(d)** is open.

Q23 **Answer: (d)**

Four figures are the same figure turned through different angles. Figure **(d)** is its mirror image, which no turn can produce.

Q24 **Answer: (e)**

Four figures are the same figure turned through different angles. Figure **(e)** is its mirror image, which no turn can produce.

Q25 **Answer: (a)**

Four figures are the same figure turned through different angles. Figure **(a)** is its mirror image, which no turn can produce.

Find the property shared by most figures (sides, count, direction, symmetry, open or closed, position, shading) and confirm that exactly one figure fails it. When all figures look like turned copies, look for the mirror image.

MODULE 06 · BEGINNER

Mirror images

Minimum duration **1 day** | Level **Beginner**

Mirror-image questions test reflecting figures, letters and numbers across a vertical or horizontal mirror. This module covers vertical and horizontal mirrors and mirror positions, symmetric capital letters, small letters and words, digits and number strings, clock mirror times, complex figures, and the keep-the-near-side-fixed method, followed by 25 solved questions whose answer figures were produced by actually reflecting the problem figure.

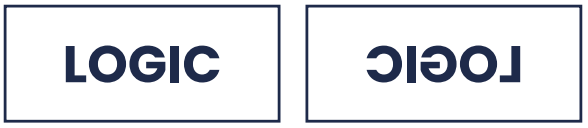
IN THIS MODULE

6.1 Concepts
6.2 Mirror images of letters
6.3 Numbers and clocks
6.4 Complex figures
6.5 Method
6.6 Practice questions
6.7 Solutions

6.1 Concepts

6.1.1 Vertical mirror

A mirror placed vertically (to the left or right of the figure) swaps **left and right**; top and bottom stay the same. Every letter is reversed and the order of letters is reversed.



A WORD AND ITS MIRROR IMAGE

6.1.2 Horizontal mirror

A mirror placed horizontally (above or below the figure) swaps **top and bottom**; left and right stay the same. This is the same as a water image (Module 7).

6.1.3 Mirror on the right, left, above or below

For a vertical mirror it does not matter whether it is on the left or the right: the image is the same shape, only its position differs. The same holds for above and below.

6.2 Mirror images of letters

6.2.1 Symmetric capital letters

Letters with a vertical line of symmetry look the same in a vertical mirror: **A, H, I, M, O, T, U, V, W, X, Y**. A word made only of these letters (such as TOMATO) is simply written in reverse order in the mirror.



MATH CONTAINS ONLY SYMMETRIC LETTERS, SO ITS MIRROR IMAGE READS HTAM WITH UNCHANGED LETTERS

6.2.2 Small letters and words

Small letters are rarely symmetric: b ↔ d and p ↔ q swap in a vertical mirror. Reverse every letter and their order.

6.3 Numbers and clocks

6.3.1 Digits and number strings

Only 0, 1 and 8 look the same in a vertical mirror; the others reverse. The order of digits also reverses.

6.3.2 Clock mirror time

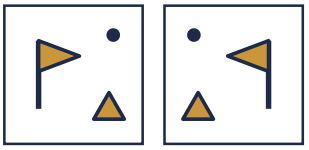
Mirror time = 11:60 − actual time (for times between 1:00 and 11:00; for 12:xx use 23:60 − time)

A clock showing 4:20 appears as 7:40 in a mirror. The rule also works backwards.

6.4 Complex figures

6.4.1 Several elements

Reflect every element: its position moves to the opposite side and its own shape flips. Elements on the mirror line stay where they are.



EVERY ELEMENT MOVES AND FLIPS

6.5 Method

6.5.1 Keep the side next to the mirror fixed

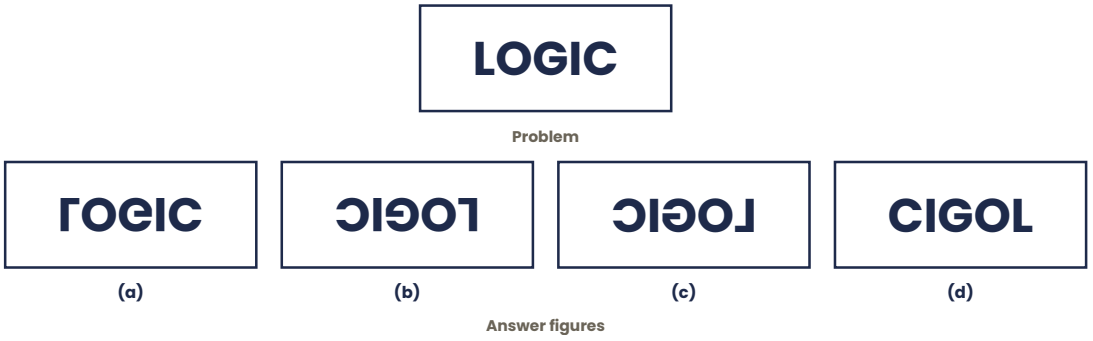
The part of the figure nearest the mirror stays nearest the mirror in the image; the far side stays far. Work across the figure from the mirror side, reversing each element.

SHORTCUT TRICK
Pick one asymmetric element and check it alone: its direction must be reversed left–right (vertical mirror). This usually eliminates all but one option.

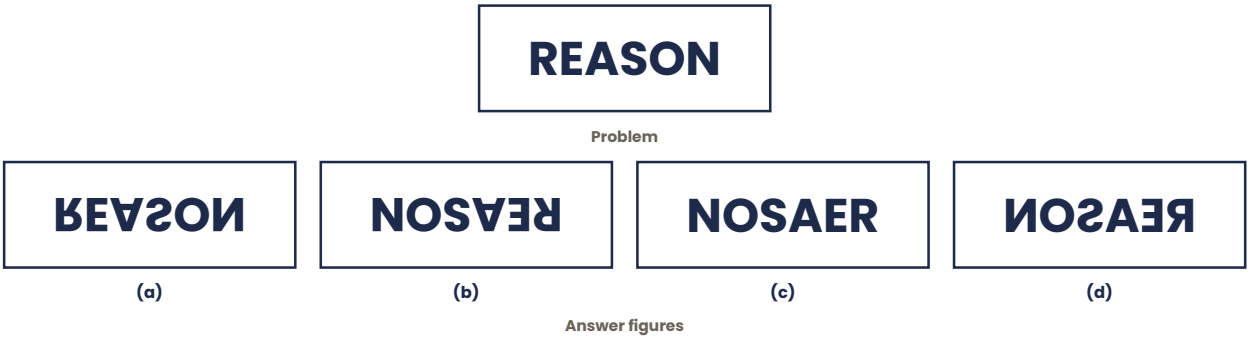
6.6 Practice questions

6.6.1 Basic level

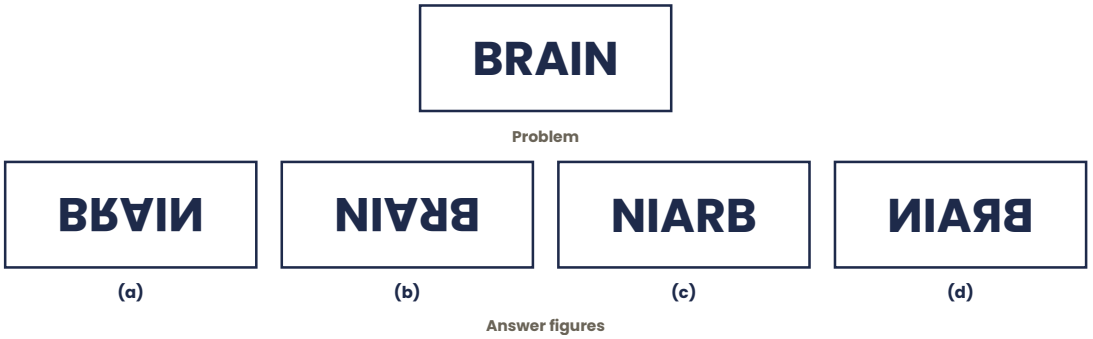
Q1 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



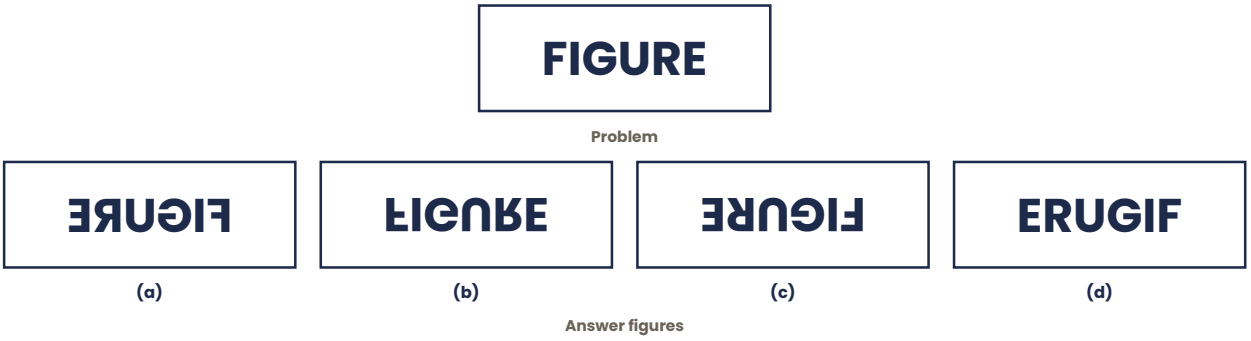
Q2 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



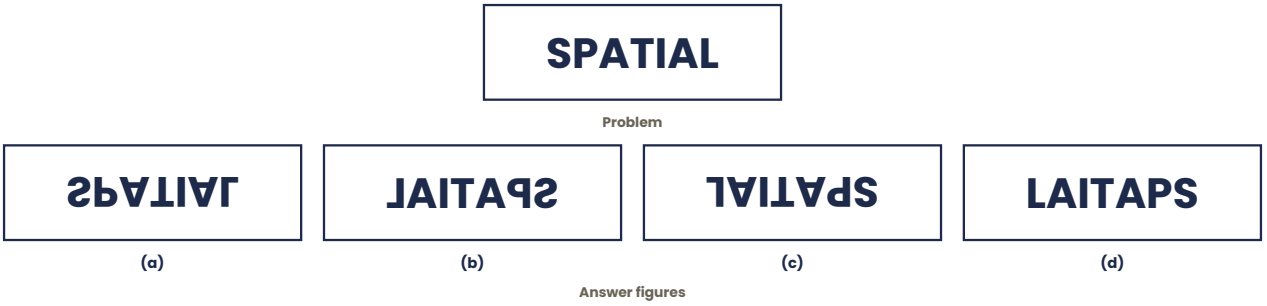
Q3 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



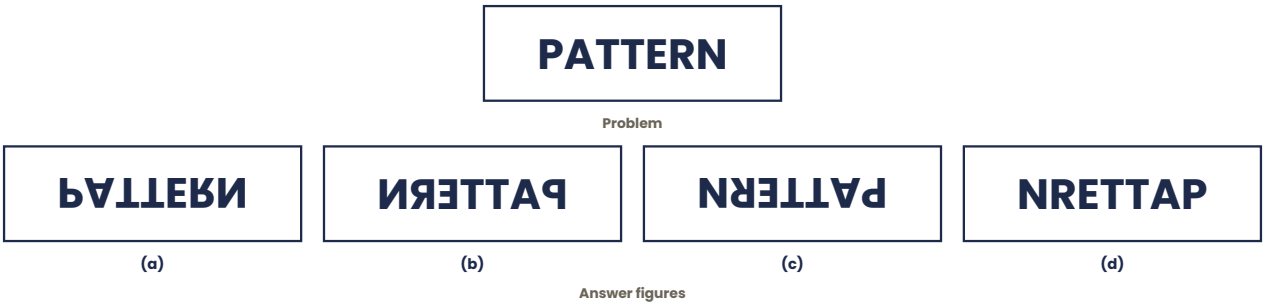
Q4 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



Q5 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



Q6 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



Q7 A clock shows 4:20. What time does its mirror image show?

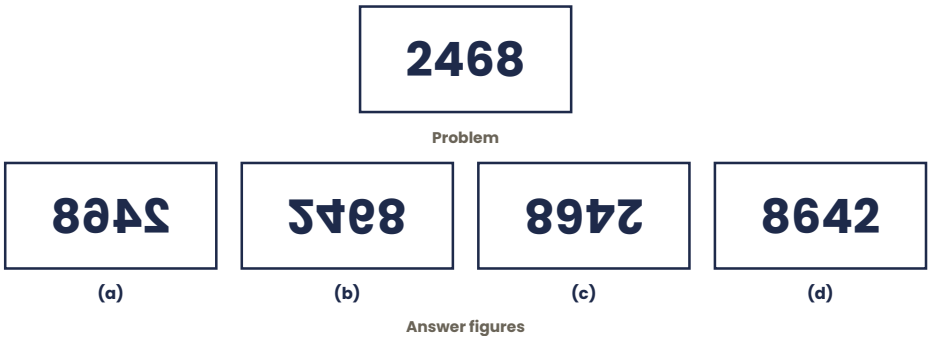
(a) 5:40 **(b)** 7:20 **(c)** 4:20 **(d)** 7:40

Q8 A clock shows 2:15. What time does its mirror image show?

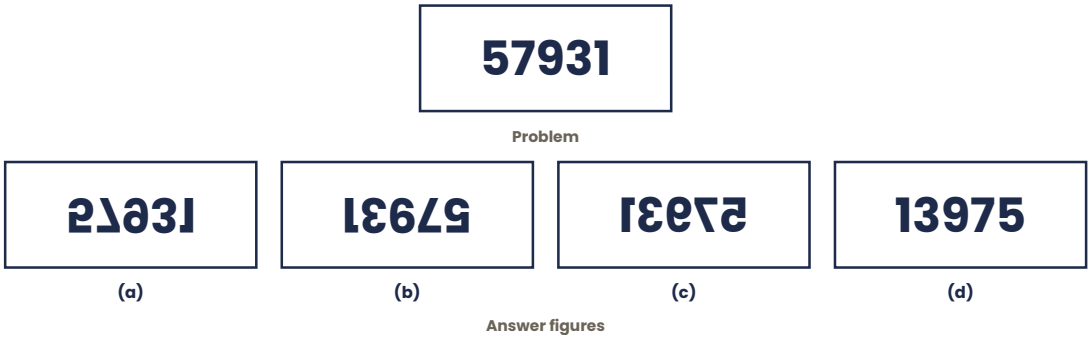
(a) 3:45 **(b)** 9:15 **(c)** 9:45 **(d)** 2:15

6.6.2 Intermediate level

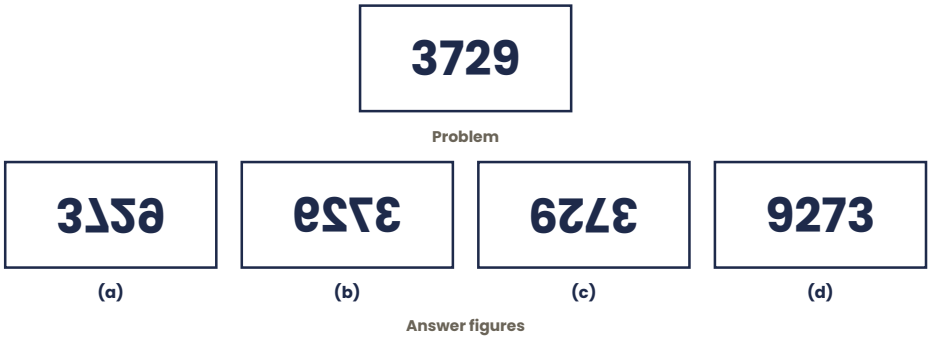
Q9 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



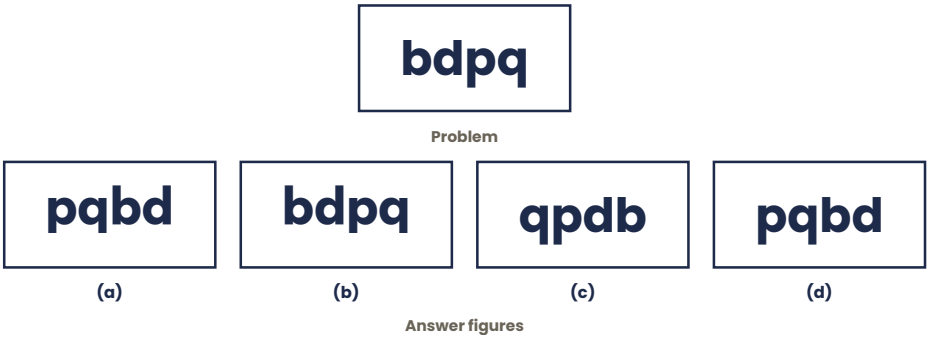
Q10 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



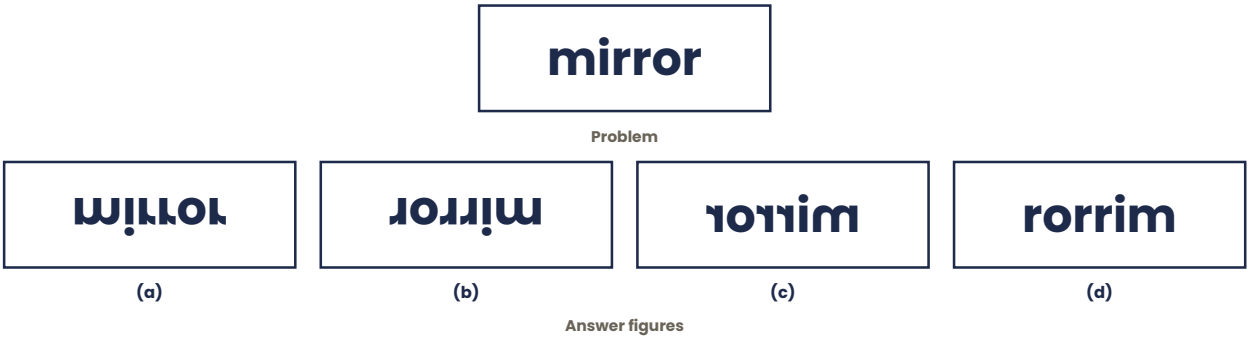
Q11 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



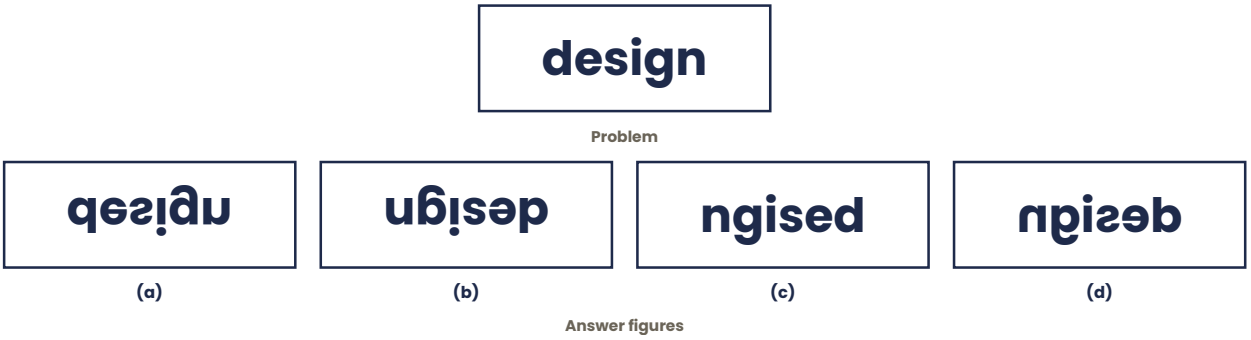
Q12 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



Q13 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



Q14 Which answer figure is the mirror image (mirror placed vertically on the right) of the problem?



Q15 A clock shows 10:10. What time does its mirror image show?

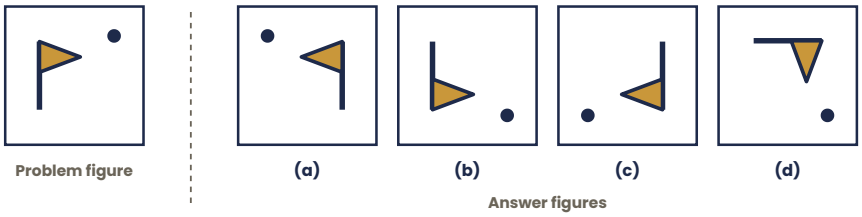
(a) 1:50 **(b)** 11:50 **(c)** 1:10 **(d)** 10:10

Q16 A clock shows 8:25. What time does its mirror image show?

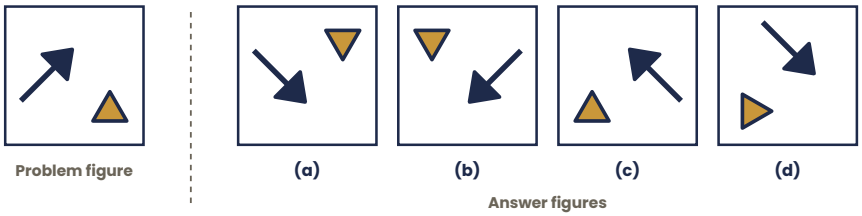
(a) 9:35 **(b)** 3:35 **(c)** 3:25 **(d)** 8:25

6.6.3 Advanced level

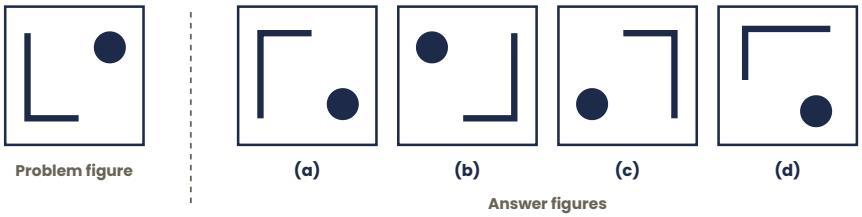
Q17 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?



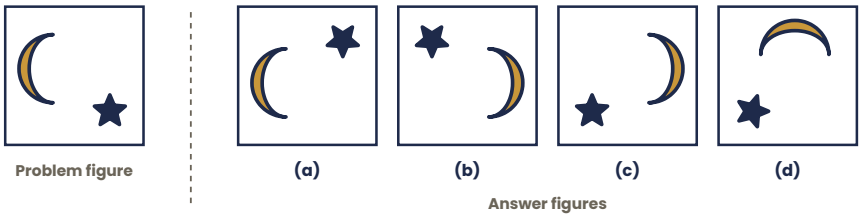
Q18 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?



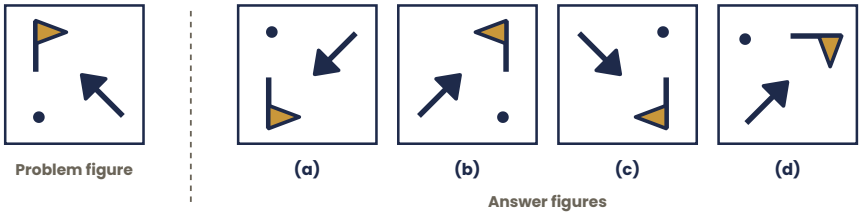
Q19 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?



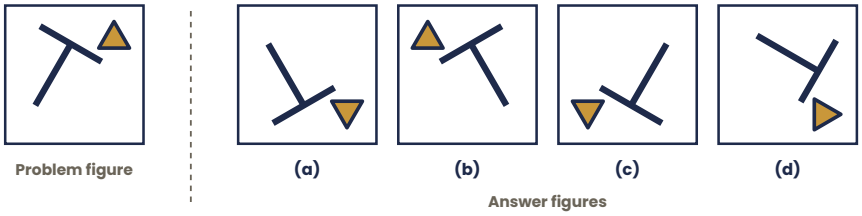
Q20 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?



Q21 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?



Q22 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?



Q23 A clock shows 5:55. What time does its mirror image show?

(a) 6:55 **(b)** 5:55 **(c)** 7:05 **(d)** 6:05

Q24 A clock shows 3:40. What time does its mirror image show?

(a) 4:20 **(b)** 8:40 **(c)** 8:20 **(d)** 3:40

Q25 A clock shows 11:30. What time does its mirror image show?

(a) 11:30 **(b)** 12:30 **(c)** 1:30 **(d)** 1:30

6.7 Solutions

Q1 **Answer: (c)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(c)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q2 **Answer: (d)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(d)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q3 **Answer: (d)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(d)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q4 **Answer: (a)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(a)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q5 **Answer: (b)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(b)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q6 **Answer: (b)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(b)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q7 **Answer: (d) 7:40**

11:60 - 4:20 = **7:40**.

Q8 **Answer: (c) 9:45**

11:60 - 2:15 = **9:45**.

Q9 **Answer: (a)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(a)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q10 **Answer: (c)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(c)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q11 **Answer: (b)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(b)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q12 **Answer: (d)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(d)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q13 **Answer: (c)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(c)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q14 **Answer: (d)**

A vertical mirror reverses left and right and reverses every letter, but keeps top and bottom: answer **(d)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q15 **Answer: (a) 1:50**

11:60 - 10:10 = **1:50**.

Q16 **Answer: (b) 3:35**

11:60 - 8:25 = **3:35**.

Q17 **Answer: (a)**

A vertical mirror swaps left and right (every element moves to the opposite side and flips) but keeps top and bottom: answer **(a)**.

Q18 **Answer: (c)**

A vertical mirror swaps left and right (every element moves to the opposite side and flips) but keeps top and bottom: answer **(c)**.

Q19 **Answer: (b)**

A vertical mirror swaps left and right (every element moves to the opposite side and flips) but keeps top and bottom: answer **(b)**.

Q20 **Answer: (c)**

A vertical mirror swaps left and right (every element moves to the opposite side and flips) but keeps top and bottom: answer **(c)**.

Q21 **Answer: (b)**

A vertical mirror swaps left and right (every element moves to the opposite side and flips) but keeps top and bottom: answer **(b)**.

Q22 **Answer: (b)**

A vertical mirror swaps left and right (every element moves to the opposite side and flips) but keeps top and bottom: answer **(b)**.

Q23 **Answer: (d) 6:05**

11:60 - 5:55 = **6:05**.

Q24 **Answer: (c) 8:20**

11:60 - 3:40 = **8:20**.

Q25 **Answer: (b) 12:30**

11:60 - 11:30 = **12:30**.

A vertical mirror reverses left and right (letters flip and their order reverses); a horizontal mirror reverses top and bottom. Letters A, H, I, M, O, T, U, V, W, X, Y are unchanged in a vertical mirror. Mirror clock time = 11:60 - actual time. Check one asymmetric element first.

MODULE 07 · BEGINNER

Water images

Minimum duration **1 day** | Level **Beginner**

Water images are reflections across a horizontal line, as if the figure stands on water. This module covers the top-bottom reversal, how water images differ from mirror images, letters that stay unchanged, numbers and words, clock water times, and combined mirror-and-water questions, followed by 25 solved questions whose answer figures were produced by reflecting the problem figure.

IN THIS MODULE

7.1 Concepts
7.2 Letters, numbers and words
7.3 Figures and clocks
7.4 Combined mirror and water images
7.5 Practice questions
7.6 Solutions

7.1 Concepts

7.1.1 Top–bottom reversal

In a water image the figure is flipped upside down about a horizontal line: top and bottom swap, while left and right stay the same. The order of letters in a word does **not** change.



A WORD AND ITS WATER IMAGE

7.1.2 Water image vs mirror image

Image	Swaps	Keeps	Letter order
Mirror (vertical)	left and right	top and bottom	reversed
Water (horizontal)	top and bottom	left and right	unchanged
Mirror then water	both	nothing	reversed; equals a 180° turn

7.2 Letters, numbers and words

7.2.1 Letters unchanged in water images

Capital letters with a horizontal line of symmetry look the same in water: **B, C, D, E, H, I, K, O, X**. A word made only of these letters (such as CHOICE or BOX) is unchanged in its water image.



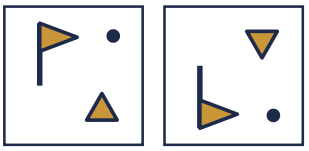
CODE CONTAINS ONLY LETTERS SYMMETRIC ABOUT A HORIZONTAL LINE

Among digits, 0, 1, 3 and 8 look the same in water (in the usual printed form); others change.

7.3 Figures and clocks

7.3.1 Water image of clock times

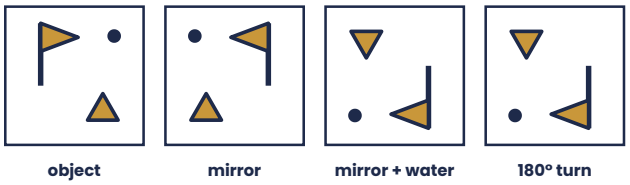
Exams use the rule **water time = 18:30 - actual time** (subtract hours from 18 and minutes from 30, borrowing 60 minutes when needed). For example 3:40 becomes 18:30 - 3:40 = 14:50, read as 2:50.



EVERY ELEMENT MOVES TO THE OPPOSITE SIDE OF THE HORIZONTAL LINE AND FLIPS

7.4 Combined mirror and water images

Taking the mirror image and then the water image (or the other way round) reverses both directions, which is the same as turning the figure through 180°.



MIRROR FOLLOWED BY WATER EQUALS A HALF TURN

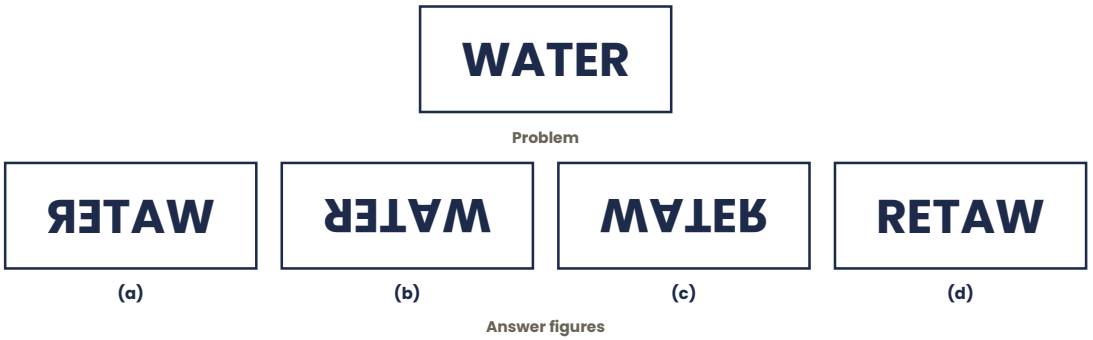
SHORTCUT TRICK

To check a water image quickly, look at the element nearest the top: in the water image it must be nearest the bottom, on the same side (left or right).

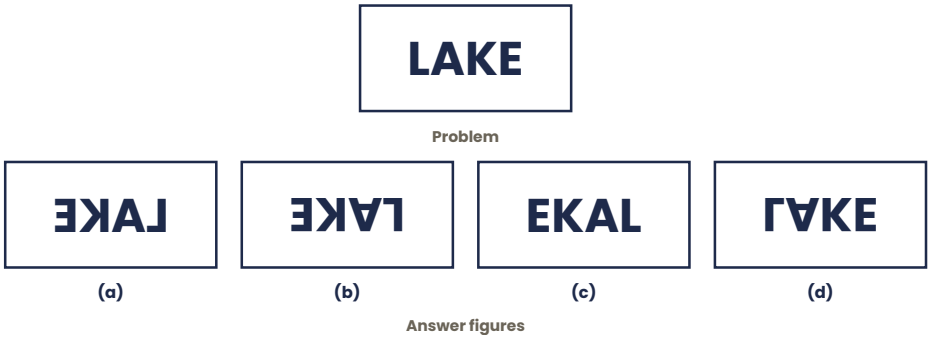
7.5 Practice questions

7.5.1 Basic level

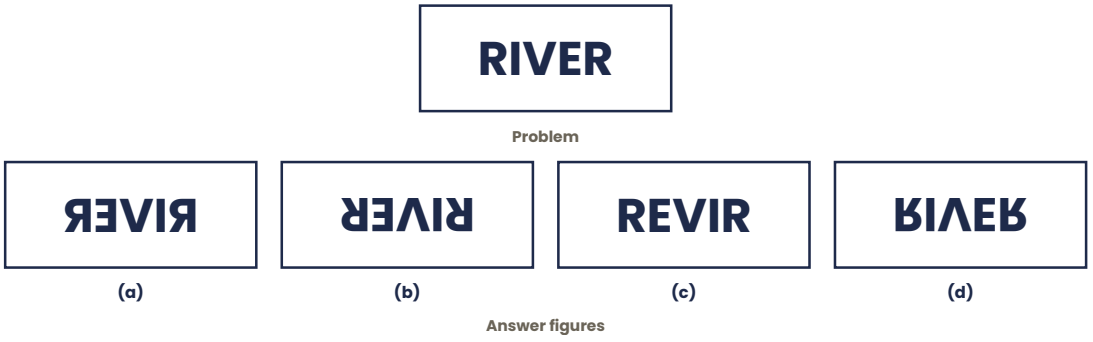
Q1 Which answer figure is the water image of the problem?



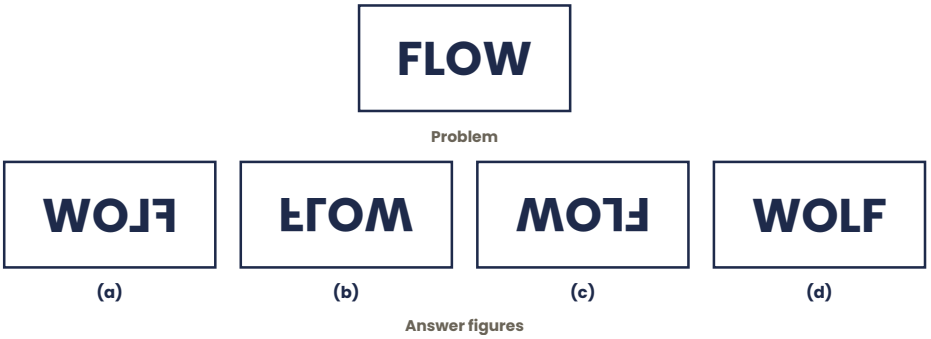
Q2 Which answer figure is the water image of the problem?



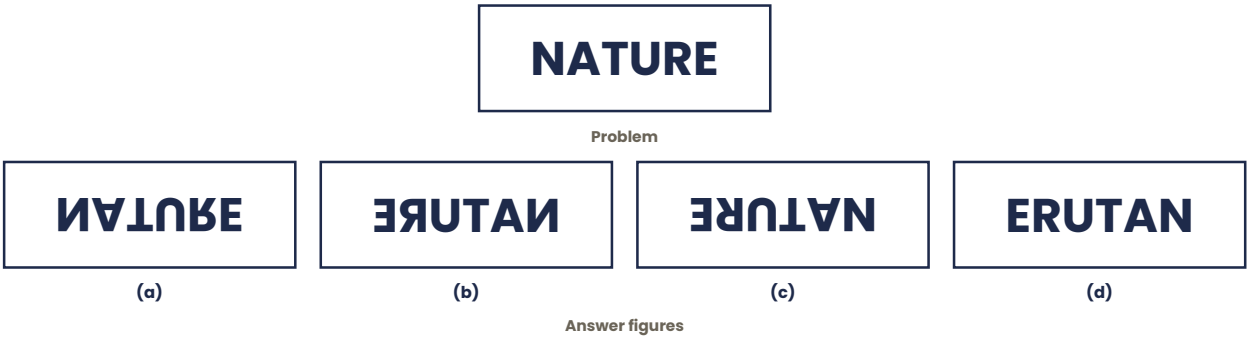
Q3 Which answer figure is the water image of the problem?



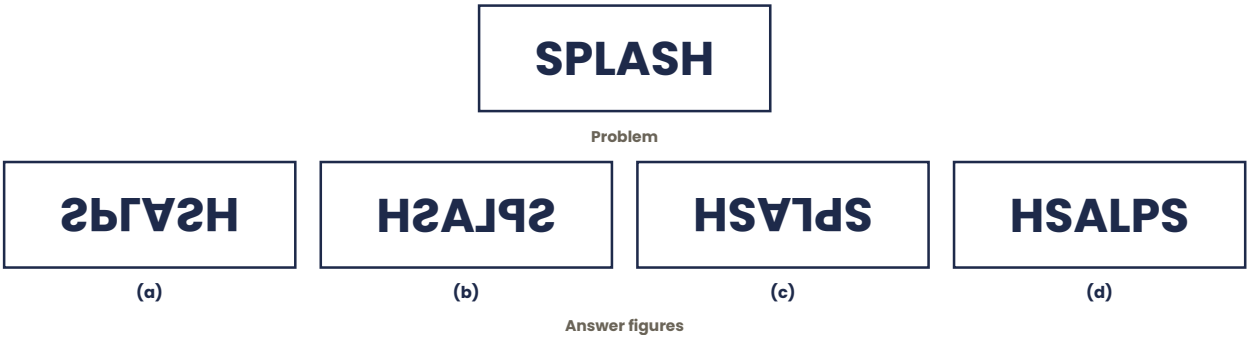
Q4 Which answer figure is the water image of the problem?



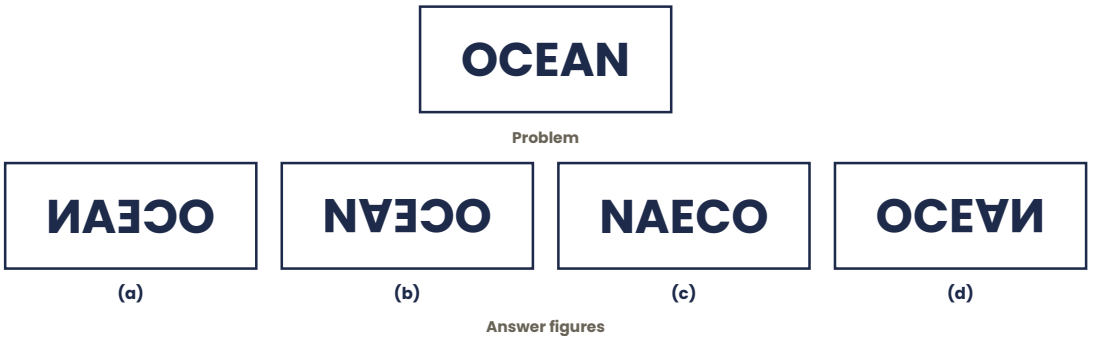
Q5 Which answer figure is the water image of the problem?



Q6 Which answer figure is the water image of the problem?



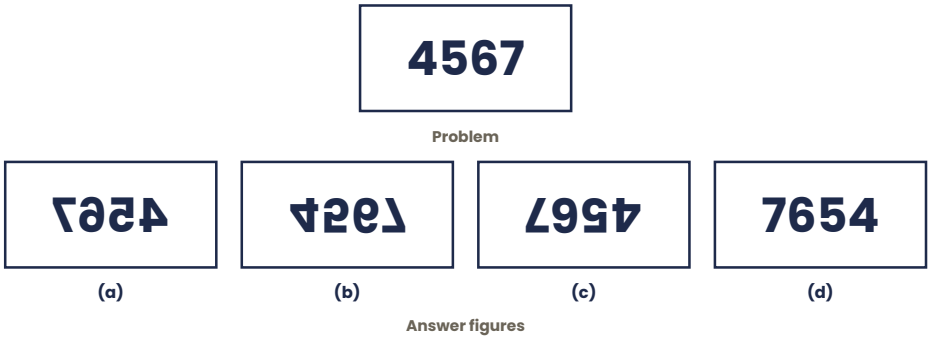
Q7 Which answer figure is the water image of the problem?



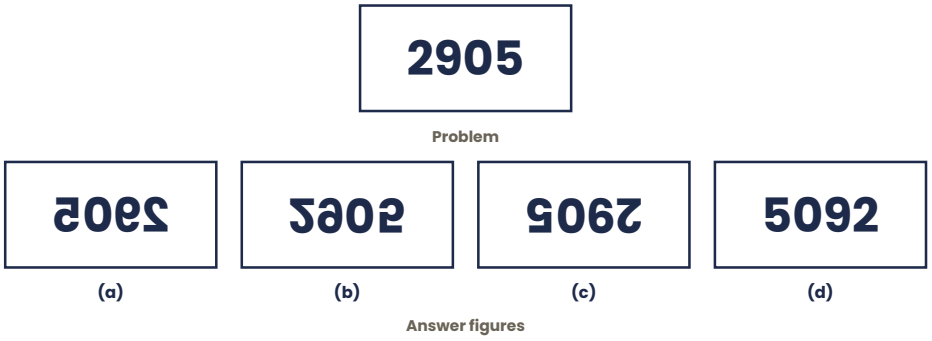
Q8 Which word looks exactly the same in its water image? Options: (a) CHOICE, (b) WATER, (c) LIGHT, (d) MIRROR.

7.5.2 Intermediate level

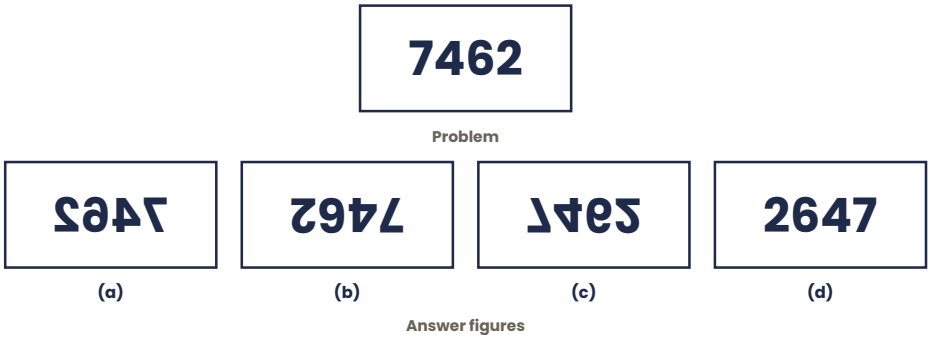
Q9 Which answer figure is the water image of the problem?



Q10 Which answer figure is the water image of the problem?



Q11 Which answer figure is the water image of the problem?



Q12 Using the exam rule (18:30 - time), what is the water image of the time 3:40?

(a) 8:20 **(b)** 3:50 **(c)** 2:50 **(d)** 3:40

Q13 Using the exam rule (18:30 - time), what is the water image of the time 2:15?

(a) 9:45 **(b)** 5:15 **(c)** 4:15 **(d)** 2:15

Q14 Using the exam rule (18:30 - time), what is the water image of the time 7:20?

(a) 4:40 **(b)** 12:10 **(c)** 7:20 **(d)** 11:10

Q15 Using the exam rule (18:30 - time), what is the water image of the time 10:05?

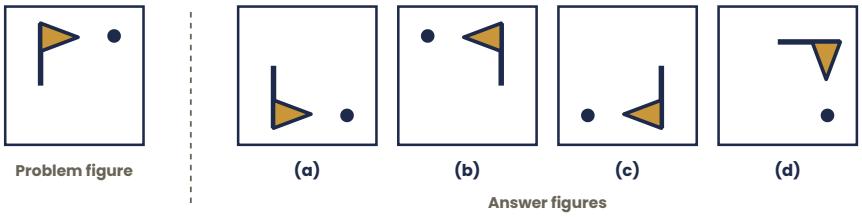
(a) 1:55 **(b)** 8:25 **(c)** 9:25 **(d)** 10:05

Q16 Using the exam rule (18:30 - time), what is the water image of the time 5:50?

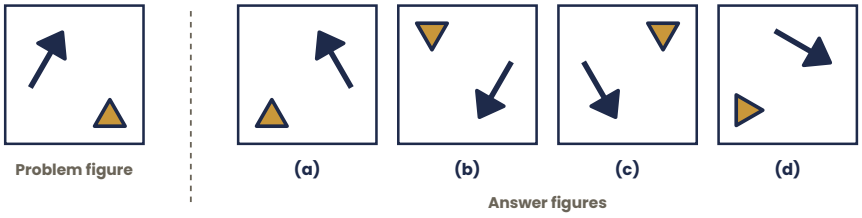
(a) 6:10 **(b)** 1:40 **(c)** 12:40 **(d)** 5:50

7.5.3 Advanced level

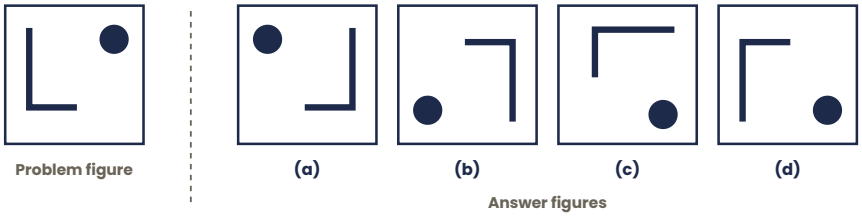
Q17 Which answer figure is the water image of the problem figure?



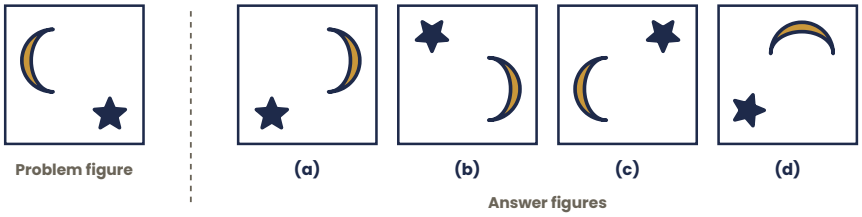
Q18 Which answer figure is the water image of the problem figure?



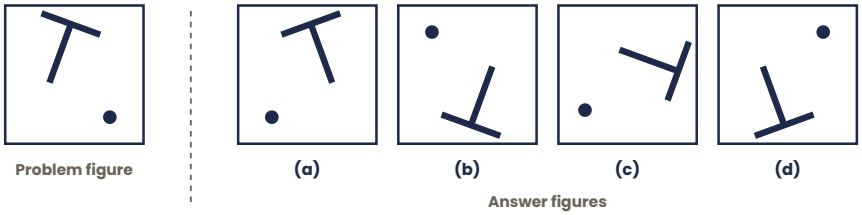
Q19 Which answer figure is the water image of the problem figure?



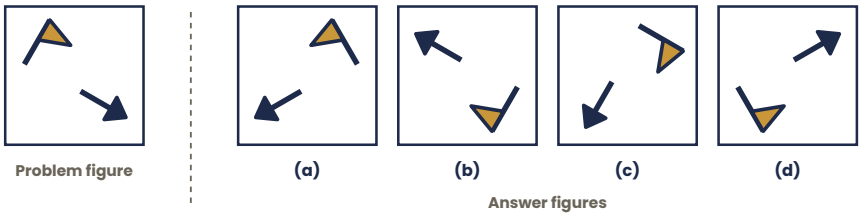
Q20 Which answer figure is the water image of the problem figure?



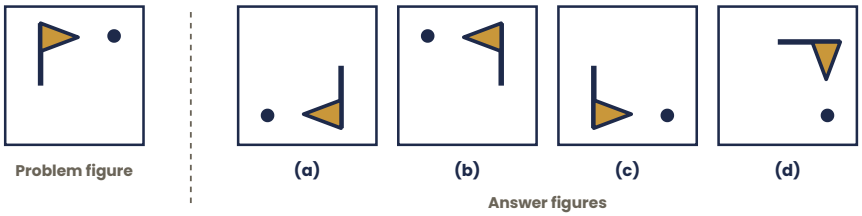
Q21 Which answer figure is the water image of the problem figure?



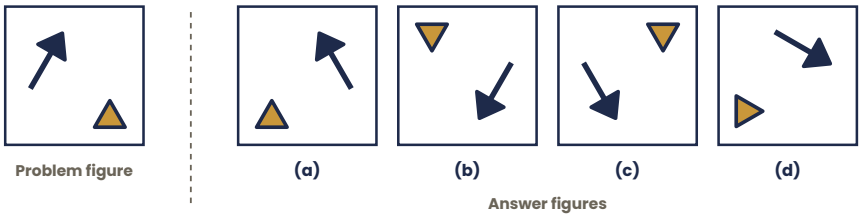
Q22 Which answer figure is the water image of the problem figure?



Q23 The mirror image (vertical mirror) of the problem figure is taken, and then the water image of that. Which answer figure results?



Q24 The mirror image (vertical mirror) of the problem figure is taken, and then the water image of that. Which answer figure results?



Q25 Which letter looks the same in both a vertical mirror and a water image? Options: (a) B, (b) A, (c) X, (d) E.

7.6 Solutions

Q1 **Answer: (c)**

A water image turns the figure upside down but keeps left and right: answer **(c)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q2 **Answer: (d)**

A water image turns the figure upside down but keeps left and right: answer **(d)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q3 **Answer: (d)**

A water image turns the figure upside down but keeps left and right: answer **(d)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q4 **Answer: (b)**

A water image turns the figure upside down but keeps left and right: answer **(b)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q5 **Answer: (a)**

A water image turns the figure upside down but keeps left and right: answer **(a)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q6 **Answer: (a)**

A water image turns the figure upside down but keeps left and right: answer **(a)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q7 **Answer: (d)**

A water image turns the figure upside down but keeps left and right: answer **(d)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q8 **Answer: (a)**

C, H, O, I, E are all symmetric about a horizontal line, so **(a) CHOICE** is unchanged.

Q9 **Answer: (b)**

A water image turns the figure upside down but keeps left and right: answer **(b)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q10 **Answer: (b)**

A water image turns the figure upside down but keeps left and right: answer **(b)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q11 **Answer: (c)**

A water image turns the figure upside down but keeps left and right: answer **(c)**. Note that simply writing the letters in reverse order (without reversing each letter) is a trap.

Q12 **Answer: (c) 2:50**

18:30 - 3:40 gives **2:50** (reading the hours on a 12-hour clock).

Q13 **Answer: (c) 4:15**

18:30 - 2:15 gives **4:15** (reading the hours on a 12-hour clock).

Q14 **Answer: (d) 11:10**

18:30 - 7:20 gives **11:10** (reading the hours on a 12-hour clock).

Q15 **Answer: (b) 8:25**

18:30 - 10:05 gives **8:25** (reading the hours on a 12-hour clock).

Q16 **Answer: (c) 12:40**

18:30 - 5:50 gives **12:40** (reading the hours on a 12-hour clock).

Q17 **Answer: (a)**

A water image swaps top and bottom (each element moves to the opposite side of the horizontal line and flips) but keeps left and right: answer **(a)**.

Q18 **Answer: (c)**

A water image swaps top and bottom (each element moves to the opposite side of the horizontal line and flips) but keeps left and right: answer **(c)**.

Q19 **Answer: (d)**

A water image swaps top and bottom (each element moves to the opposite side of the horizontal line and flips) but keeps left and right: answer **(d)**.

Q20 **Answer: (c)**

A water image swaps top and bottom (each element moves to the opposite side of the horizontal line and flips) but keeps left and right: answer **(c)**.

Q21 **Answer: (d)**

A water image swaps top and bottom (each element moves to the opposite side of the horizontal line and flips) but keeps left and right: answer **(d)**.

Q22 **Answer: (d)**

A water image swaps top and bottom (each element moves to the opposite side of the horizontal line and flips) but keeps left and right: answer **(d)**.

Q23 **Answer: (a)**

Mirror then water reverses both directions, which equals a 180° turn: answer **(a)**.

Q24 **Answer: (b)**

Mirror then water reverses both directions, which equals a 180° turn: answer **(b)**.

Q25 **Answer: (c)**

X has both a vertical and a horizontal line of symmetry: **(c) X**.

A water image swaps top and bottom and keeps left and right; letter order does not change. B, C, D, E, H, I, K, O, X are unchanged in water. Exam water time = 18:30 - time. Mirror followed by water equals a 180° turn.

LEVEL 2

Intermediate level

Find embedded figures, complete patterns, solve paper folding and cutting, count figures, crack figure matrices and form figures from pieces

8 MODULES · 9 DAYS MINIMUM

Module 08 · Embedded figures	1 day
Module 09 · Completion of figures and pattern completion	1 day
Module 10 · Paper folding	1 day
Module 11 · Paper cutting and punching	1 day
Module 12 · Counting figures	2 days
Module 13 · Figure matrix	1 day
Module 14 · Grouping of figures	1 day
Module 15 · Dot situation, figure formation and figure analysis	1 day

MODULE 08 · INTERMEDIATE

Embedded figures

Minimum duration **1 day** | Level **Intermediate**

Embedded-figure questions ask you to find a simple figure hidden inside a complex one, with the same size and orientation. This module covers the rules (exact size and orientation, rotations usually not allowed), the find-a-distinctive-corner-then-trace method and reverse questions, followed by 25 solved questions in which a program checked that exactly one option is hidden in each figure.

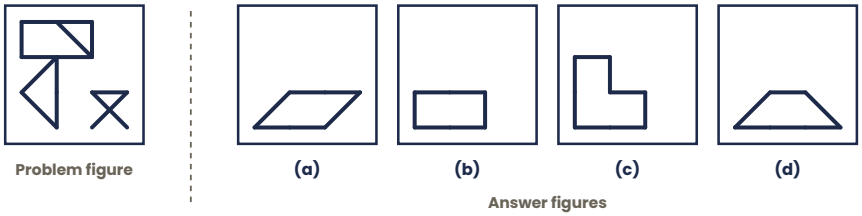
IN THIS MODULE

8.1 Concepts
8.2 Method
8.3 Reverse questions
8.4 Practice questions
8.5 Solutions

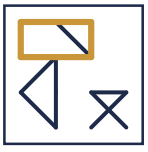
8.1 Concepts

8.1.1 Exact size and orientation

The simple figure must appear in the complex figure **exactly**: same shape, same size and same orientation. Extra lines crossing it do not matter, but every side of the simple figure must be present.



FIND THE FIGURE HIDDEN IN THE COMPLEX FIGURE



THE HIDDEN FIGURE TRACED IN GOLD

8.1.2 Rotated versions do not count

WATCH OUT

Unless the question says otherwise, a rotated or mirrored copy of the simple figure does not count. A right triangle with its right angle at the bottom-left is different from one with its right angle at the bottom-right.

8.2 Method

8.2.1 Find a distinctive corner or line first

Look for the most unusual feature of the simple figure: a sharp angle, a slanted side, a long side. Find where that feature occurs in the complex figure.

8.2.2 Trace the rest of the outline

From that starting point, follow the sides of the simple figure one by one. If any side is missing, try the next candidate position.

> **SHORTCUT TRICK**
> Count lengths in grid steps: a side two steps long cannot be matched by a side one step long, which removes many candidate positions quickly.

8.3 Reverse questions

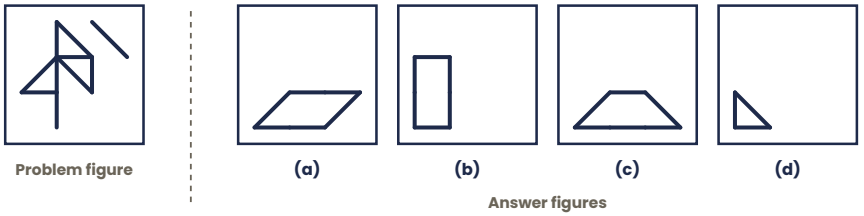
8.3.1 Which complex figure contains the given figure?

Here the simple figure is given and the options are complex figures. Apply the same method to each option and stop at the first that contains the figure completely.

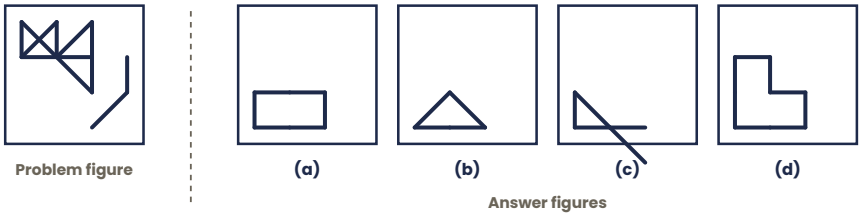
8.4 Practice questions

8.4.1 Basic level

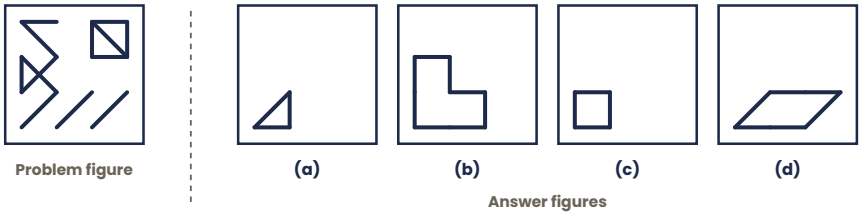
Q1 Which answer figure is hidden in the problem figure (same size and orientation)?



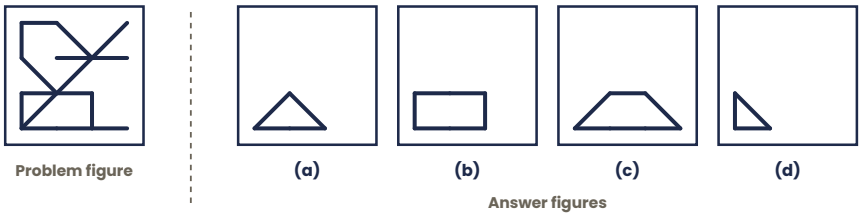
Q2 Which answer figure is hidden in the problem figure (same size and orientation)?



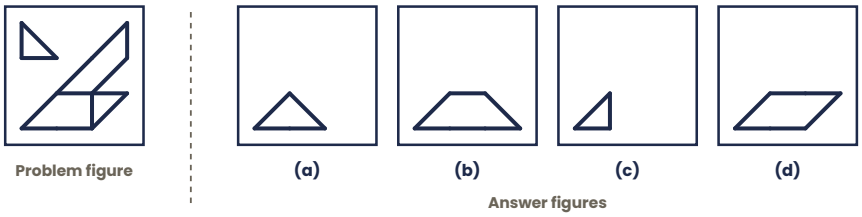
Q3 Which answer figure is hidden in the problem figure (same size and orientation)?



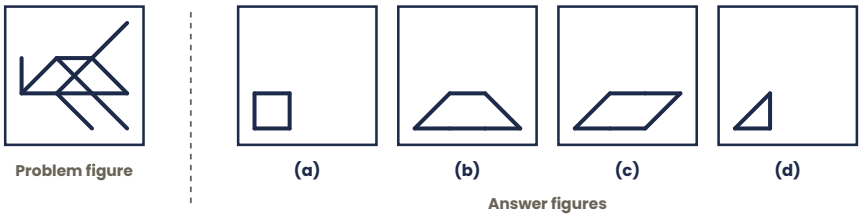
Q4 Which answer figure is hidden in the problem figure (same size and orientation)?



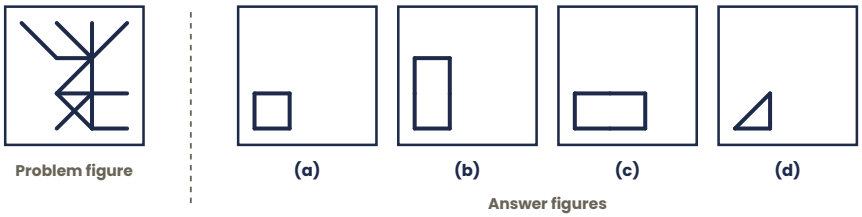
Q5 Which answer figure is hidden in the problem figure (same size and orientation)?



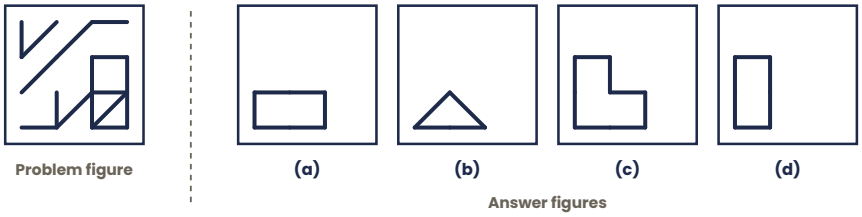
Q6 Which answer figure is hidden in the problem figure (same size and orientation)?



Q7 Which answer figure is hidden in the problem figure (same size and orientation)?

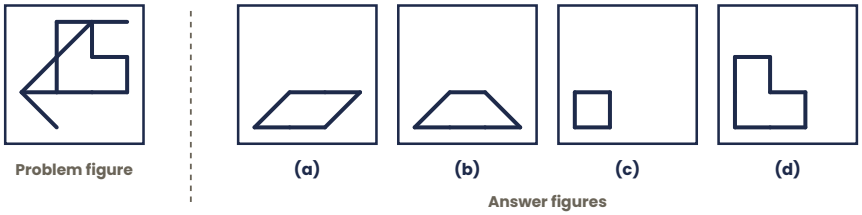


Q8 Which answer figure is hidden in the problem figure (same size and orientation)?

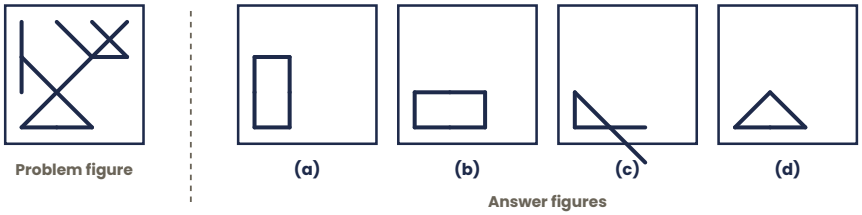


8.4.2 Intermediate level

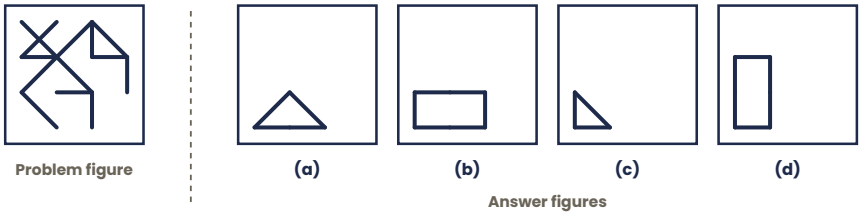
Q9 Which answer figure is hidden in the problem figure (same size and orientation)?



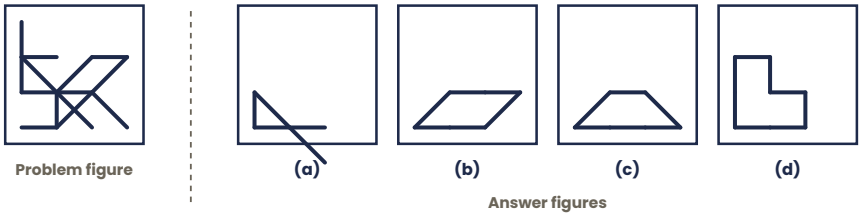
Q10 Which answer figure is hidden in the problem figure (same size and orientation)?



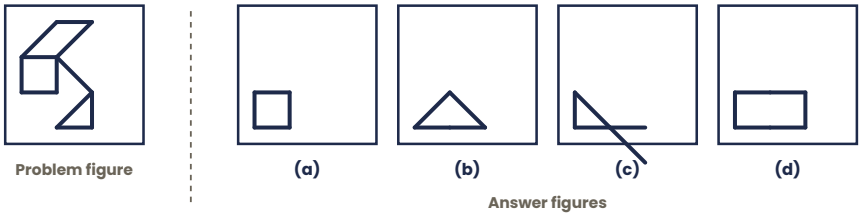
Q11 Which answer figure is hidden in the problem figure (same size and orientation)?



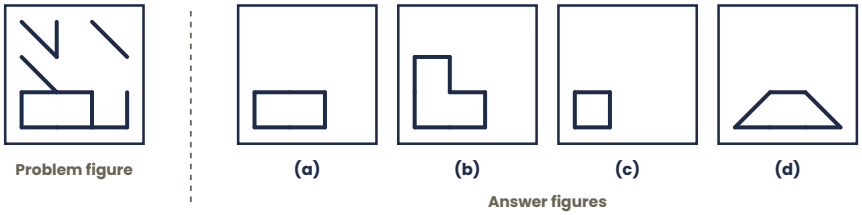
Q12 Which answer figure is hidden in the problem figure (same size and orientation)?



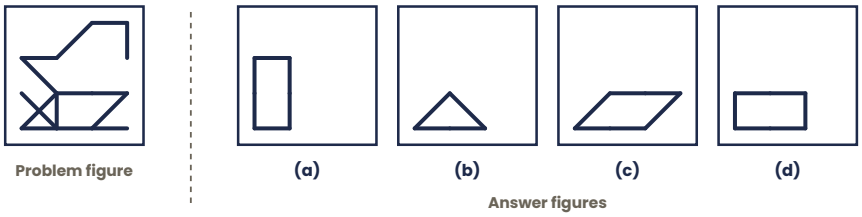
Q13 Which answer figure is hidden in the problem figure (same size and orientation)?



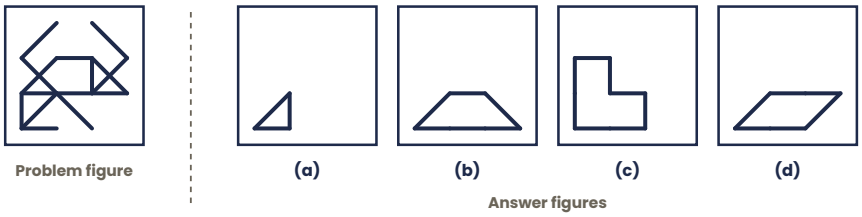
Q14 Which answer figure is hidden in the problem figure (same size and orientation)?



Q15 Which answer figure is hidden in the problem figure (same size and orientation)?

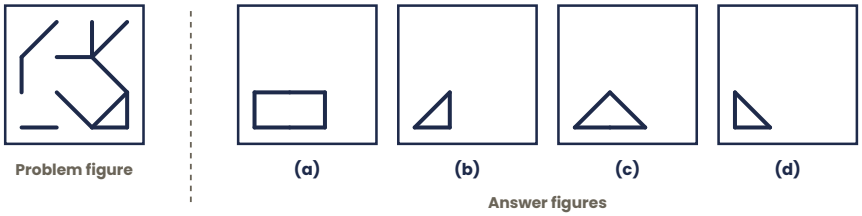


Q16 Which answer figure is hidden in the problem figure (same size and orientation)?

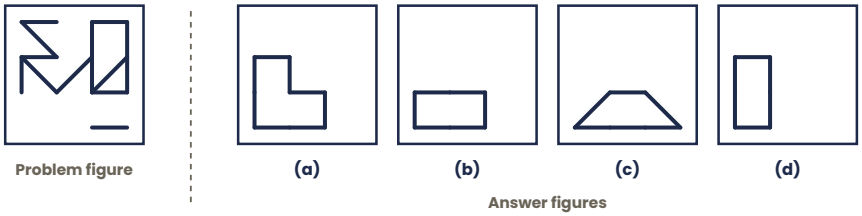


8.4.3 Advanced level

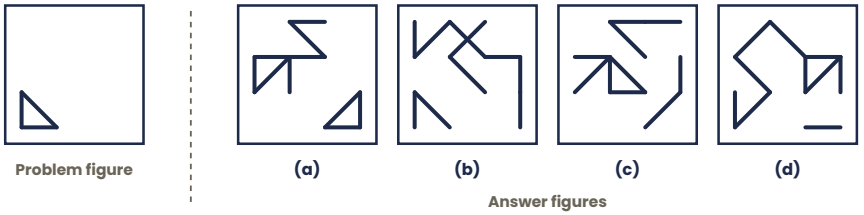
Q17 Which answer figure is hidden in the problem figure (same size and orientation)?



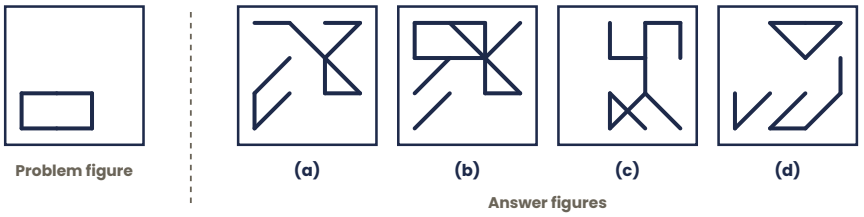
Q18 Which answer figure is hidden in the problem figure (same size and orientation)?



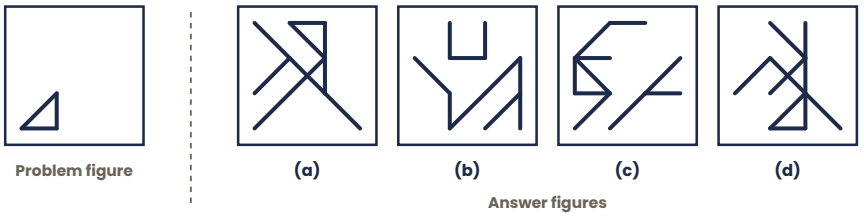
Q19 Which answer figure contains the problem figure (same size and orientation)?



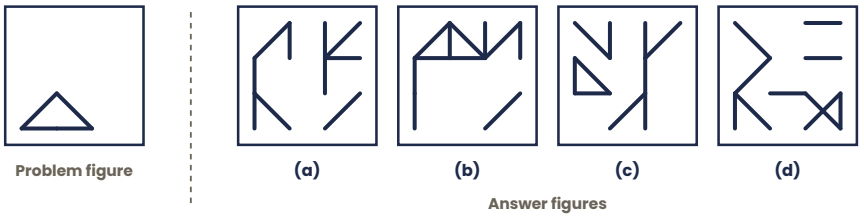
Q20 Which answer figure contains the problem figure (same size and orientation)?



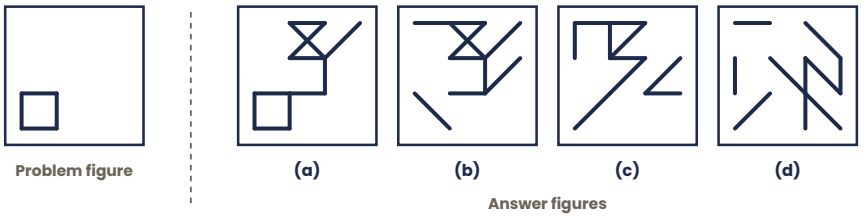
Q21 Which answer figure contains the problem figure (same size and orientation)?



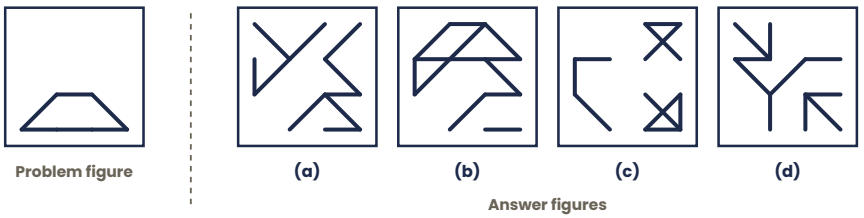
Q22 Which answer figure contains the problem figure (same size and orientation)?



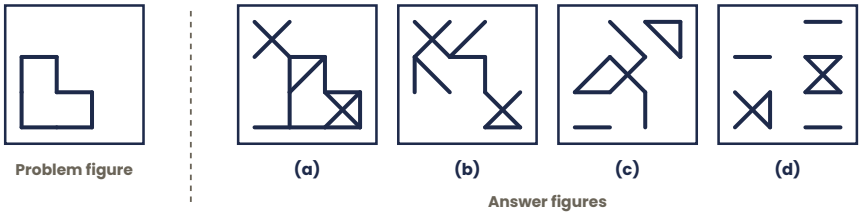
Q23 Which answer figure contains the problem figure (same size and orientation)?



Q24 Which answer figure contains the problem figure (same size and orientation)?



Q25 Which answer figure contains the problem figure (same size and orientation)?



8.5 Solutions

Q1 **Answer: (d)**

The small triangle can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(d)**.



Q2 **Answer: (c)**

The right triangle (2 × 1) can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(c)**.



Q3 **Answer: (c)**

The unit square can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(c)**.



Q4 **Answer: (b)**

The rectangle 2 × 1 can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(b)**.



traced

Q5 **Answer: (d)**

The parallelogram can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(d)**.



traced

Q6 **Answer: (b)**

The trapezium can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(b)**.



traced

Q7 **Answer: (d)**

The flipped triangle can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(d)**.



traced

Q8 **Answer: (d)**

The tall rectangle 1 × 2 can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(d)**.



traced

Q9 **Answer: (d)**

The L-hexagon can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(d)**.



traced

Q10 **Answer: (d)**

The kite triangle can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(d)**.



traced

Q11 **Answer: (c)**

The small triangle can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(c)**.



traced

Q12 **Answer: (a)**

The right triangle (2 × 1) can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(a)**.



traced

Q13 **Answer: (a)**

The unit square can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(a)**.



traced

Q14 **Answer: (a)**

The rectangle 2 × 1 can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(a)**.



traced

Q15 **Answer: (c)**

The parallelogram can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(c)**.



traced

Q16 **Answer: (b)**

The trapezium can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(b)**.



traced

Q17 **Answer: (b)**

The flipped triangle can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(b)**.



traced

Q18 **Answer: (d)**

The tall rectangle 1 × 2 can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(d)**.



traced

Q19 **Answer: (c)**

Only figure **(c)** contains the small triangle with the same size and orientation (traced in gold).



traced

Q20 **Answer: (b)**

Only figure **(b)** contains the rectangle 2 × 1 with the same size and orientation (traced in gold).



traced

Q21 **Answer: (d)**

Only figure **(d)** contains the flipped triangle with the same size and orientation (traced in gold).



traced

Q22 **Answer: (b)**

Only figure **(b)** contains the kite triangle with the same size and orientation (traced in gold).



traced

Q23 **Answer: (a)**

Only figure **(a)** contains the unit square with the same size and orientation (traced in gold).



traced

Q24 **Answer: (b)**

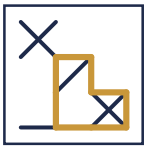
Only figure **(b)** contains the trapezium with the same size and orientation (traced in gold).



traced

Q25 **Answer: (a)**

Only figure **(a)** contains the L-hexagon with the same size and orientation (traced in gold).



traced

The hidden figure must appear with the same size and orientation, and every one of its sides must be present. Start from its most distinctive corner or side, trace the outline, and count lengths in grid steps to reject candidate positions quickly.

MODULE 09 · INTERMEDIATE

Completion of figures and pattern completion

Minimum duration **1 day** | Level **Intermediate**

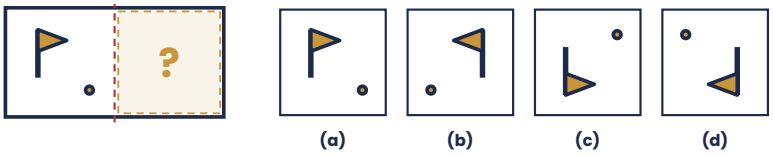
These questions ask which piece completes a figure or pattern. This module covers symmetry-based completion, continuing lines and shapes, the missing quarter of a square pattern, repeating tile patterns and rotational patterns, and the use-the-symmetry-lines method, followed by 25 solved questions whose patterns and answer pieces were generated by a program.

IN THIS MODULE

9.1 Completion of incomplete figures
9.2 Pattern completion
9.3 Method
9.4 Practice questions
9.5 Solutions

9.1 Completion of incomplete figures

9.1.1 Symmetry-based completion



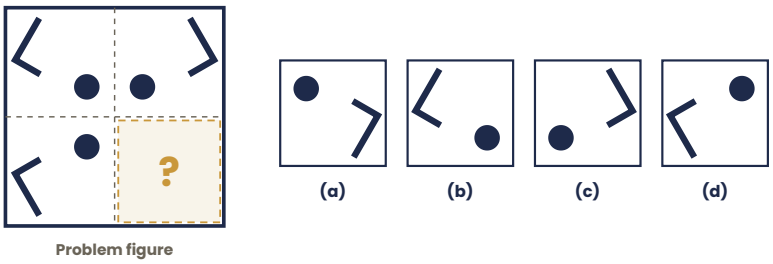
COMPLETE THE FIGURE ABOUT ITS LINE OF SYMMETRY

9.1.2 Continuing lines and shapes

Lines that run up to the gap must continue across it with the same direction and spacing; shapes cut by the gap must be completed with their missing part.

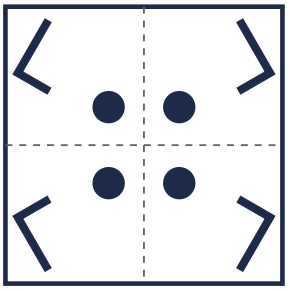
9.2 Pattern completion

9.2.1 Missing quarter of a square pattern



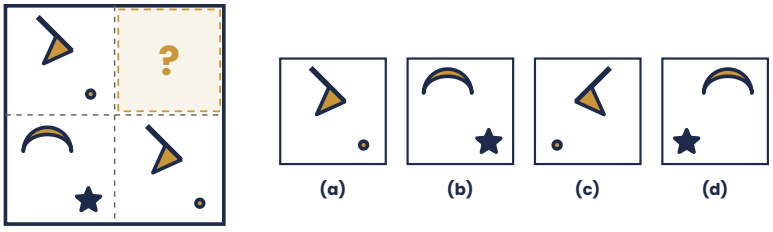
Problem figure

A PATTERN SYMMETRIC ABOUT BOTH CENTRE LINES



THE COMPLETED PATTERN

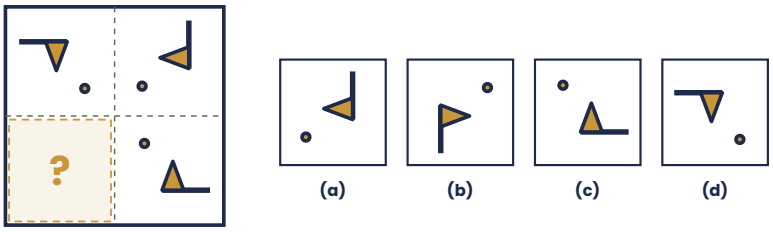
9.2.2 Repeating tile patterns



Problem figure

DIAGONALLY OPPOSITE QUARTERS ARE IDENTICAL

9.2.3 Rotational patterns



Problem figure

EACH QUARTER IS THE PREVIOUS ONE TURNED 90° CLOCKWISE

9.3 Method

9.3.1 Use symmetry lines and edges

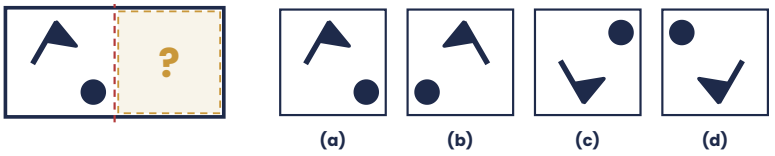
First decide the type of pattern: mirror (reflection across the centre lines), rotational (quarter turns) or tiled (repeats). Then predict the missing part from its neighbours and check that every element meets the edges of the gap correctly.

> **SHORTCUT TRICK**
> In a mirror pattern, an element near the centre in one quarter must also be near the centre in the missing quarter. Positions relative to the centre lines are the fastest check.

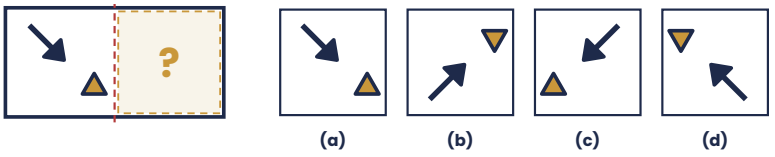
9.4 Practice questions

9.4.1 Basic level

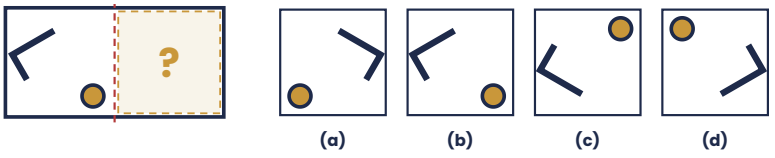
Q1 The dashed red line is a line of symmetry. Which answer figure completes the figure?



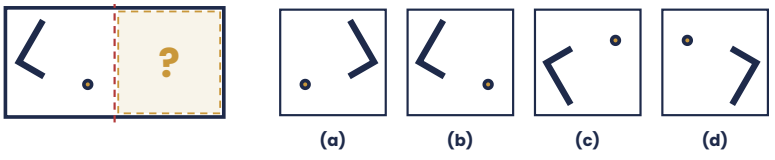
Q2 The dashed red line is a line of symmetry. Which answer figure completes the figure?



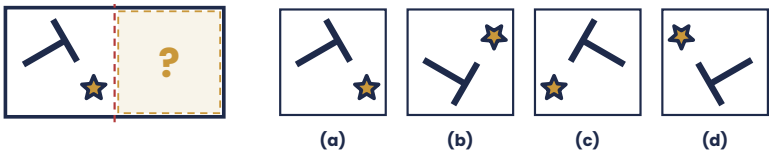
Q3 The dashed red line is a line of symmetry. Which answer figure completes the figure?



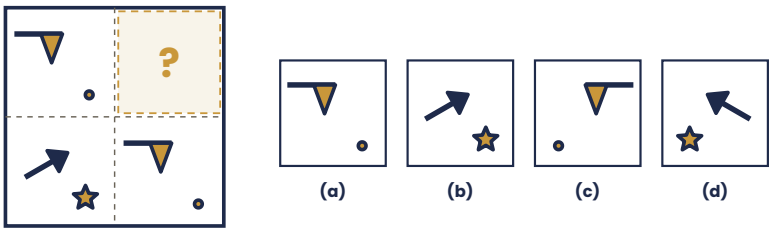
Q4 The dashed red line is a line of symmetry. Which answer figure completes the figure?



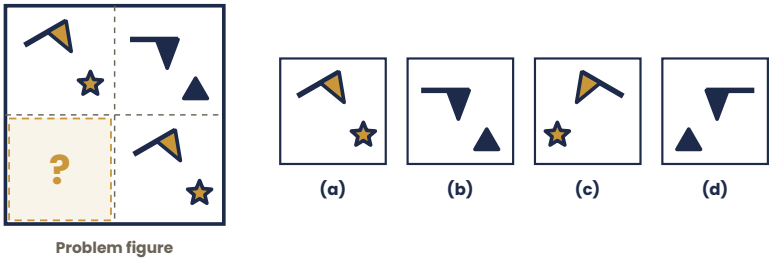
Q5 The dashed red line is a line of symmetry. Which answer figure completes the figure?



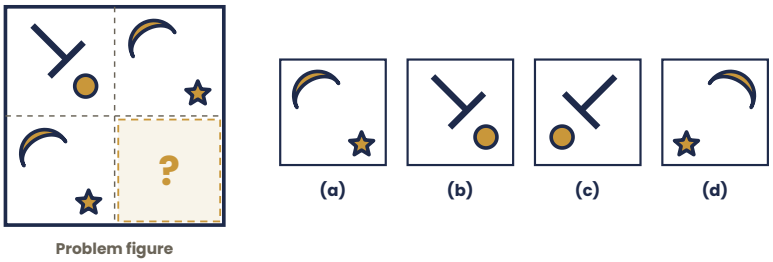
Q6 Which answer figure completes the pattern?



Q7 Which answer figure completes the pattern?

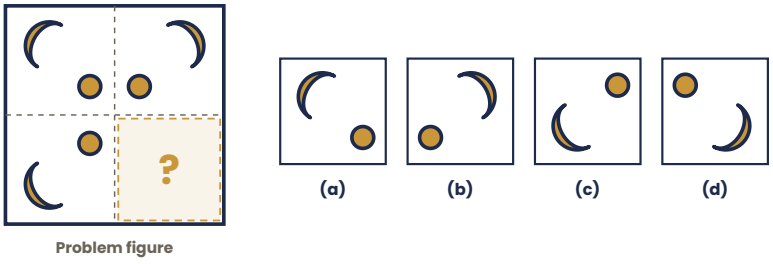


Q8 Which answer figure completes the pattern?

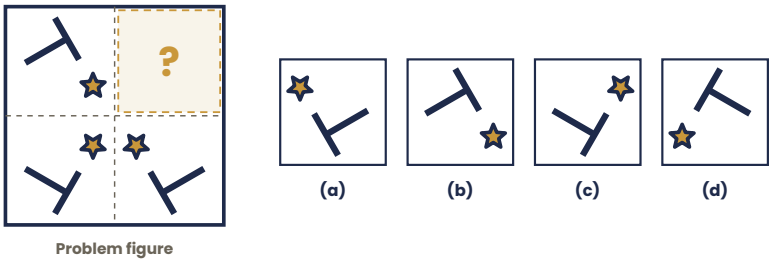


9.4.2 Intermediate level

Q9 Which answer figure completes the pattern?

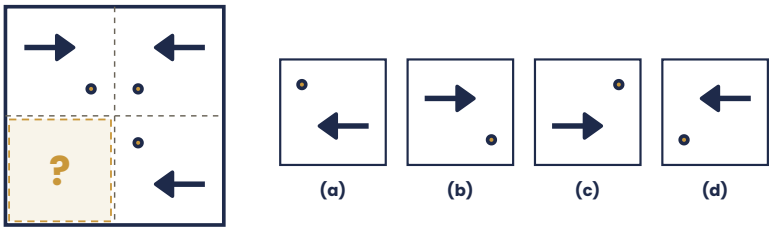


Q10 Which answer figure completes the pattern?



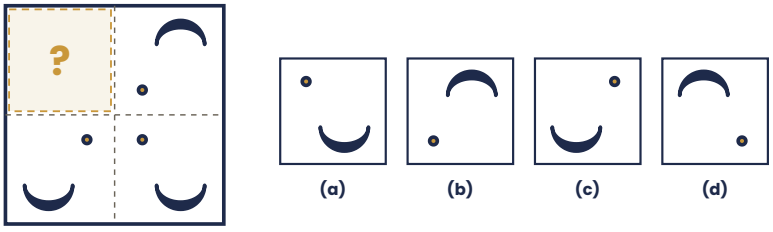
Problem figure

Q11 Which answer figure completes the pattern?



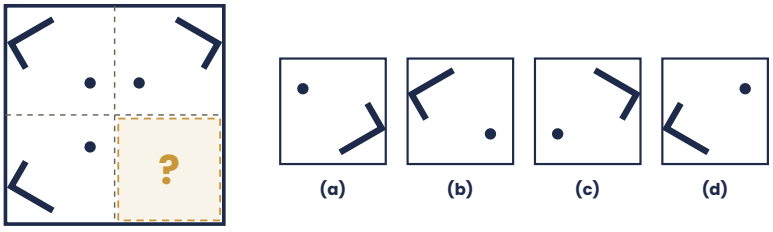
Problem figure

Q12 Which answer figure completes the pattern?



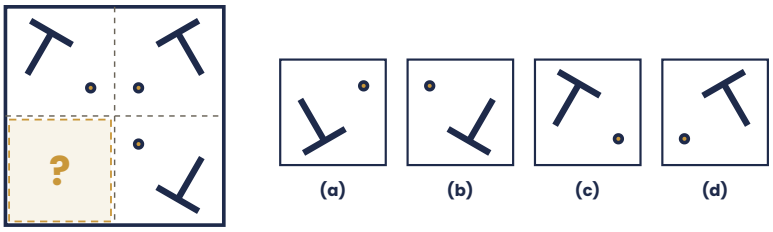
Problem figure

Q13 Which answer figure completes the pattern?



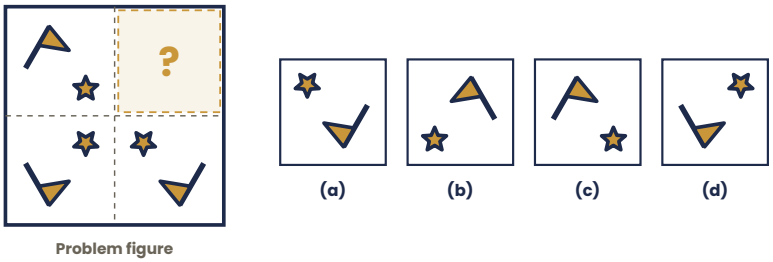
Problem figure

Q14 Which answer figure completes the pattern?



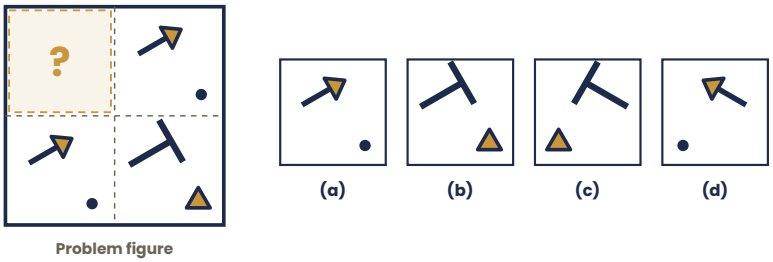
Problem figure

Q15 Which answer figure completes the pattern?



Problem figure

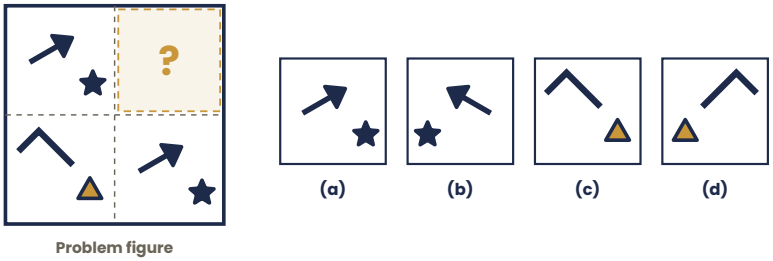
Q16 Which answer figure completes the pattern?



Problem figure

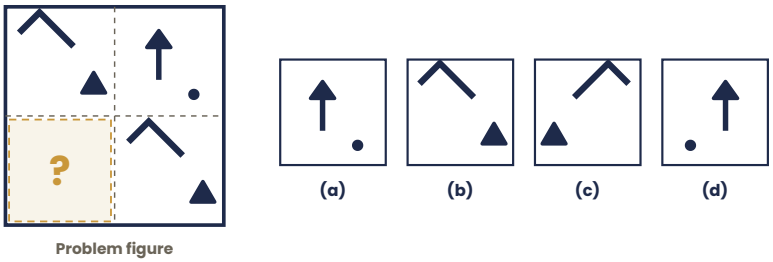
9.4.3 Advanced level

Q17 Which answer figure completes the pattern?



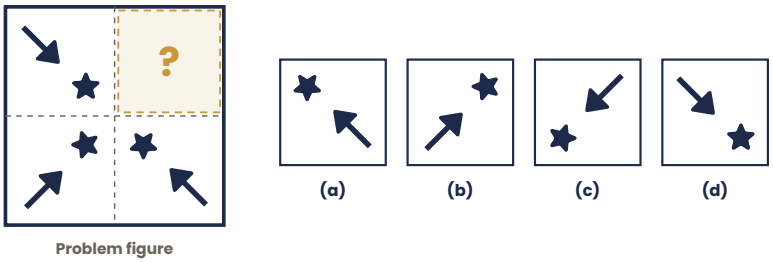
Problem figure

Q18 Which answer figure completes the pattern?



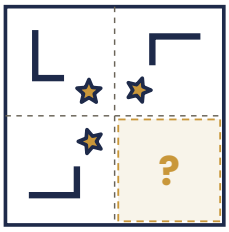
Problem figure

Q19 Which answer figure completes the pattern?



Problem figure

Q20 Which answer figure completes the pattern?

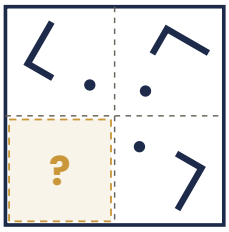


Problem figure

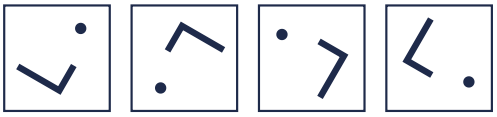


(a) (b) (c) (d)

Q21 Which answer figure completes the pattern?

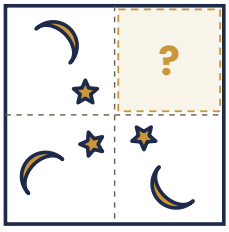


Problem figure



(a) (b) (c) (d)

Q22 Which answer figure completes the pattern?

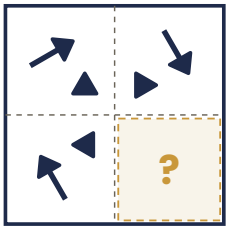


Problem figure

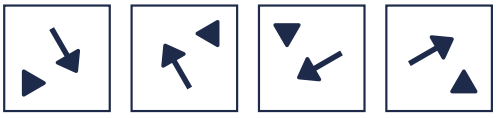


(a) (b) (c) (d)

Q23 Which answer figure completes the pattern?

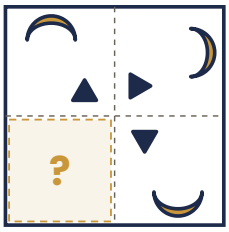


Problem figure



(a) (b) (c) (d)

Q24 Which answer figure completes the pattern?

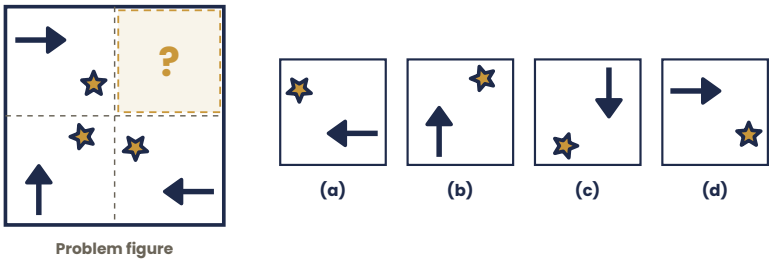


Problem figure



(a) (b) (c) (d)

Q25 Which answer figure completes the pattern?



Problem figure

9.5 Solutions

Q1 **Answer: (b)**

The right half must be the mirror image of the left half across the red line: **(b)**.

Q2 **Answer: (c)**

The right half must be the mirror image of the left half across the red line: **(c)**.

Q3 **Answer: (a)**

The right half must be the mirror image of the left half across the red line: **(a)**.

Q4 **Answer: (a)**

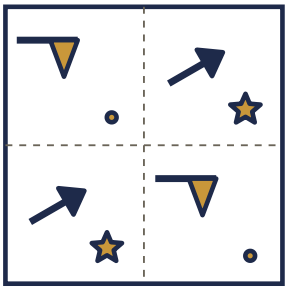
The right half must be the mirror image of the left half across the red line: **(a)**.

Q5 **Answer: (c)**

The right half must be the mirror image of the left half across the red line: **(c)**.

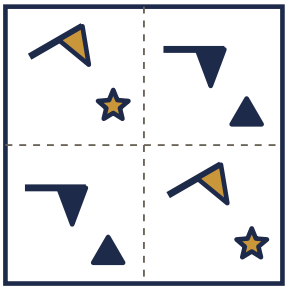
Q6 **Answer: (b)**

The pattern repeats two tiles like a chessboard: the diagonally opposite quarters are identical. The missing quarter equals the one diagonally opposite it: **(b)**.



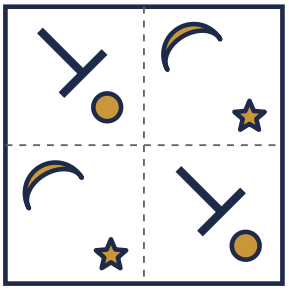
Q7 **Answer: (b)**

The pattern repeats two tiles like a chessboard: the diagonally opposite quarters are identical. The missing quarter equals the one diagonally opposite it: **(b)**.



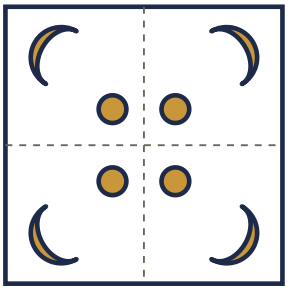
Q8 **Answer: (b)**

The pattern repeats two tiles like a chessboard: the diagonally opposite quarters are identical. The missing quarter equals the one diagonally opposite it: **(b)**.



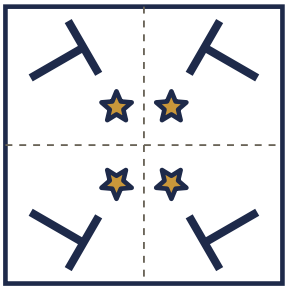
Q9 **Answer: (d)**

The pattern is symmetric about both the vertical and horizontal centre lines: each quarter is the mirror image of its neighbours. The missing quarter must be the reflection of the quarter beside it: **(d)**.



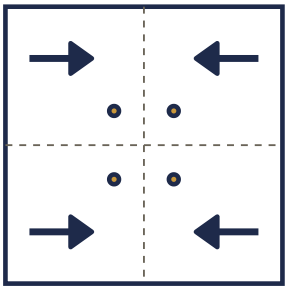
Q10 **Answer: (d)**

The pattern is symmetric about both the vertical and horizontal centre lines: each quarter is the mirror image of its neighbours. The missing quarter must be the reflection of the quarter beside it: **(d)**.



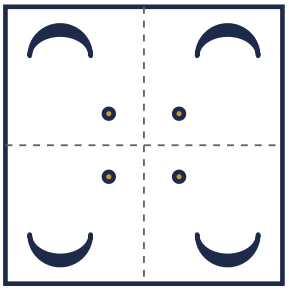
Q11 **Answer: (c)**

The pattern is symmetric about both the vertical and horizontal centre lines: each quarter is the mirror image of its neighbours. The missing quarter must be the reflection of the quarter beside it: **(c)**.



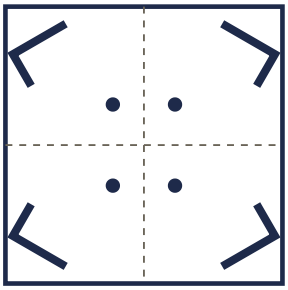
Q12 **Answer: (d)**

The pattern is symmetric about both the vertical and horizontal centre lines: each quarter is the mirror image of its neighbours. The missing quarter must be the reflection of the quarter beside it: **(d)**.



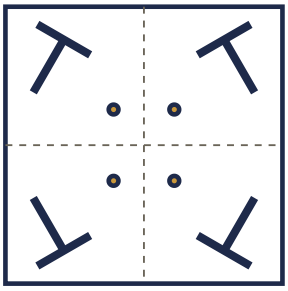
Q13 **Answer: (a)**

The pattern is symmetric about both the vertical and horizontal centre lines: each quarter is the mirror image of its neighbours. The missing quarter must be the reflection of the quarter beside it: **(a)**.



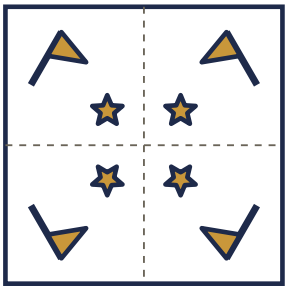
Q14 **Answer: (a)**

The pattern is symmetric about both the vertical and horizontal centre lines: each quarter is the mirror image of its neighbours. The missing quarter must be the reflection of the quarter beside it: **(a)**.



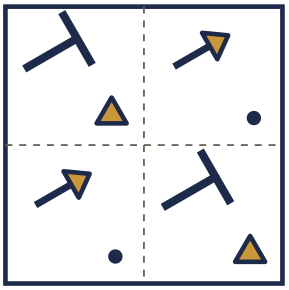
Q15 **Answer: (b)**

The pattern is symmetric about both the vertical and horizontal centre lines: each quarter is the mirror image of its neighbours. The missing quarter must be the reflection of the quarter beside it: **(b)**.



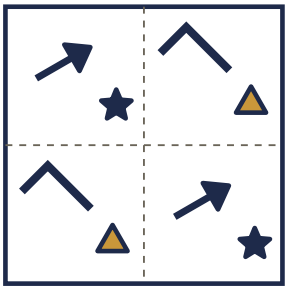
Q16 **Answer: (b)**

The pattern repeats two tiles like a chessboard: the diagonally opposite quarters are identical. The missing quarter equals the one diagonally opposite it: **(b)**.



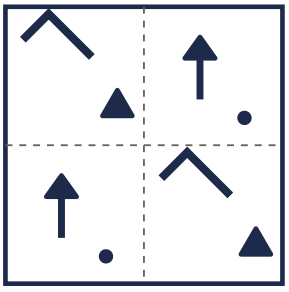
Q17 **Answer: (c)**

The pattern repeats two tiles like a chessboard: the diagonally opposite quarters are identical. The missing quarter equals the one diagonally opposite it: **(c)**.



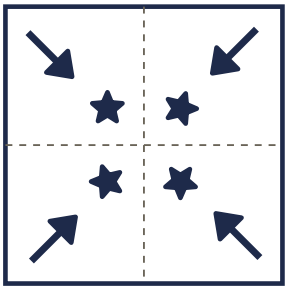
Q18 **Answer: (a)**

The pattern repeats two tiles like a chessboard: the diagonally opposite quarters are identical. The missing quarter equals the one diagonally opposite it: **(a)**.



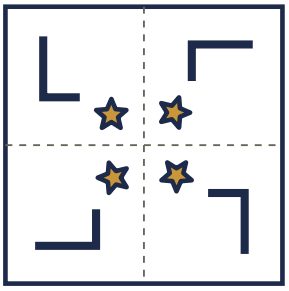
Q19 **Answer: (c)**

Each quarter is the previous one turned 90° clockwise as you go round the square (top-left → top-right → bottom-right → bottom-left). The missing quarter continues the turn: **(c)**.



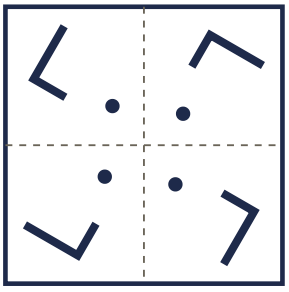
Q20 **Answer: (a)**

Each quarter is the previous one turned 90° clockwise as you go round the square (top-left → top-right → bottom-right → bottom-left). The missing quarter continues the turn: **(a)**.



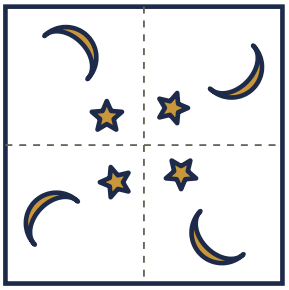
Q21 **Answer: (a)**

Each quarter is the previous one turned 90° clockwise as you go round the square (top-left → top-right → bottom-right → bottom-left). The missing quarter continues the turn: **(a)**.



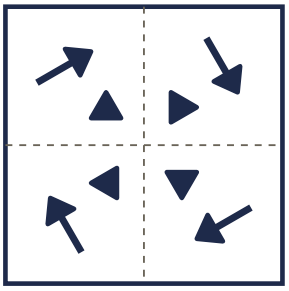
Q22 **Answer: (d)**

Each quarter is the previous one turned 90° clockwise as you go round the square (top-left → top-right → bottom-right → bottom-left). The missing quarter continues the turn: **(d)**.



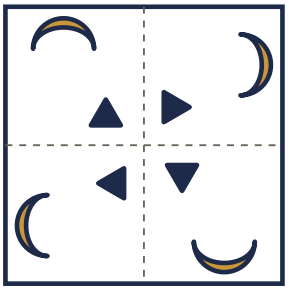
Q23 **Answer: (c)**

Each quarter is the previous one turned 90° clockwise as you go round the square (top-left → top-right → bottom-right → bottom-left). The missing quarter continues the turn: **(c)**.



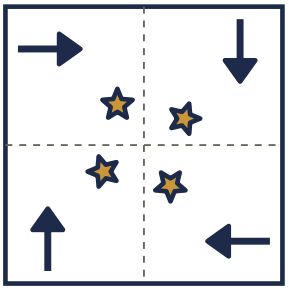
Q24 **Answer: (d)**

Each quarter is the previous one turned 90° clockwise as you go round the square (top-left → top-right → bottom-right → bottom-left). The missing quarter continues the turn: **(d)**.



Q25 **Answer: (c)**

Each quarter is the previous one turned 90° clockwise as you go round the square (top-left → top-right → bottom-right → bottom-left). The missing quarter continues the turn: **(c)**.



Identify the pattern type (mirror, rotational or tiled), predict the missing part from its neighbours, and confirm that every element meets the edges of the gap correctly. In mirror patterns check distances from the centre lines; in rotational patterns check the direction of each quarter turn.

MODULE 10 · INTERMEDIATE

Paper folding

Minimum duration **1 day** | Level **Intermediate**

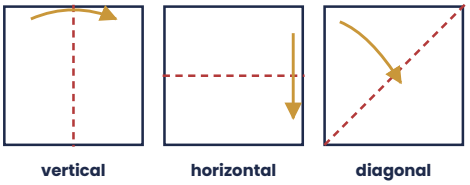
Paper-folding questions ask how a folded sheet looks, or what it looks like after unfolding. This module covers folding along vertical, horizontal and diagonal lines, why the folded part lands as a reflection, appearance after one and several folds, folding transparent sheets with patterns, and the reflect-step-by-step method, followed by 25 solved questions computed by reflecting every mark across each fold.

IN THIS MODULE

10.1 Concepts
10.2 Question types
10.3 Method
10.4 Practice questions
10.5 Solutions

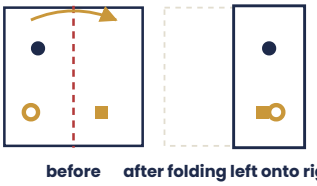
10.1 Concepts

10.1.1 Folding lines



FOLDING ALONG VERTICAL, HORIZONTAL AND DIAGONAL LINES

10.1.2 The folded part lands as a reflection

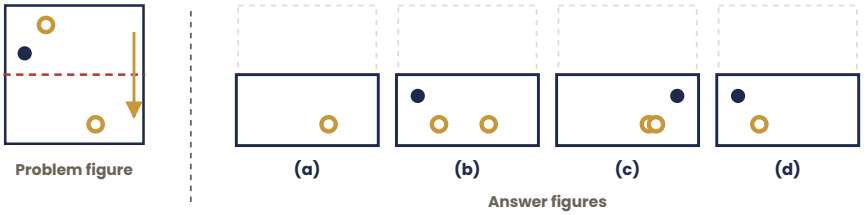


EACH MARK ON THE FOLDED HALF MOVES TO ITS MIRROR POSITION

A mark at distance d from the fold line moves to the same distance d on the other side, on the same perpendicular to the fold.

10.2 Question types

10.2.1 Appearance after folding



ONE FOLD

10.2.2 Multiple folds

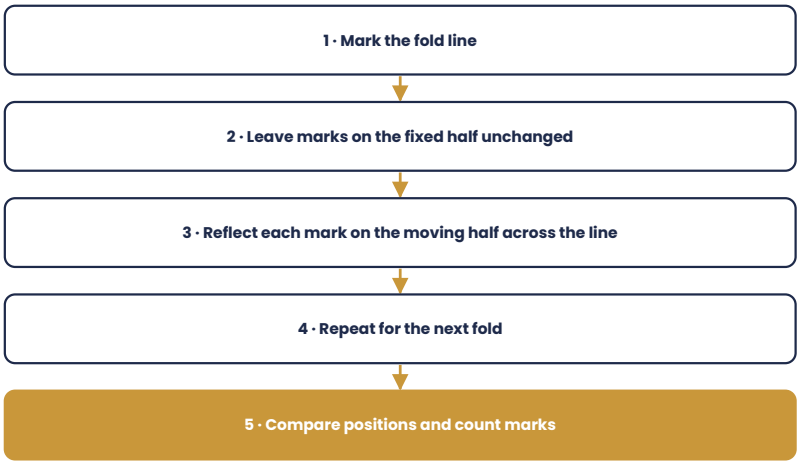
Apply the folds one at a time. After the first fold, only the remaining half exists; the second fold acts on that half.

10.2.3 Transparent sheets with patterns

With a transparent sheet, marks on both layers remain visible after folding. With opaque paper, only the marks on the top layer show; read the question carefully.

10.3 Method

10.3.1 Reflect step by step



THE FOLDING METHOD

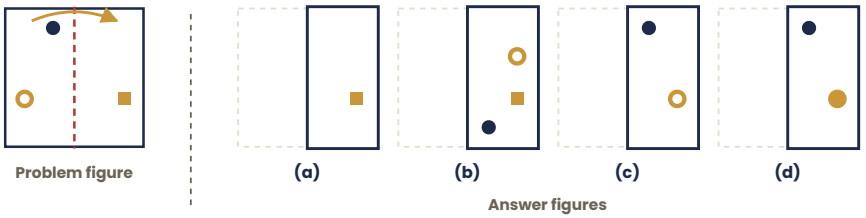
10.3.2 Practise with real paper

> **TIP**
> Fold a square of tracing paper and draw marks with a pen; checking ten questions physically builds the mental picture faster than any rule.

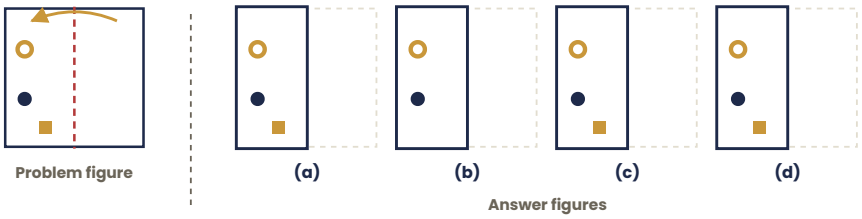
10.4 Practice questions

10.4.1 Basic level

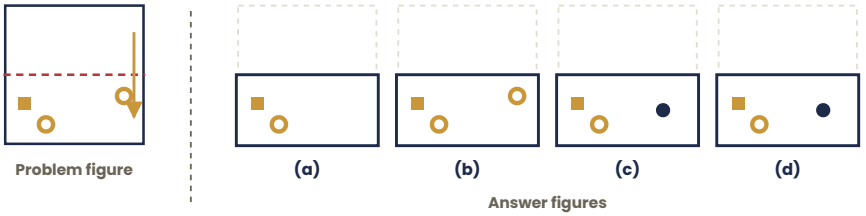
Q1 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



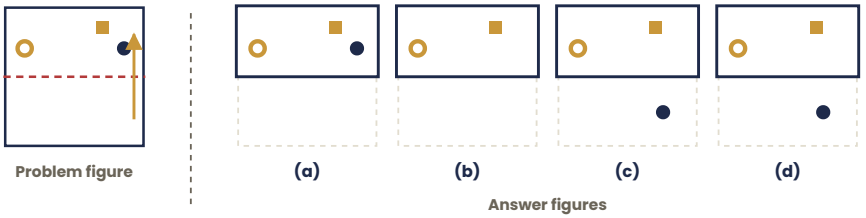
Q2 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



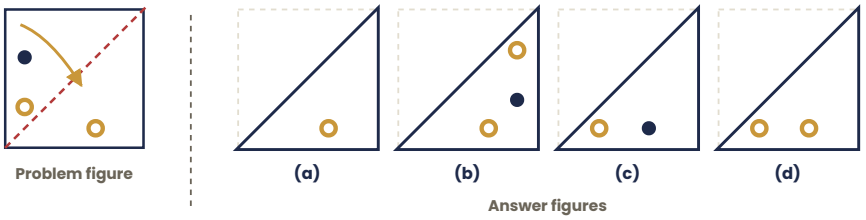
Q3 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



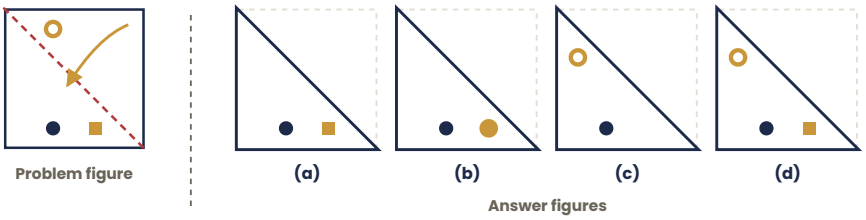
Q4 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



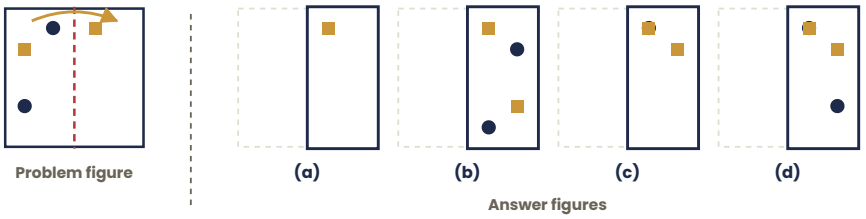
Q5 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



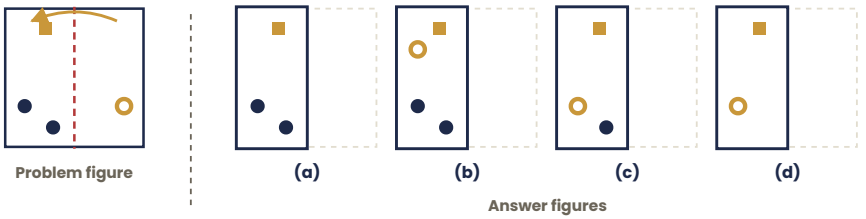
Q6 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



Q7 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?

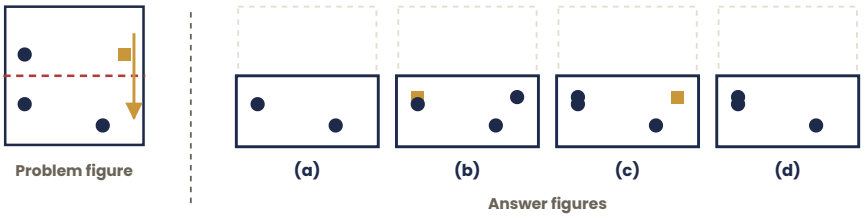


Q8 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?

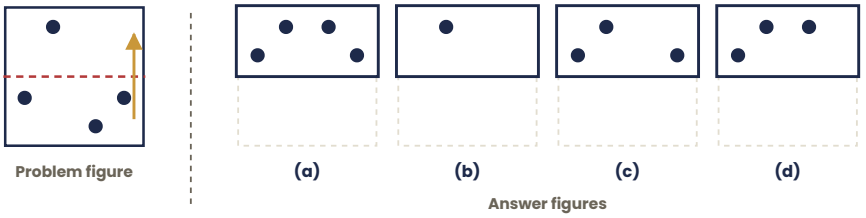


10.4.2 Intermediate level

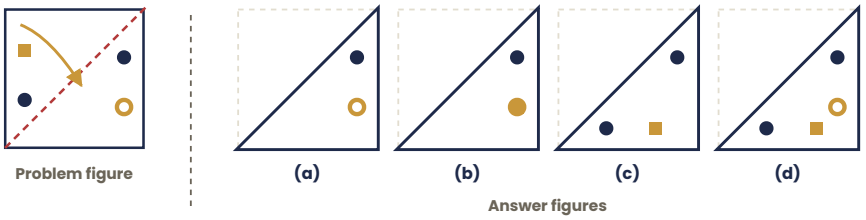
Q9 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



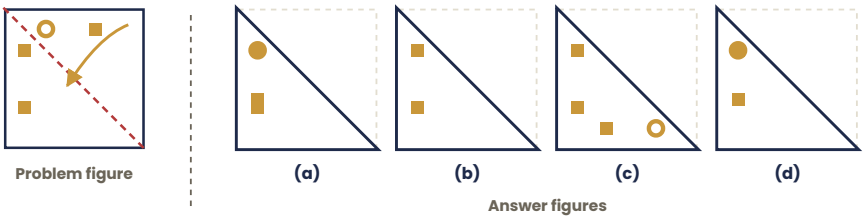
Q10 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



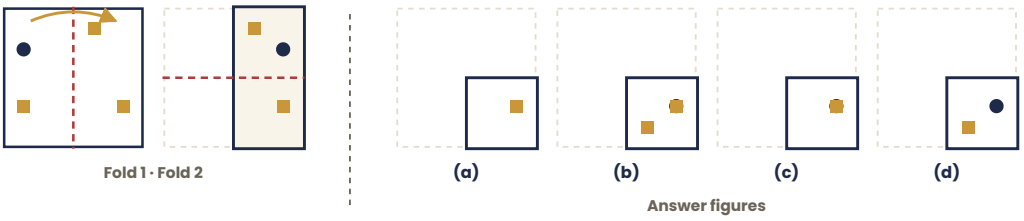
Q11 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



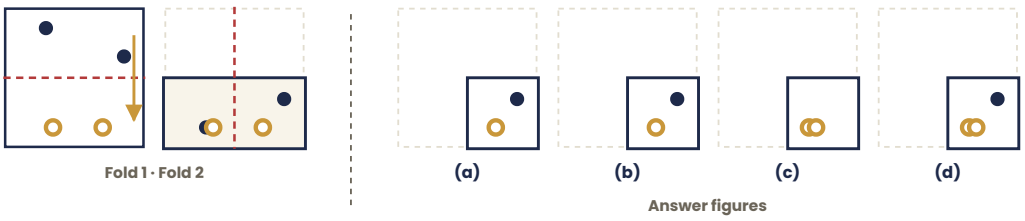
Q12 A transparent sheet with the marks shown is folded along the dashed line in the direction of the arrow. How does it look after folding?



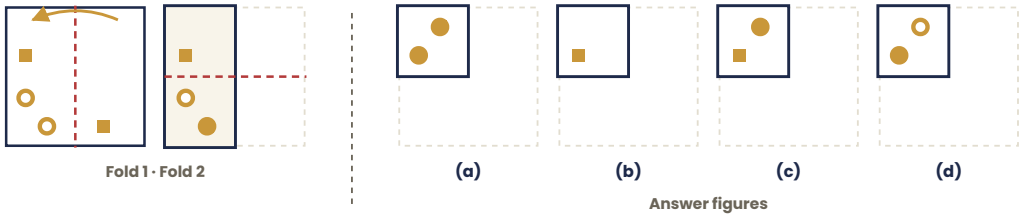
Q13 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



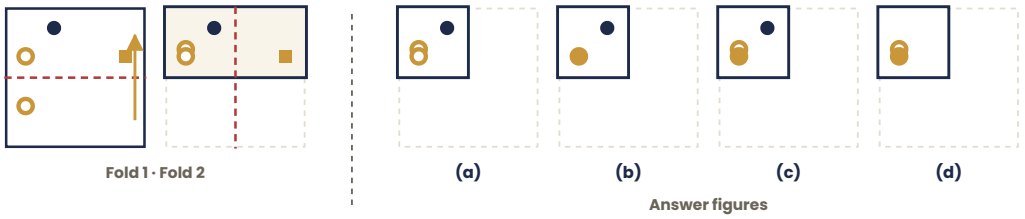
Q14 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



Q15 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?

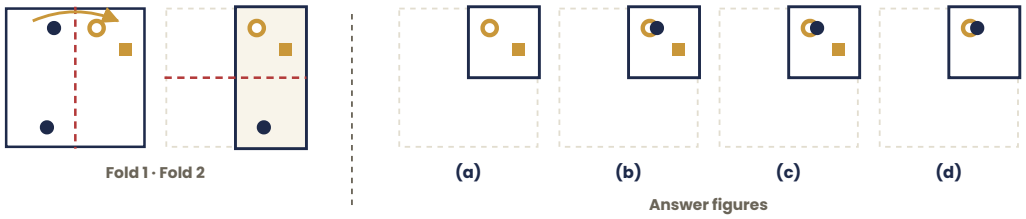


Q16 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?

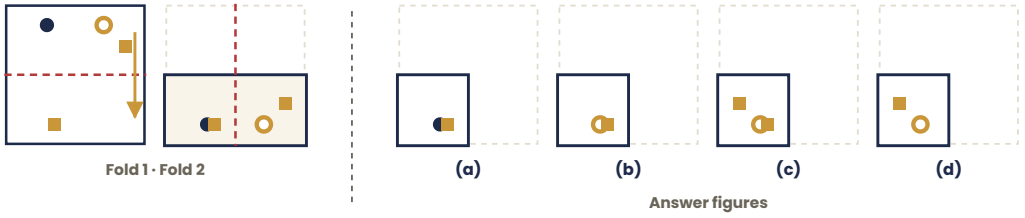


10.4.3 Advanced level

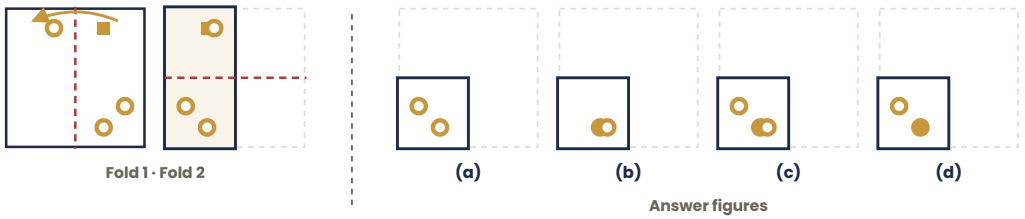
Q17 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



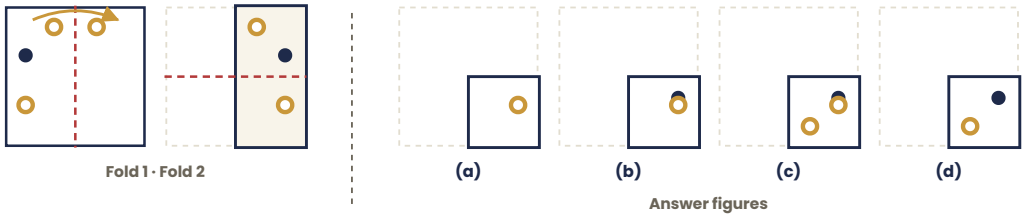
Q18 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



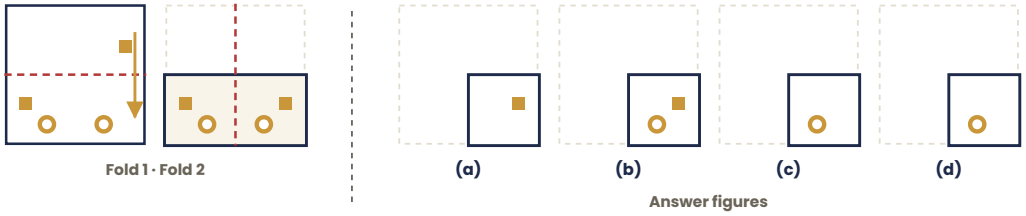
Q19 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



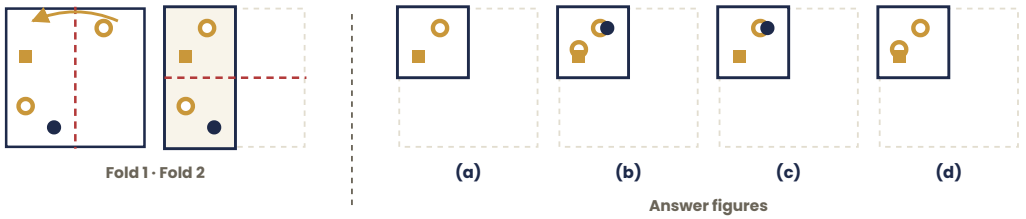
Q20 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



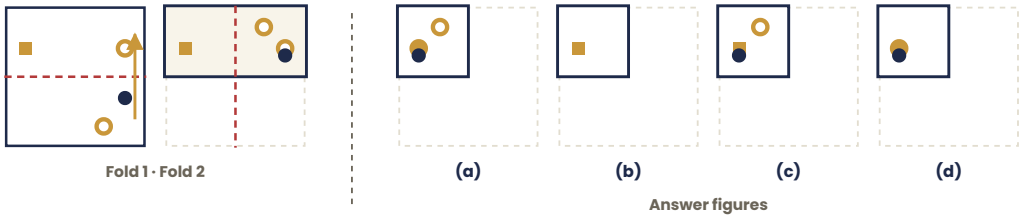
Q21 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



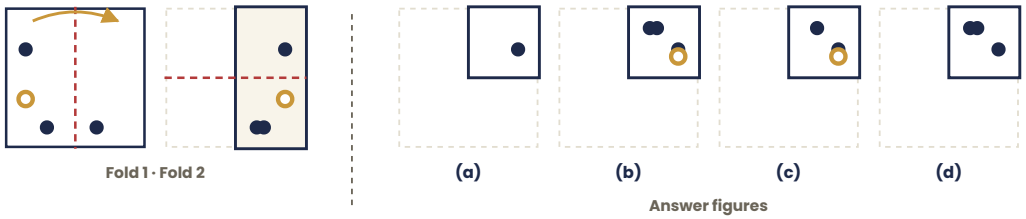
Q22 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



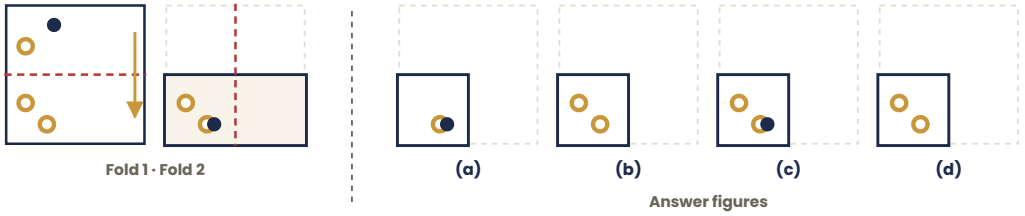
Q23 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



Q24 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



Q25 A transparent sheet is folded twice as shown (first fold, then second fold). How does it look after both folds?



10.5 Solutions

Q1 **Answer: (d)**

The folded part (left onto right) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(d)**.

Q2 **Answer: (a)**

The folded part (right onto left) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(a)**.

Q3 **Answer: (b)**

The folded part (top onto bottom) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(b)**.

Q4 **Answer: (a)**

The folded part (bottom onto top) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(a)**.

Q5 **Answer: (d)**

The folded part (top-left corner onto bottom-right) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(d)**.

Q6 **Answer: (d)**

The folded part (top-right corner onto bottom-left) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(d)**.

Q7 **Answer: (d)**

The folded part (left onto right) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(d)**.

Q8 **Answer: (c)**

The folded part (right onto left) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(c)**.

Q9 **Answer: (c)**

The folded part (top onto bottom) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(c)**.

Q10 **Answer: (a)**

The folded part (bottom onto top) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(a)**.

Q11 **Answer: (d)**

The folded part (top-left corner onto bottom-right) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(d)**.

Q12 **Answer: (a)**

The folded part (top-right corner onto bottom-left) lands as a mirror image across the fold line, so each of its marks moves to its reflection and joins the marks already on the other half: answer **(a)**.

Q13 **Answer: (b)**

Reflect the marks across the first fold (left onto right), then reflect everything outside the second fold across it (top onto bottom). Only **(b)** shows every mark in its reflected place.

Q14 **Answer: (d)**

Reflect the marks across the first fold (top onto bottom), then reflect everything outside the second fold across it (left onto right). Only **(d)** shows every mark in its reflected place.

Q15 **Answer: (a)**

Reflect the marks across the first fold (right onto left), then reflect everything outside the second fold across it (bottom onto top). Only **(a)** shows every mark in its reflected place.

Q16 **Answer: (c)**

Reflect the marks across the first fold (bottom onto top), then reflect everything outside the second fold across it (right onto left). Only **(c)** shows every mark in its reflected place.

Q17 **Answer: (b)**

Reflect the marks across the first fold (left onto right), then reflect everything outside the second fold across it (bottom onto top). Only **(b)** shows every mark in its reflected place.

Q18 **Answer: (c)**

Reflect the marks across the first fold (top onto bottom), then reflect everything outside the second fold across it (right onto left). Only **(c)** shows every mark in its reflected place.

Q19 **Answer: (c)**

Reflect the marks across the first fold (right onto left), then reflect everything outside the second fold across it (top onto bottom). Only **(c)** shows every mark in its reflected place.

Q20 **Answer: (c)**

Reflect the marks across the first fold (left onto right), then reflect everything outside the second fold across it (top onto bottom). Only **(c)** shows every mark in its reflected place.

Q21 **Answer: (b)**

Reflect the marks across the first fold (top onto bottom), then reflect everything outside the second fold across it (left onto right). Only **(b)** shows every mark in its reflected place.

Q22 **Answer: (b)**

Reflect the marks across the first fold (right onto left), then reflect everything outside the second fold across it (bottom onto top). Only **(b)** shows every mark in its reflected place.

Q23 **Answer: (a)**

Reflect the marks across the first fold (bottom onto top), then reflect everything outside the second fold across it (right onto left). Only **(a)** shows every mark in its reflected place.

Q24 **Answer: (b)**

Reflect the marks across the first fold (left onto right), then reflect everything outside the second fold across it (bottom onto top). Only **(b)** shows every mark in its reflected place.

Q25 **Answer: (c)**

Reflect the marks across the first fold (top onto bottom), then reflect everything outside the second fold across it (right onto left). Only **(c)** shows every mark in its reflected place.

The folded part lands as a mirror image across the fold line: marks on the fixed half stay, marks on the moving half move to their reflections. Apply several folds one at a time, and count the marks in each option to eliminate quickly.

MODULE 11 · INTERMEDIATE

Paper cutting and punching

Minimum duration **1 day** | Level **Intermediate**

These questions show a folded sheet being punched or cut, and ask how it looks when unfolded. This module covers why each fold doubles the holes, reflecting holes across every fold line when unfolding, single and multiple folds, cuts on edges and corners, diagonal folds, and the unfold-in-reverse method with hole counting, followed by 25 solved questions computed by unfolding every hole.

IN THIS MODULE

11.1 Concepts
11.2 Question types
11.3 Method
11.4 Practice questions
11.5 Solutions

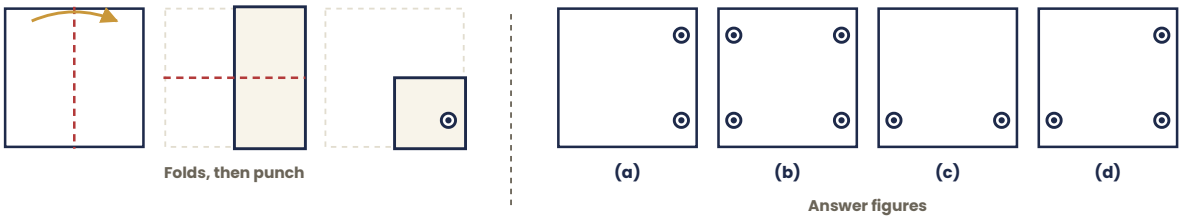
11.1 Concepts

11.1.1 Each fold doubles the holes

A sheet folded once has 2 layers; folded twice, 4 layers; three times, 8. A single punch through all layers therefore makes 2, 4 or 8 holes (fewer only if a hole lies exactly on a fold line).

Folds	Layers	Holes from one punch
1	2	2
2	4	4
3	8	8

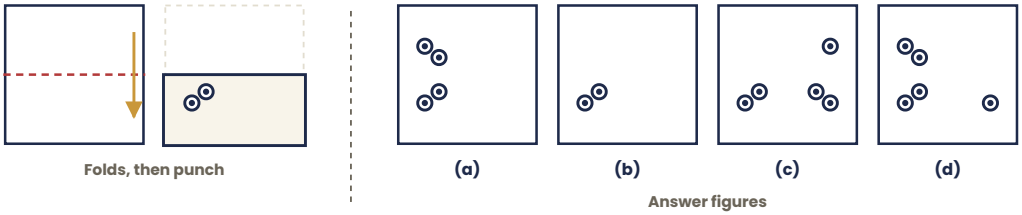
11.1.2 Holes are reflected across every fold when unfolding



FOLD TWICE, PUNCH ONCE, UNFOLD: FOUR HOLES

11.2 Question types

11.2.1 Single and multiple folds

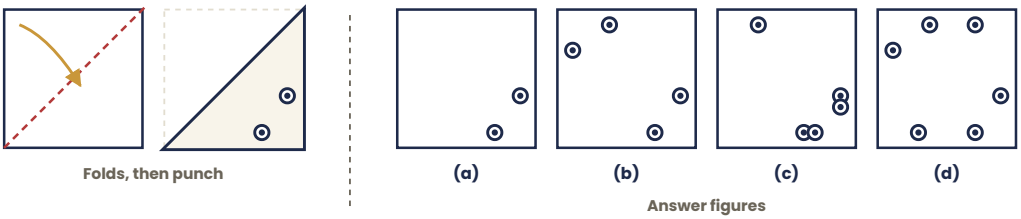


ONE FOLD, TWO PUNCHES

11.2.2 Cuts on edges and corners

A cut on a folded edge opens into a symmetric notch across the fold; a cut at a corner where two folds meet becomes a shape in the middle of the sheet. Treat a cut exactly like a hole: reflect it across each fold.

11.2.3 Diagonal folds



A DIAGONAL FOLD REFLECTS HOLES ACROSS THE DIAGONAL

11.3 Method

11.3.1 Unfold in reverse order

Undo the last fold first: reflect the holes across that fold line. Then undo the previous fold, reflecting all the holes you now have. Continue until the sheet is fully open.

11.3.2 Count holes to eliminate options

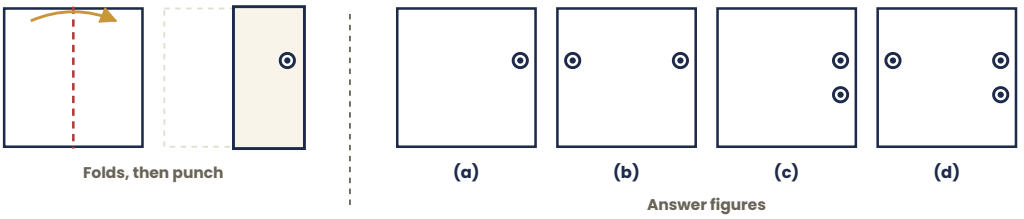
SHORTCUT TRICK

Count first: with k folds and one punch away from the fold lines there must be 2^k holes. Options with the wrong count go immediately; then check the symmetry about each fold line.

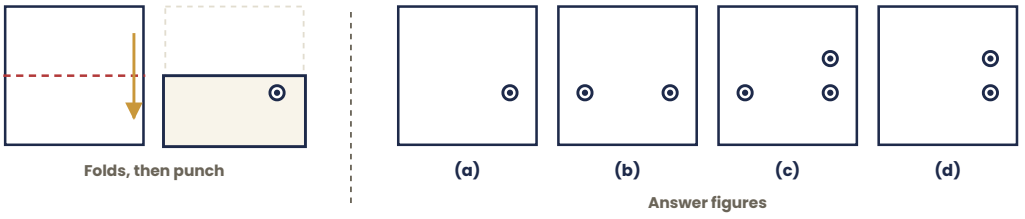
11.4 Practice questions

11.4.1 Basic level

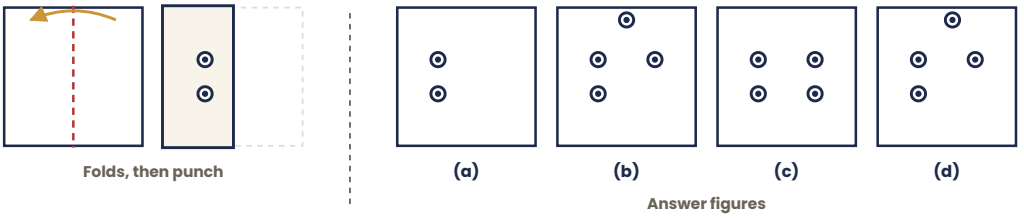
Q1 A square sheet is folded as shown and then punched. How does it look when unfolded?



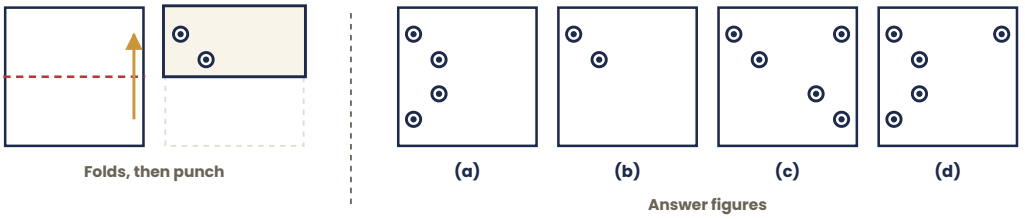
Q2 A square sheet is folded as shown and then punched. How does it look when unfolded?



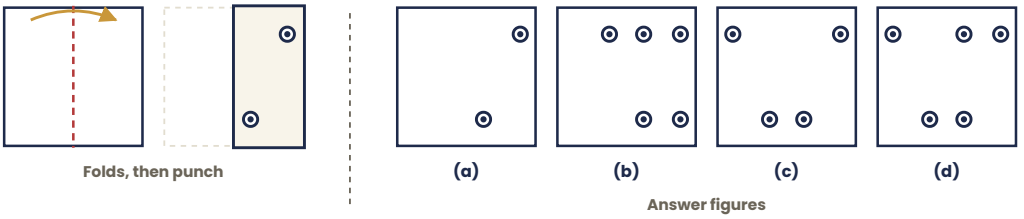
Q3 A square sheet is folded as shown and then punched. How does it look when unfolded?



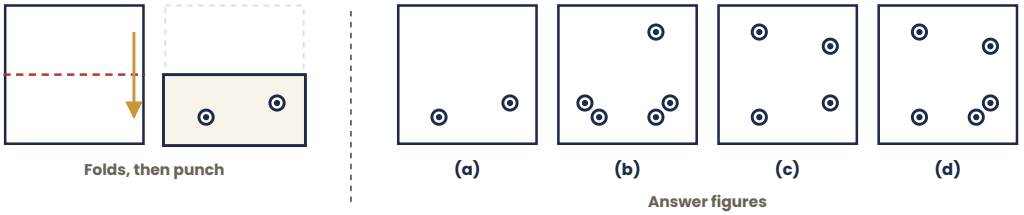
Q4 A square sheet is folded as shown and then punched. How does it look when unfolded?



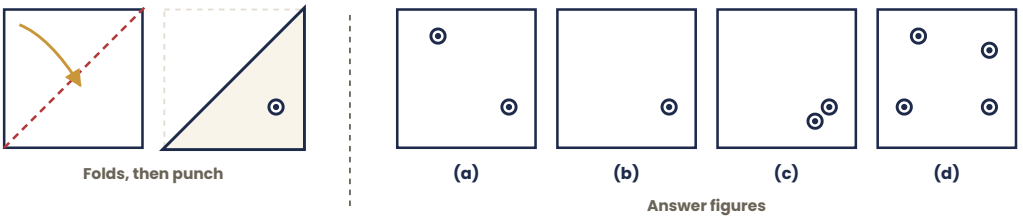
Q5 A square sheet is folded as shown and then punched. How does it look when unfolded?



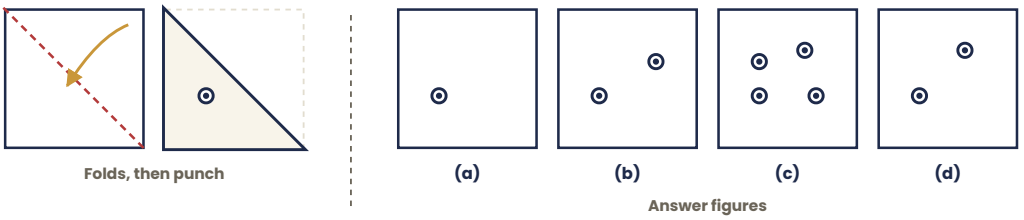
Q6 A square sheet is folded as shown and then punched. How does it look when unfolded?



Q7 A square sheet is folded as shown and then punched. How does it look when unfolded?

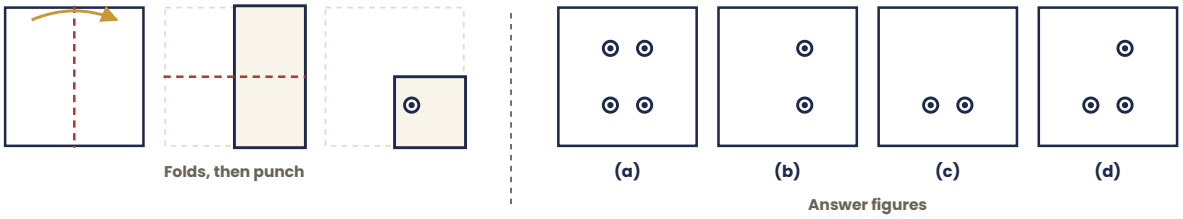


Q8 A square sheet is folded as shown and then punched. How does it look when unfolded?

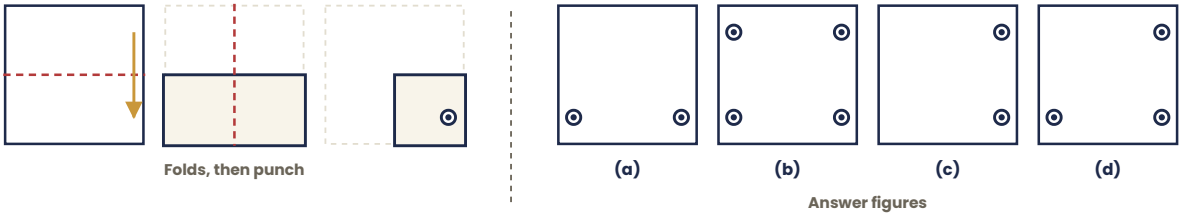


11.4.2 Intermediate level

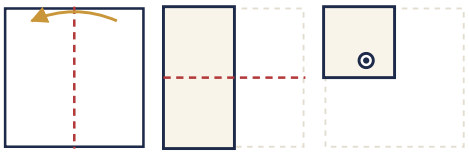
Q9 A square sheet is folded as shown and then punched. How does it look when unfolded?



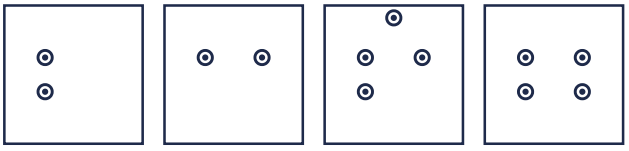
Q10 A square sheet is folded as shown and then punched. How does it look when unfolded?



Q11 A square sheet is folded as shown and then punched. How does it look when unfolded?



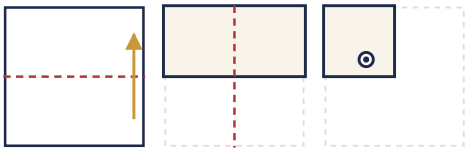
Folds, then punch



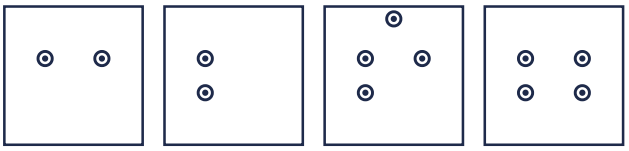
(a) (b) (c) (d)

Answer figures

Q12 A square sheet is folded as shown and then punched. How does it look when unfolded?



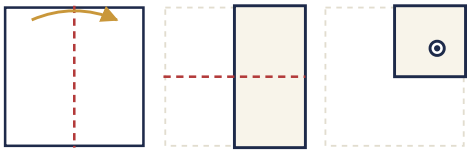
Folds, then punch



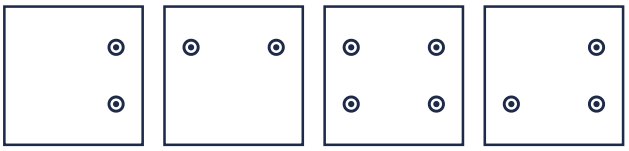
(a) (b) (c) (d)

Answer figures

Q13 A square sheet is folded as shown and then punched. How does it look when unfolded?



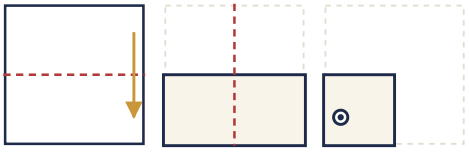
Folds, then punch



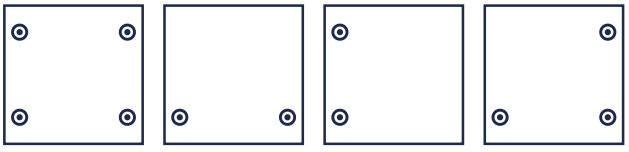
(a) (b) (c) (d)

Answer figures

Q14 A square sheet is folded as shown and then punched. How does it look when unfolded?



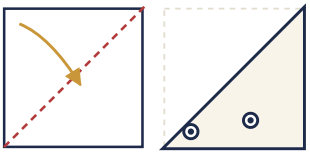
Folds, then punch



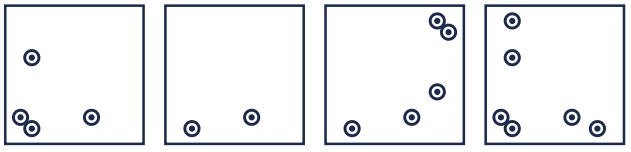
(a) (b) (c) (d)

Answer figures

Q15 A square sheet is folded as shown and then punched. How does it look when unfolded?



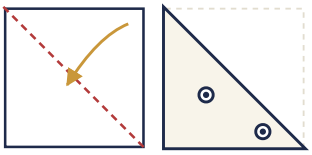
Folds, then punch



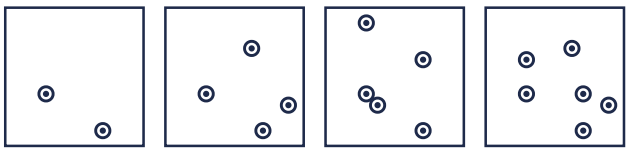
(a) (b) (c) (d)

Answer figures

Q16 A square sheet is folded as shown and then punched. How does it look when unfolded?



Folds, then punch

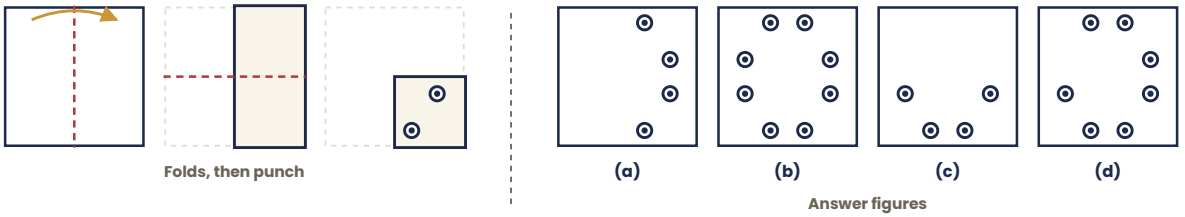


(a) (b) (c) (d)

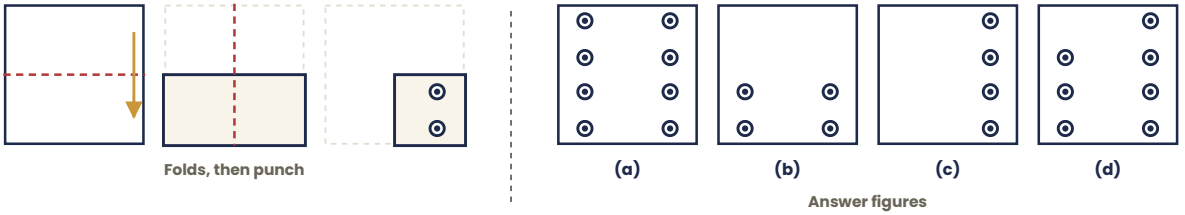
Answer figures

11.4.3 Advanced level

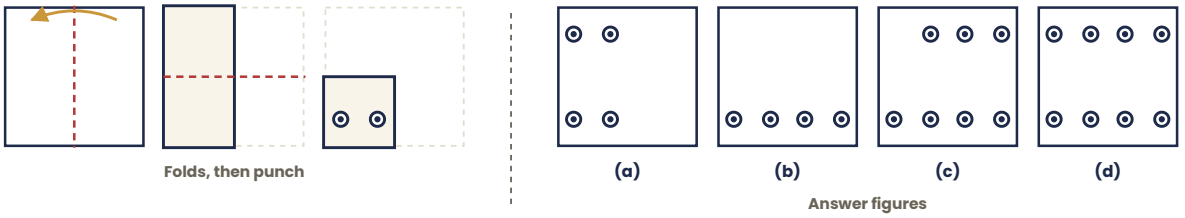
Q17 A square sheet is folded as shown and then punched. How does it look when unfolded?



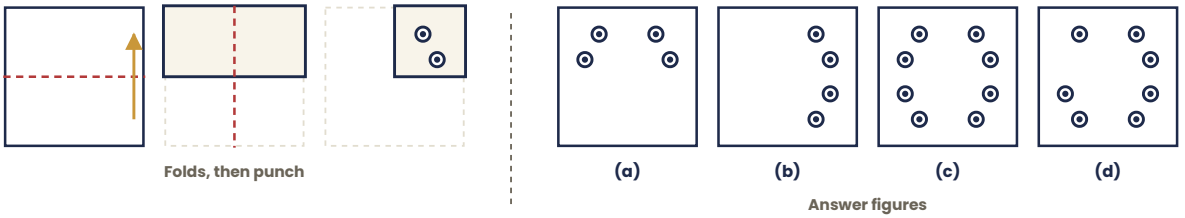
Q18 A square sheet is folded as shown and then punched. How does it look when unfolded?



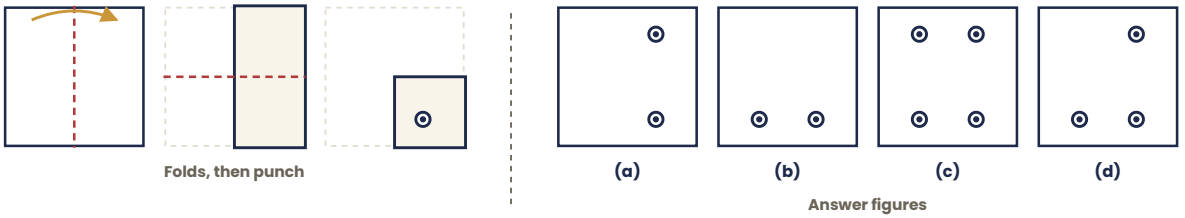
Q19 A square sheet is folded as shown and then punched. How does it look when unfolded?



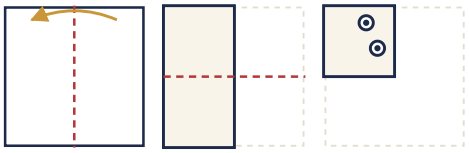
Q20 A square sheet is folded as shown and then punched. How does it look when unfolded?



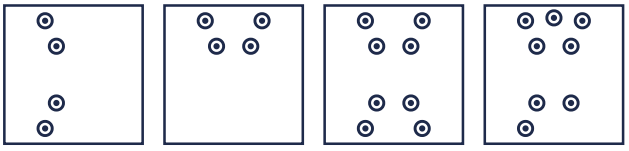
Q21 A square sheet is folded as shown and then punched. How does it look when unfolded?



Q22 A square sheet is folded as shown and then punched. How does it look when unfolded?



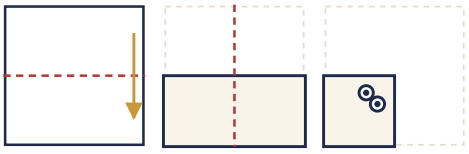
Folds, then punch



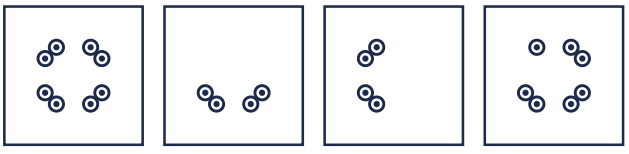
(a) (b) (c) (d)

Answer figures

Q23 A square sheet is folded as shown and then punched. How does it look when unfolded?



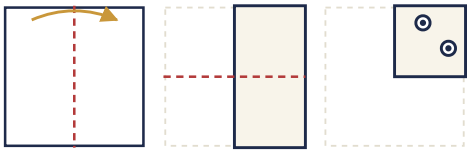
Folds, then punch



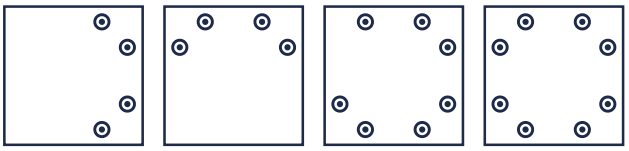
(a) (b) (c) (d)

Answer figures

Q24 A square sheet is folded as shown and then punched. How does it look when unfolded?



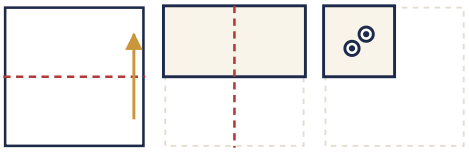
Folds, then punch



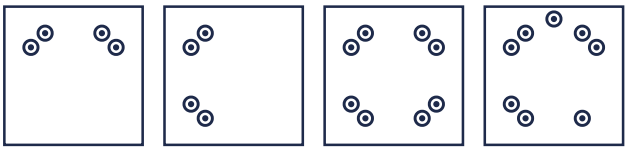
(a) (b) (c) (d)

Answer figures

Q25 A square sheet is folded as shown and then punched. How does it look when unfolded?



Folds, then punch



(a) (b) (c) (d)

Answer figures

11.5 Solutions

Q1 **Answer: (b)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 1 punch produce 2 holes: answer **(b)**.

Q2 **Answer: (d)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 1 punch produce 2 holes: answer **(d)**.

Q3 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 2 punches produce 4 holes: answer **(c)**.

Q4 **Answer: (a)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 2 punches produce 4 holes: answer **(a)**.

Q5 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 2 punches produce 4 holes: answer **(c)**.

Q6 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 2 punches produce 4 holes: answer **(c)**.

Q7 **Answer: (a)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 1 punch produce 2 holes: answer **(a)**.

Q8 **Answer: (d)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 1 punch produce 2 holes: answer **(d)**.

Q9 **Answer: (a)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(a)**.

Q10 **Answer: (b)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(b)**.

Q11 **Answer: (d)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(d)**.

Q12 **Answer: (d)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(d)**.

Q13 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(c)**.

Q14 **Answer: (a)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(a)**.

Q15 **Answer: (a)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 2 punches produce 4 holes: answer **(a)**.

Q16 **Answer: (b)**

Unfold in reverse order, reflecting every hole across each fold line. 1 fold gives up to 2 layers, so the 2 punches produce 4 holes: answer **(b)**.

Q17 **Answer: (b)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(b)**.

Q18 **Answer: (a)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(a)**.

Q19 **Answer: (d)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(d)**.

Q20 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(c)**.

Q21 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(c)**.

Q22 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(c)**.

Q23 **Answer: (a)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(a)**.

Q24 **Answer: (d)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(d)**.

Q25 **Answer: (c)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 2 punches produce 8 holes: answer **(c)**.

Each fold doubles the layers, so one punch makes 2, 4 or 8 holes. Unfold in reverse order, reflecting every hole (or cut) across each fold line, and use the hole count and the symmetry about each fold to eliminate options.

MODULE 12 · INTERMEDIATE

Counting figures

Minimum duration **2 days** | Level **Intermediate**

Counting questions ask how many triangles, squares, rectangles or lines a figure contains. This module covers the systematic labelling method, formulas for standard patterns (lines from a vertex, n × n grids, m × n grids), counting straight lines, and other shapes, followed by 25 solved questions in which every count was produced by a program that finds all intersection points and tests every combination.

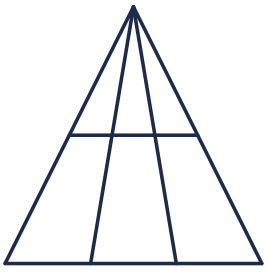
IN THIS MODULE

12.1 Counting triangles
12.2 Counting squares and rectangles
12.3 Counting straight lines
12.4 Counting other shapes
12.5 Method
12.6 Practice questions
12.7 Solutions

12.1 Counting triangles

12.1.1 Systematic labelling method

Label every corner and every intersection with a letter. Then count triangles in an organised way: first the smallest (single regions), then those made of two regions, three regions and so on; or list all triangles that use vertex A, then those using B but not A, and so on.



LABEL THE POINTS, THEN COUNT BY SIZE

12.1.2 Formulas for standard patterns

A triangle divided by lines from one vertex into n small triangles contains n(n + 1)/2 triangles; with one horizontal line across it, double that

Figure	Triangles
triangle with 1 lines from the apex	3
triangle with 2 lines from the apex	6

Figure	Triangles
triangle with 3 lines from the apex	10
triangle with 4 lines from the apex	15
same with one line across (k = 1)	6
same with one line across (k = 2)	12
same with one line across (k = 3)	20
square with both diagonals	8
square with diagonals and midlines	16

12.2 Counting squares and rectangles

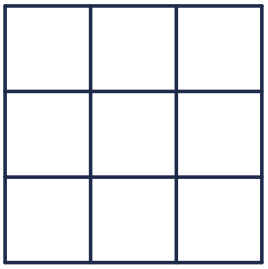
12.2.1 Squares in an n × n grid

Squares in an n × n grid = $1^2 + 2^2 + \ldots + n^2 = n(n + 1)(2n + 1)/6$

12.2.2 Rectangles in an m × n grid

Rectangles (including squares) in an m × n grid = $C(m + 1, 2) \times C(n + 1, 2)$

Grid	Squares	Rectangles
2 × 2	5	9
3 × 3	14	36
4 × 4	30	100
3 × 2	8	18
4 × 2	11	30



A 3 × 3 GRID: 14 SQUARES, 36 RECTANGLES

12.3 Counting straight lines

12.3.1 Horizontal, vertical and slanting lines

Count horizontal lines, then vertical lines, then each family of slanting lines. A straight line that passes through several points counts once, however many segments it is drawn as.

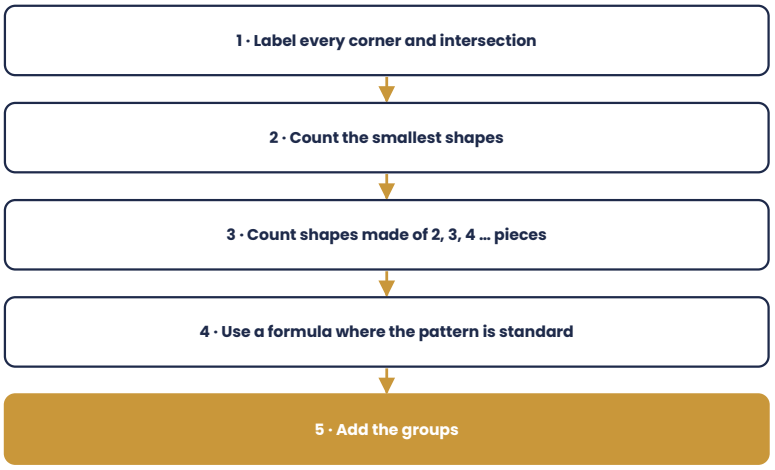
12.4 Counting other shapes

12.4.1 Circles, parallelograms and trapeziums

Use the same labelling method. For parallelograms, pick pairs of parallel line families: in a figure with p lines in one direction and q in another, crossing each other, there are C(p, 2) × C(q, 2) parallelograms.

12.5 Method

12.5.1 Label and count systematically



THE COUNTING METHOD

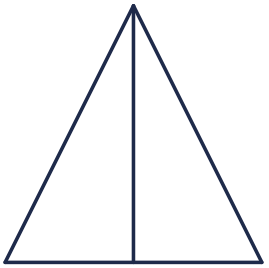
WATCH OUT
Guessing is the main cause of errors here. A systematic list may take a minute longer but is almost always right.

12.6 Practice questions

12.6.1 Basic level

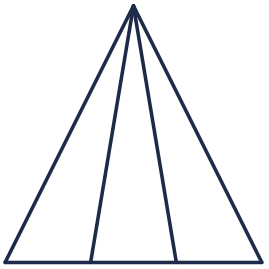
Q1 How many triangles are there in the figure?

(a) 1 **(b)** 5 **(c)** 7 **(d)** 3



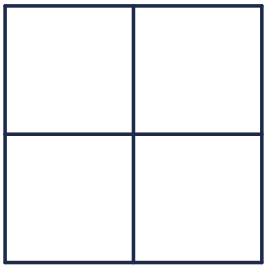
Q2 How many triangles are there in the figure?

(a) 6 **(b)** 4 **(c)** 8 **(d)** 10



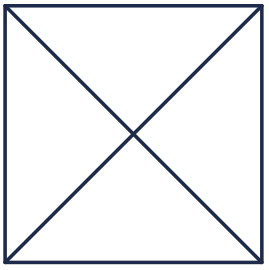
Q3 How many squares are there in the figure?

(a) 3 **(b)** 7 **(c)** 9 **(d)** 5



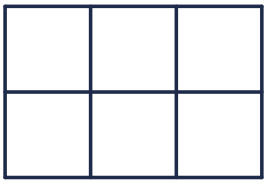
Q4 How many triangles are there in the figure?

(a) 6 **(b)** 10 **(c)** 12 **(d)** 8



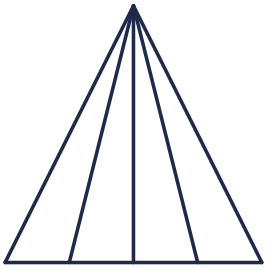
Q5 How many rectangles are there in the figure?

(a) 16 **(b)** 20 **(c)** 22 **(d)** 18



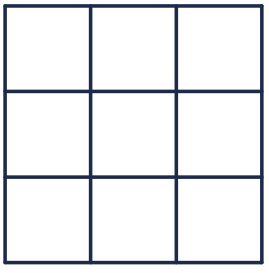
Q6 How many triangles are there in the figure?

(a) 8 **(b)** 12 **(c)** 14 **(d)** 10



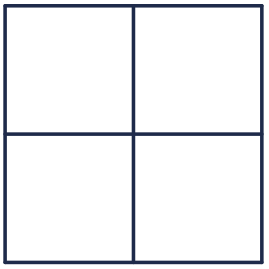
Q7 How many squares are there in the figure?

(a) 12 **(b)** 16 **(c)** 14 **(d)** 18



Q8 How many straight lines are there in the figure?

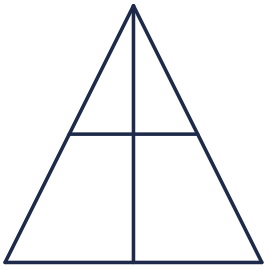
(a) 6 **(b)** 4 **(c)** 8 **(d)** 10



12.6.2 Intermediate level

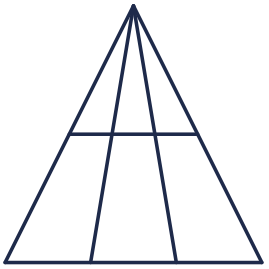
Q9 How many triangles are there in the figure?

(a) 4 **(b)** 8 **(c)** 10 **(d)** 6



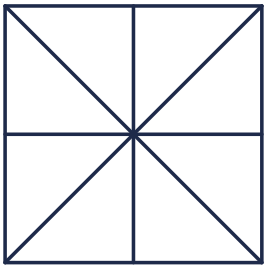
Q10 How many triangles are there in the figure?

(a) 10 **(b)** 12 **(c)** 14 **(d)** 16



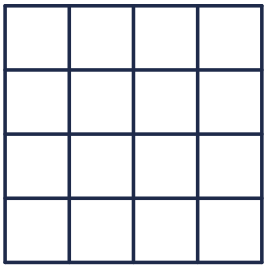
Q11 How many triangles are there in the figure?

(a) 14 **(b)** 16 **(c)** 18 **(d)** 20



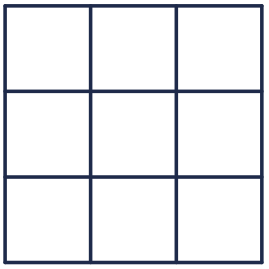
Q12 How many squares are there in the figure?

(a) 28 **(b)** 32 **(c)** 30 **(d)** 34



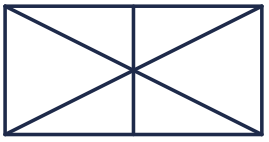
Q13 How many rectangles are there in the figure?

(a) 34 **(b)** 38 **(c)** 36 **(d)** 40



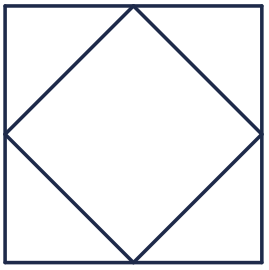
Q14 How many triangles are there in the figure?

(a) 10 **(b)** 14 **(c)** 16 **(d)** 12



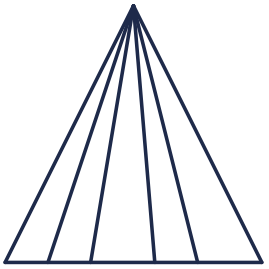
Q15 How many triangles are there in the figure?

(a) 4 **(b)** 2 **(c)** 6 **(d)** 8



Q16 How many triangles are there in the figure?

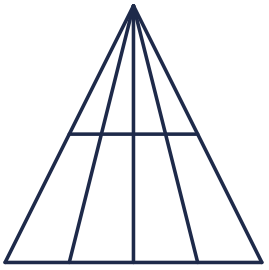
(a) 15 **(b)** 13 **(c)** 17 **(d)** 19



12.6.3 Advanced level

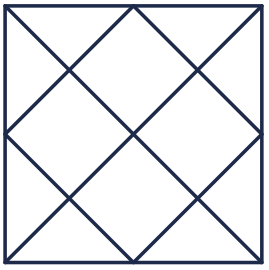
Q17 How many triangles are there in the figure?

(a) 18 **(b)** 22 **(c)** 20 **(d)** 24



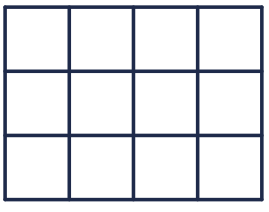
Q18 How many triangles are there in the figure?

(a) 18 **(b)** 22 **(c)** 24 **(d)** 20



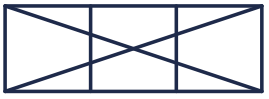
Q19 How many rectangles are there in the figure?

(a) 58 **(b)** 62 **(c)** 64 **(d)** 60



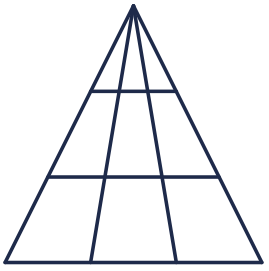
Q20 How many triangles are there in the figure?

(a) 18 **(b)** 16 **(c)** 20 **(d)** 22



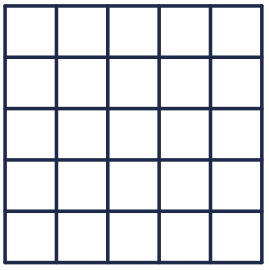
Q21 How many triangles are there in the figure?

(a) 18 **(b)** 16 **(c)** 20 **(d)** 22



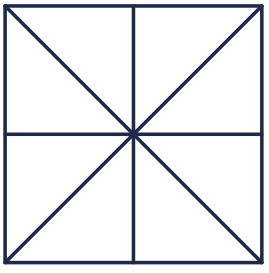
Q22 How many squares are there in the figure?

(a) 53 **(b)** 55 **(c)** 57 **(d)** 59



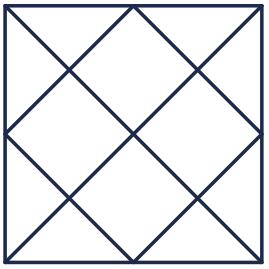
Q23 How many squares are there in the figure?

(a) 3 **(b)** 5 **(c)** 7 **(d)** 9



Q24 How many straight lines are there in the figure?

(a) 8 **(b)** 12 **(c)** 14 **(d)** 10



Q25 How many rectangles are there in the figure?

(a) 30 **(b)** 28 **(c)** 32 **(d)** 34



12.7 Solutions

Q1 **Answer: (d) 3**

The 1 line from the apex split the base into 2 parts, so $n(n + 1)/2 = 2\times3/2 = 3$ triangles. Labelling every point and testing every combination gives **3** triangles.

Q2 **Answer: (a) 6**

The 2 lines from the apex split the base into 3 parts, so $n(n + 1)/2 = 3\times4/2 = 6$ triangles. Labelling every point and testing every combination gives **6** triangles.

Q3 **Answer: (d) 5**

Squares in a 2 × 2 grid = $1^2 + 2^2 = 5$. Labelling every point and testing every combination gives **5** squares.

Q4 **Answer: (d) 8**

The two diagonals cut the square into 4 small triangles; each half of the square (cut by one diagonal) gives 4 more: 4 + 4 = 8. Labelling every point and testing every combination gives **8** triangles.

Q5 **Answer: (d) 18**

Rectangles = $C(4, 2) \times C(3, 2) = 6 \times 3 = 18$. Labelling every point and testing every combination gives **18** rectangles.

Q6 **Answer: (d) 10**

The 3 lines from the apex split the base into 4 parts, so $n(n + 1)/2 = 4 \times 5/2 = 10$ triangles. Labelling every point and testing every combination gives **10** triangles.

Q7 **Answer: (c) 14**

Squares in a 3 × 3 grid = $1^2 + 2^2 + 3^2 = 14$. Labelling every point and testing every combination gives **14** squares.

Q8 **Answer: (a) 6**

3 horizontal + 3 vertical lines. Labelling every point and testing every combination gives **6** straight lines.

Q9 **Answer: (d) 6**

Each of the 2 horizontal levels (base and 1 bar) forms its own fan of 3 triangles: 2 × 3 = 6. Labelling every point and testing every combination gives **6** triangles.

Q10 **Answer: (b) 12**

Each of the 2 horizontal levels (base and 1 bar) forms its own fan of 6 triangles: 2 × 6 = 12. Labelling every point and testing every combination gives **12** triangles.

Q11 **Answer: (b) 16**

8 one-piece triangles, 4 two-piece triangles standing on each side of the square, and 4 half-square triangles: 8 + 4 + 4 = 16. Labelling every point and testing every combination gives **16** triangles.

Q12 **Answer: (c) 30**

Squares in a 4 × 4 grid = $1^2 + 2^2 + 3^2 + 4^2 = 30$. Labelling every point and testing every combination gives **30** squares.

Q13 **Answer: (c) 36**

Rectangles = $C(4, 2) \times C(4, 2) = 6 \times 6 = 36$. Labelling every point and testing every combination gives **36** rectangles.

Q14 **Answer: (d) 12**

Count the small triangles round each crossing point, then triangles made of two or more pieces, working from left to right. Labelling every point and testing every combination gives **12** triangles.

Q15 **Answer: (a) 4**

Only the 4 corner triangles are triangles; the inner diamond is a square standing on its corner. Labelling every point and testing every combination gives **4** triangles.

Q16 **Answer: (a) 15**

The 4 lines from the apex split the base into 5 parts, so $n(n + 1)/2 = 5 \times 6/2 = 15$ triangles. Labelling every point and testing every combination gives **15** triangles.

Q17 **Answer: (c) 20**

Each of the 2 horizontal levels (base and 1 bar) forms its own fan of 10 triangles: 2 × 10 = 20. Labelling every point and testing every combination gives **20** triangles.

Q18 **Answer: (d) 20**

Count by size: smallest pieces first, then triangles made of 2 pieces, then 4 pieces, then the half-square triangles cut by each diagonal. Labelling every point and testing every combination gives **20** triangles.

Q19 **Answer: (d) 60**

Rectangles = C(5, 2) × C(4, 2) = 10 × 6 = 60. Labelling every point and testing every combination gives **60** rectangles.

Q20 **Answer: (a) 18**

Count the small triangles round each crossing point, then triangles made of two or more pieces, working from left to right. Labelling every point and testing every combination gives **18** triangles.

Q21 **Answer: (a) 18**

Each of the 3 horizontal levels (base and 2 bars) forms its own fan of 6 triangles: 3 × 6 = 18. Labelling every point and testing every combination gives **18** triangles.

Q22 **Answer: (b) 55**

Squares in a 5 × 5 grid = $1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$. Labelling every point and testing every combination gives **55** squares.

Q23 **Answer: (b) 5**

The midlines give 4 small squares, plus the whole square: 4 + 1 = 5. Labelling every point and testing every combination gives **5** squares.

Q24 **Answer: (d) 10**

Lines: 4 sides of the square, 4 sides of the diamond, and 2 diagonals, each counted once. Labelling every point and testing every combination gives **10** straight lines.

Q25 **Answer: (a) 30**

Rectangles = C(5, 2) × C(3, 2) = 10 × 3 = 30. Labelling every point and testing every combination gives **30** rectangles.

Label every point and count systematically by size. Lines from a vertex dividing a triangle into n parts give $n(n + 1)/2$ triangles. An n × n grid has Σk^2 squares; an m × n grid has C(m + 1, 2) × C(n + 1, 2) rectangles. Count each straight line once.

MODULE 13 · INTERMEDIATE

Figure matrix

Minimum duration **1 day** | Level **Intermediate**

A figure matrix arranges figures in a 3 × 3 grid in which the rows, the columns or the diagonals follow rules; one cell is missing. This module covers row, column and diagonal rules, the common rule types (each element once per row, overlay addition and cancellation, progressive rotation or movement, counting rules), Raven-style matrices that combine two rules, and the rows-then-columns-then-diagonals method, followed by 25 solved questions generated from their rules.

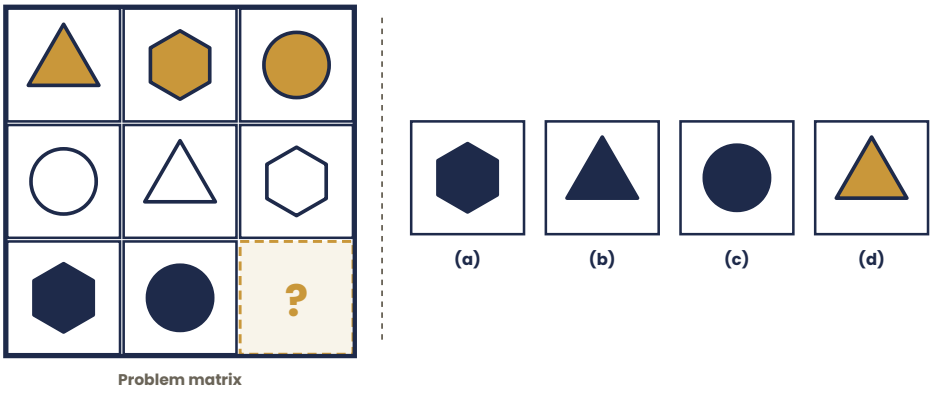
IN THIS MODULE

13.1 Matrix rules
13.2 Common rule types
13.3 Raven's Progressive Matrices style questions
13.4 Method
13.5 Practice questions
13.6 Solutions

13.1 Matrix rules

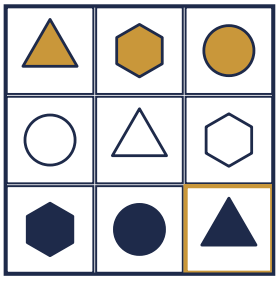
13.1.1 Row-wise rules

Most matrices are built row by row: the three figures in each row follow the same rule (for example, each shape once, or figure 3 = figure 1 + figure 2). Check row 1 and row 2; the rule that fits both completes row 3.



Problem matrix

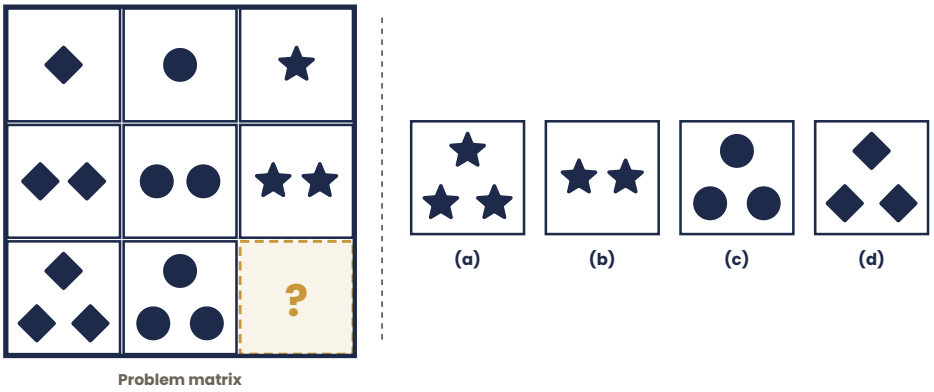
EACH ROW CONTAINS EACH SHAPE ONCE, WITH ONE SHADING PER ROW



COMPLETED MATRIX (MISSING CELL IN GOLD)

13.1.2 Column-wise rules

If the rows do not show a pattern, read down the columns. The same rule types appear: a fixed shape per column, a count that grows down the column, and so on.



Problem matrix

SHAPE FIXED IN EACH COLUMN, COUNT INCREASING DOWN THE COLUMNS

13.1.3 Diagonal rules

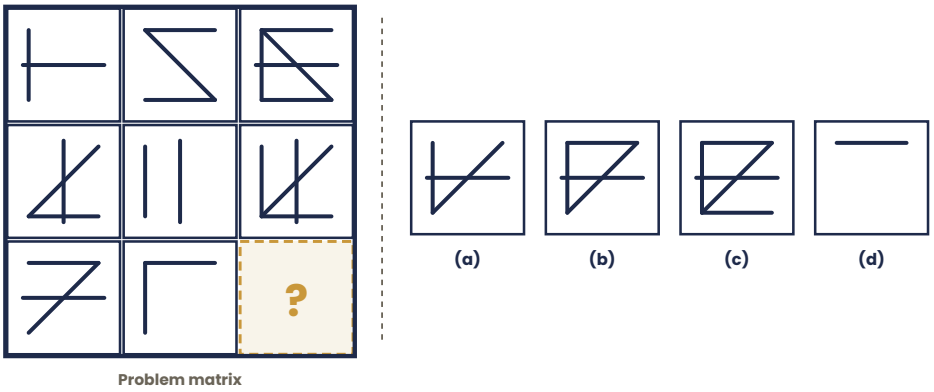
Sometimes identical figures lie along diagonals. Look at the three cells of the main diagonal (top-left, centre, bottom-right): if they match, the missing bottom-right cell copies them.

13.2 Common rule types

13.2.1 Each element appears once per row (Latin square)

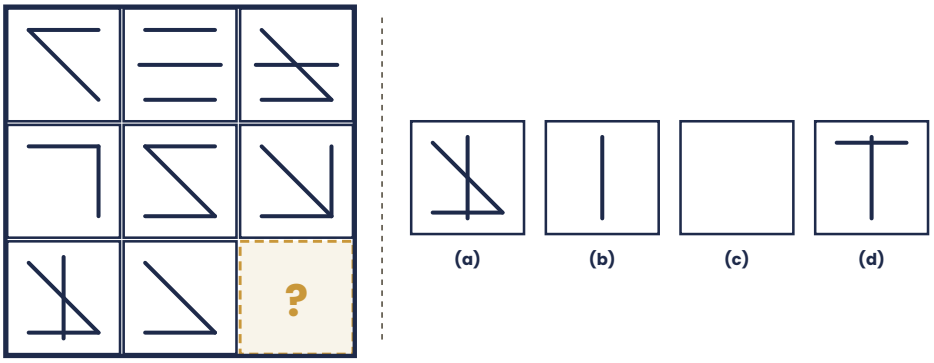
In a Latin-square matrix every element (shape, shading, count) appears exactly once in each row and each column. The missing cell is whichever element row 3 (and column 3) still lacks.

13.2.2 Addition and subtraction of figures (overlay)



Problem matrix

ADDITION: FIGURE 3 = FIGURE 1 + FIGURE 2



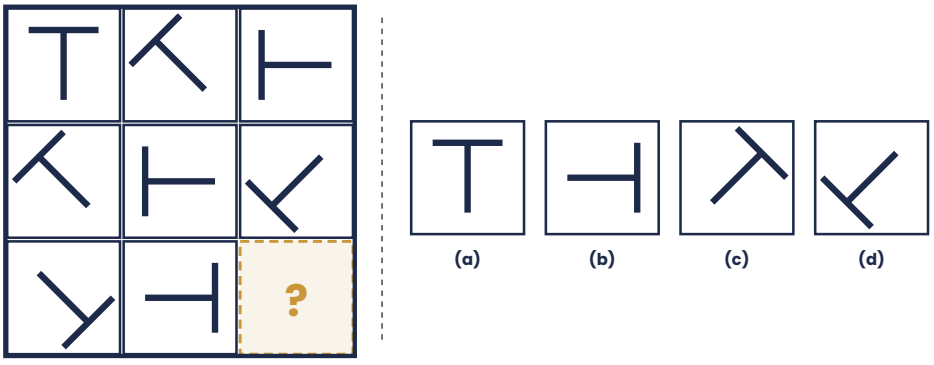
Problem matrix

CANCELLATION: COMMON LINES DISAPPEAR

SHORTCUT TRICK

For overlay matrices, check a single line at a time: if it is in figure 1 and figure 2, is it in figure 3? That one test tells you whether the rule is addition (yes) or cancellation (no).

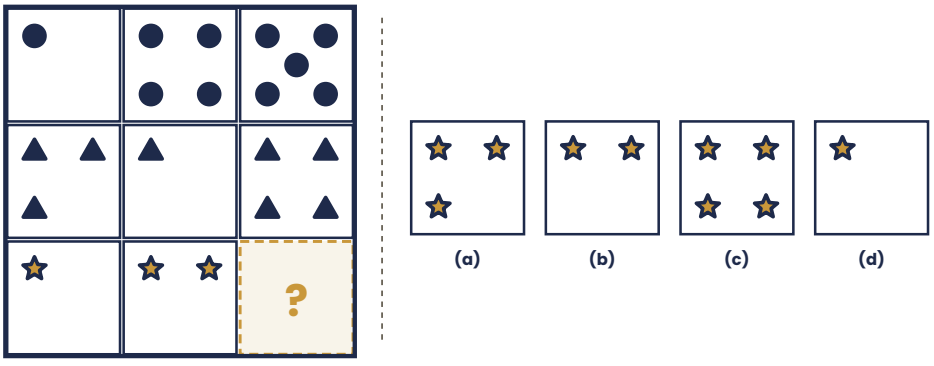
13.2.3 Progressive rotation or change



Problem matrix

THE ARROW TURNS BY THE SAME ANGLE ALONG EACH ROW

13.2.4 Counting-based rules

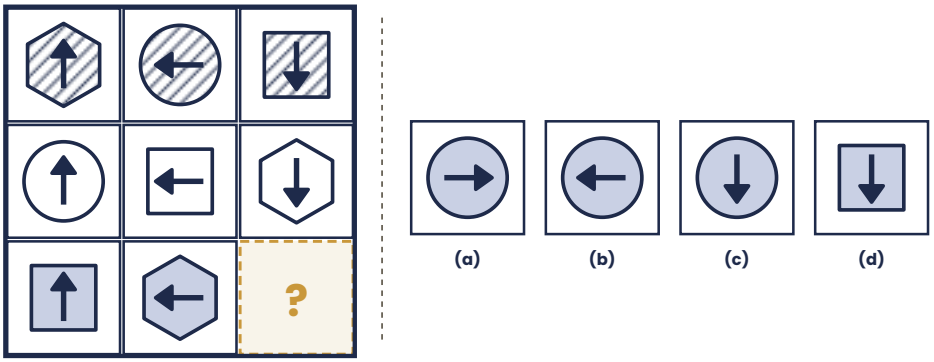


Problem matrix

COUNT IN COLUMN 3 = COLUMN 1 + COLUMN 2

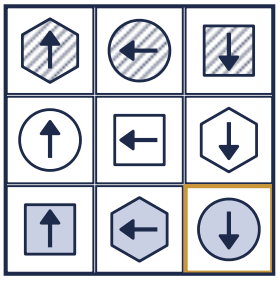
13.3 Raven's Progressive Matrices style questions

Raven's matrices combine two or three independent rules in one grid: for instance one rule for the outer shape, another for its shading and a third for an inner element. Separate the attributes and solve each one on its own.



Problem matrix

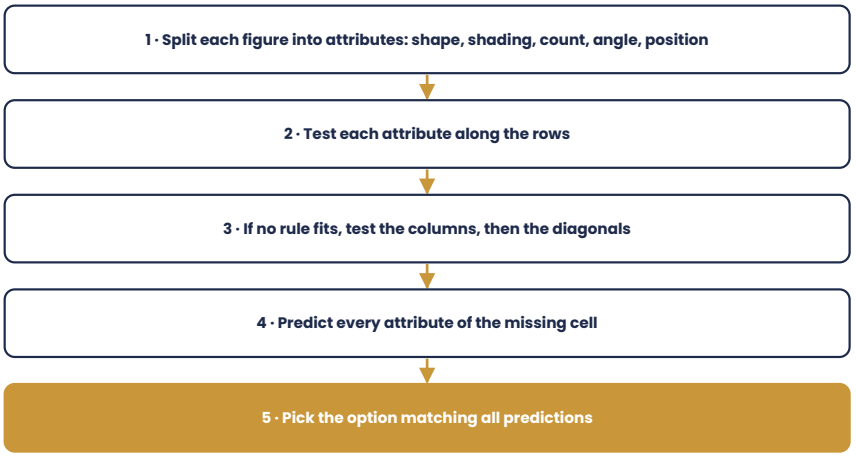
THREE INDEPENDENT RULES



COMPLETED MATRIX

13.4 Method

13.4.1 Test rows first, then columns, then diagonals



THE MATRIX METHOD

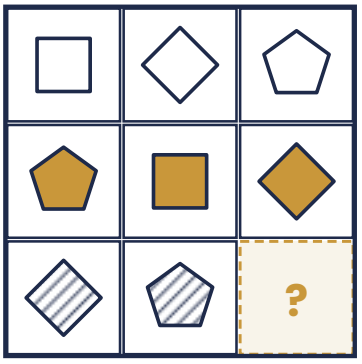
WATCH OUT

Do not stop at the first rule you find: an option may match the shape rule but break the shading or rotation rule. Check every attribute against the chosen option.

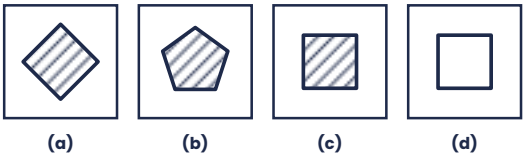
13.5 Practice questions

13.5.1 Basic level

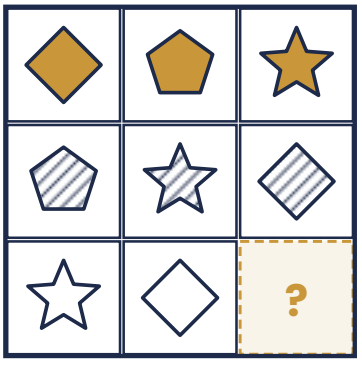
Q1 Which answer figure completes the matrix?



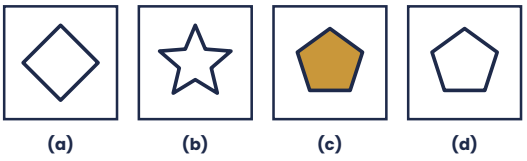
Problem matrix



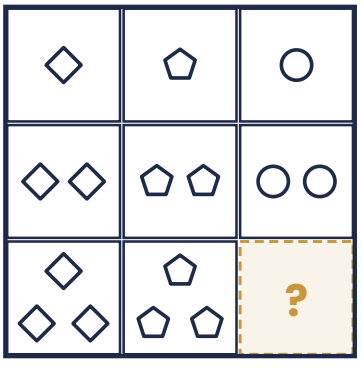
Q2 Which answer figure completes the matrix?



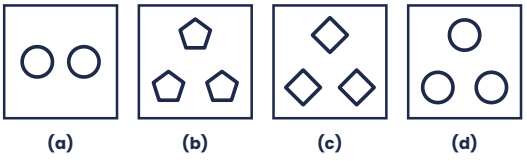
Problem matrix



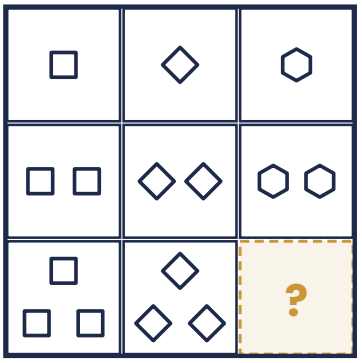
Q3 Which answer figure completes the matrix?



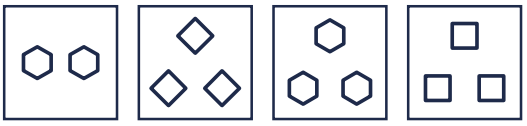
Problem matrix



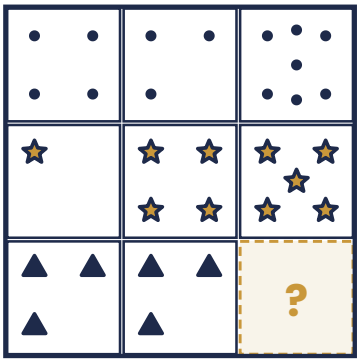
Q4 Which answer figure completes the matrix?



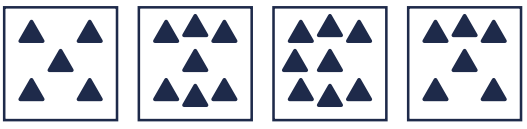
Problem matrix



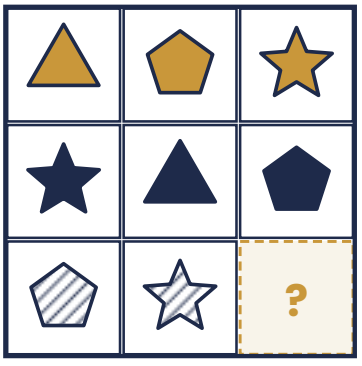
Q5 Which answer figure completes the matrix?



Problem matrix



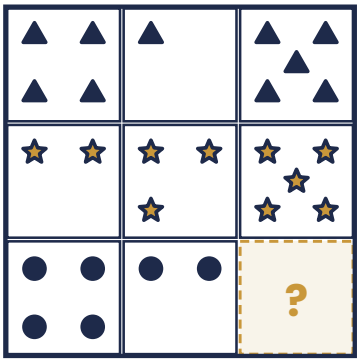
Q6 Which answer figure completes the matrix?



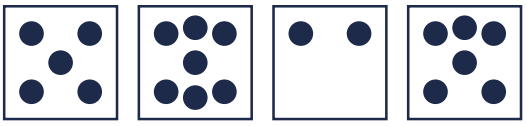
Problem matrix



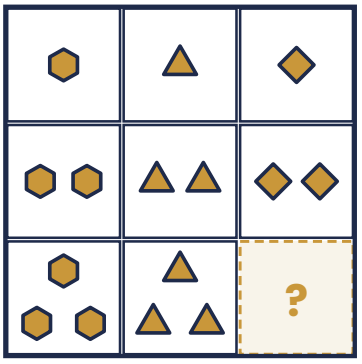
Q7 Which answer figure completes the matrix?



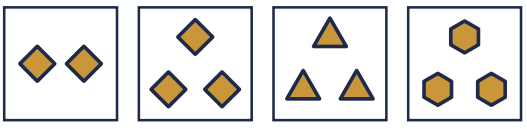
Problem matrix



Q8 Which answer figure completes the matrix?

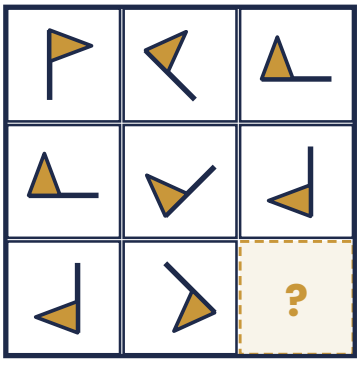


Problem matrix



13.5.2 Intermediate level

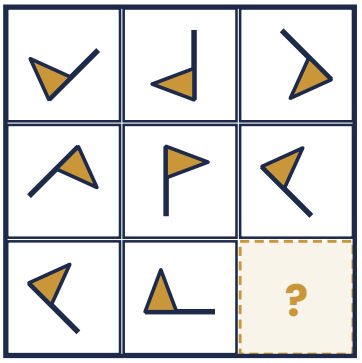
Q9 Which answer figure completes the matrix?



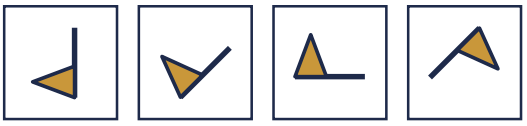
Problem matrix



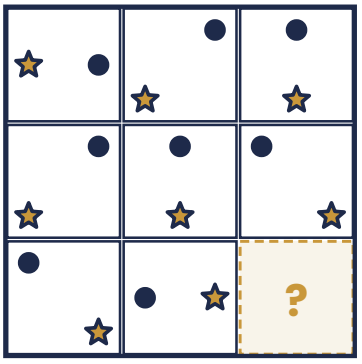
Q10 Which answer figure completes the matrix?



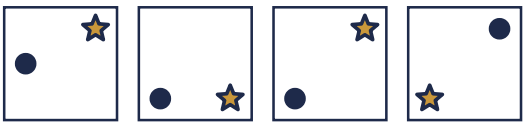
Problem matrix



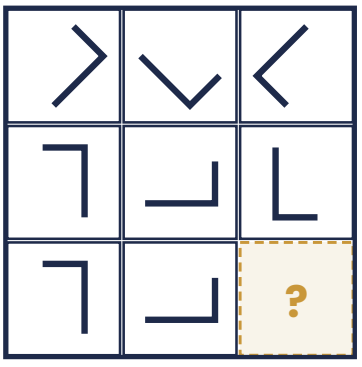
Q11 Which answer figure completes the matrix?



Problem matrix



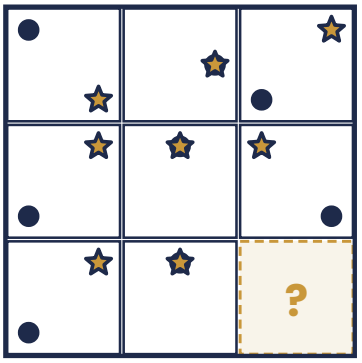
Q12 Which answer figure completes the matrix?



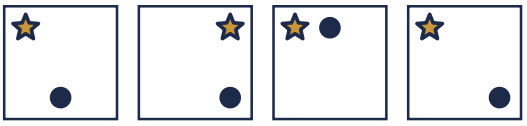
Problem matrix



Q13 Which answer figure completes the matrix?

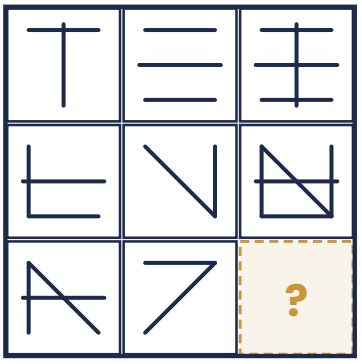


Problem matrix

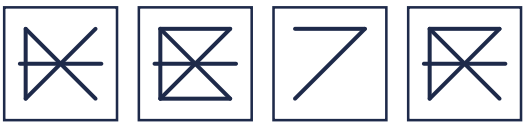


(a) (b) (c) (d)

Q14 Which answer figure completes the matrix?

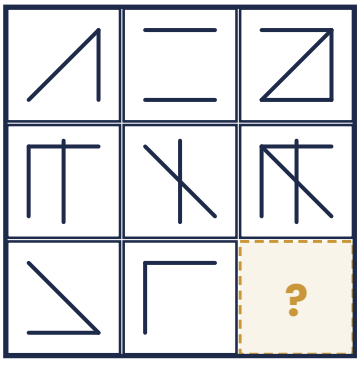


Problem matrix

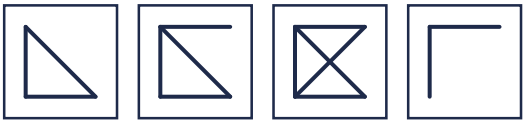


(a) (b) (c) (d)

Q15 Which answer figure completes the matrix?

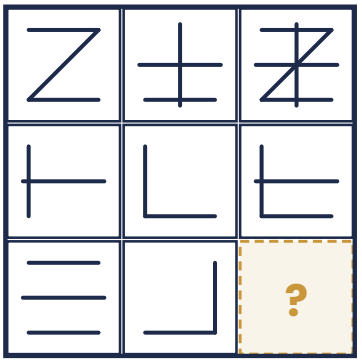


Problem matrix

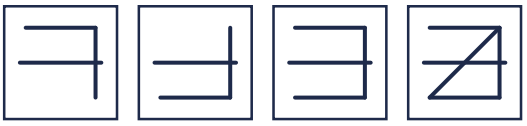


(a) (b) (c) (d)

Q16 Which answer figure completes the matrix?

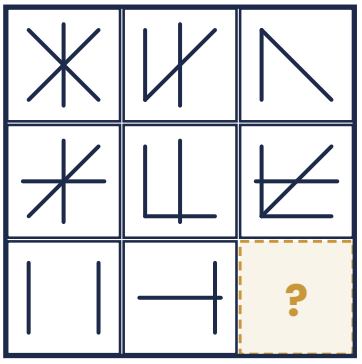


Problem matrix

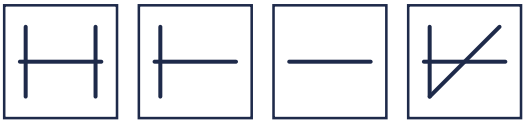


13.5.3 Advanced level

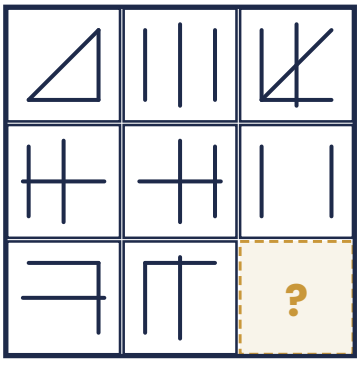
Q17 Which answer figure completes the matrix?



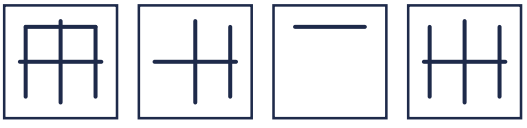
Problem matrix



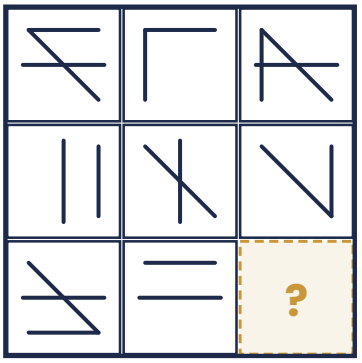
Q18 Which answer figure completes the matrix?



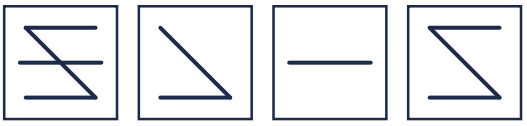
Problem matrix



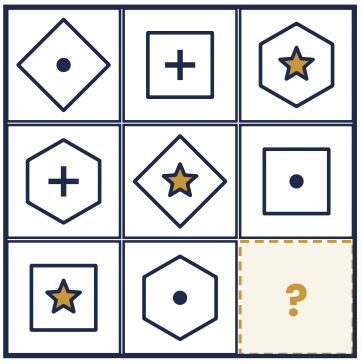
Q19 Which answer figure completes the matrix?



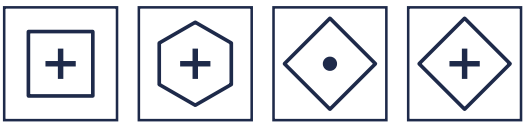
Problem matrix



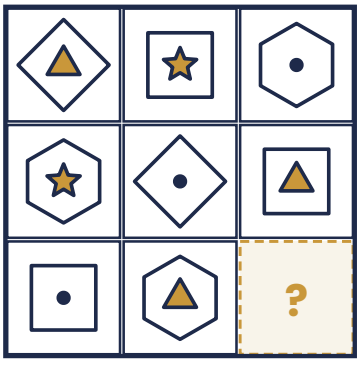
Q20 Which answer figure completes the matrix?



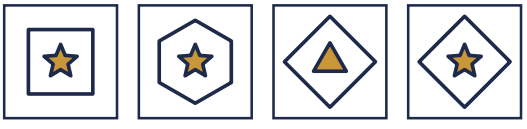
Problem matrix



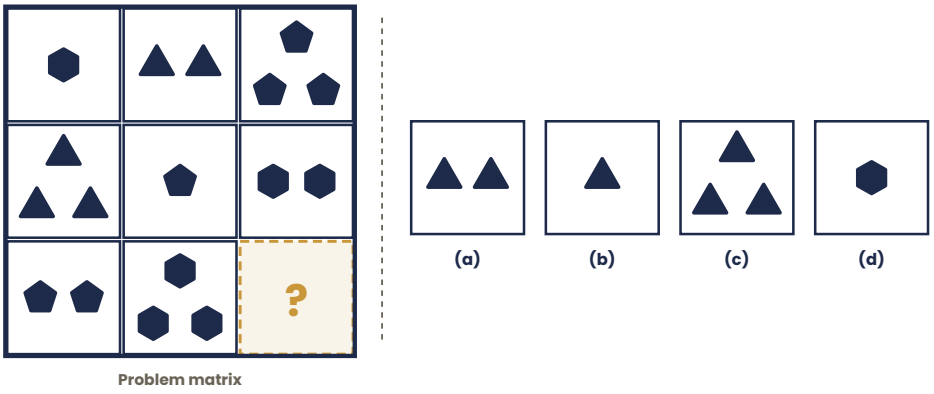
Q21 Which answer figure completes the matrix?



Problem matrix

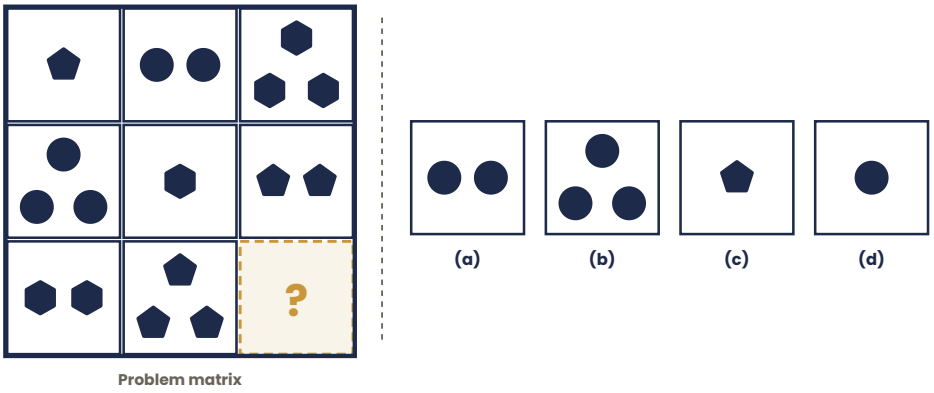


Q22 Which answer figure completes the matrix?



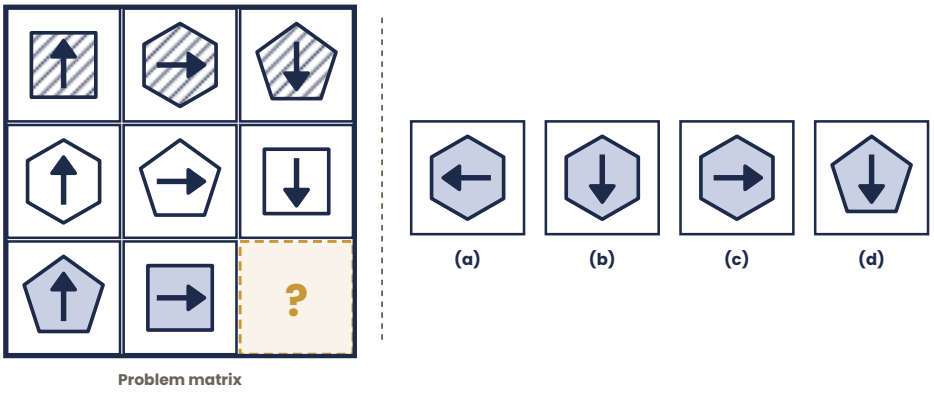
Problem matrix

Q23 Which answer figure completes the matrix?



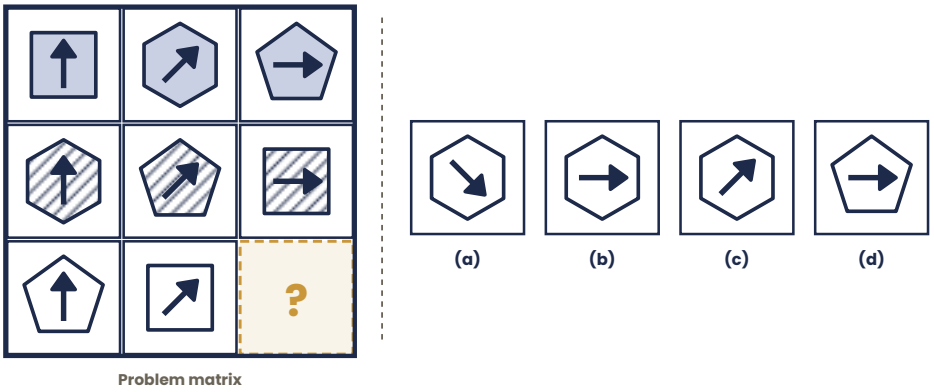
Problem matrix

Q24 Which answer figure completes the matrix?



Problem matrix

Q25 Which answer figure completes the matrix?

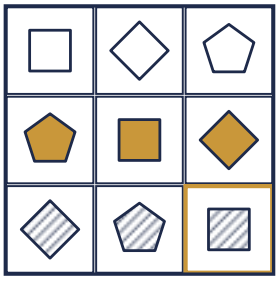


Problem matrix

13.6 Solutions

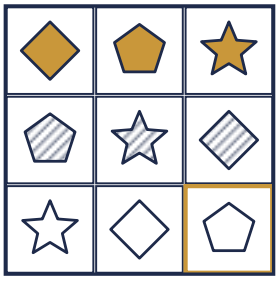
Q1 **Answer: (c)**

Each row contains a square, a diamond and a pentagon exactly once, and all the figures in a row have the same shading. Row 3 is missing the square, shaded striped: **(c)**.



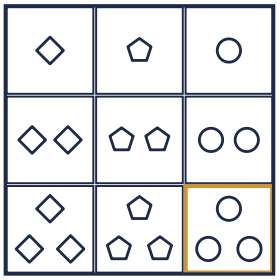
Q2 **Answer: (d)**

Each row contains a diamond, a pentagon and a star exactly once, and all the figures in a row have the same shading. Row 3 is missing the pentagon, shaded white: **(d)**.



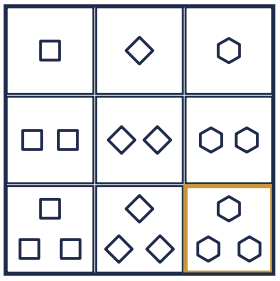
Q3 **Answer: (d)**

Work down the columns: each column keeps one shape (diamond, pentagon, circle), and the number of copies is 1 in row 1, 2 in row 2 and 3 in row 3. The missing cell has three circles: **(d)**.



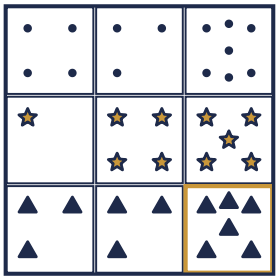
Q4 **Answer: (c)**

Work down the columns: each column keeps one shape (square, diamond, hexagon), and the number of copies is 1 in row 1, 2 in row 2 and 3 in row 3. The missing cell has three hexagons: **(c)**.



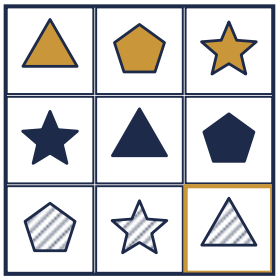
Q5 **Answer: (d)**

In each row the third figure has as many elements as the first two together: 4 + 3 = 7, 1 + 4 = 5. So the missing cell has 3 + 3 = 6: **(d)**.



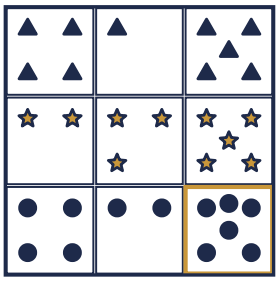
Q6 **Answer: (b)**

Each row contains a triangle, a pentagon and a star exactly once, and all the figures in a row have the same shading. Row 3 is missing the triangle, shaded striped: **(b)**.



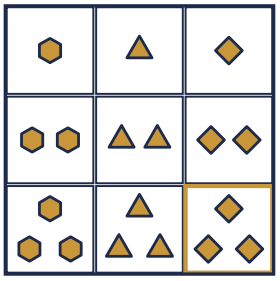
Q7 **Answer: (d)**

In each row the third figure has as many elements as the first two together: 4 + 1 = 5, 2 + 3 = 5. So the missing cell has 4 + 2 = 6: **(d)**.



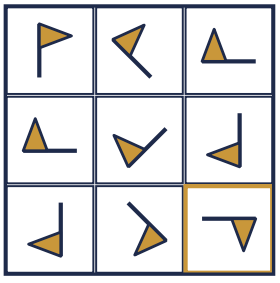
Q8 **Answer: (b)**

Work down the columns: each column keeps one shape (hexagon, triangle, diamond), and the number of copies is 1 in row 1, 2 in row 2 and 3 in row 3. The missing cell has three diamonds: **(b)**.



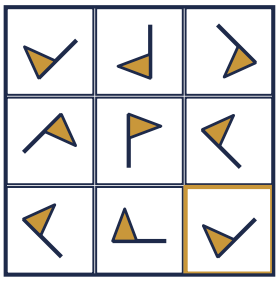
Q9 **Answer: (b)**

Along every row the flag turns 45° anticlockwise from one cell to the next. Turning the second figure of row 3 once more gives **(b)**.



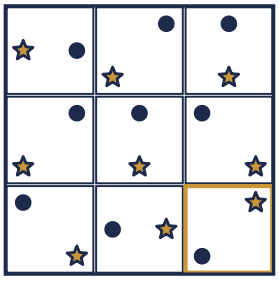
Q10 **Answer: (b)**

Along every row the flag turns 45° anticlockwise from one cell to the next. Turning the second figure of row 3 once more gives **(b)**.



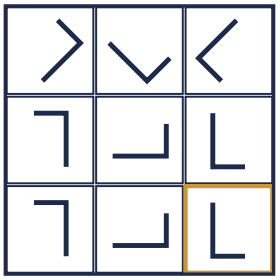
Q11 **Answer: (c)**

Along each row the black dot moves 1 place anticlockwise round the frame and the gold star moves 1 place anticlockwise. Continuing both movements in row 3 gives **(c)**.



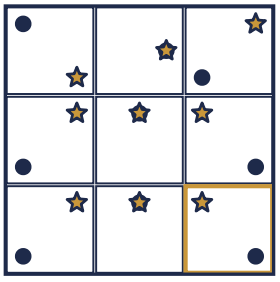
Q12 **Answer: (d)**

Along every row the L turns 90° clockwise from one cell to the next. Turning the second figure of row 3 once more gives **(d)**.



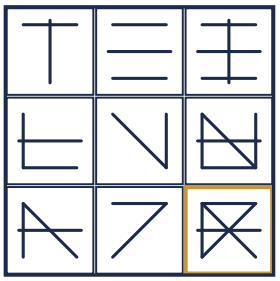
Q13 **Answer: (d)**

Along each row the black dot moves 3 places clockwise round the frame and the gold star moves 1 place anticlockwise. Continuing both movements in row 3 gives **(d)**.



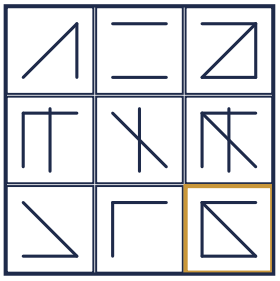
Q14 **Answer: (d)**

In each row the third figure is the first two figures placed on top of each other: every line in figure 1 or figure 2 appears in figure 3. Overlaying the first two figures of row 3 gives **(d)**.



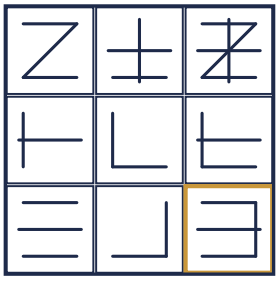
Q15 **Answer: (b)**

In each row the third figure is the first two figures placed on top of each other: every line in figure 1 or figure 2 appears in figure 3. Overlaying the first two figures of row 3 gives **(b)**.



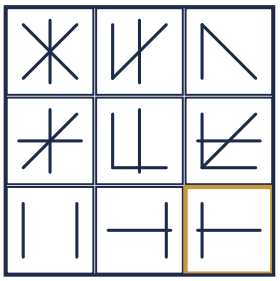
Q16 **Answer: (c)**

In each row the third figure is the first two figures placed on top of each other: every line in figure 1 or figure 2 appears in figure 3. Overlaying the first two figures of row 3 gives **(c)**.



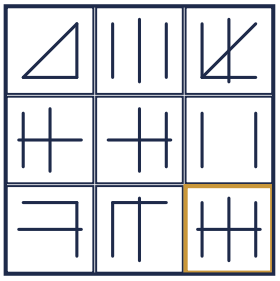
Q17 **Answer: (b)**

In each row the first two figures are overlaid, and lines that appear in **both** cancel out; lines in only one figure remain. Doing this for row 3 gives **(b)**.



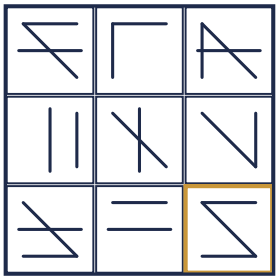
Q18 **Answer: (d)**

In each row the first two figures are overlaid, and lines that appear in **both** cancel out; lines in only one figure remain. Doing this for row 3 gives **(d)**.



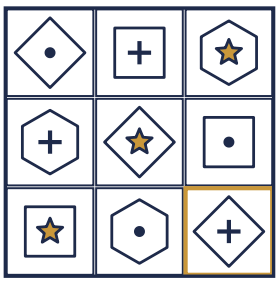
Q19 **Answer: (d)**

In each row the first two figures are overlaid, and lines that appear in **both** cancel out; lines in only one figure remain. Doing this for row 3 gives **(d)**.



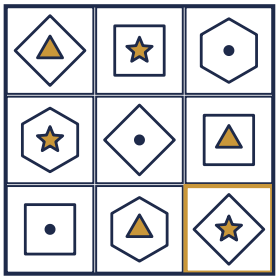
Q20 **Answer: (d)**

Outer shapes are constant along each diagonal running down to the right: the main diagonal (top-left, centre, bottom-right) is all diamonds. Inner shapes are constant along diagonals running down to the left, so each inner shape appears once in every row; row 3 already has the star and the dot, so the missing inner shape is the plus. Answer **(d)**.



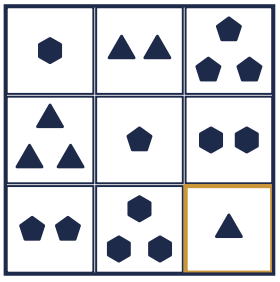
Q21 **Answer: (d)**

Outer shapes are constant along each diagonal running down to the right: the main diagonal (top-left, centre, bottom-right) is all diamonds. Inner shapes are constant along diagonals running down to the left, so each inner shape appears once in every row; row 3 already has the dot and the triangle, so the missing inner shape is the star. Answer **(d)**.



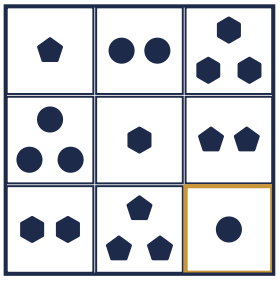
Q22 **Answer: (b)**

Two separate rules: each shape (hexagon, triangle, pentagon) appears once in every row and column, and each count (1, 2, 3) appears once in every row and column. Row 3 lacks the triangle and the count 1: **(b)**.



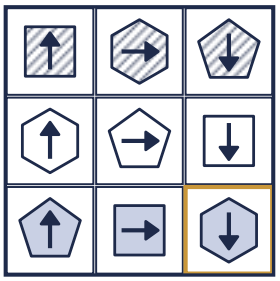
Q23 **Answer: (d)**

Two separate rules: each shape (pentagon, circle, hexagon) appears once in every row and column, and each count (1, 2, 3) appears once in every row and column. Row 3 lacks the circle and the count 1: **(d)**.



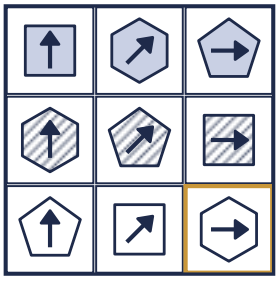
Q24 **Answer: (b)**

Three rules work together: (1) each outer shape appears once per row; (2) the shading of the outer shape is fixed for each row; (3) the arrow turns 90° clockwise from column to column (pointing up in column 1). Row 3 needs the hexagon, shaded like the rest of row 3, with the arrow turned 180°: **(b)**.



Q25 **Answer: (b)**

Three rules work together: (1) each outer shape appears once per row; (2) the shading of the outer shape is fixed for each row; (3) the arrow turns 45° clockwise from column to column (pointing up in column 1). Row 3 needs the hexagon, shaded like the rest of row 3, with the arrow turned 90°: **(b)**.



Break every figure into attributes and test each one along rows, then columns, then diagonals. The usual rules are: each element once per row (Latin square), overlay addition or cancellation, progressive rotation or movement, and counting. Raven-style matrices combine two or three of these.

MODULE 14 · INTERMEDIATE

Grouping of figures

Minimum duration **1 day** | Level **Intermediate**

Grouping questions show nine figures and ask you to divide them into three groups of three, each group sharing a common property. This module covers the properties used for grouping (shape, number of sides, number of elements, shading, curved or straight lines, open or closed, position of a small element, direction) and the find-the-obvious-property-then-check-every-figure method, followed by 25 solved questions whose groups were generated from a hidden property.

IN THIS MODULE

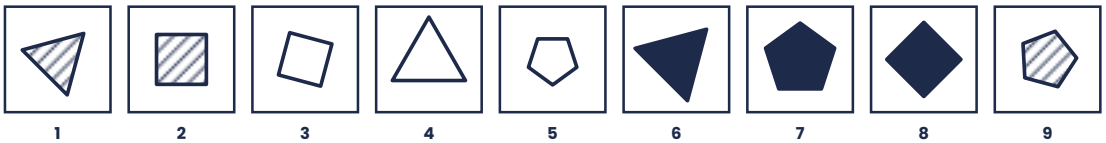
14.1 Concepts
14.2 Method
14.3 Practice questions
14.4 Solutions

14.1 Concepts

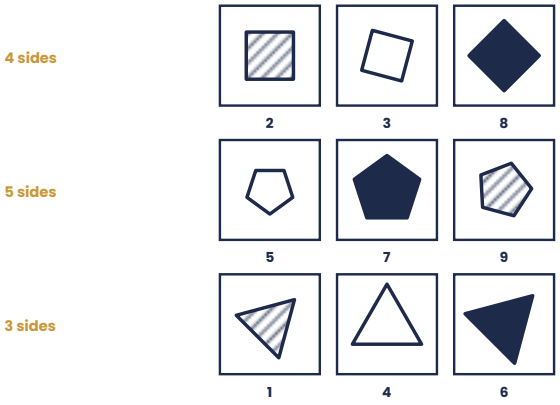
14.1.1 Grouping by shape, number of sides, shading, symmetry, open or closed

Nine figures are numbered 1 to 9. You must split them into three groups of three so that the figures in each group share one property that the other groups do not. The options list possible groupings such as "1, 4, 7; 2, 5, 8; 3, 6, 9".

Property	What to look at
Number of sides	triangles, quadrilaterals, pentagons … whatever their size or rotation
Number of elements	1, 2 or 3 parts in each figure
Shading	white, black, striped
Curved or straight	circles and arcs versus straight-sided shapes
Open or closed	L, T, + shapes versus closed outlines
Position	a small element inside, outside or on the boundary
Direction	which way an arrow or pointer faces
Symmetry	symmetric figures versus non-symmetric ones



GROUP THESE NINE FIGURES



THE GROUPS: NUMBER OF SIDES

14.2 Method

14.2.1 Find the most obvious shared property first

Look across all nine figures and ask what changes from figure to figure: shape, count, shading, position, direction. The property that takes exactly three values, each three times, is usually the grouping property.

14.2.2 Check that every figure fits exactly one group

Once you have a candidate property, write the group number under each figure. Each group must have exactly three figures and no figure may fit two groups.

SHORTCUT TRICK
Use the options. Take the first group in each option and test whether its three figures share a property. Usually only one option passes this test for its first group; check its other two groups to confirm.

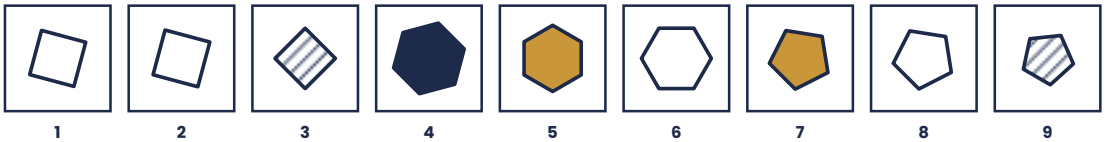
WATCH OUT
Size, rotation and a small decorative element are often varied on purpose to distract you. A triangle stays a triangle whether it is large, small, turned or shaded.

14.3 Practice questions

14.3.1 Basic level

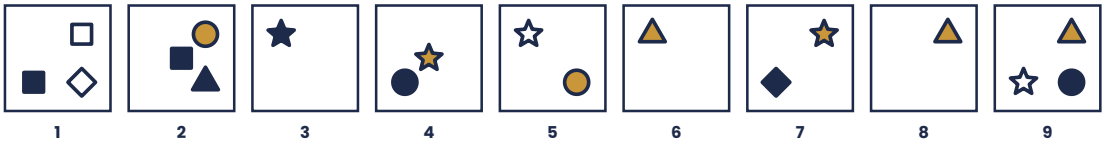
Q1 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 2, 7; 3, 8, 9; 4, 5, 6
(b) 1, 2, 3; 4, 5, 6; 7, 8, 9
(c) 1, 3, 9; 2, 4, 5; 6, 7, 8
(d) 1, 2, 3; 4, 5, 9; 6, 7, 8



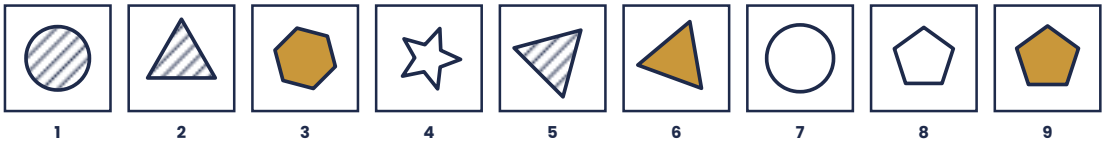
Q2 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 5, 7; 2, 4, 9; 3, 6, 8
(b) 1, 2, 9; 3, 4, 5; 6, 7, 8
(c) 1, 2, 9; 3, 4, 7; 5, 6, 8
(d) 1, 2, 9; 3, 6, 8; 4, 5, 7



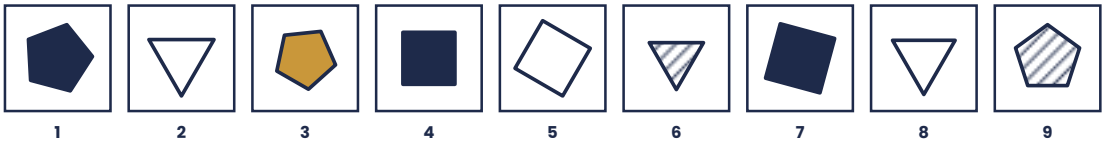
Q3 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 2, 5; 3, 6, 9; 4, 7, 8
(b) 1, 5, 6; 2, 7, 9; 3, 4, 8
(c) 1, 4, 9; 2, 7, 8; 3, 5, 6
(d) 1, 2, 4; 3, 6, 9; 5, 7, 8



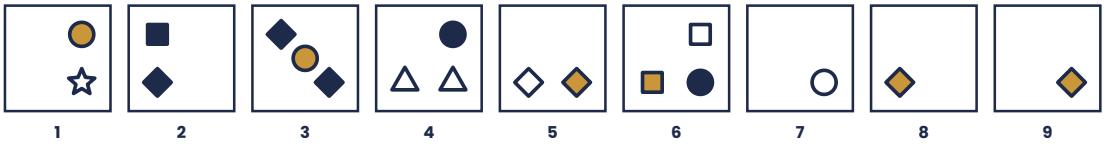
Q4 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 3, 7; 2, 6, 8; 4, 5, 9
(b) 1, 5, 9; 2, 4, 6; 3, 7, 8
(c) 1, 3, 9; 2, 6, 8; 4, 5, 7
(d) 1, 3, 9; 2, 6, 7; 4, 5, 8



Q5 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 2, 5; 3, 4, 6; 7, 8, 9
(b) 1, 3, 4; 2, 5, 6; 7, 8, 9
(c) 1, 2, 6; 3, 4, 5; 7, 8, 9
(d) 1, 3, 6; 2, 7, 9; 4, 5, 8



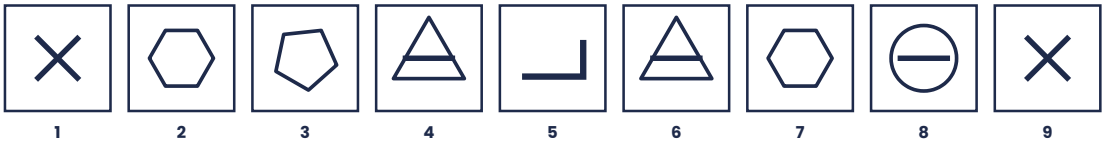
Q6 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 5, 6; 2, 3, 7; 4, 8, 9
(b) 1, 5, 6; 2, 3, 4; 7, 8, 9
(c) 1, 5, 6; 2, 4, 7; 3, 8, 9
(d) 1, 5, 6; 2, 7, 9; 3, 4, 8



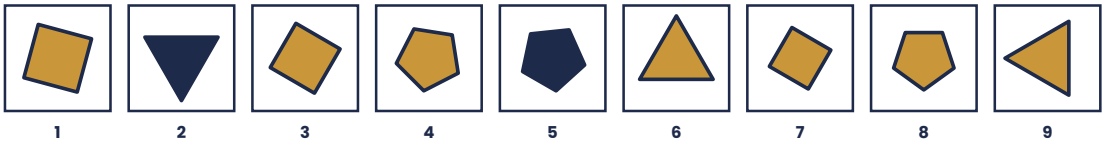
Q7 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 8, 9; 2, 3, 7; 4, 5, 6
(b) 1, 5, 9; 2, 3, 7; 4, 6, 8
(c) 1, 3, 9; 2, 5, 7; 4, 6, 8
(d) 1, 5, 9; 2, 4, 6; 3, 7, 8



Q8 Divide the nine figures into three groups of three, each group sharing a common property.

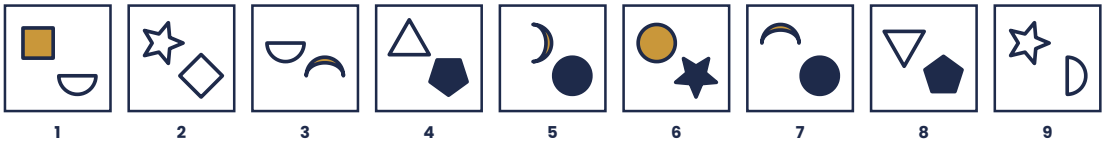
(a) 1, 3, 7; 2, 6, 9; 4, 5, 8
(b) 1, 4, 5; 2, 8, 9; 3, 6, 7
(c) 1, 2, 3; 4, 5, 8; 6, 7, 9
(d) 1, 2, 5; 3, 7, 8; 4, 6, 9



14.3.2 Intermediate level

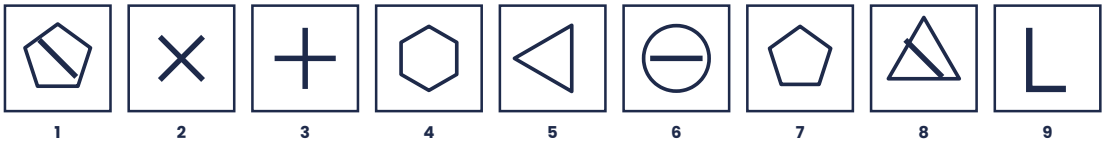
Q9 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 3, 7; 2, 4, 8; 5, 6, 9
(b) 1, 2, 6; 3, 4, 9; 5, 7, 8
(c) 1, 6, 9; 2, 7, 8; 3, 4, 5
(d) 1, 6, 9; 2, 4, 8; 3, 5, 7



Q10 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 6, 8; 2, 3, 9; 4, 5, 7
(b) 1, 3, 6; 2, 8, 9; 4, 5, 7
(c) 1, 4, 6; 2, 3, 9; 5, 7, 8
(d) 1, 6, 8; 2, 3, 4; 5, 7, 9



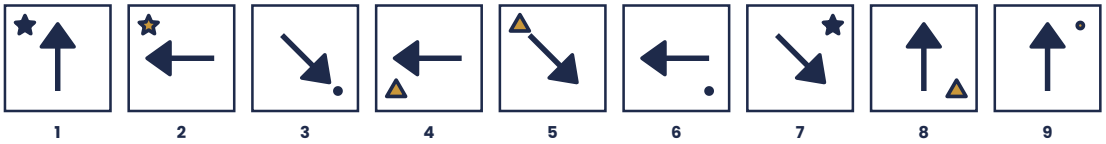
Q11 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 3, 5; 2, 6, 7; 4, 8, 9
(b) 1, 7, 9; 2, 4, 6; 3, 5, 8
(c) 1, 3, 5; 2, 4, 6; 7, 8, 9
(d) 1, 3, 5; 2, 6, 9; 4, 7, 8



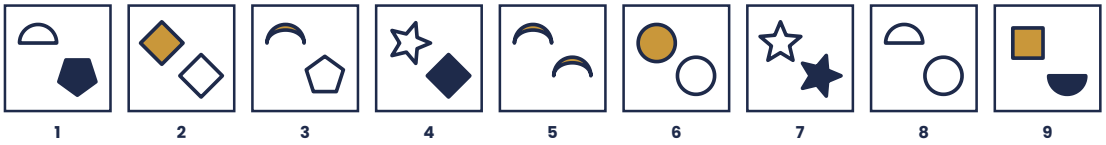
Q12 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 8, 9; 2, 4, 6; 3, 5, 7
(b) 1, 6, 9; 2, 4, 8; 3, 5, 7
(c) 1, 8, 9; 2, 6, 7; 3, 4, 5
(d) 1, 3, 9; 2, 4, 6; 5, 7, 8



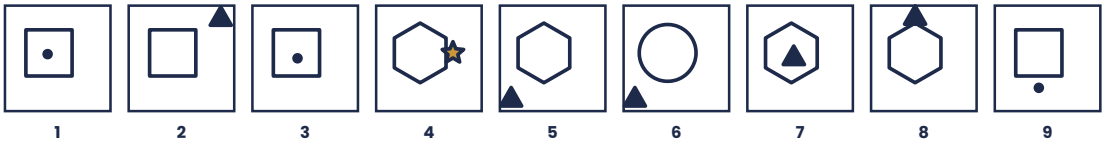
Q13 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 3, 8; 2, 5, 6; 4, 7, 9
(b) 1, 3, 4; 2, 7, 9; 5, 6, 8
(c) 1, 2, 4; 3, 7, 9; 5, 6, 8
(d) 1, 3, 9; 2, 4, 7; 5, 6, 8



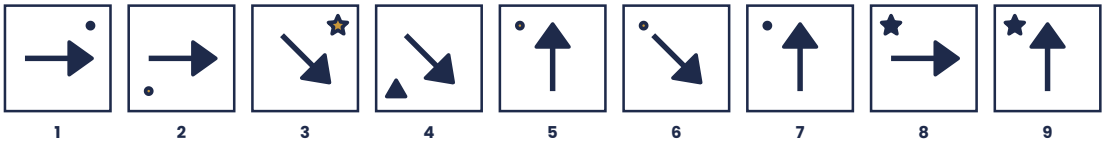
Q14 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 5, 7; 2, 3, 6; 4, 8, 9
(b) 1, 4, 7; 2, 6, 8; 3, 5, 9
(c) 1, 2, 5; 3, 7, 8; 4, 6, 9
(d) 1, 3, 7; 2, 5, 6; 4, 8, 9



Q15 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 5, 7; 2, 8, 9; 3, 4, 6
(b) 1, 2, 8; 3, 4, 6; 5, 7, 9
(c) 1, 2, 3; 4, 6, 8; 5, 7, 9
(d) 1, 5, 9; 2, 4, 8; 3, 6, 7



Q16 Divide the nine figures into three groups of three, each group sharing a common property.

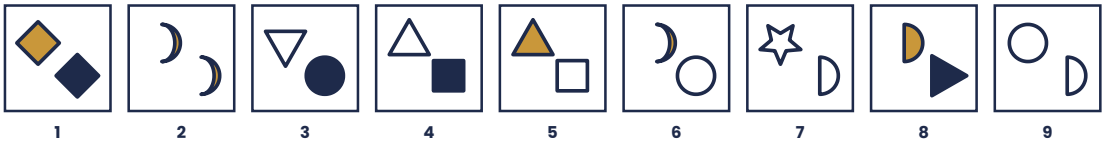
(a) 1, 8, 9; 2, 4, 7; 3, 5, 6
(b) 1, 2, 5; 3, 6, 7; 4, 8, 9
(c) 1, 3, 8; 2, 4, 5; 6, 7, 9
(d) 1, 8, 9; 2, 4, 5; 3, 6, 7



14.3.3 Advanced level

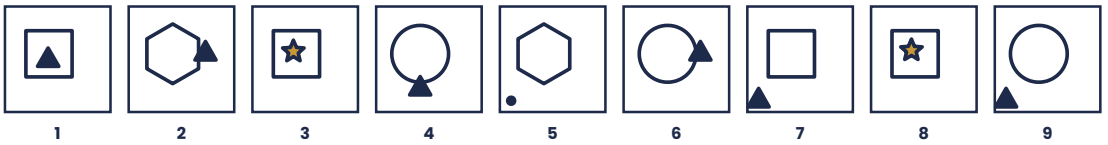
Q17 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 4, 5; 2, 6, 9; 3, 7, 8
(b) 1, 4, 5; 2, 6, 8; 3, 7, 9
(c) 1, 4, 5; 2, 7, 8; 3, 6, 9
(d) 1, 3, 4; 2, 6, 9; 5, 7, 8



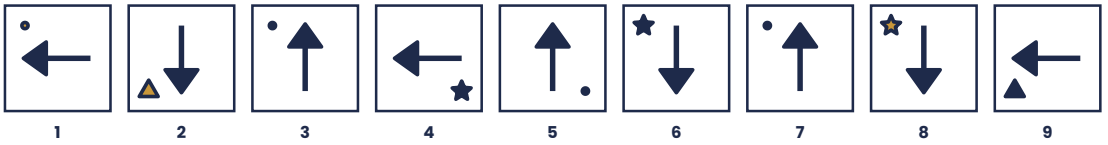
Q18 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 2, 8; 3, 4, 6; 5, 7, 9
(b) 1, 5, 8; 2, 4, 6; 3, 7, 9
(c) 1, 3, 8; 2, 4, 6; 5, 7, 9
(d) 1, 2, 6; 3, 4, 8; 5, 7, 9



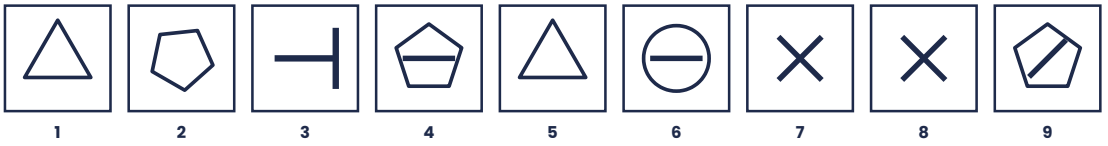
Q19 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 4, 9; 2, 6, 8; 3, 5, 7
(b) 1, 4, 7; 2, 6, 9; 3, 5, 8
(c) 1, 3, 9; 2, 6, 8; 4, 5, 7
(d) 1, 4, 6; 2, 8, 9; 3, 5, 7



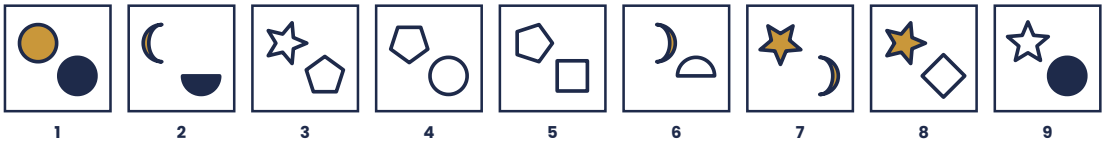
Q20 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 2, 5; 3, 7, 8; 4, 6, 9
(b) 1, 7, 8; 2, 3, 5; 4, 6, 9
(c) 1, 5, 6; 2, 3, 4; 7, 8, 9
(d) 1, 2, 8; 3, 5, 7; 4, 6, 9



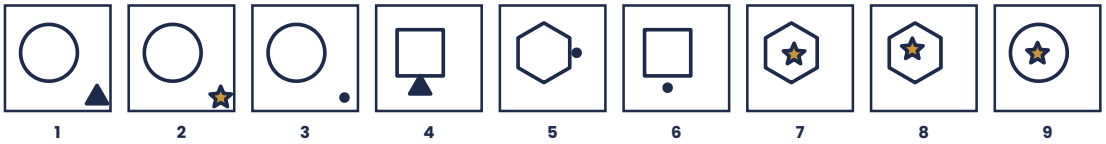
Q21 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 5, 7; 2, 4, 6; 3, 8, 9
(b) 1, 2, 6; 3, 7, 8; 4, 5, 9
(c) 1, 5, 6; 2, 3, 7; 4, 8, 9
(d) 1, 2, 6; 3, 5, 8; 4, 7, 9



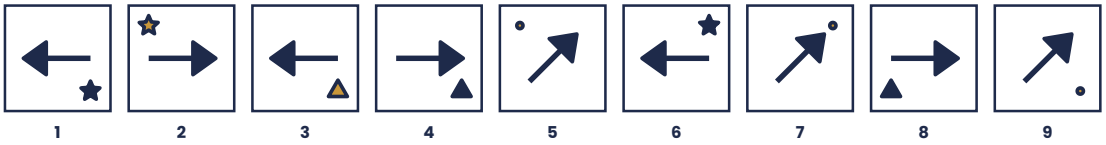
Q22 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 2, 3; 4, 5, 6; 7, 8, 9
(b) 1, 3, 6; 2, 4, 5; 7, 8, 9
(c) 1, 4, 6; 2, 3, 5; 7, 8, 9
(d) 1, 2, 8; 3, 7, 9; 4, 5, 6



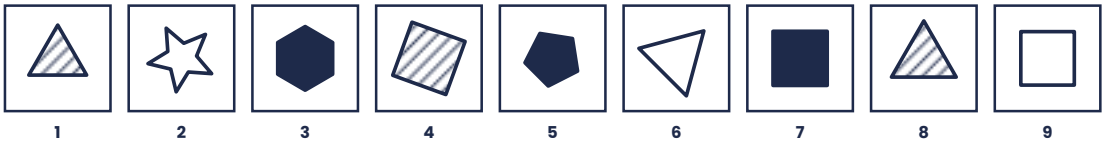
Q23 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 3, 6; 2, 4, 5; 7, 8, 9
(b) 1, 3, 6; 2, 4, 8; 5, 7, 9
(c) 1, 2, 6; 3, 4, 8; 5, 7, 9
(d) 1, 6, 9; 2, 3, 4; 5, 7, 8



Q24 Divide the nine figures into three groups of three, each group sharing a common property.

(a) 1, 6, 8; 2, 4, 9; 3, 5, 7
(b) 1, 4, 8; 2, 6, 7; 3, 5, 9
(c) 1, 3, 4; 2, 8, 9; 5, 6, 7
(d) 1, 4, 8; 2, 6, 9; 3, 5, 7



Q25 Divide the nine figures into three groups of three, each group sharing a common property.

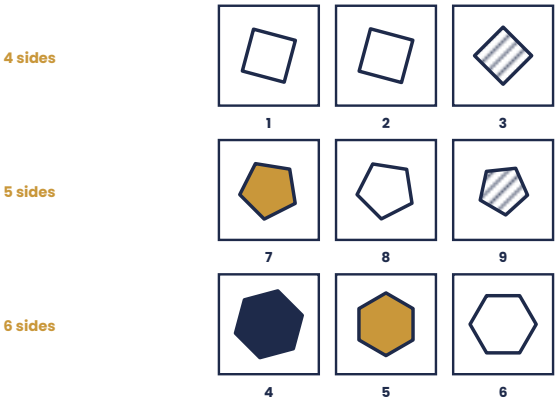
(a) 1, 4, 7; 2, 5, 8; 3, 6, 9
(b) 1, 4, 8; 2, 5, 7; 3, 6, 9
(c) 1, 4, 7; 2, 3, 6; 5, 8, 9
(d) 1, 2, 8; 3, 6, 9; 4, 5, 7



14.4 Solutions

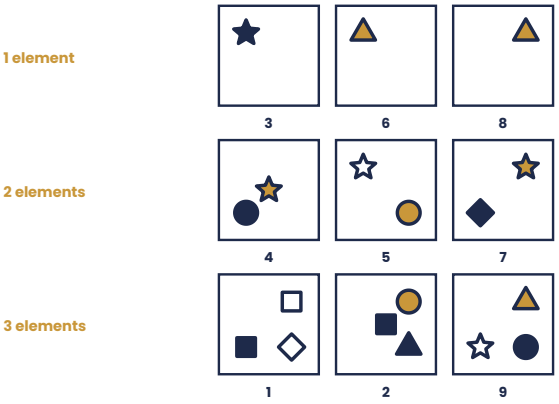
Q1 **Answer: (b) 1, 2, 3; 4, 5, 6; 7, 8, 9**

The figures group by the number of sides of the figure (size, rotation and shading vary and do not matter). 4 sides: 1, 2, 3; 5 sides: 7, 8, 9; 6 sides: 4, 5, 6. Answer **(b)**.



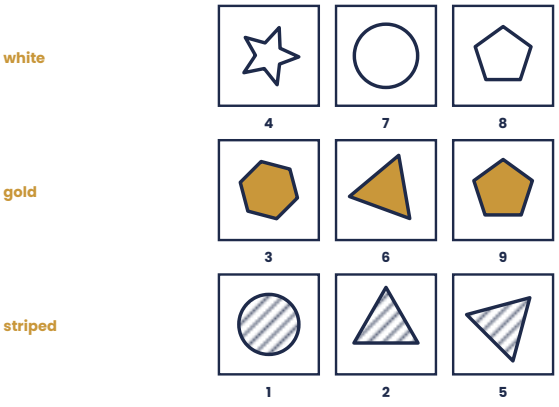
Q2 **Answer: (d) 1, 2, 9; 3, 6, 8; 4, 5, 7**

The figures group by the number of elements in each figure, whatever their shape or shading. 1 element: 3, 6, 8; 2 elements: 4, 5, 7; 3 elements: 1, 2, 9. Answer **(d)**.



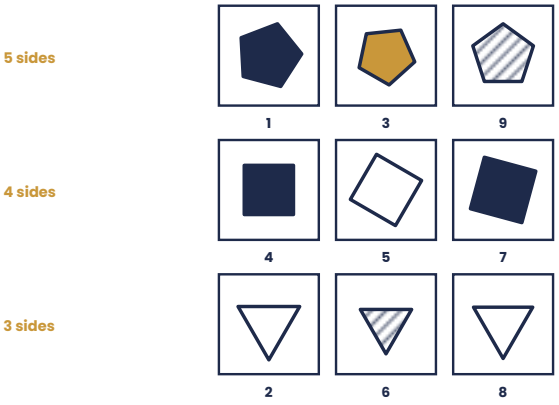
Q3 **Answer: (a) 1, 2, 5; 3, 6, 9; 4, 7, 8**

The figures group by the shading of the figure, whatever its shape. white: 4, 7, 8; gold: 3, 6, 9; striped: 1, 2, 5. Answer **(a)**.



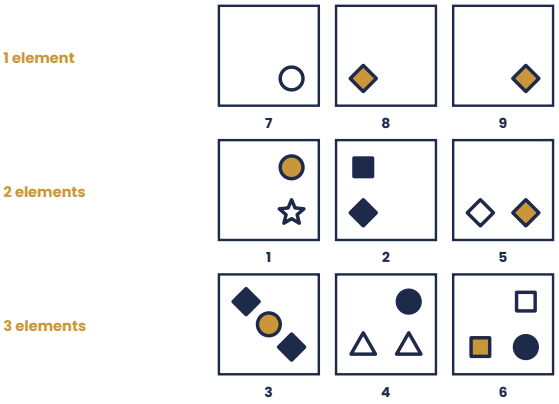
Q4 **Answer: (c) 1, 3, 9; 2, 6, 8; 4, 5, 7**

The figures group by the number of sides of the figure (size, rotation and shading vary and do not matter). 5 sides: 1, 3, 9; 4 sides: 4, 5, 7; 3 sides: 2, 6, 8. Answer **(c)**.



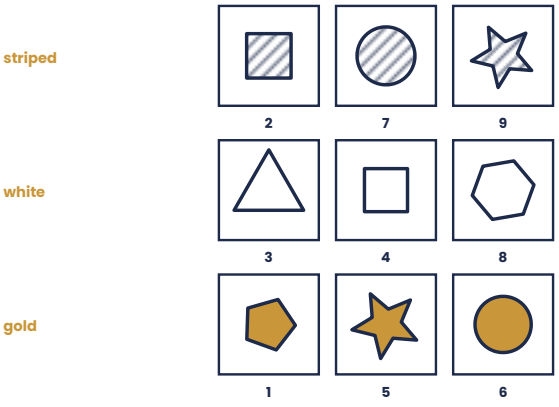
Q5 **Answer: (a) 1, 2, 5; 3, 4, 6; 7, 8, 9**

The figures group by the number of elements in each figure, whatever their shape or shading. 1 element: 7, 8, 9; 2 elements: 1, 2, 5; 3 elements: 3, 4, 6. Answer **(a)**.



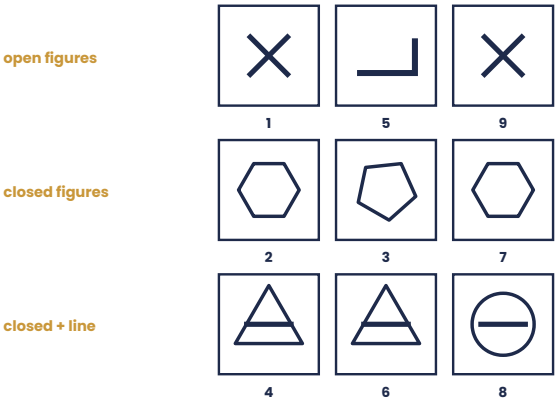
Q6 **Answer: (d) 1, 5, 6; 2, 7, 9; 3, 4, 8**

The figures group by the shading of the figure, whatever its shape. striped: 2, 7, 9; white: 3, 4, 8; gold: 1, 5, 6. Answer **(d)**.



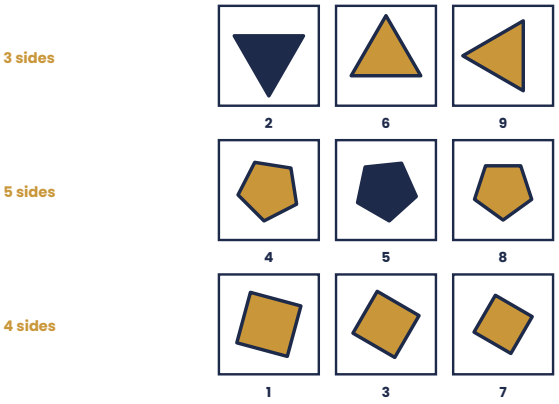
Q7 **Answer: (b) 1, 5, 9; 2, 3, 7; 4, 6, 8**

The figures group by open figures, closed figures, and closed figures crossed by a line. open figures: 1, 5, 9; closed figures: 2, 3, 7; closed + line: 4, 6, 8. Answer **(b)**.



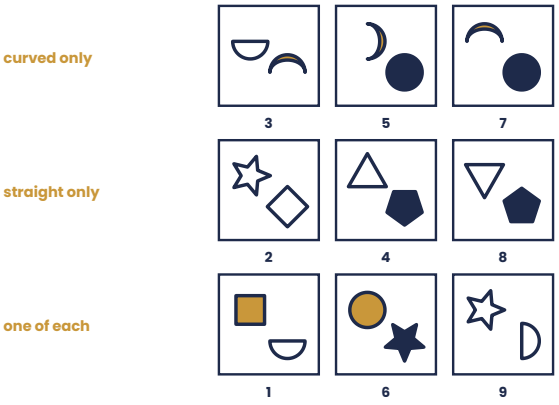
Q8 **Answer: (a) 1, 3, 7; 2, 6, 9; 4, 5, 8**

The figures group by the number of sides of the figure (size, rotation and shading vary and do not matter). 3 sides: 2, 6, 9; 5 sides: 4, 5, 8; 4 sides: 1, 3, 7. Answer **(a)**.



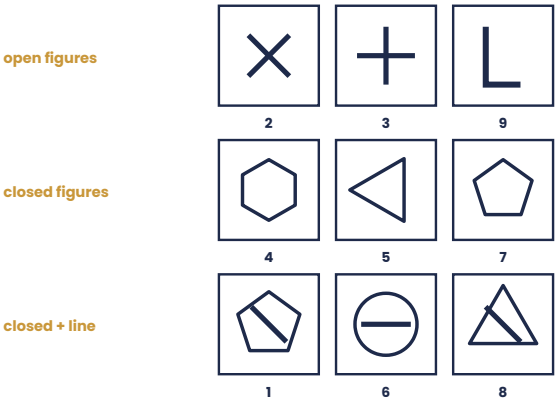
Q9 **Answer: (d) 1, 6, 9; 2, 4, 8; 3, 5, 7**

The figures group by whether the two parts are both curved, both straight-sided, or one of each. curved only: 3, 5, 7; straight only: 2, 4, 8; one of each: 1, 6, 9. Answer **(d)**.



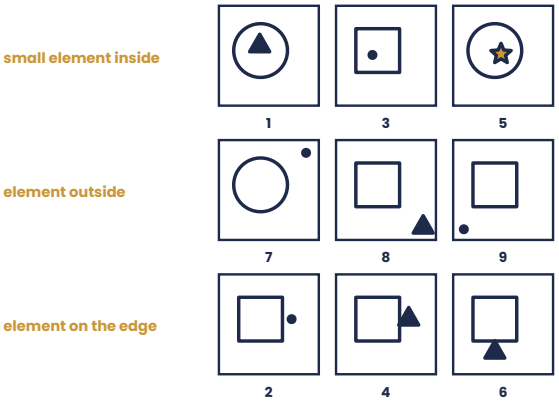
Q10 **Answer: (a) 1, 6, 8; 2, 3, 9; 4, 5, 7**

The figures group by open figures, closed figures, and closed figures crossed by a line. open figures: 2, 3, 9; closed figures: 4, 5, 7; closed + line: 1, 6, 8. Answer **(a)**.



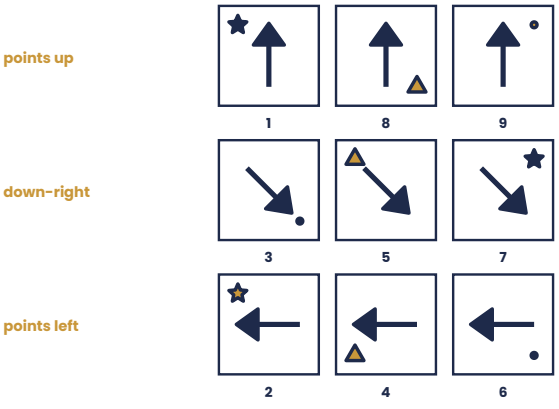
Q11 **Answer: (c) 1, 3, 5; 2, 4, 6; 7, 8, 9**

The figures group by where the small element lies: inside the large shape, outside it, or on its boundary. small element inside: 1, 3, 5; element outside: 7, 8, 9; element on the edge: 2, 4, 6. Answer **(c)**.



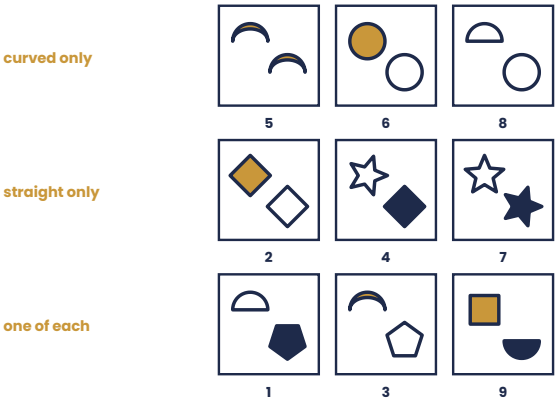
Q12 **Answer: (a) 1, 8, 9; 2, 4, 6; 3, 5, 7**

The figures group by the direction in which the arrow points; the small corner element is a distraction. points up: 1, 8, 9; down-right: 3, 5, 7; points left: 2, 4, 6. Answer **(a)**.



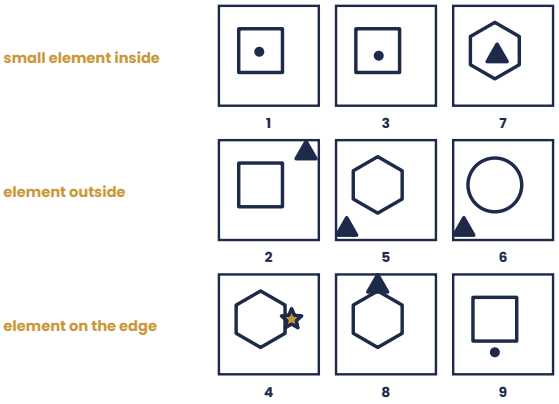
Q13 **Answer: (d) 1, 3, 9; 2, 4, 7; 5, 6, 8**

The figures group by whether the two parts are both curved, both straight-sided, or one of each. curved only: 5, 6, 8; straight only: 2, 4, 7; one of each: 1, 3, 9. Answer **(d)**.



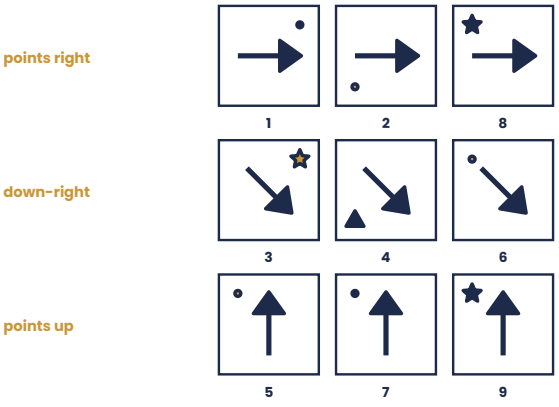
Q14 **Answer: (d) 1, 3, 7; 2, 5, 6; 4, 8, 9**

The figures group by where the small element lies: inside the large shape, outside it, or on its boundary. small element inside: 1, 3, 7; element outside: 2, 5, 6; element on the edge: 4, 8, 9. Answer **(d)**.



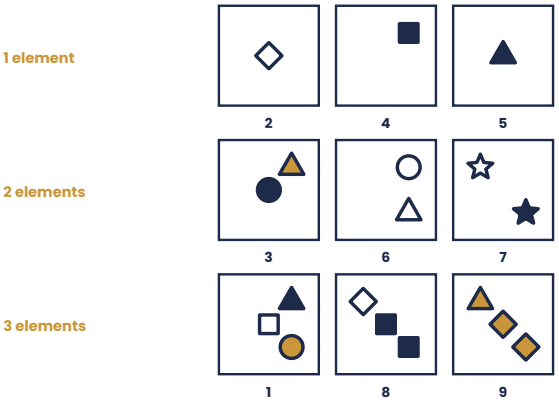
Q15 **Answer: (b) 1, 2, 8; 3, 4, 6; 5, 7, 9**

The figures group by the direction in which the arrow points; the small corner element is a distraction. points right: 1, 2, 8; down-right: 3, 4, 6; points up: 5, 7, 9. Answer **(b)**.



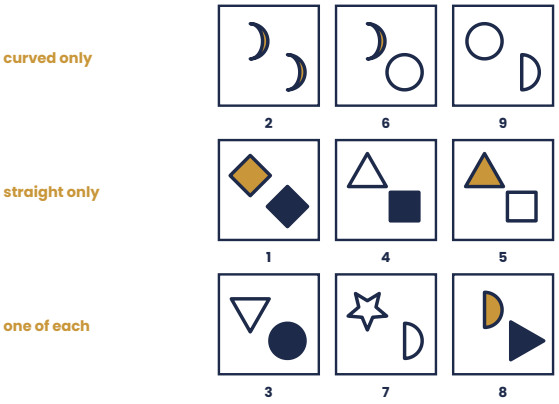
Q16 **Answer: (d) 1, 8, 9; 2, 4, 5; 3, 6, 7**

The figures group by the number of elements in each figure, whatever their shape or shading. 1 element: 2, 4, 5; 2 elements: 3, 6, 7; 3 elements: 1, 8, 9. Answer **(d)**.



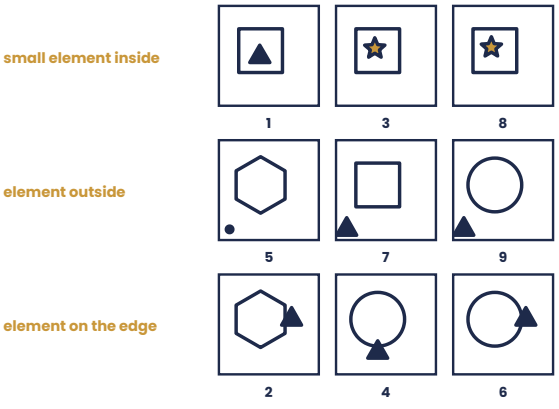
Q17 **Answer: (a) 1, 4, 5; 2, 6, 9; 3, 7, 8**

The figures group by whether the two parts are both curved, both straight-sided, or one of each. curved only: 2, 6, 9; straight only: 1, 4, 5; one of each: 3, 7, 8. Answer **(a)**.



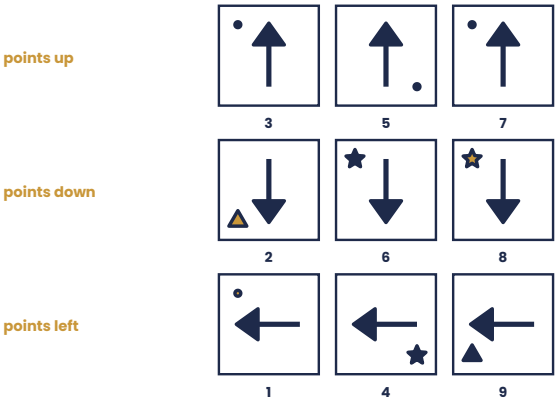
Q18 **Answer: (c) 1, 3, 8; 2, 4, 6; 5, 7, 9**

The figures group by where the small element lies: inside the large shape, outside it, or on its boundary. small element inside: 1, 3, 8; element outside: 5, 7, 9; element on the edge: 2, 4, 6. Answer **(c)**.



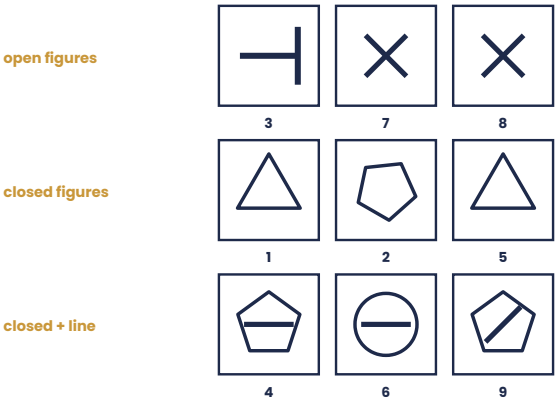
Q19 **Answer: (a) 1, 4, 9; 2, 6, 8; 3, 5, 7**

The figures group by the direction in which the arrow points; the small corner element is a distraction. points up: 3, 5, 7; points down: 2, 6, 8; points left: 1, 4, 9. Answer **(a)**.



Q20 **Answer: (a) 1, 2, 5; 3, 7, 8; 4, 6, 9**

The figures group by open figures, closed figures, and closed figures crossed by a line. open figures: 3, 7, 8; closed figures: 1, 2, 5; closed + line: 4, 6, 9. Answer **(a)**.



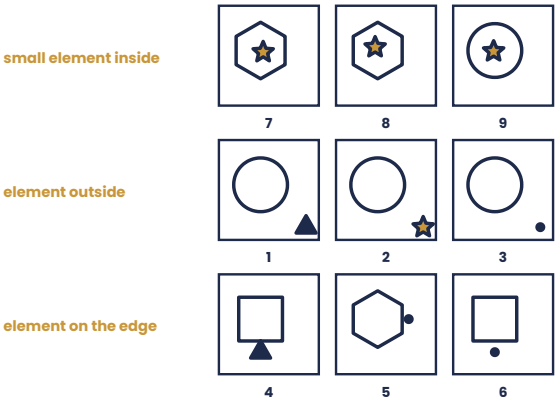
Q21 **Answer: (d) 1, 2, 6; 3, 5, 8; 4, 7, 9**

The figures group by whether the two parts are both curved, both straight-sided, or one of each. curved only: 1, 2, 6; straight only: 3, 5, 8; one of each: 4, 7, 9. Answer **(d)**.



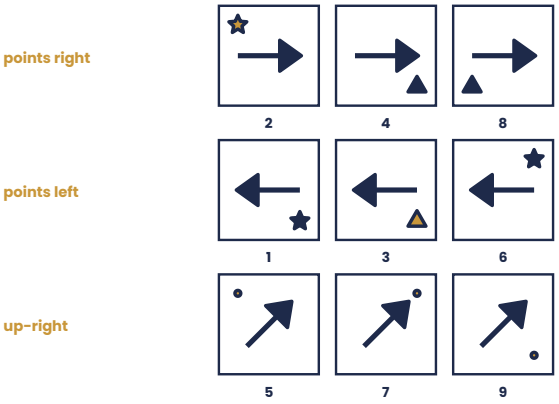
Q22 **Answer: (a) 1, 2, 3; 4, 5, 6; 7, 8, 9**

The figures group by where the small element lies: inside the large shape, outside it, or on its boundary. small element inside: 7, 8, 9; element outside: 1, 2, 3; element on the edge: 4, 5, 6. Answer **(a)**.



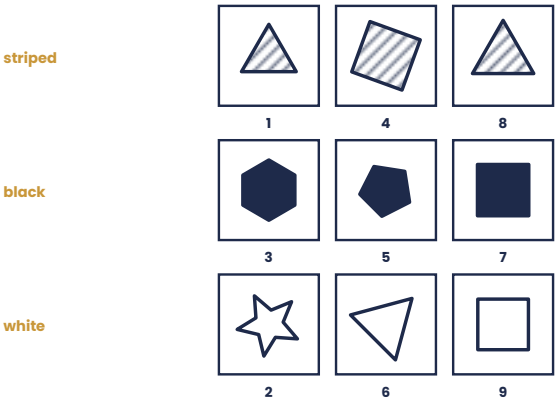
Q23 **Answer: (b) 1, 3, 6; 2, 4, 8; 5, 7, 9**

The figures group by the direction in which the arrow points; the small corner element is a distraction. points right: 2, 4, 8; points left: 1, 3, 6; up-right: 5, 7, 9. Answer **(b)**.



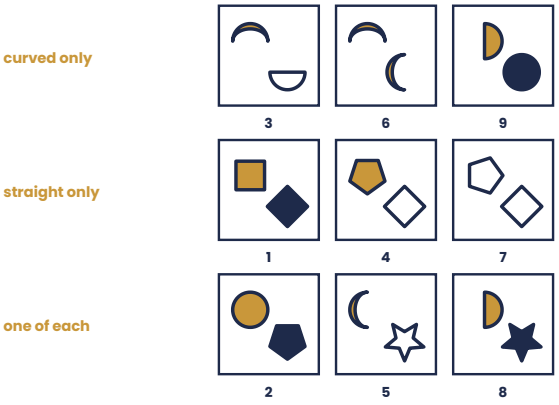
Q24 **Answer: (d) 1, 4, 8; 2, 6, 9; 3, 5, 7**

The figures group by the shading of the figure, whatever its shape. striped: 1, 4, 8; black: 3, 5, 7; white: 2, 6, 9. Answer **(d)**.



Q25 **Answer: (a) 1, 4, 7; 2, 5, 8; 3, 6, 9**

The figures group by whether the two parts are both curved, both straight-sided, or one of each. curved only: 3, 6, 9; straight only: 1, 4, 7; one of each: 2, 5, 8. Answer **(a)**.



Find the property that takes exactly three values, each on three figures: sides, count, shading, curved or straight, open or closed, position, direction or symmetry. Ignore size, rotation and decoration, and confirm that every figure belongs to exactly one group. Testing the first group of each option is the fastest route.

MODULE 15 · INTERMEDIATE

Dot situation, figure formation and figure analysis

Minimum duration **1 day** | Level **Intermediate**

These questions test analysing positions inside overlapping shapes and combining pieces into a whole. This module covers dot-situation questions (naming the region in which a dot lies and finding an option with the same region), figure formation (which pieces make a figure, and which figure the pieces make), construction of squares and triangles from cut pieces, and superimposing transparent sheets, followed by 25 solved questions in which every region and every piece was computed geometrically.

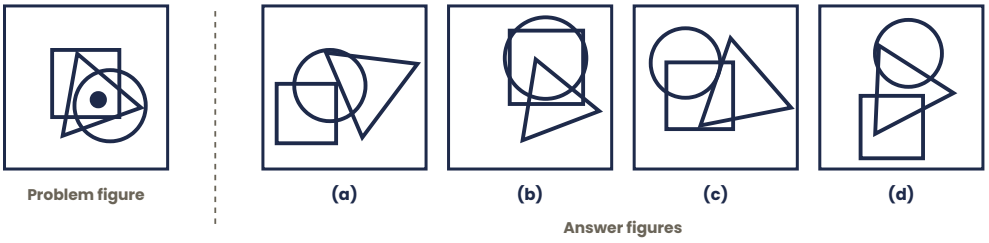
IN THIS MODULE

15.1 Dot situation
15.2 Figure formation
15.3 Construction of squares and triangles
15.4 Shape overlay and transparency
15.5 Practice questions
15.6 Solutions

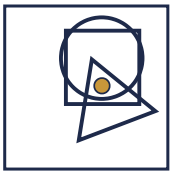
15.1 Dot situation

15.1.1 Identifying regions where a dot lies in overlapping shapes

Overlapping shapes divide the figure into regions. Describe the region holding the dot by listing every shape it is **inside** and every shape it is **outside**, for example "inside the circle and the square but outside the triangle".



FIND A FIGURE WITH THE SAME REGION



(b)

THE MATCHING REGION, WITH THE DOT PLACED IN GOLD

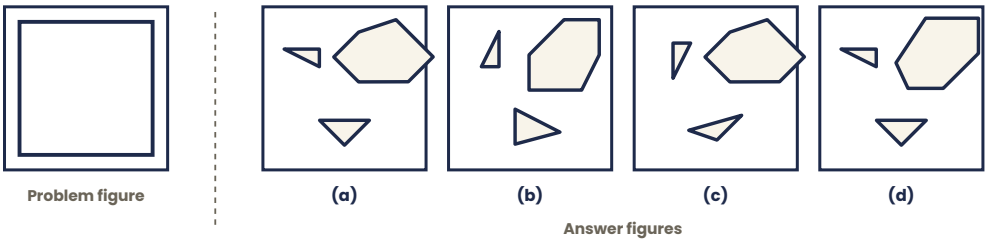
15.1.2 Finding an option with the same region conditions

Check each answer figure for a region that satisfies **all** the conditions at once. Shapes may be moved, resized or overlap differently; only the inside/outside conditions matter.

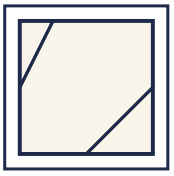
> **SHORTCUT TRICK**
> Start with the rarest condition, usually the overlap of the most shapes (for example, inside all three). If an option has no such overlap, reject it at once.

15.2 Figure formation

15.2.1 Which pieces form a given figure



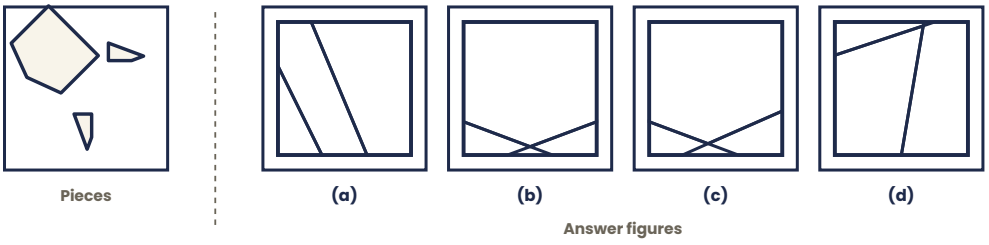
CHOOSE THE PIECES THAT MAKE THE SQUARE



HOW THE PIECES FIT

Pieces may be turned but not overlapped, and together they must fill the figure exactly. Compare corners first: a square needs four right angles at its corners, so the pieces must supply exactly those.

15.2.2 Which figure can be formed from given pieces

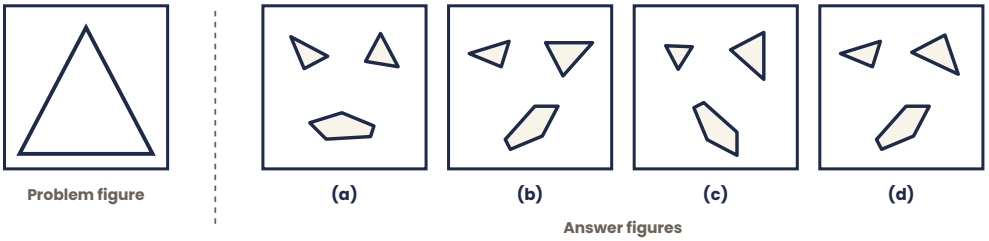


WHICH DIVISION OF THE SQUARE USES EXACTLY THESE PIECES?

15.3 Construction of squares and triangles

15.3.1 Fitting cut pieces together

Look for edges of equal length that can meet, and for angles that add up correctly: angles meeting at a point inside the figure total 360°, and those meeting at a point on a straight edge total 180°.



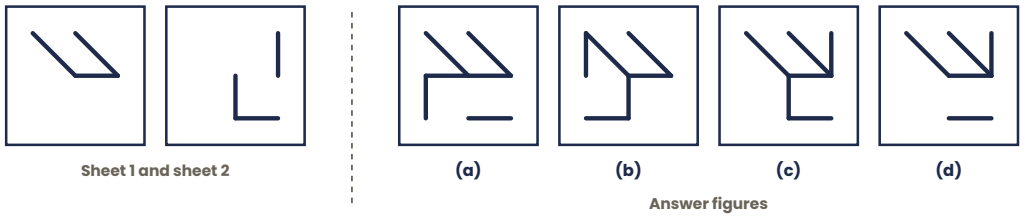
WHICH PIECES FORM THE TRIANGLE?

WATCH OUT
An option with the right number of pieces can still be wrong: check one piece at a time for a side or angle that has no partner in the figure.

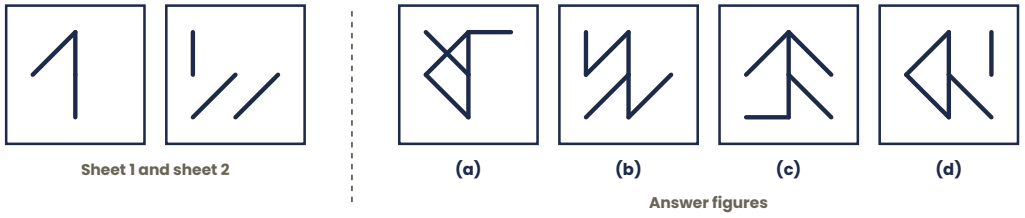
15.4 Shape overlay and transparency

15.4.1 Superimposing transparent figures

When transparent sheets are laid on top of each other, every line on every sheet shows in the result. If a sheet is turned or flipped first, do that transformation before combining.



TWO TRANSPARENT SHEETS PLACED TOGETHER

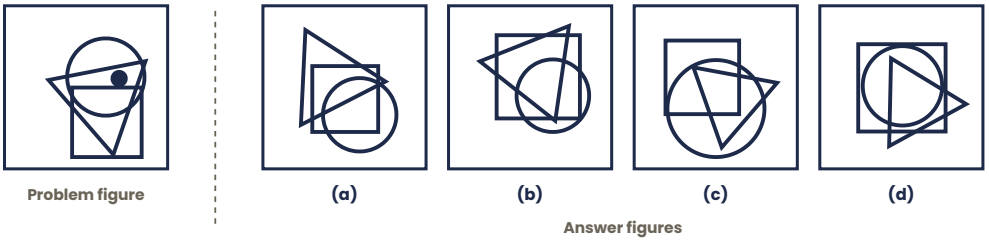


SHEET 2 IS TURNED 90° CLOCKWISE FIRST

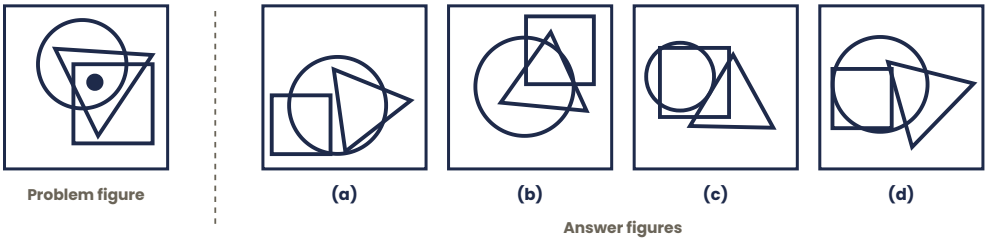
15.5 Practice questions

15.5.1 Basic level

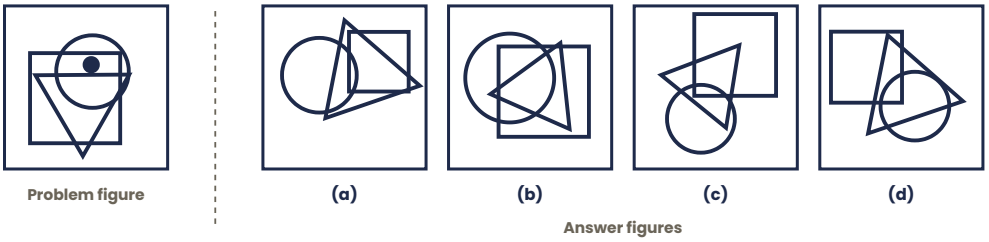
Q1 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



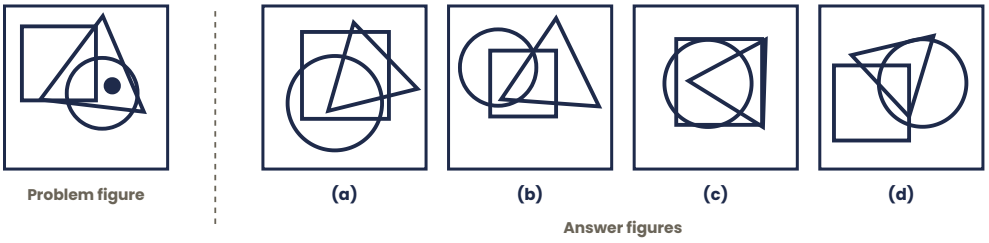
Q2 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



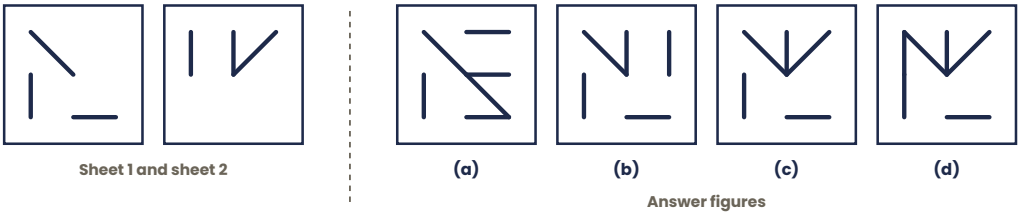
Q3 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



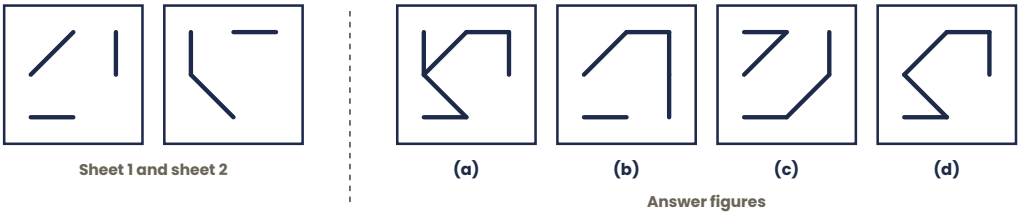
Q4 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



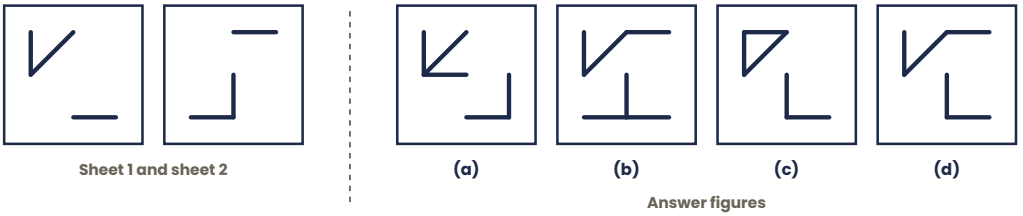
Q5 Transparent sheets 1 and 2 are placed exactly on top of each other. Which answer figure shows the result?



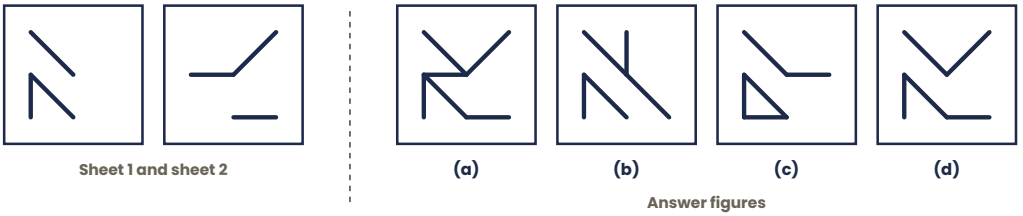
Q6 Transparent sheets 1 and 2 are placed exactly on top of each other. Which answer figure shows the result?



Q7 Transparent sheets 1 and 2 are placed exactly on top of each other. Which answer figure shows the result?

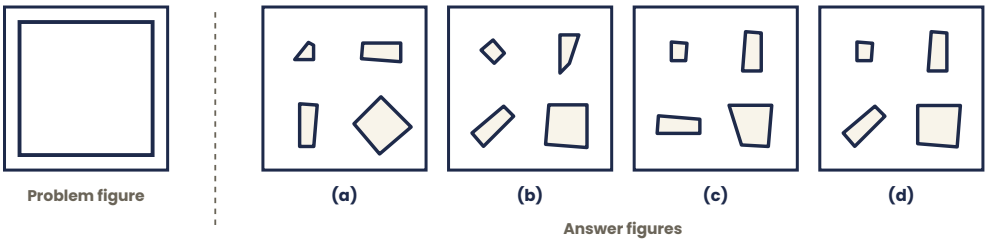


Q8 Transparent sheets 1 and 2 are placed exactly on top of each other. Which answer figure shows the result?

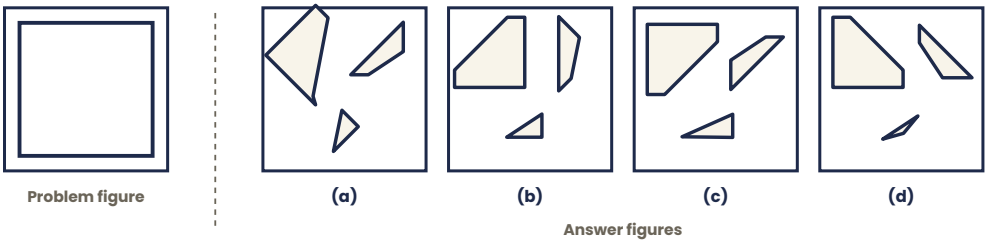


15.5.2 Intermediate level

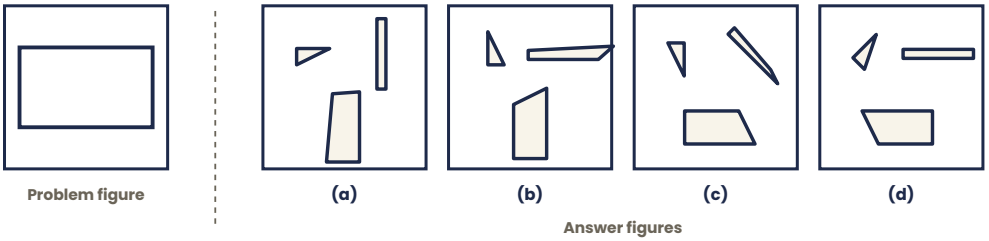
Q9 Which set of pieces can be fitted together to form the square in the problem figure?



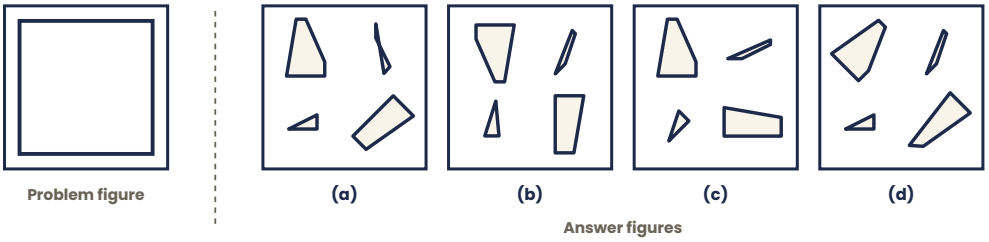
Q10 Which set of pieces can be fitted together to form the square in the problem figure?



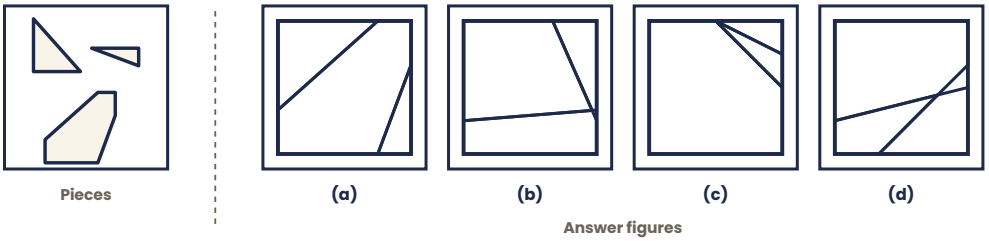
Q11 Which set of pieces can be fitted together to form the rectangle in the problem figure?



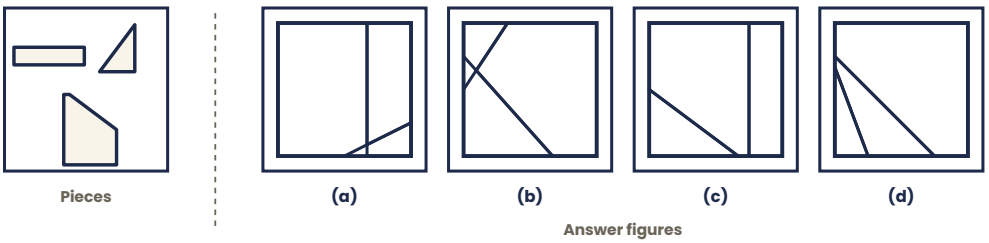
Q12 Which set of pieces can be fitted together to form the square in the problem figure?



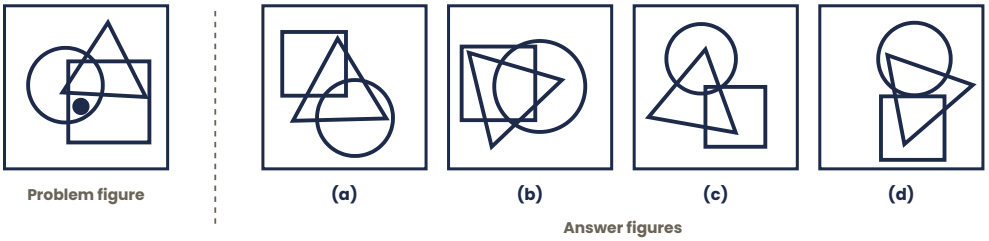
Q13 The pieces in the problem figure are fitted together without overlapping. Which answer figure shows how they make a square?



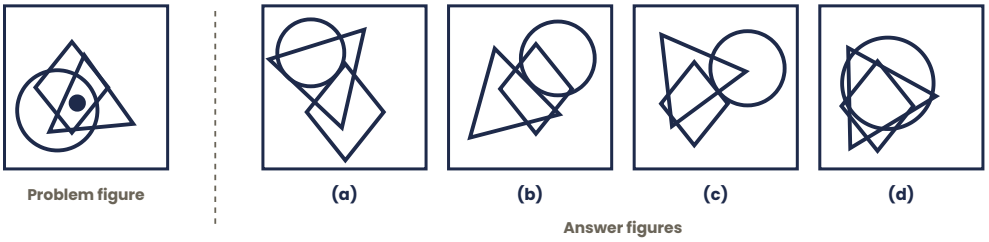
Q14 The pieces in the problem figure are fitted together without overlapping. Which answer figure shows how they make a square?



Q15 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?

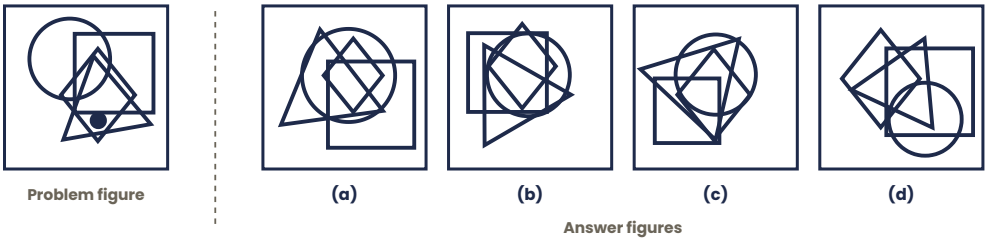


Q16 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?

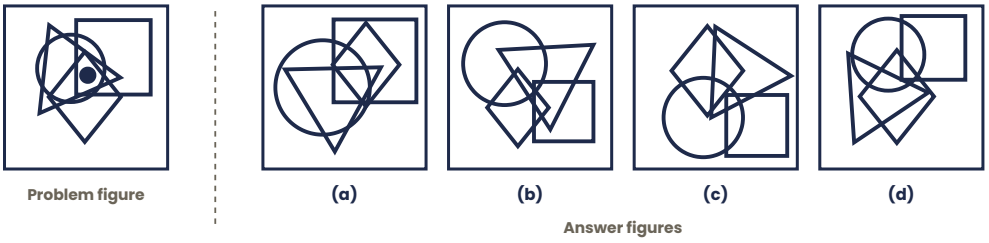


15.5.3 Advanced level

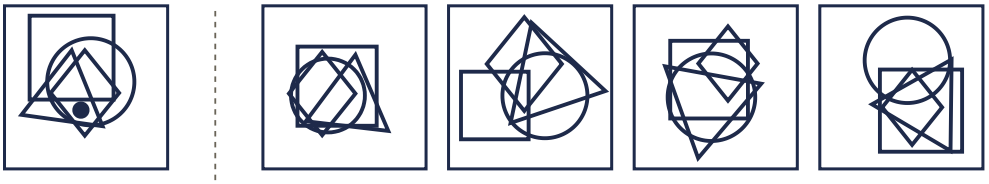
Q17 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



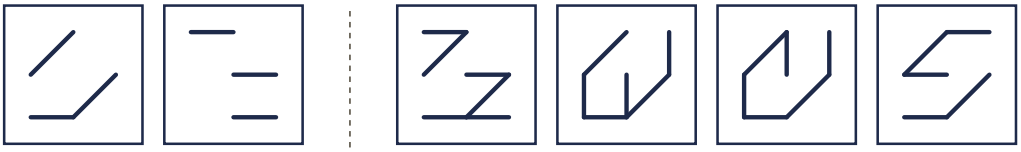
Q18 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



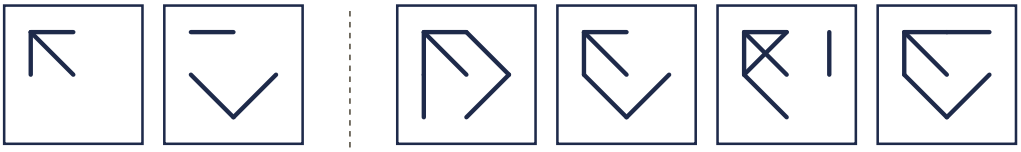
Q19 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



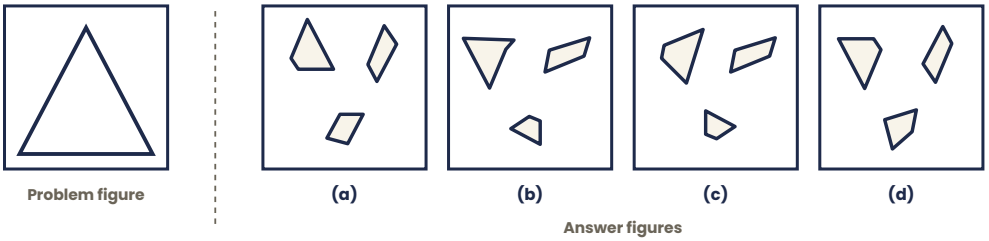
Q20 Transparent sheets 1 and 2 are placed exactly on top of each other, with sheet 2 first turned 90° clockwise. Which answer figure shows the result?



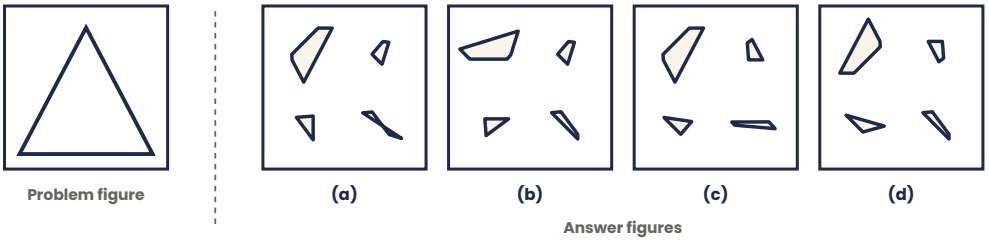
Q21 Transparent sheets 1 and 2 are placed exactly on top of each other, with sheet 2 first turned 90° anticlockwise. Which answer figure shows the result?



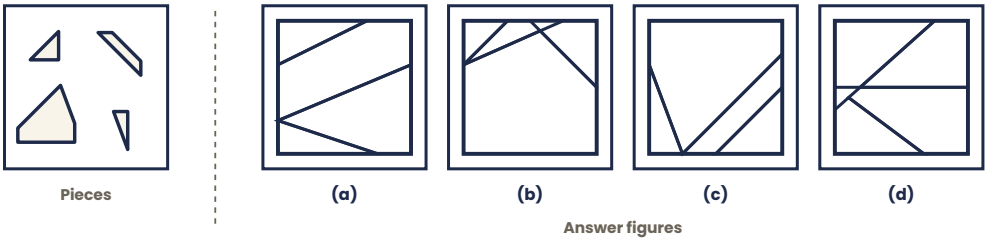
Q22 Which set of pieces can be fitted together to form the triangle in the problem figure?



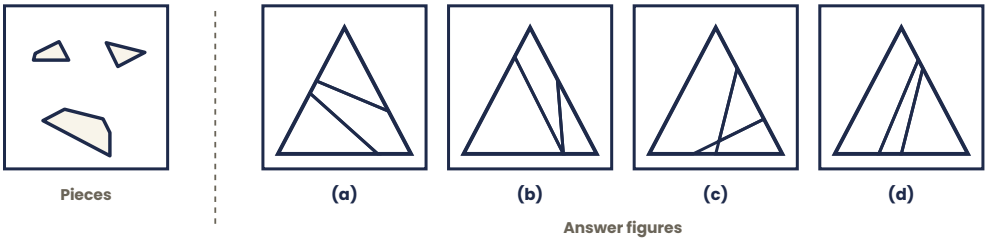
Q23 Which set of pieces can be fitted together to form the triangle in the problem figure?



Q24 The pieces in the problem figure are fitted together without overlapping. Which answer figure shows how they make a square?



Q25 The pieces in the problem figure are fitted together without overlapping. Which answer figure shows how they make a triangle?



15.6 Solutions

Q1 **Answer: (c)**

The dot lies inside the circle and the triangle but outside the square. Only figure (c) has a region that is inside the circle and the triangle but outside the square; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(c)**.



(c)

Q2 **Answer: (b)**

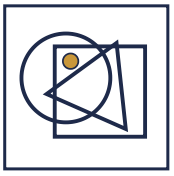
The dot lies inside the circle and the square and the triangle. Only figure (b) has a region that is inside the circle and the square and the triangle; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(b)**.



(b)

Q3 **Answer: (b)**

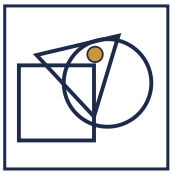
The dot lies inside the circle and the square but outside the triangle. Only figure (b) has a region that is inside the circle and the square but outside the triangle; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(b)**.



(b)

Q4 **Answer: (d)**

The dot lies inside the circle and the triangle but outside the square. Only figure (d) has a region that is inside the circle and the triangle but outside the square; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(d)**.



(d)

Q5 **Answer: (d)**

Then every line of both sheets appears in the result (sheet 1 in navy, sheet 2 in gold below). Answer **(d)**.



result

Q6 **Answer: (a)**

Then every line of both sheets appears in the result (sheet 1 in navy, sheet 2 in gold below). Answer **(a)**.



result

Q7 **Answer: (b)**

Then every line of both sheets appears in the result (sheet 1 in navy, sheet 2 in gold below). Answer **(b)**.



result

Q8 **Answer: (a)**

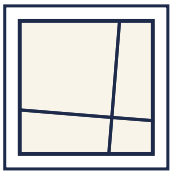
Then every line of both sheets appears in the result (sheet 1 in navy, sheet 2 in gold below). Answer **(a)**.



result

Q9 **Answer: (d)**

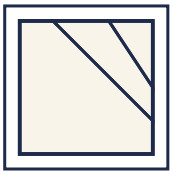
The square can be cut into the 4 pieces of figure (d) as shown; every other set contains a piece whose angle or side length does not fit. Answer **(d)**.



assembled

Q10 **Answer: (b)**

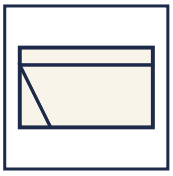
The square can be cut into the 3 pieces of figure (b) as shown; every other set contains a piece whose angle or side length does not fit. Answer **(b)**.



assembled

Q11 **Answer: (d)**

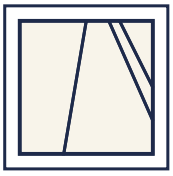
The rectangle can be cut into the 3 pieces of figure (d) as shown; every other set contains a piece whose angle or side length does not fit. Answer **(d)**.



assembled

Q12 **Answer: (c)**

The square can be cut into the 4 pieces of figure (c) as shown; every other set contains a piece whose angle or side length does not fit. Answer **(c)**.



assembled

Q13 **Answer: (a)**

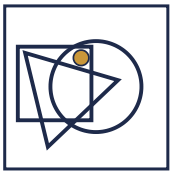
Match the pieces one by one: the largest piece and the pieces with the sharpest angles fix the layout. Only figure (a) divides the square into exactly these pieces. Answer **(a)**.

Q14 **Answer: (c)**

Match the pieces one by one: the largest piece and the pieces with the sharpest angles fix the layout. Only figure (c) divides the square into exactly these pieces. Answer **(c)**.

Q15 **Answer: (b)**

The dot lies inside the circle and the square but outside the triangle. Only figure (b) has a region that is inside the circle and the square but outside the triangle; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(b)**.



(b)

Q16 **Answer: (d)**

The dot lies inside the circle and the triangle and the diamond. Only figure (d) has a region that is inside the circle and the triangle and the diamond; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(d)**.



(d)

Q17 **Answer: (d)**

The dot lies inside the triangle and the diamond but outside the circle and the square. Only figure (d) has a region that is inside the triangle and the diamond but outside the circle and the square; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(d)**.



(d)

Q18 **Answer: (a)**

The dot lies inside the circle and the square and the triangle and the diamond. Only figure (a) has a region that is inside the circle and the square and the triangle and the diamond; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(a)**.



(a)

Q19 **Answer: (b)**

The dot lies inside the circle and the triangle and the diamond but outside the square. Only figure (b) has a region that is inside the circle and the triangle and the diamond but outside the square; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(b)**.



(b)

Q20 **Answer: (b)**

First turn sheet 2: turned 90° clockwise. Then every line of both sheets appears in the result (sheet 1 in navy, sheet 2 in gold below). Answer **(b)**.



result

Q21 **Answer: (a)**

First turn sheet 2: turned 90° anticlockwise. Then every line of both sheets appears in the result (sheet 1 in navy, sheet 2 in gold below). Answer **(a)**.



result

Q22 **Answer: (c)**

The triangle can be cut into the 3 pieces of figure (c) as shown; every other set contains a piece whose angle or side length does not fit. Answer **(c)**.



assembled

Q23 **Answer: (b)**

The triangle can be cut into the 4 pieces of figure (b) as shown; every other set contains a piece whose angle or side length does not fit. Answer **(b)**.



assembled

Q24 **Answer: (c)**

Match the pieces one by one: the largest piece and the pieces with the sharpest angles fix the layout. Only figure (c) divides the square into exactly these pieces. Answer **(c)**.

Q25 **Answer: (c)**

Match the pieces one by one: the largest piece and the pieces with the sharpest angles fix the layout. Only figure (c) divides the triangle into exactly these pieces. Answer **(c)**.

Dot situation: list every shape the dot is inside and outside, then find an option with a region meeting all those conditions, starting with the rarest. Figure formation: match corners, side lengths and angles piece by piece; pieces may turn but not overlap. Transparent overlay: transform a sheet first if told to, then keep every line from every sheet.

LEVEL 3

Advanced level

Master painted cubes, dice, 3D nets and views, abstract rule-based puzzles, visual Venn diagrams and diagrammatic operator tests, and perform in company-specific assessments

7 MODULES · 14 DAYS MINIMUM

Module 16 · Cubes and cuboids	**2 days**
Module 17 · Dice	**2 days**
Module 18 · 3D spatial visualisation	**2 days**
Module 19 · Abstract and rule-based reasoning	**2 days**
Module 20 · Venn diagrams and visual relationships	**1 day**
Module 21 · Diagrammatic and flowchart reasoning	**2 days**
Module 22 · Exam strategy, mock tests and company-specific preparation	**3 days**

MODULE 16 · ADVANCED

Cubes and cuboids

Minimum duration **2 days** | Level **Advanced**

Cube questions ask you to picture a large painted cube or cuboid cut into small cubes and count the pieces with a given number of painted faces. This module covers faces, edges and corners, the four painted-cube formulas, cubes with different colours on different faces, cuboids with unequal sides, removing cubes, the number of cuts needed, and counting cubes in stacks, followed by 25 solved questions in which every count was checked by examining each small cube one by one.

IN THIS MODULE

16.1 Basics
16.2 Painted cube formulas
16.3 Variations
16.4 Cube counting in stacks
16.5 Practice questions
16.6 Solutions

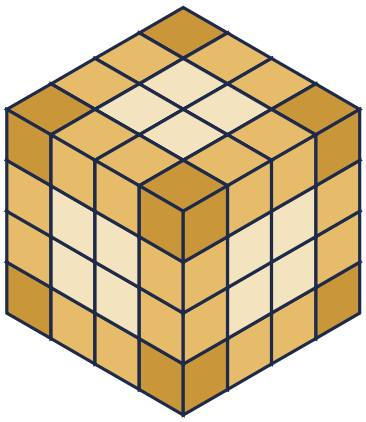
16.1 Basics

16.1.1 Faces, edges and corners of a cube

Part	How many
Faces	6
Edges	12
Corners (vertices)	8

16.1.2 Cutting into n × n × n smaller cubes

If each edge of a cube is divided into n equal parts and the cube is cut along those lines, it splits into n × n × n = n^3 small cubes. This needs (n - 1) cuts in each of the three directions, so 3(n - 1) cuts in total.



A 4 × 4 × 4 CUBE PAINTED ON THE OUTSIDE: CORNERS (3 FACES) DARK GOLD, EDGES (2 FACES) GOLD, FACES (1 FACE) LIGHT, INNER CUBES UNPAINTED

16.2 Painted cube formulas

A cube painted on all six faces and cut into n^3 small cubes gives four kinds of pieces:

Painted faces	Where they are	Formula	n = 3	n = 4	n = 5
3	corners	8	8	8	8
2	edges (not corners)	$12(n - 2)$	12	24	36
1	faces (not edges)	$6(n - 2)^2$	6	24	54
0	inside	$(n - 2)^3$	1	8	27

16.2.1 Three faces painted: 8 corners

Only the corner cubes touch three faces, and a cube has 8 corners, so the answer is 8 for every $n \geq 2$.

16.2.2 Two faces painted: 12(n - 2)

Each of the 12 edges holds n cubes; removing the 2 corner cubes leaves $n - 2$ with exactly two painted faces.

16.2.3 One face painted: $6(n - 2)^2$

Each of the 6 faces has an $n \times n$ layer; removing the border leaves an $(n - 2) \times (n - 2)$ block of cubes with one painted face.

16.2.4 No face painted: $(n - 2)^3$

Peeling one layer off every face leaves an inner cube of side $n - 2$.

> **SHORTCUT TRICK**
> Check your work: $8 + 12(n - 2) + 6(n - 2)^2 + (n - 2)^3 = n^3$. For $n = 4$: $8 + 24 + 24 + 8 = 64$.

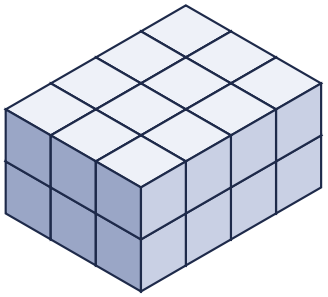
16.3 Variations

16.3.1 Different colours on opposite or adjacent faces

When pairs of faces have different colours, think about where two particular colours meet. If red and blue are on **adjacent** faces, they share one edge, holding n cubes (n - 2 of them plus 2 corners). If they are on **opposite** faces, no small cube can carry both colours.

16.3.2 Cuboids with unequal dimensions

For an a × b × c cuboid (all sides ≥ 2): 3 faces = 8; 2 faces = 4[(a - 2) + (b - 2) + (c - 2)]; 1 face = 2[(a - 2)(b - 2) + (b - 2)(c - 2) + (c - 2)(a - 2)]; none = (a - 2)(b - 2)(c - 2)



A 4 × 3 × 2 CUBOID

16.3.3 Removing cubes and counting visible faces

If a corner cube is removed, three new faces of neighbouring cubes become exposed, so the total visible surface does not change. Think about which faces become uncovered, cube by cube.

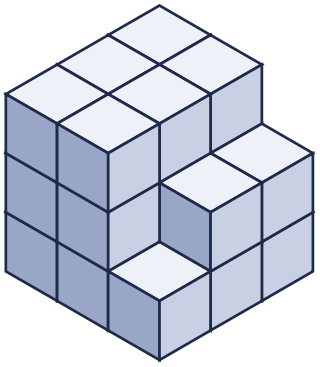
16.3.4 Number of cuts needed

To get n^3 identical cubes you need (n - 1) cuts in each direction: 3(n - 1) cuts in total. With k cuts in total, the greatest number of pieces comes from splitting the cuts as evenly as possible among the three directions: pieces = (p + 1)(q + 1)(r + 1) with p + q + r = k.

16.4 Cube counting in stacks

16.4.1 Counting cubes in 3D stacks, including hidden ones

Count column by column. Every visible cube rests on cubes below it, even when they are hidden, so a column whose top is at height h contains h cubes.



COLUMN HEIGHTS ADD UP TO 23 CUBES

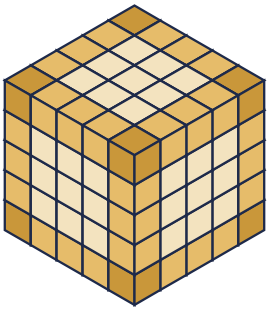
WATCH OUT
Do not count only the cubes you can see. A cube on the top layer means there are cubes under it all the way to the floor.

16.5 Practice questions

16.5.1 Basic level

Q1 A cube is painted on all faces and cut into 125 identical small cubes. How many small cubes have exactly three painted faces?

(a) 8 **(b)** 6 **(c)** 12 **(d)** 24



Q2 A cube painted on all faces is cut into 64 identical cubes. How many have exactly two painted faces?

(a) 12 **(b)** 36 **(c)** 32 **(d)** 24

Q3 A cube painted on all faces is cut into 125 identical cubes. How many have exactly one painted face?

(a) 36 **(b)** 27 **(c)** 54 **(d)** 48

Q4 A cube painted on all faces is cut into 64 identical cubes. How many have no painted face?

(a) 8 **(b)** 4 **(c)** 16 **(d)** 27

Q5 A cube is cut into small cubes by making 3 cuts parallel to each face (9 cuts in all). How many small cubes are formed?

(a) 64 **(b)** 27 **(c)** 48 **(d)** 81

Q6 A cube painted on all faces is cut into 216 identical cubes. How many have exactly two painted faces?

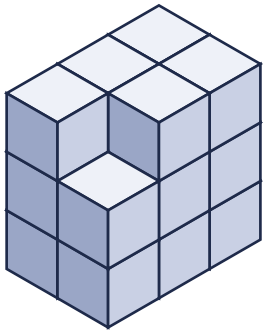
(a) 36 **(b)** 48 **(c)** 64 **(d)** 24

Q7 A cube painted on all faces is cut into 27 identical cubes. How many have at least one painted face?

(a) 20 **(b)** 24 **(c)** 8 **(d)** 26

Q8 How many cubes are there in the stack? (Cubes that support visible cubes are present even if hidden.)

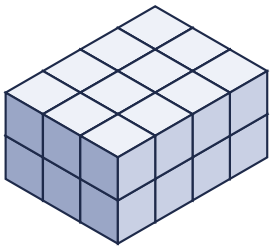
(a) 17 **(b)** 15 **(c)** 19 **(d)** 6



16.5.2 Intermediate level

Q9 A 4 cm × 3 cm × 2 cm cuboid is painted on all faces and cut into 1 cm cubes. How many cubes have exactly two painted faces?

(a) 8 **(b)** 16 **(c)** 12 **(d)** 13



Q10 A 5 × 4 × 3 cuboid painted on all faces is cut into unit cubes. How many cubes have no painted face?

(a) 12 **(b)** 4 **(c)** 6 **(d)** 24

Q11 A 5 × 4 × 3 cuboid painted on all faces is cut into unit cubes. How many cubes have exactly one painted face?

(a) 26 **(b)** 22 **(c)** 24 **(d)** 18

Q12 What is the least number of cuts needed to divide a cube into 64 identical smaller cubes?

(a) 12 **(b)** 8 **(c)** 9 **(d)** 16

Q13 What is the greatest number of pieces into which a cube can be cut with 7 straight cuts, each parallel to a face?

(a) 27 **(b)** 36 **(c)** 24 **(d)** 32

Q14 A cube has opposite faces painted the same colour: two red, two blue, two green. It is cut into 64 identical cubes. How many small cubes have only red and blue on them (no green)?

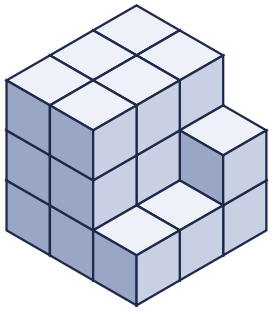
(a) 4 **(b)** 16 **(c)** 8 **(d)** 12

Q15 In the same cube (opposite faces the same colour, cut into 64), how many small cubes are painted green only?

(a) 4 **(b)** 12 **(c)** 8 **(d)** 16

Q16 How many cubes are there in the stack?

(a) 19 **(b)** 24 **(c)** 21 **(d)** 22



16.5.3 Advanced level

Q17 A cube with opposite faces painted the same colour (two red, two blue, two green) is cut into 125 identical cubes. How many cubes show all three colours?

(a) 8 **(b)** 4 **(c)** 12 **(d)** 6

Q18 A cube has its top and bottom painted red, front blue, back green, and left and right black. It is cut into 64 identical cubes. How many cubes are painted blue only?

(a) 9 **(b)** 4 **(c)** 8 **(d)** 2

Q19 A 4 × 4 × 4 painted cube has one corner cube removed. How many small faces of the remaining solid are painted? (Faces exposed by the removal are unpainted.)

(a) 96 **(b)** 90 **(c)** 94 **(d)** 93

Q20 A cube painted on all faces is cut into 343 identical cubes. How many have exactly one painted face?

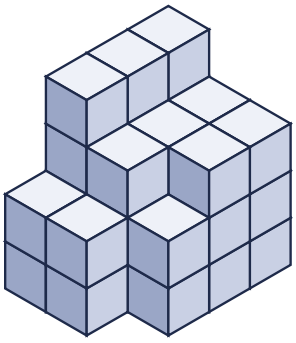
(a) 125 **(b)** 150 **(c)** 108 **(d)** 60

Q21 A cube painted on all faces is cut into small identical cubes, and 8 of them have no painted face. How many small cubes have exactly two painted faces?

(a) 12 **(b)** 24 **(c)** 36 **(d)** 48

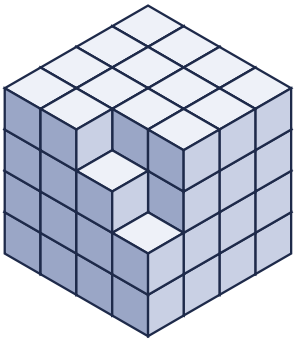
Q22 How many more cubes are needed to turn the stack into a solid 4 × 3 × 4 cuboid?

(a) 17 **(b)** 13 **(c)** 15 **(d)** 33



Q23 How many cubes are there in the stack?

(a) 57 **(b)** 64 **(c)** 61 **(d)** 59



Q24 A 6 × 4 × 2 cuboid painted on all faces is cut into unit cubes. How many cubes have exactly three painted faces?

(a) 20 **(b)** 4 **(c)** 8 **(d)** 9

Q25 A large cube is cut into 27 identical cubes. If the large cube was painted only on its top and bottom faces, how many small cubes have no paint?

(a) 9 **(b)** 1 **(c)** 3 **(d)** 18

16.6 Solutions

Q1 **Answer: (a) 8**

Only the 8 corner cubes have three painted faces, whatever the size of the cube: **8**.

Q2 **Answer: (d) 24**

Two faces: $12(n - 2) = 12 \times 2 =$ **24**.

Q3 **Answer: (c) 54**

$n = 5$. One face: $6(n - 2)^2 = 6 \times 9 =$ **54**.

Q4 **Answer: (a) 8**

No face: $(n - 2)^3 = 2^3 =$ **8**.

Q5 **Answer: (a) 64**

Three cuts in a direction give 4 slices, so there are $4 \times 4 \times 4 =$ **64** cubes.

Q6 **Answer: (b) 48**

$n = 6$. Two faces: $12(n - 2) = 12 \times 4 =$ **48**.

Q7 **Answer: (d) 26**

Only the single centre cube is unpainted: $27 - 1 =$ **26**.

Q8 **Answer: (a) 17**

Column heights: 2, 3, 3, 3, 3, 3; total **17**.

Q9 **Answer: (c) 12**

With $c = 2$, every cube touches the top or bottom. Two faces: $4[(4 - 2) + (3 - 2) + (2 - 2)] = 4 \times 3 =$ **12** (check: corners 8, two faces 12, one face 4, none 0; total 24).

Q10 **Answer: (c) 6**

Inner block: $(5 - 2)(4 - 2)(3 - 2) = 3 \times 2 \times 1 =$ **6**.

Q11 **Answer: (b) 22**

One face: $2[(3)(2) + (2)(1) + (1)(3)] = 2 \times 11 = \mathbf{22}$.

Q12 **Answer: (c) 9**

Each direction needs $4 - 1 = 3$ cuts: $3 \times 3 = \mathbf{9}$ cuts.

Q13 **Answer: (b) 36**

Split the 7 cuts as evenly as possible: 2, 2, 3. Pieces $= 3 \times 3 \times 4 = \mathbf{36}$.

Q14 **Answer: (c) 8**

Red and blue meet along the 4 vertical edges (not touching the green top or bottom). Each such edge holds $n - 2 = 2$ cubes without green: $4 \times 2 = \mathbf{8}$.

Q15 **Answer: (c) 8**

Green faces are the top and bottom. Each has $(n - 2)^2 = 4$ cubes with only that face painted: $2 \times 4 = \mathbf{8}$.

Q16 **Answer: (d) 22**

Column heights: 1, 3, 3, 1, 3, 3, 2, 3, 3; total **22**.

Q17 **Answer: (a) 8**

Only corner cubes touch three different faces, and each corner touches one red, one blue and one green face: **8**.

Q18 **Answer: (b) 4**

The front face is 4×4; its border cubes also touch another face. The inner $(4 - 2)^2 = \mathbf{4}$ cubes are blue only.

Q19 **Answer: (d) 93**

The original surface has $6 \times 16 = 96$ painted small faces. The removed corner cube carried 3 of them, and the 3 newly exposed faces are unpainted, so $96 - 3 = \mathbf{93}$ painted faces remain.

Q20 **Answer: (b) 150**

$n = 7$. One face: $6(n - 2)^2 = 6 \times 25 = \mathbf{150}$.

Q21 **Answer: (b) 24**

$(n - 2)^3 = 8$ gives $n = 4$. Two faces: $12 \times 2 = \mathbf{24}$.

Q22 **Answer: (c) 15**

The stack holds $0 + 2 + 2 + 2 + 3 + 4 + 3 + 3 + 4 + 3 + 3 + 4 = 33$ cubes. A full cuboid holds $4 \times 3 \times 4 = 48$, so **15** more are needed.

Q23 **Answer: (c) 61**

Column heights: 2, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4; total **61**.

Q24 **Answer: (c) 8**

Even with a height of only 2, the only cubes touching three faces are the 8 corner cubes; the other edge cubes touch just two. Answer **8**.

Q25 **Answer: (a) 9**

The 27 cubes form three layers of 9. Top and bottom layers are painted, so the 9 cubes of the middle layer have no paint: **9**.

Painted n-cube: 3 faces 8, 2 faces $12(n - 2)$, 1 face $6(n - 2)^2$, none $(n - 2)^3$; these add up to n^3. For $a \times b \times c$ use $(a - 2)$, $(b - 2)$, $(c - 2)$ in the same pattern. n^3 cubes need $3(n - 1)$ cuts. In stacks, every visible top cube has cubes under it.

MODULE 17 · ADVANCED

Dice

Minimum duration **2 days** | Level **Advanced**

Dice questions show one or more views of a cube (or its net) and ask which faces are opposite, adjacent or hidden. This module covers standard dice (opposite faces sum to 7), ordinary dice, the one-common-face and two-common-faces rules, elimination, the clockwise-order method, nets, dice with symbols, and rolled dice, followed by 25 solved questions in which every answer was confirmed by testing all 720 possible numberings of a cube against the views shown.

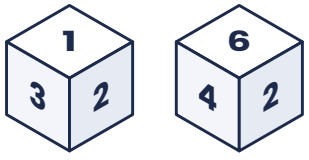
IN THIS MODULE

17.1 Basics
17.2 Finding opposite faces
17.3 Unfolded dice (nets)
17.4 Other variations
17.5 Practice questions
17.6 Solutions

17.1 Basics

17.1.1 Standard dice (opposite faces sum to 7)

On a standard die the opposite faces are 1–6, 2–5 and 3–4, each pair adding up to 7. Two adjacent faces never add up to 7.



A STANDARD DIE FROM TWO POSITIONS: 1 TOP, 6 BOTTOM

17.1.2 Ordinary dice with arbitrary numbering

Ordinary dice can carry any arrangement of numbers, letters or symbols. Then nothing can be assumed: the opposite pairs must be worked out from the views or from the net.

17.2 Finding opposite faces

17.2.1 One common face rule

If two views share exactly one face, read the other faces clockwise starting from the common face in each view. Faces in the same position in the two readings are opposite.

17.2.2 Two common faces rule

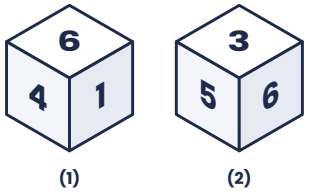
If two views share two faces, the remaining (third) faces of the two views are opposite each other.

17.2.3 Elimination when no common faces exist

Any two faces seen together in a view are adjacent, so they cannot be opposite. A face adjacent to four others must be opposite the sixth.

17.2.4 Clockwise order method

Around any corner of a die the three faces appear in a fixed clockwise order. If a view shows the same three faces in the reverse order, it cannot be the same die.

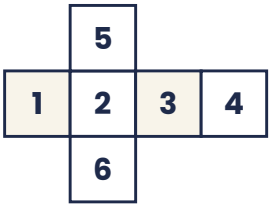


FIND THE FACE OPPOSITE A GIVEN FACE FROM TWO VIEWS

17.3 Unfolded dice (nets)

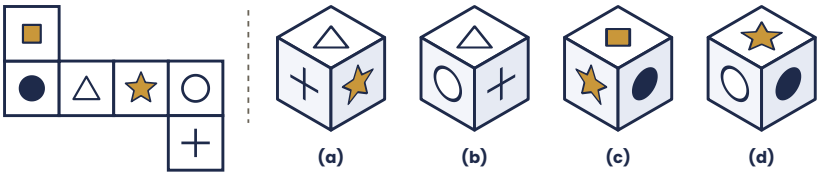
17.3.1 Identifying opposite faces from a net

In a net, faces in a straight line with one face between them become opposite. In the common cross-shaped net, the two ends of the long row of four are opposite, as are the two middle-row faces with one gap, and the faces sticking out above and below.



FACES 1 AND 3 ARE SEPARATED BY ONE FACE: THEY ARE OPPOSITE

17.3.2 Which die can be formed from a net



CHOOSE THE CUBE THAT THIS NET FOLDS INTO

SHORTCUT TRICK
Rule out options in two passes: first any view showing two opposite faces together, then any view whose clockwise order is the mirror image of the net.

17.4 Other variations

17.4.1 Dice with symbols and patterns

Symbol dice follow the same rules as numbered dice. When symbols are directional (arrows, half-shaded faces), their orientation on the cube must also match, which makes the clockwise check even more important.

17.4.2 Rolled dice and hidden faces

When a die is rolled one turn towards a side face, that face goes to the bottom and the face opposite it comes to the top; the faces on the axis of rolling do not move.

17.5 Practice questions

17.5.1 Basic level

Q1 On a standard die, opposite faces add up to 7. Which number is on the bottom face of the die shown?

(a) 1 **(b)** 2 **(c)** 5 **(d)** 3



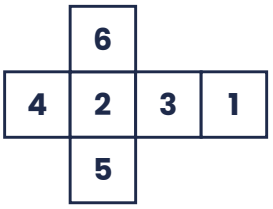
Q2 On a standard die, opposite faces add up to 7. Which number is on the bottom face of the die shown?

(a) 2 **(b)** 1 **(c)** 3 **(d)** 4



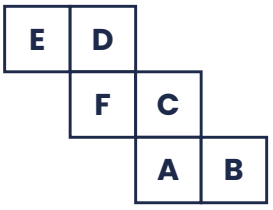
Q3 The net is folded to form a cube. Which face is opposite **1**?

(a) 4 **(b)** 3 **(c)** 6 **(d)** 2



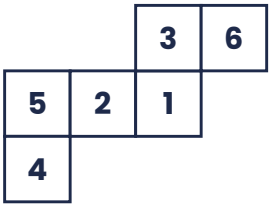
Q4 The net is folded to form a cube. Which face is opposite **E**?

(a) D **(b)** F **(c)** A **(d)** C



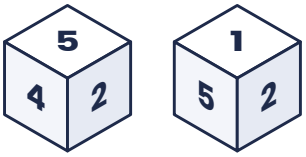
Q5 The net is folded to form a cube. Which face is opposite **2**?

(a) 5 **(b)** 6 **(c)** 1 **(d)** 3



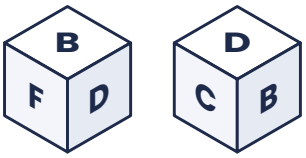
Q6 2 different positions of the same die are shown. Which number is opposite **1**?

(a) 4 **(b)** 3 **(c)** 5 **(d)** 2



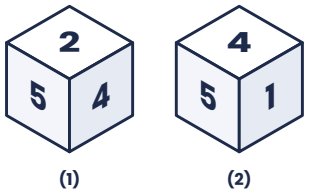
Q7 2 different positions of the same die are shown. Which letter is opposite **F**?

(a) A **(b)** D **(c)** B **(d)** C



Q8 2 different positions of the same die are shown. Which number is opposite **2**?

(a) 3 **(b)** 1 **(c)** 5 **(d)** 6

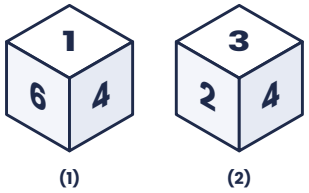


(1) (2)

17.5.2 Intermediate level

Q9 2 different positions of the same die are shown. Which number is opposite **6**?

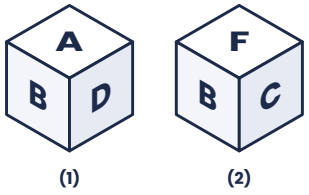
(a) 4 **(b)** 3 **(c)** 1 **(d)** 2



(1) (2)

Q10 2 different positions of the same die are shown. Which letter is opposite **D**?

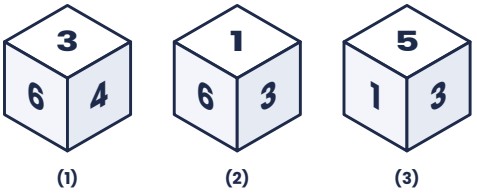
(a) E **(b)** B **(c)** A **(d)** C



(1) (2)

Q11 3 different positions of the same die are shown. Which number is opposite **4**?

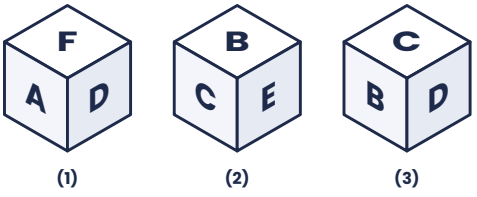
(a) 3 **(b)** 2 **(c)** 6 **(d)** 1



(1) (2) (3)

Q12 3 different positions of the same die are shown. Which letter is opposite **C**?

(a) A **(b)** B **(c)** F **(d)** E



(1) (2) (3)

Q13 3 different positions of the same die are shown. Which number is opposite **1**?

(a) 6 **(b)** 2 **(c)** 3 **(d)** 4



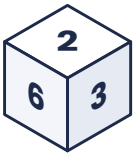
Q14 The standard die shown (opposite faces add up to 7) is rolled one turn towards the right-hand face. Which number is now on top?

(a) 1 **(b)** 5 **(c)** 2 **(d)** 3

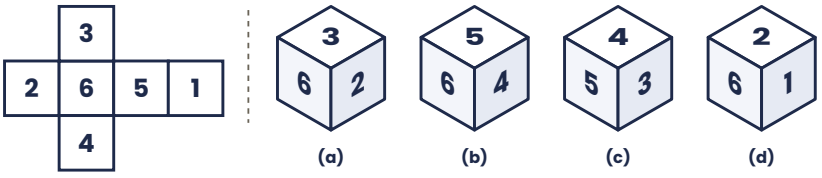


Q15 The standard die shown (opposite faces add up to 7) is rolled one turn towards the right-hand face and then one turn towards the left-hand face. Which number is now on top?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

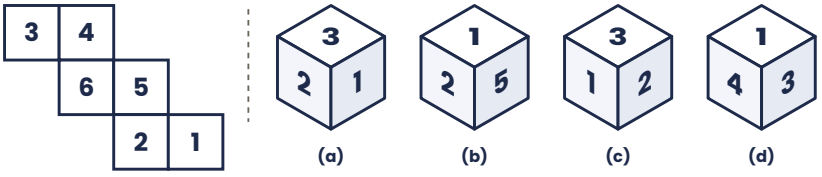


Q16 The net is folded into a cube. Which of the answer figures shows that cube?

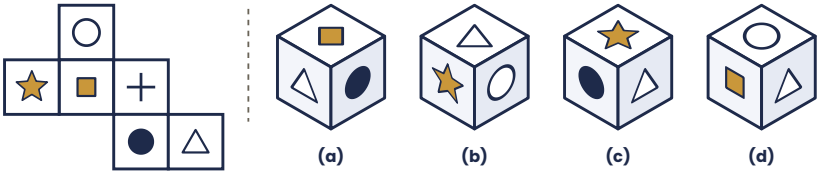


17.5.3 Advanced level

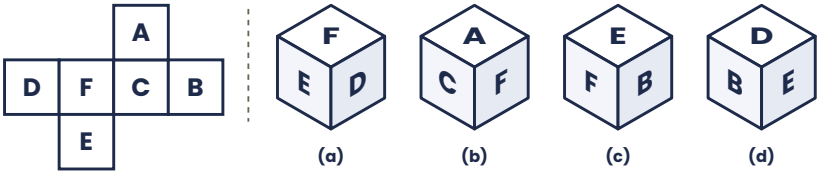
Q17 The net is folded into a cube. Which of the answer figures shows that cube?



Q18 The net is folded into a cube. Which of the answer figures shows that cube?

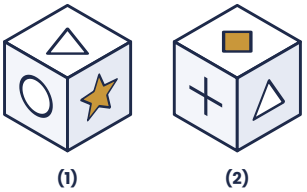


Q19 The net is folded into a cube. Which of the answer figures shows that cube?



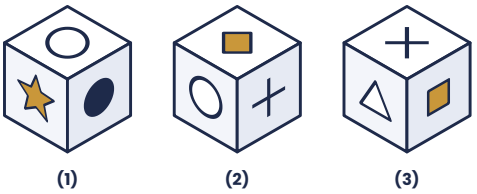
Q20 2 different positions of the same die are shown. Which symbol is opposite ✚?

(a) ■ **(b)** ○ **(c)** ★ **(d)** ●



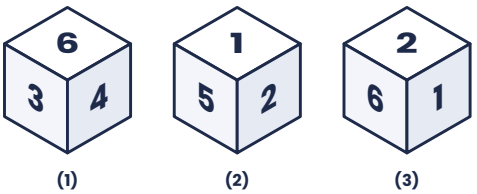
Q21 3 different positions of the same die are shown. Which symbol is opposite ▲?

(a) ○ **(b)** + **(c)** ■ **(d)** ●



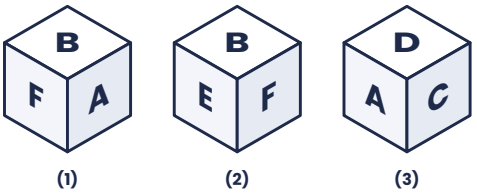
Q22 Three positions of the same die are shown. Which face is **not** adjacent to 1?

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4



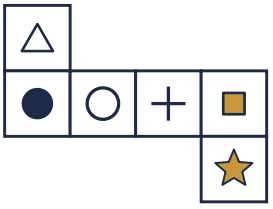
Q23 Three positions of the same die are shown. Which face is **not** adjacent to D?

(a) C **(b)** F **(c)** B **(d)** A



Q24 The net is folded to form a cube. Which face is opposite ●?

(a) + **(b)** ▲ **(c)** ○ **(d)** ■



Q25 The standard die shown (opposite faces add up to 7) is rolled one turn towards the right-hand face and then one turn towards the right-hand face and then one turn towards the left-hand face. Which number is now on top?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 1



17.6 Solutions

Q1 **Answer: (d) 3**

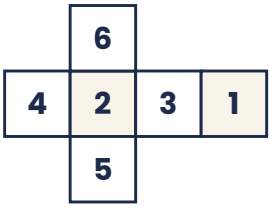
Top shows 4, so the bottom shows 7 - 4 = **3**.

Q2 **Answer: (a) 2**

Top shows 5, so the bottom shows 7 - 5 = **2**.

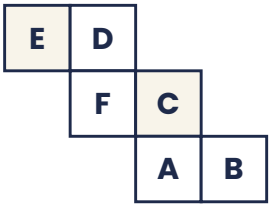
Q3 **Answer: (d) 2**

In a net, two faces in the same row or column with exactly one face between them become opposite faces; for the remaining faces, fold step by step. Folding this net, 1 ends up opposite **2** (answer (d)).



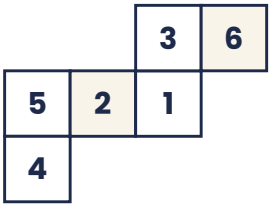
Q4 **Answer: (d) C**

In a net, two faces in the same row or column with exactly one face between them become opposite faces; for the remaining faces, fold step by step. Folding this net, E ends up opposite **C** (answer (d)).



Q5 **Answer: (b) 6**

In a net, two faces in the same row or column with exactly one face between them become opposite faces; for the remaining faces, fold step by step. Folding this net, 2 ends up opposite **6** (answer (b)).



Q6 **Answer: (a) 4**

Two faces are common to both views; the remaining faces of the two views are therefore opposite each other. Here 1 is opposite **4**. Answer **(a) 4**.

Q7 **Answer: (d) C**

Two faces are common to both views; the remaining faces of the two views are therefore opposite each other. Here F is opposite **C**. Answer **(d) C**.

Q8 **Answer: (b) 1**

Two faces are common to both views; the remaining faces of the two views are therefore opposite each other. Here 2 is opposite **1**. Answer **(b) 1**.

Q9 **Answer: (d) 2**

The face 4 appears in both views. Reading the other two faces clockwise from 4 in each view and matching positions pairs each face with the one opposite it. The checked result: 6 is opposite 2. Answer **(d) 2**.

Q10 **Answer: (d) C**

The face B appears in both views. Reading the other two faces clockwise from B in each view and matching positions pairs each face with the one opposite it. The checked result: D is opposite C. Answer **(d) C**.

Q11 **Answer: (d) 1**

Faces seen together on one die are adjacent, never opposite. In the views, 4 appears beside 3, 6, so these cannot be opposite it. That leaves 1, 2, 5; matching the clockwise order of the faces around 4 in the views rules out all but 1. Answer **(d) 1**.

Q12 **Answer: (c) F**

Faces seen together on one die are adjacent, never opposite. In the views, C appears beside B, D, E, so these cannot be opposite it. That leaves A, F; matching the clockwise order of the faces around C in the views rules out all but F. Answer **(c) F**.

Q13 **Answer: (d) 4**

Faces seen together on one die are adjacent, never opposite. In the views, 1 appears beside 3, 5, 6, so these cannot be opposite it. That leaves 2, 4; matching the clockwise order of the faces around 1 in the views rules out all but 4. Answer **(d) 4**.

Q14 **Answer: (b) 5**

Rolling towards a side face tips that face to the bottom and brings the face opposite it to the top. Following the turns step by step, the top face becomes **5** (answer (b)).

Q15 **Answer: (a) 1**

Rolling towards a side face tips that face to the bottom and brings the face opposite it to the top. Following the turns step by step, the top face becomes **1** (answer (a)).

Q16 **Answer: (b)**

Check opposite pairs first: two faces that are opposite in the net can never be seen together. Then check the clockwise order around a corner. Only (b) passes both tests.

Q17 **Answer: (c)**

Check opposite pairs first: two faces that are opposite in the net can never be seen together. Then check the clockwise order around a corner. Only (c) passes both tests.

Q18 **Answer: (b)**

Check opposite pairs first: two faces that are opposite in the net can never be seen together. Then check the clockwise order around a corner. Only (b) passes both tests.

Q19 **Answer: (d)**

Check opposite pairs first: two faces that are opposite in the net can never be seen together. Then check the clockwise order around a corner. Only (d) passes both tests.

Q20 **Answer: (c) ★**

The face ▲ appears in both views. Reading the other two faces clockwise from ▲ in each view and matching positions pairs each face with the one opposite it. The checked result: + is opposite ★. Answer **(c) ★**.

Q21 **Answer: (a) ○**

Faces seen together on one die are adjacent, never opposite. In the views, ▲ appears beside +, ■, so these cannot be opposite it. That leaves ●, ○, ★; matching the clockwise order of the faces around ▲ in the views rules out all but ○. Answer **(a)** ○.

Q22 **Answer: (c) 3**

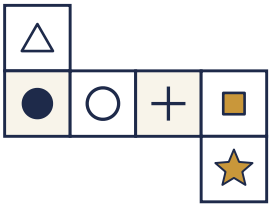
The only face that is not adjacent to 1 is the face opposite it. The views show 1 beside 2, 5, 6; checking all numberings consistent with the views, the face opposite 1 is **3** (answer (c)).

Q23 **Answer: (b) F**

The only face that is not adjacent to D is the face opposite it. The views show D beside A, C; checking all numberings consistent with the views, the face opposite D is **F** (answer (b)).

Q24 **Answer: (a) +**

In a net, two faces in the same row or column with exactly one face between them become opposite faces; for the remaining faces, fold step by step. Folding this net, ● ends up opposite **+** (answer (a)).



Q25 **Answer: (d) 1**

Rolling towards a side face tips that face to the bottom and brings the face opposite it to the top. Following the turns step by step, the top face becomes **1** (answer (d)).

Standard dice: opposite faces add to 7. Two views with two common faces: the third faces are opposite. One common face: read clockwise from it and match positions. Faces seen together are adjacent; the face never seen beside X is opposite X. In nets, faces in line with one gap are opposite; check opposite pairs and clockwise order to pick the folded cube.

MODULE 18 · ADVANCED

3D spatial visualisation

Minimum duration **2 days** | Level **Advanced**

Spatial questions ask you to fold nets, read views of block structures, turn objects in your mind and picture cross-sections. This module covers the eleven nets of a cube, nets of other solids, top, front and side views, building objects from their views, 3D rotation and mirror-image traps, cross-sections and isometric drawing, followed by 25 solved questions in which every net was folded, every view projected and every rotation tested by a program.

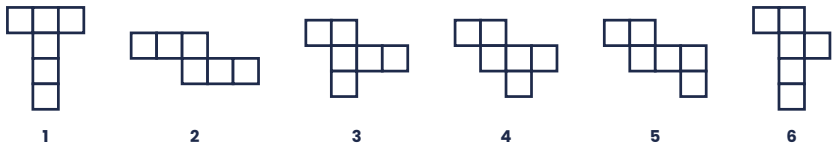
IN THIS MODULE

18.1 Nets of 3D shapes
18.2 Orthographic views
18.3 3D rotation
18.4 Cross-sections
18.5 Isometric drawing (overview)
18.6 Practice questions
18.7 Solutions

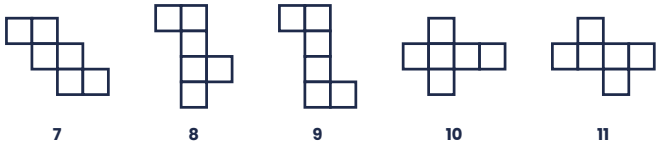
18.1 Nets of 3D shapes

18.1.1 The eleven nets of a cube

A net is a flat shape that folds into a solid. Of the 35 shapes that can be made by joining six squares edge to edge, exactly eleven fold into a cube. All eleven are shown below (found by folding every one of the 35 shapes).



CUBE NETS 1 TO 6



CUBE NETS 7 TO 11

SHORTCUT TRICK

A net is never a cube net if it contains a 2 × 2 block of squares, or a straight row of five or more squares. In a valid net, squares in a line with one square between them always become opposite faces.

18.1.2 Nets of cuboids, pyramids and prisms

Solid	Faces	Edges	Vertices	Net contains
Cube / cuboid	6	12	8	6 squares / 3 pairs of rectangles
Triangular prism	5	9	6	2 triangles + 3 rectangles
Square pyramid	5	8	5	1 square + 4 triangles
Tetrahedron	4	6	4	4 triangles
Hexagonal prism	8	18	12	2 hexagons + 6 rectangles

Euler's rule for any such solid: faces + vertices − edges = 2

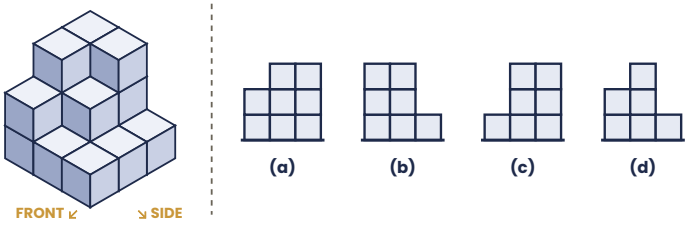
18.1.3 Folding a net into a box with patterns

When the faces carry patterns, first find the opposite pairs (they can never be seen together), then check the order of the faces around one corner, exactly as with dice in Module 17.

18.2 Orthographic views

18.2.1 Top, front and side views of block structures

A view shows the structure flat, from one direction. In a front or side view each column shows the tallest stack along that line of sight; in a top view every occupied position shows as one square, whatever its height.



FIND THE FRONT VIEW

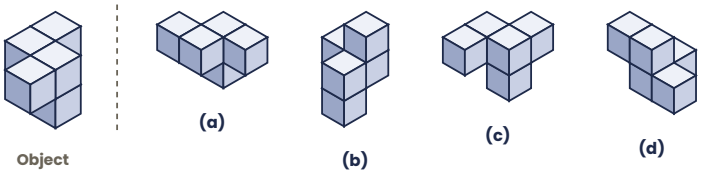
18.2.2 Building the 3D object from views

Given the front and side views, the tallest a stack can be at any position is the smaller of its two column heights. Using that height everywhere gives the greatest number of cubes; the least number uses one stack to satisfy a front column and a side column of equal height at the same time.

18.3 3D rotation

18.3.1 Identifying rotated versions of the same object

Pick a distinctive part of the object (a branch, a corner, a long arm) and follow where it goes. In a true rotation the parts keep their arrangement around each other.



ONE OPTION IS THE OBJECT TURNED; THE OTHERS ARE ITS MIRROR IMAGE

18.3.2 Mirror-image (chiral) traps

WATCH OUT
Options are often mirror images of the object. A mirror image cannot be produced by turning, however long you try. Check the direction of turns: if going along the object you turn left then up in the original, you must turn left then up in a rotated copy too.

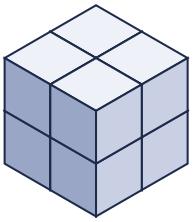
18.4 Cross-sections

18.4.1 Shapes formed by slicing solids

Solid	Slice	Cross-section
Cube	parallel to a face	square
Cube	through four vertices along a face diagonal	rectangle
Cube	cutting off a corner	triangle
Cube	through the midpoints of six edges	regular hexagon
Cylinder	parallel to the base / along the axis	circle / rectangle
Cone	parallel to the base / through the apex	circle / triangle
Cone	slanted, not through the base	ellipse
Sphere	any plane	circle
Square pyramid	parallel to the base	square

18.5 Isometric drawing (overview)

Isometric drawings show three faces of a solid at once, with the three edge directions drawn 120° apart. Vertical edges stay vertical; the other two edge directions slope at 30°. Every cube in this book is drawn this way, so each visible cube shows a top, a left and a right face.

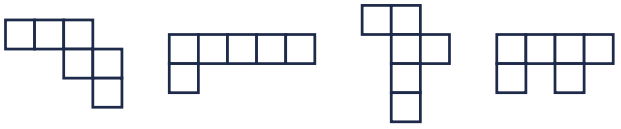


A 2 × 2 × 2 BLOCK IN ISOMETRIC VIEW

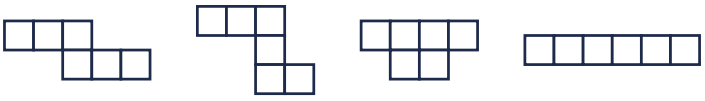
18.6 Practice questions

18.6.1 Basic level

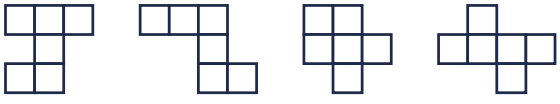
Q1 Which of the following figures can be folded to make a cube?



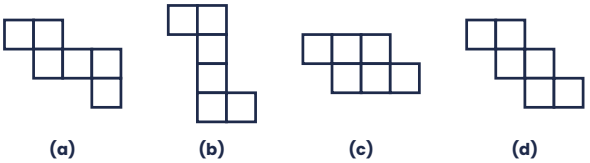
Q2 Which of the following figures can be folded to make a cube?



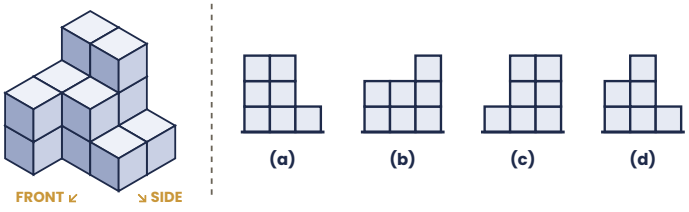
Q3 Which of the following figures can be folded to make a cube?



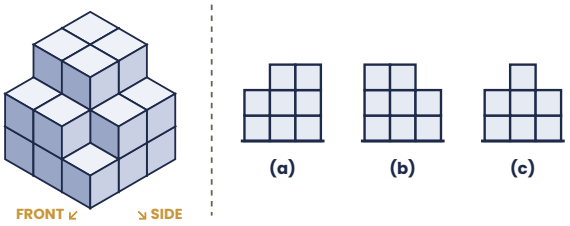
Q4 Which of the following figures can **not** be folded to make a cube?



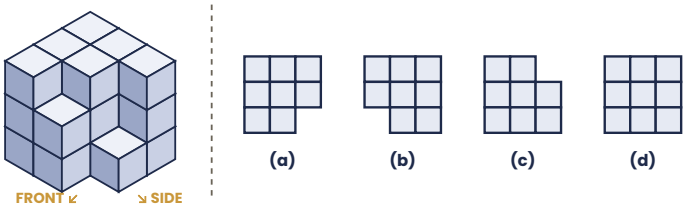
Q5 Which answer figure is the front view (looking at the face marked FRONT) of the block structure?



Q6 Which answer figure is the side view (looking at the face marked SIDE) of the block structure?



Q7 Which answer figure is the top view of the block structure? (In the top view, the FRONT edge is at the bottom.)

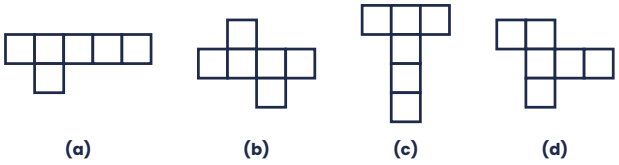


Q8 A cube is sliced by a plane parallel to one of its faces. What is the shape of the cut face?

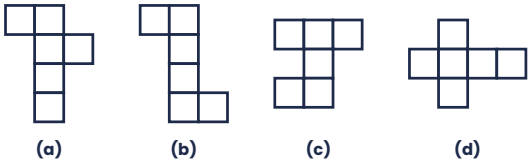
(a) rectangle **(b)** square **(c)** triangle **(d)** hexagon

18.6.2 Intermediate level

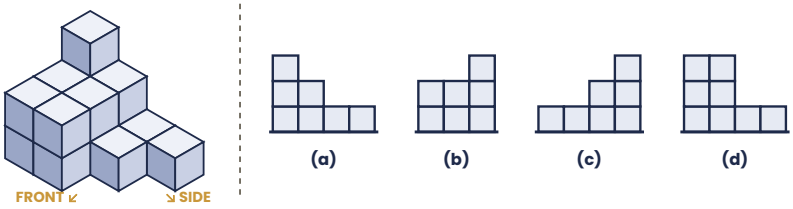
Q9 Which of the following figures can **not** be folded to make a cube?



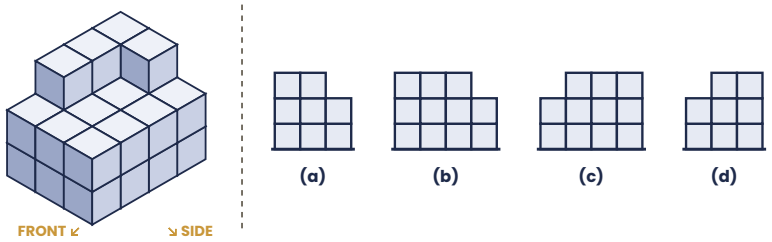
Q10 Which of the following figures can **not** be folded to make a cube?



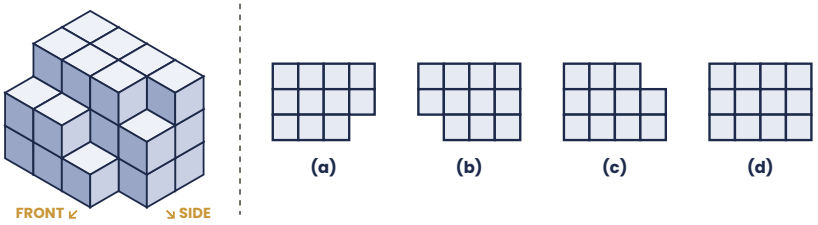
Q11 Which answer figure is the front view (looking at the face marked FRONT) of the block structure?



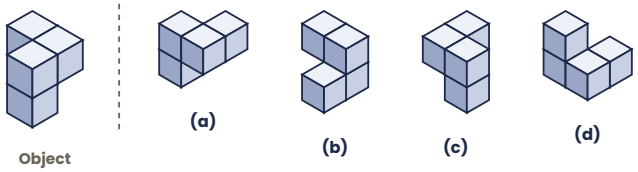
Q12 Which answer figure is the side view (looking at the face marked SIDE) of the block structure?



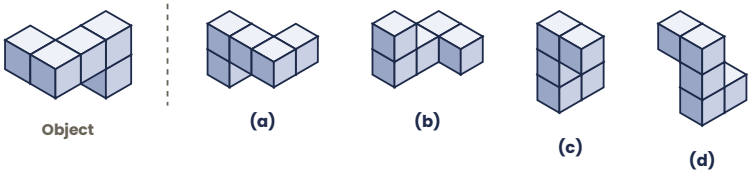
Q13 Which answer figure is the top view of the block structure? (In the top view, the FRONT edge is at the bottom.)



Q14 Which answer figure shows the same object as the one on the left, only turned to a different position?



Q15 Which answer figure shows the same object as the one on the left, only turned to a different position?

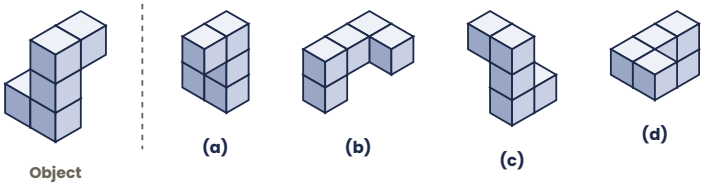


Q16 How many edges does a triangular prism have?

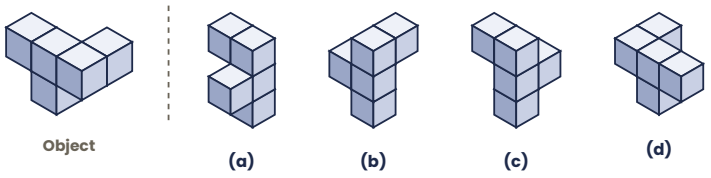
(a) 6 **(b)** 8 **(c)** 12 **(d)** 9

18.6.3 Advanced level

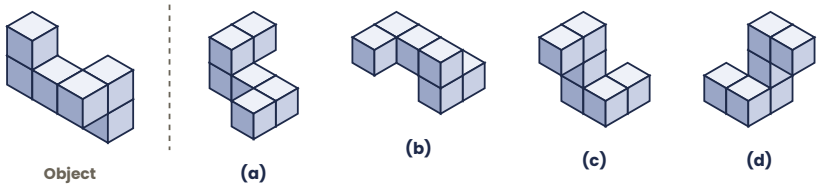
Q17 Which answer figure shows the same object as the one on the left, only turned to a different position?



Q18 Which answer figure shows the same object as the one on the left, only turned to a different position?

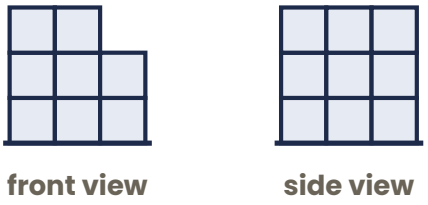


Q19 Which answer figure shows the same object as the one on the left, only turned to a different position?



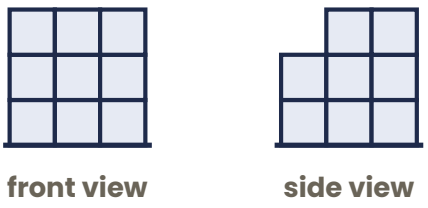
Q20 A structure is built from identical cubes on a 3 × 3 base. Its front and side views are shown. What is the greatest number of cubes it can contain?

(a) 11 **(b)** 26 **(c)** 24 **(d)** 22

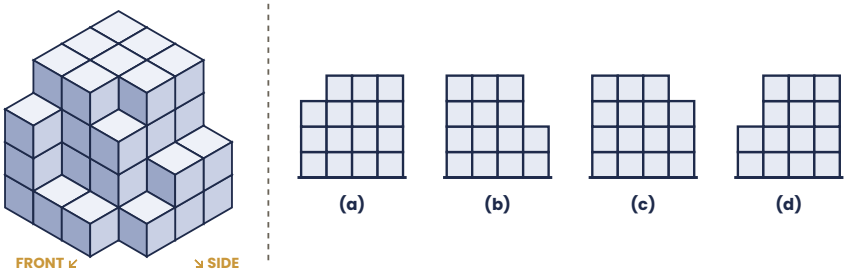


Q21 A structure is built from identical cubes on a 3 × 3 base. Its front and side views are shown. What is the least number of cubes it can contain?

(a) 24 **(b)** 11 **(c)** 13 **(d)** 9



Q22 Which answer figure is the side view (looking at the face marked SIDE) of the block structure?



Q23 A cube is sliced by a plane passing through the midpoints of six of its edges, cutting all six faces. What is the cross-section?

(a) square **(b)** regular hexagon
(c) equilateral triangle **(d)** pentagon

Q24 A cone is cut by a vertical plane passing through its apex. What is the shape of the cross-section?

(a) circle **(b)** ellipse **(c)** triangle **(d)** rectangle

Q25 A square pyramid has how many faces, edges and vertices?

(a) 5, 8, 5 **(b)** 4, 8, 5 **(c)** 5, 9, 6 **(d)** 5, 8, 4

18.7 Solutions

Q1 **Answer: (c)**

Only (c) folds into a closed cube: every face lands on a different side. In each of the others two squares fold onto the same side of the cube, leaving one side open. Answer **(c)**.

Q2 **Answer: (a)**

Only (a) folds into a closed cube: every face lands on a different side. In each of the others two squares fold onto the same side of the cube, leaving one side open. Answer **(a)**.

Q3 **Answer: (d)**

Only (d) folds into a closed cube: every face lands on a different side. In each of the others two squares fold onto the same side of the cube, leaving one side open. Answer **(d)**.

Q4 **Answer: (c)**

(c) cannot be folded into a cube: two of its squares fall on the same side, leaving another side open. The other three are among the eleven cube nets. Answer **(c)**.

Q5 **Answer: (a)**

In a front view each column shows the tallest stack in that line of sight. Reading the columns from left to right gives heights 3, 3, 1: answer **(a)**.

Q6 **Answer: (a)**

In a side view each column shows the tallest stack in that line of sight. Reading the columns from left to right gives heights 2, 3, 3: answer **(a)**.

Q7 **Answer: (a)**

From above you see one square for every column that has at least one cube, whatever its height. With the front at the bottom, the occupied squares match (a) only.

Q8 **Answer: (b) square**

Every slice parallel to a face of a cube copies that face. Answer **(b) square**.

Q9 **Answer: (a)**

(a) cannot be folded into a cube: two of its squares fall on the same side, leaving another side open. The other three are among the eleven cube nets. Answer **(a)**.

Q10 **Answer: (c)**

(c) cannot be folded into a cube: two of its squares fall on the same side, leaving another side open. The other three are among the eleven cube nets. Answer **(c)**.

Q11 **Answer: (a)**

In a front view each column shows the tallest stack in that line of sight. Reading the columns from left to right gives heights 3, 2, 1, 1: answer **(a)**.

Q12 **Answer: (c)**

In a side view each column shows the tallest stack in that line of sight. Reading the columns from left to right gives heights 2, 3, 3, 3: answer **(c)**.

Q13 **Answer: (a)**

From above you see one square for every column that has at least one cube, whatever its height. With the front at the bottom, the occupied squares match (a) only.

Q14 **Answer: (a)**

Figure (a) is the object turned in space. The other three are its **mirror image** in various positions: however they are turned, they can never be made to match the original, just as a left hand cannot be turned into a right hand.

Q15 **Answer: (c)**

Figure (c) is the object turned in space. The other three are its **mirror image** in various positions: however they are turned, they can never be made to match the original, just as a left hand cannot be turned into a right hand.

Q16 **Answer: (d) 9**

Two triangles give 3 + 3 = 6 edges, and 3 more edges join them: 9. (Check with Euler: 5 faces + 6 vertices − 9 edges = 2.) Answer **(d) 9**.

Q17 **Answer: (b)**

Figure (b) is the object turned in space. The other three are its **mirror image** in various positions: however they are turned, they can never be made to match the original, just as a left hand cannot be turned into a right hand.

Q18 **Answer: (c)**

Figure (c) is the object turned in space. The other three are its **mirror image** in various positions: however they are turned, they can never be made to match the original, just as a left hand cannot be turned into a right hand.

Q19 **Answer: (c)**

Figure (c) is the object turned in space. The other three are its **mirror image** in various positions: however they are turned, they can never be made to match the original, just as a left hand cannot be turned into a right hand.

Q20 **Answer: (c) 24**

For the greatest number, every position can be as tall as both views allow: the height at each position is the smaller of its front-view and side-view column. Adding these gives **24**.

Q21 **Answer: (b) 11**

For the least number, use as few stacks as possible: one stack can show a column in the front view and a column in the side view at the same time when the two heights are equal. Checking every arrangement of up to 3 cubes per position, the fewest that give both views is **11**.

Q22 **Answer: (a)**

In a side view each column shows the tallest stack in that line of sight. Reading the columns from left to right gives heights 3, 4, 4, 4: answer **(a)**.

Q23 **Answer: (b) regular hexagon**

A plane cutting all six faces produces a six-sided section; through the edge midpoints all its sides are equal. Answer **(b) regular hexagon**.

Q24 **Answer: (c) triangle**

A plane through the apex and the base cuts two straight slant lines and the base diameter. Answer **(c) triangle**.

Q25 **Answer: (a) 5, 8, 5**

One square base and four triangles make 5 faces; 4 base edges + 4 slant edges = 8; 4 base corners + the apex = 5 vertices. Answer **(a) 5, 8, 5**.

Eleven of the 35 six-square shapes fold into a cube; a 2 × 2 block or a row of five rules a shape out. Front and side views show the tallest stack in each line of sight; a top view shows occupied positions. A turned object keeps its arrangement; a mirror image never matches. Slices parallel to a face copy the face.

MODULE 19 · ADVANCED

Abstract and rule-based reasoning

Minimum duration **2 days** | Level **Advanced**

Modern assessments use abstract puzzles in which you discover a hidden rule from a few examples. This module covers the Set A / Set B / Neither format, abstract sequences, which-figure-belongs questions, the main rule types (counting, position, shading, conditional rules) and the hypothesis-test-revise method with distractor elements, followed by 25 solved questions in which every example figure was generated to obey its set rule.

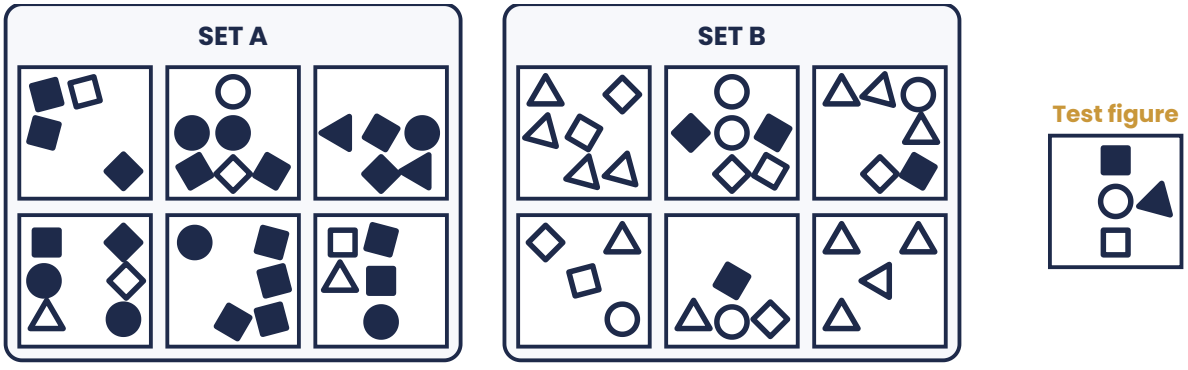
IN THIS MODULE

19.1 Abstract reasoning formats
19.2 Rule types
19.3 Method
19.4 Practice questions
19.5 Solutions

19.1 Abstract reasoning formats

19.1.1 Set A / Set B / Neither questions

Six figures in Set A share one rule; six figures in Set B share a related but different rule. You decide whether a test figure belongs to Set A, Set B or neither. The two rules are usually opposites (3 shapes versus 5, black outnumbering white versus white outnumbering black), and "neither" covers figures that obey neither rule.

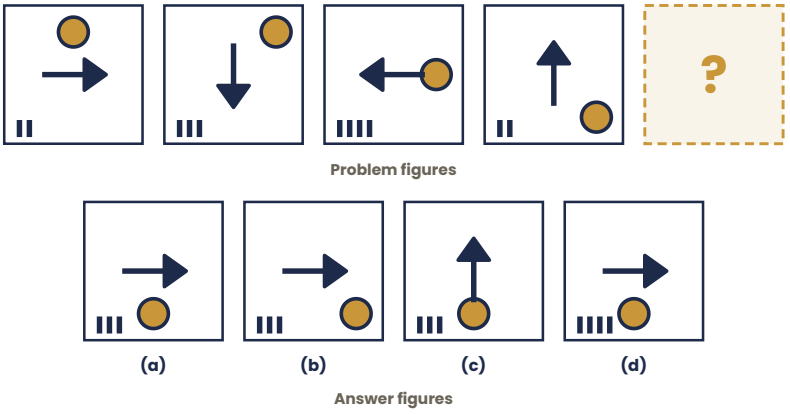


SET A, SET B AND A TEST FIGURE

Rule: In Set A black shapes outnumber white ones; in Set B white shapes outnumber black ones.

19.1.2 Next figure in an abstract sequence

Abstract sequences usually run two or three independent changes at once: something moves, something turns, something is counted. Track each element separately.



THREE INDEPENDENT CHANGES RUN TOGETHER

19.1.3 Statement-based abstract questions

Some tests give the two sets and ask which option belongs to Set A (or completes a statement such as "this figure belongs to Set B"). Find the rule first, then test each option against it.

19.2 Rule types

19.2.1 Counting rules (sides, intersections, enclosed areas)

Count shapes, sides, right angles, intersections, enclosed areas or curved versus straight lines. Rules often compare two counts: more curved than straight, odd number of sides, and so on.

19.2.2 Position and alignment rules

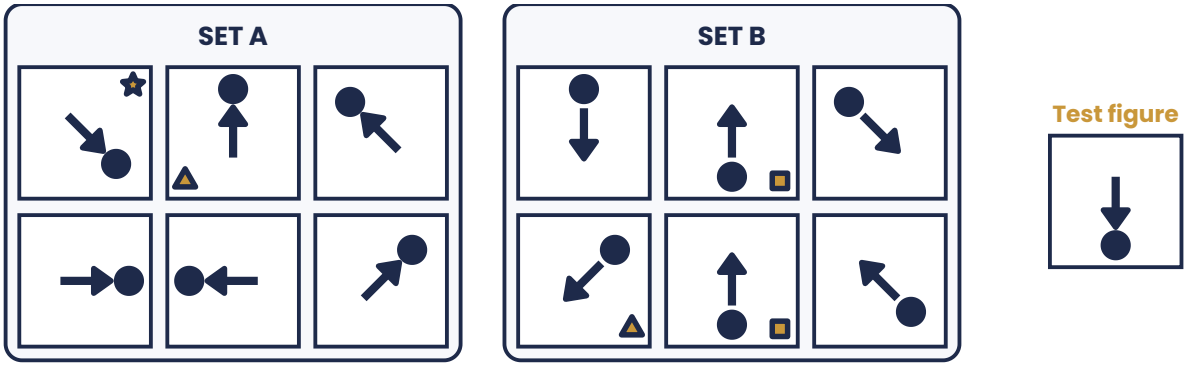
Look at where elements sit: left or right half, touching the frame, lined up diagonally, inside or outside another shape.

19.2.3 Colour and shading rules

Count black, white and patterned shapes. Compare the numbers, or check whether a particular shape is always shaded.

19.2.4 Conditional rules (if a circle is present, the arrow points up)

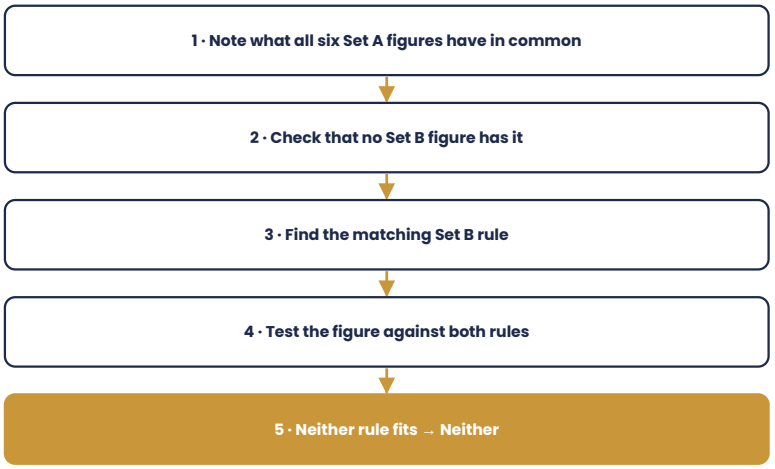
A conditional rule links two features: "the arrow always points at the circle" or "if there is a triangle, it is black". Test such a rule on every figure, including those where the condition does not apply.



A CONDITIONAL RULE: WHERE DOES THE ARROW POINT?

19.3 Method

19.3.1 Form a hypothesis, test it on every example, revise



HYPOTHESIS, TEST, REVISE

19.3.2 Separating distractor elements from rule elements

WATCH OUT

Many elements are random decoration: extra shapes, colours, rotations. If a feature varies freely within Set A, it cannot be the rule. Focus on what stays constant within a set and differs between sets.

SHORTCUT TRICK

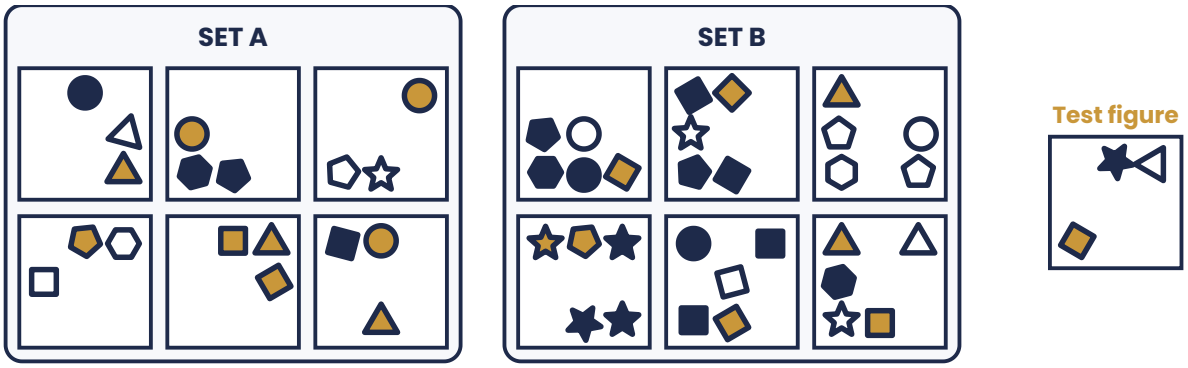
Start with the simplest figure in each set: it has the fewest distractors, so the rule shows most clearly.

19.4 Practice questions

19.4.1 Basic level

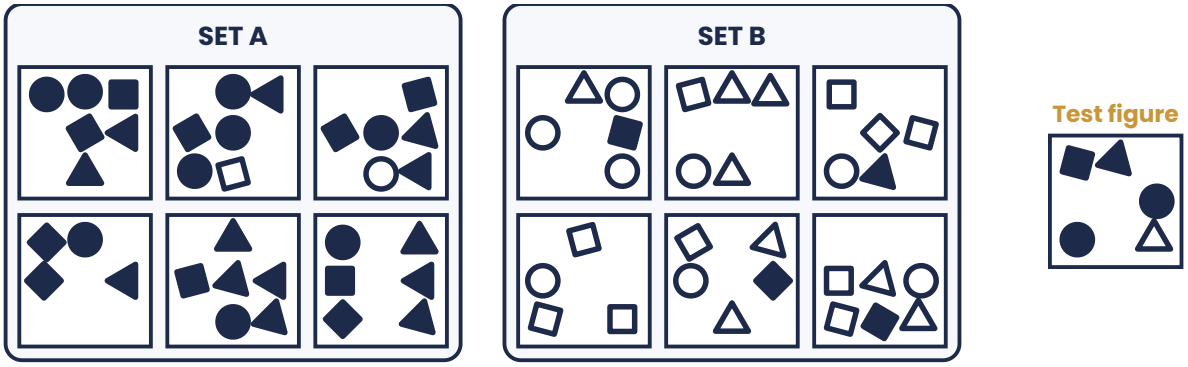
Q1 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



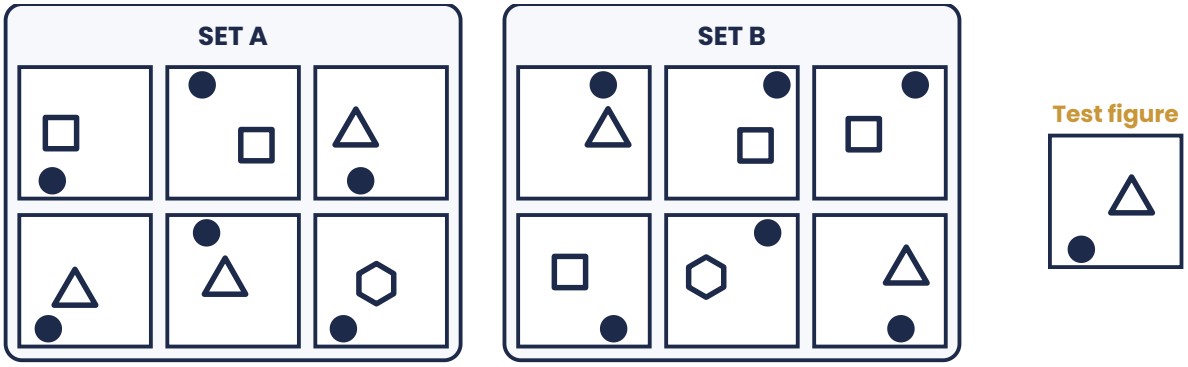
Q2 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



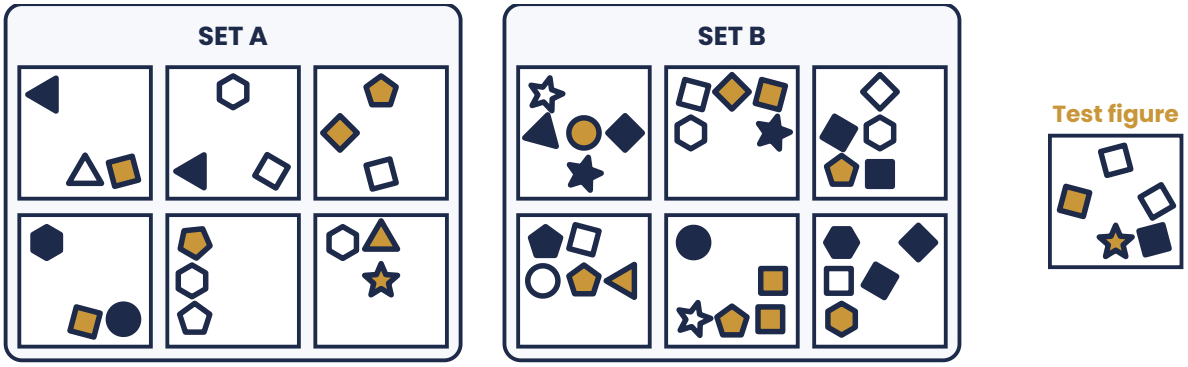
Q3 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither

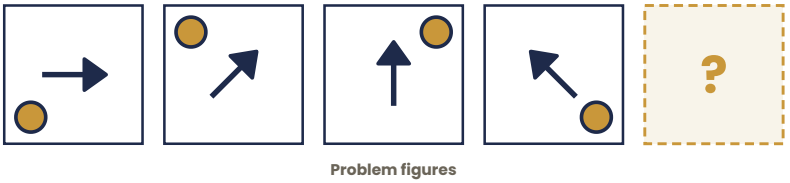


Q4 Does the test figure belong to Set A, Set B or neither?

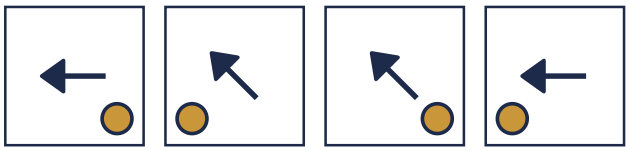
(a) Set A **(b)** Set B **(c)** Neither



Q5 Which answer figure continues the sequence?



Problem figures

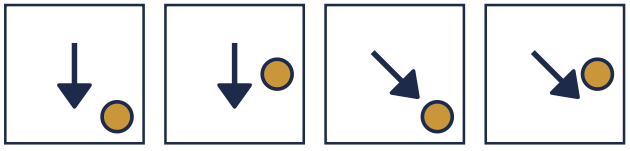


Answer figures

Q6 Which answer figure continues the sequence?

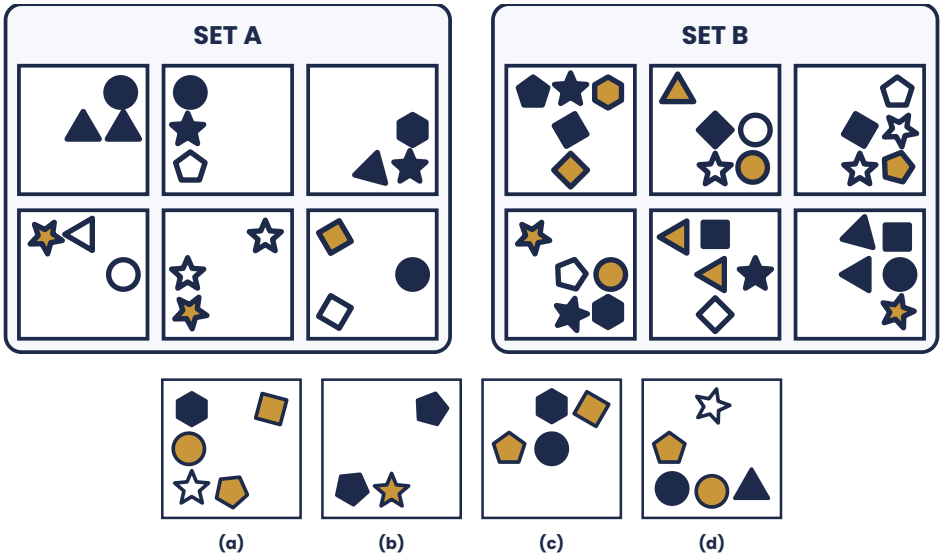


Problem figures

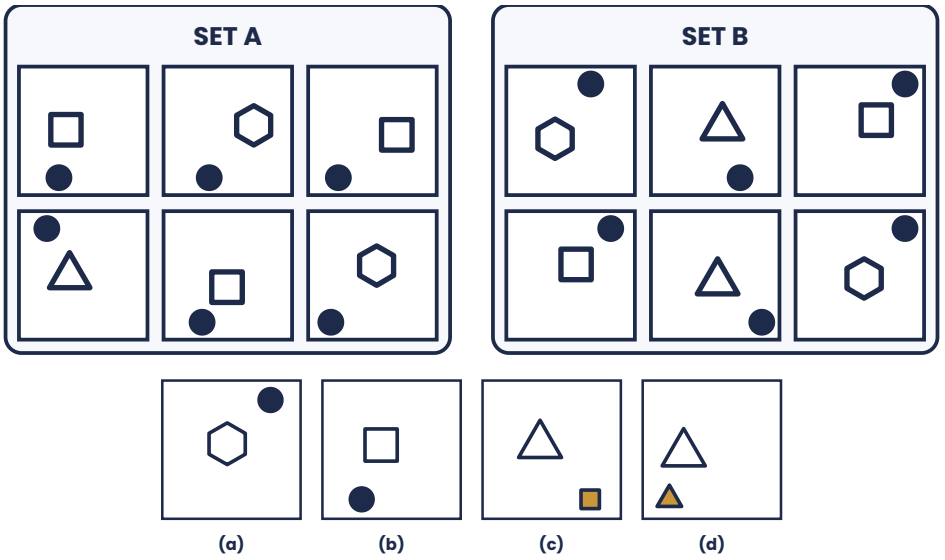


Answer figures

Q7 Which answer figure belongs to Set A?



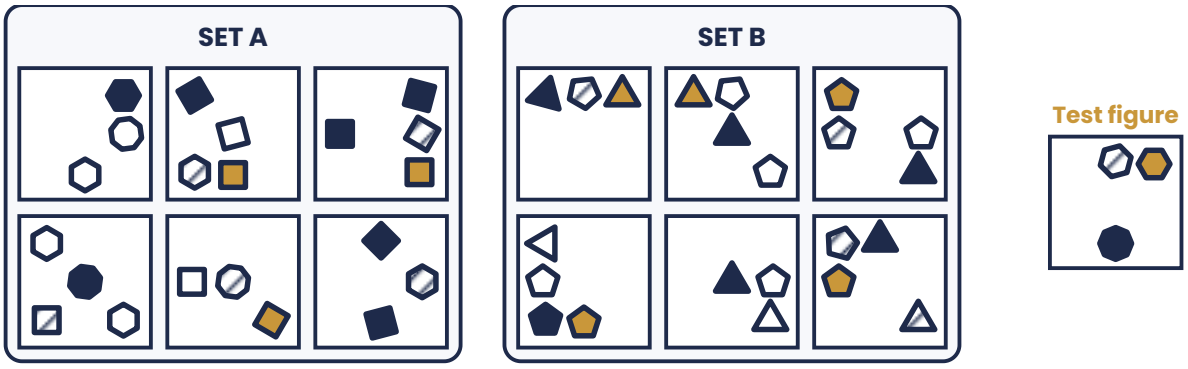
Q8 Which answer figure belongs to Set A?



19.4.2 Intermediate level

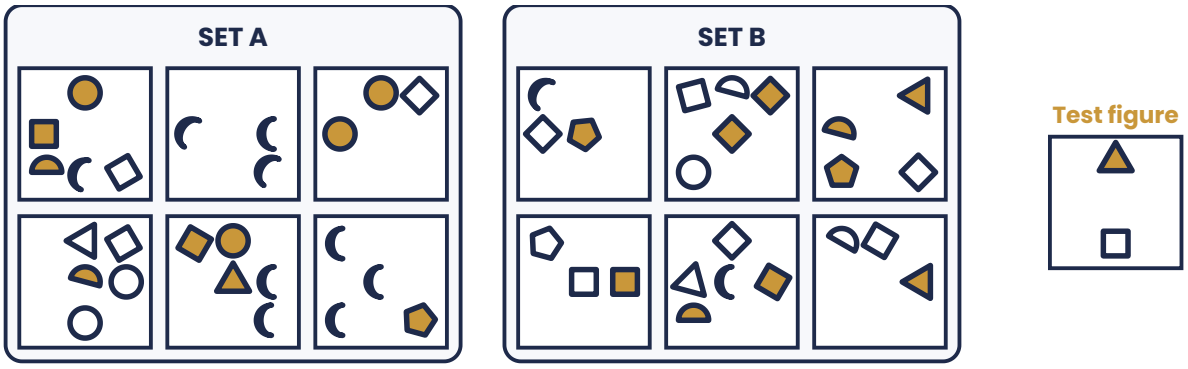
Q9 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



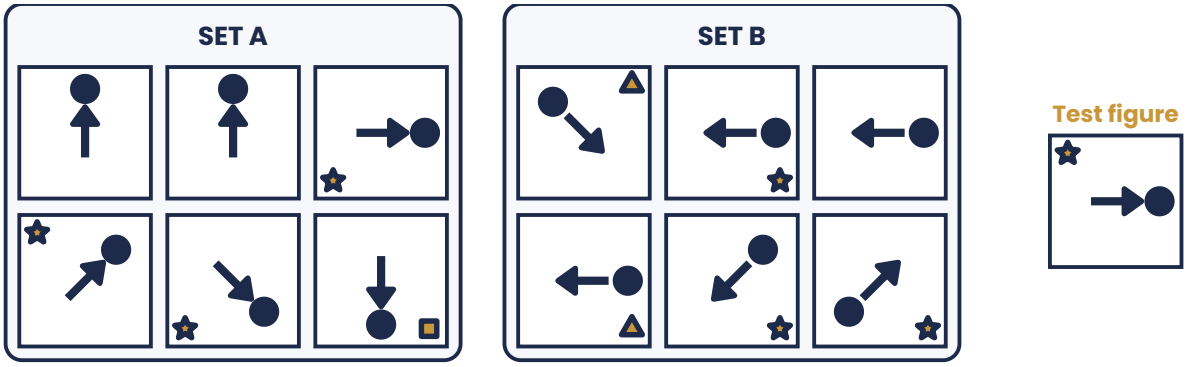
Q10 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



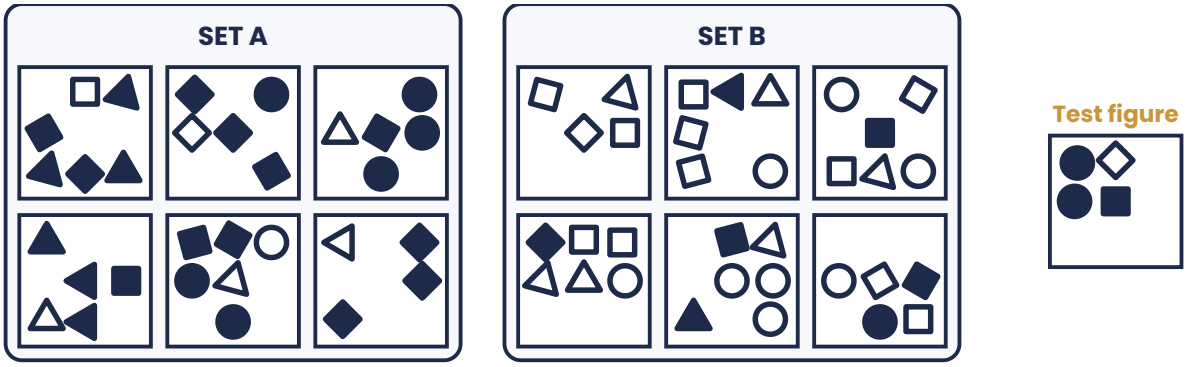
Q11 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither

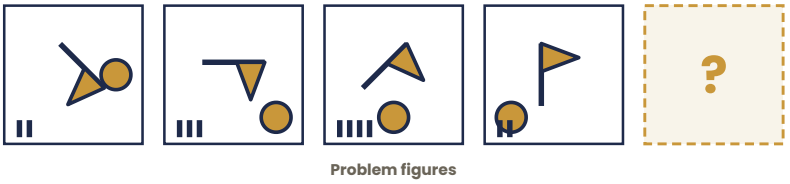


Q12 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



Q13 Which answer figure continues the sequence?



Problem figures



Answer figures

Q14 Which answer figure continues the sequence?



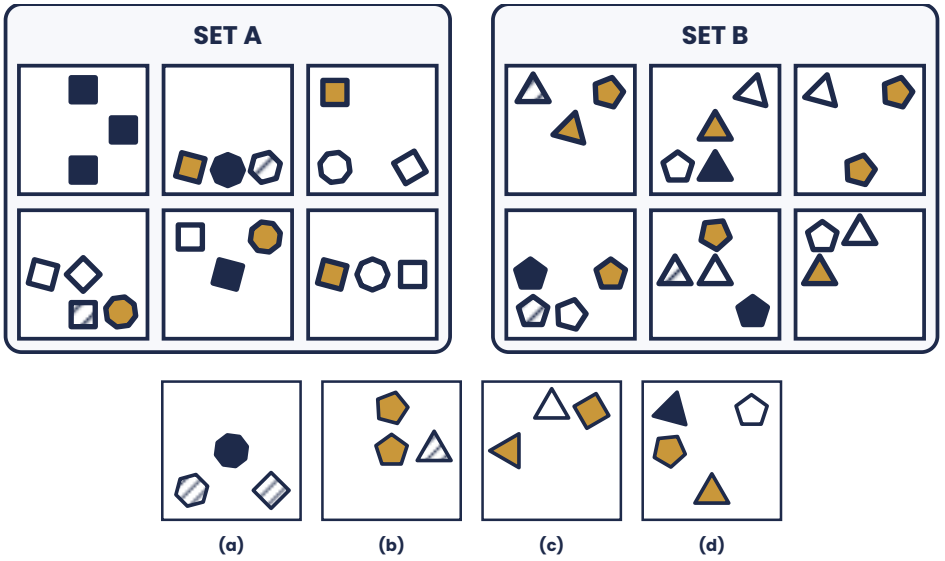
Problem figures



(a) (b) (c) (d)

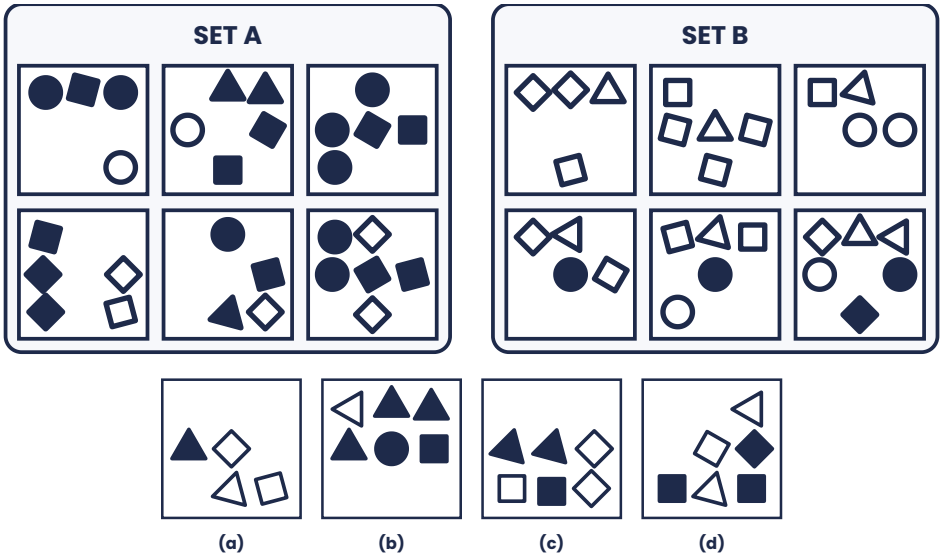
Answer figures

Q15 Which answer figure belongs to Set A?



(a) (b) (c) (d)

Q16 Which answer figure belongs to Set A?

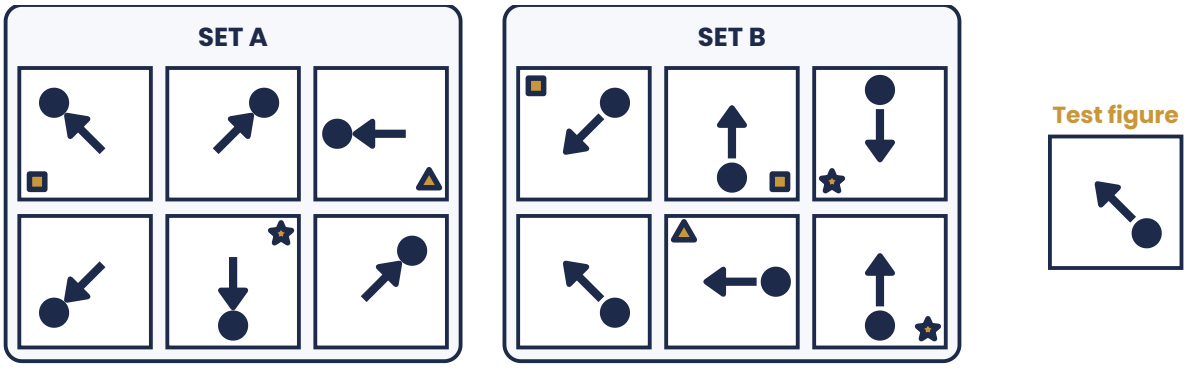


(a) (b) (c) (d)

19.4.3 Advanced level

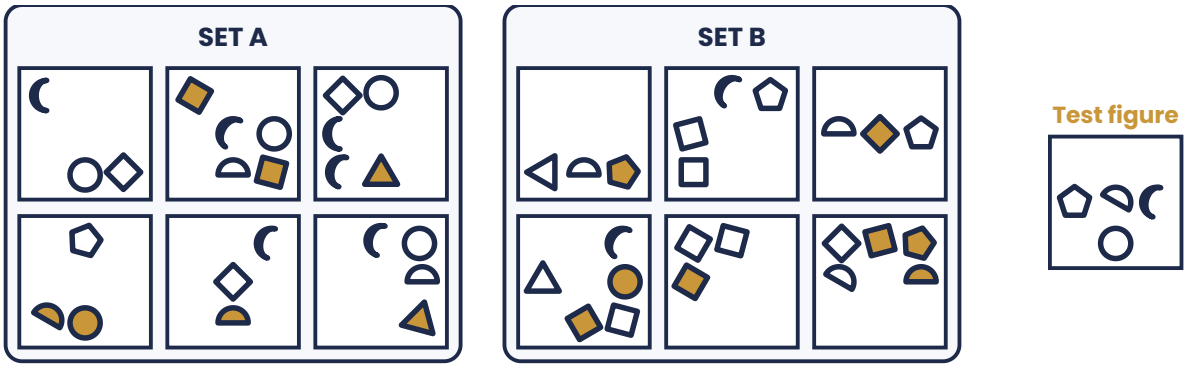
Q17 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



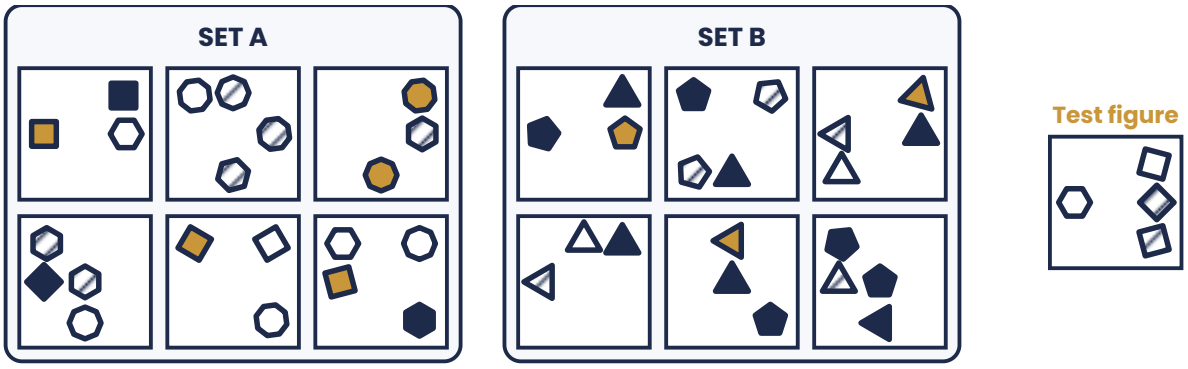
Q18 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



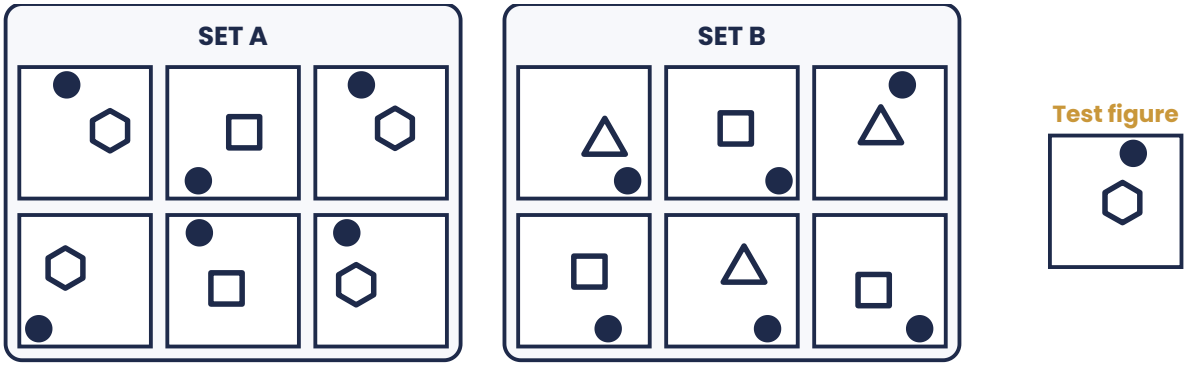
Q19 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



Q20 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither



Q21 Which answer figure continues the sequence?



Problem figures

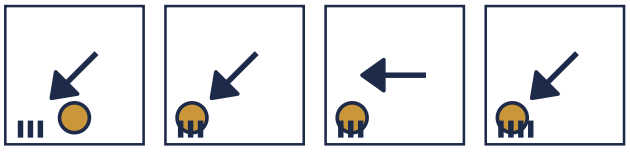


Answer figures

Q22 Which answer figure continues the sequence?



Problem figures

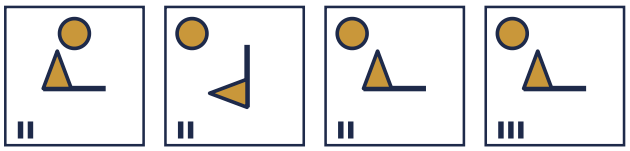


Answer figures

Q23 Which answer figure continues the sequence?



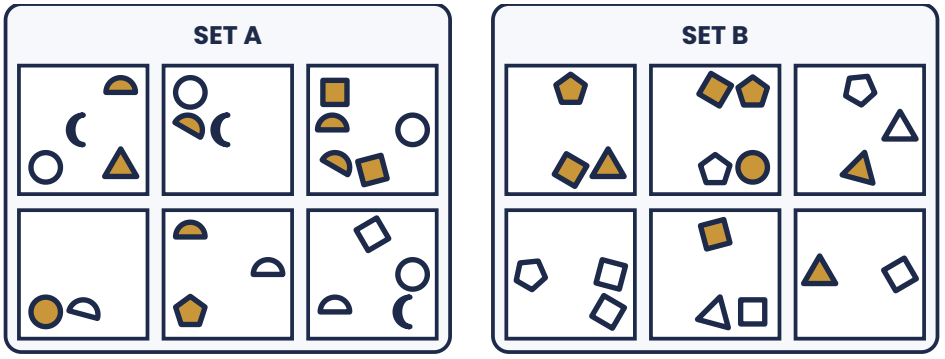
Problem figures

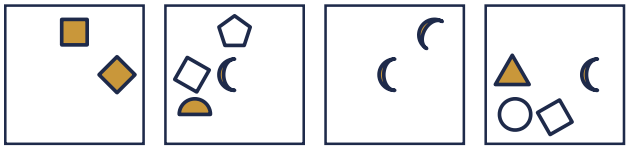


(a) (b) (c) (d)

Answer figures

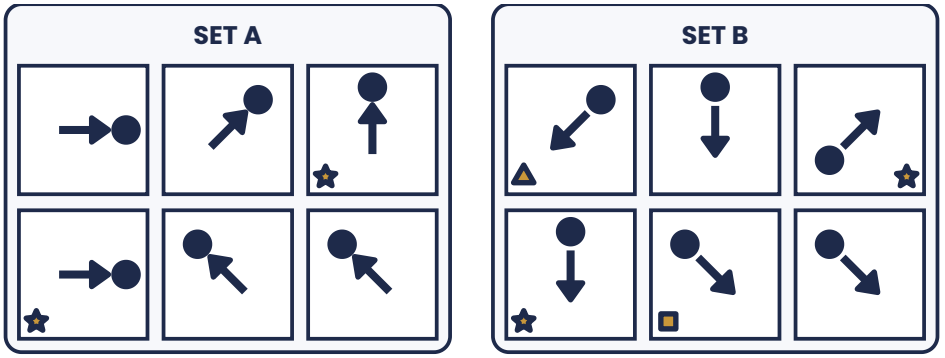
Q24 Which answer figure belongs to Set A?

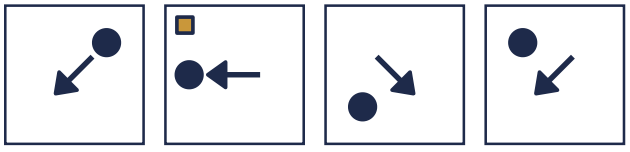




(a) (b) (c) (d)

Q25 Which answer figure belongs to Set A?





(a) (b) (c) (d)

19.5 Solutions

Q1 **Answer: (a) Set A**

Every Set A figure has exactly 3 shapes; every Set B figure has exactly 5. Shape and colour vary at random. The test figure belongs to **Set A**: it has 3 shapes.

Q2 **Answer: (a) Set A**

In Set A black shapes outnumber white ones; in Set B white shapes outnumber black ones. The test figure belongs to **Set A**: black shapes outnumber white.

Q3 **Answer: (a) Set A**

In Set A the small black circle is in the left part of the box; in Set B it is in the right part. The large shape is a distractor. The test figure belongs to **Set A**: the black circle is on the left.

Q4 **Answer: (b) Set B**

Every Set A figure has exactly 3 shapes; every Set B figure has exactly 5. Shape and colour vary at random. The test figure belongs to **Set B**: it has 5 shapes.

Q5 **Answer: (d)**

Three things change independently: the gold dot moves 2 places clockwise round the frame; the arrow turns 45° anticlockwise each step. Each wrong option breaks exactly one of these rules. Answer **(d)**.

Q6 **Answer: (a)**

Three things change independently: the gold dot moves 1 place clockwise round the frame; the arrow turns 45° clockwise each step. Each wrong option breaks exactly one of these rules. Answer **(a)**.

Q7 **Answer: (b)**

Every Set A figure has exactly 3 shapes; every Set B figure has exactly 5. Shape and colour vary at random. Only (b) obeys the Set A rule. Answer **(b)**.

Q8 **Answer: (b)**

In Set A the small black circle is in the left part of the box; in Set B it is in the right part. The large shape is a distractor. Only (b) obeys the Set A rule. Answer **(b)**.

Q9 **Answer: (a) Set A**

In Set A every shape has an even number of sides; in Set B every shape has an odd number of sides. Colours are distractors. The test figure belongs to **Set A**: all its shapes have an even number of sides.

Q10 **Answer: (b) Set B**

In Set A curved shapes outnumber straight-sided ones; in Set B straight-sided shapes outnumber curved ones. The test figure belongs to **Set B**: straight shapes outnumber curved ones.

Q11 **Answer: (a) Set A**

In Set A the arrow points at the black circle; in Set B it points directly away from it. The gold corner shapes are distractors. The test figure belongs to **Set A**: the arrow points at the circle.

Q12 **Answer: (a) Set A**

In Set A black shapes outnumber white ones; in Set B white shapes outnumber black ones. The test figure belongs to **Set A**: black shapes outnumber white.

Q13 **Answer: (a)**

Three things change independently: the gold dot moves 1 place clockwise round the frame; the flag turns 45° anticlockwise each step; and the number of short lines at the bottom cycles 2, 3, 4, 2, 3. Each wrong option breaks exactly one of these rules. Answer **(a)**.

Q14 **Answer: (a)**

Three things change independently: the gold dot moves 1 place anticlockwise round the frame; the flag turns 45° clockwise each step; and the number of short lines at the bottom cycles 2, 3, 4, 2, 3. Each wrong option breaks exactly one of these rules. Answer **(a)**.

Q15 **Answer: (a)**

In Set A every shape has an even number of sides; in Set B every shape has an odd number of sides. Colours are distractors. Only (a) obeys the Set A rule. Answer **(a)**.

Q16 **Answer: (b)**

In Set A black shapes outnumber white ones; in Set B white shapes outnumber black ones. Only (b) obeys the Set A rule. Answer **(b)**.

Q17 **Answer: (b) Set B**

In Set A the arrow points at the black circle; in Set B it points directly away from it. The gold corner shapes are distractors. The test figure belongs to **Set B**: the arrow points away from the circle.

Q18 **Answer: (a) Set A**

In Set A curved shapes outnumber straight-sided ones; in Set B straight-sided shapes outnumber curved ones. The test figure belongs to **Set A**: curved shapes outnumber straight ones.

Q19 **Answer: (a) Set A**

In Set A every shape has an even number of sides; in Set B every shape has an odd number of sides. Colours are distractors. The test figure belongs to **Set A**: all its shapes have an even number of sides.

Q20 **Answer: (b) Set B**

In Set A the small black circle is in the left part of the box; in Set B it is in the right part. The large shape is a distractor. The test figure belongs to **Set B**: the black circle is on the right.

Q21 **Answer: (c)**

Three things change independently: the gold dot moves 2 places clockwise round the frame; the arrow turns 45° anticlockwise each step; and the number of short lines at the bottom cycles 1, 2, 3, 1, 2. Each wrong option breaks exactly one of these rules. Answer **(c)**.

Q22 **Answer: (b)**

Three things change independently: the gold dot moves 1 place clockwise round the frame; the arrow turns 45° anticlockwise each step; and the number of short lines at the bottom cycles 2, 3, 4, 2, 3. Each wrong option breaks exactly one of these rules. Answer **(b)**.

Q23 **Answer: (c)**

Three things change independently: the gold dot moves 1 place anticlockwise round the frame; the flag turns 90° clockwise each step; and the number of short lines at the bottom cycles 1, 2, 3, 1, 2. Each wrong option breaks exactly one of these rules. Answer **(c)**.

Q24 **Answer: (c)**

In Set A curved shapes outnumber straight-sided ones; in Set B straight-sided shapes outnumber curved ones. Only (c) obeys the Set A rule. Answer **(c)**.

Q25 **Answer: (b)**

In Set A the arrow points at the black circle; in Set B it points directly away from it. The gold corner shapes are distractors. Only (b) obeys the Set A rule. Answer **(b)**.

Find what is constant within Set A and different in Set B: counts, shading balance, position, sides, curved versus straight, or a conditional link. Ignore features that vary freely. A figure that fits neither rule is Neither. In sequences, track each element on its own.

MODULE 20 · ADVANCED

Venn diagrams and visual relationships

Minimum duration **1 day** | Level **Advanced**

Venn diagram questions show relationships between categories as overlapping shapes. This module covers choosing the right diagram for two or three groups (subset, overlap and disjoint relationships), reading diagrams by region conditions, counting values in regions with the inclusion–exclusion rule, and diagrams that use triangles, squares and circles for different groups, followed by 25 solved questions in which every region total was computed from the drawn figure.

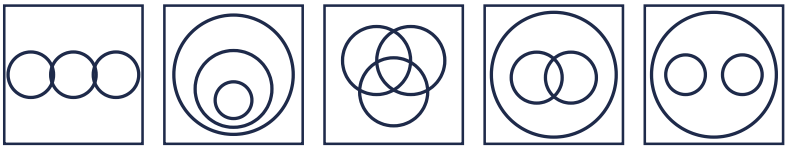
IN THIS MODULE

20.1 Choosing the right diagram
20.2 Reading Venn diagrams
20.3 Mixed-shape diagrams
20.4 Practice questions
20.5 Solutions

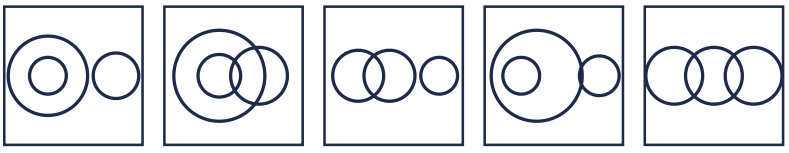
20.1 Choosing the right diagram

20.1.1 Representing relationships between two and three groups

For each pair of groups ask: is one wholly inside the other (subset), do they share some members (overlap), or do they share none (disjoint)? Three answers fix the diagram.



COMMON THREE-GROUP DIAGRAMS (1)



COMMON THREE-GROUP DIAGRAMS (2)

Diagram	Relationship	Example
D1	three separate circles: no two groups share members	Sun, Moon, Earth
D2	one circle inside another inside a third: each group is wholly contained in the next	Delhi, India, Asia
D3	three mutually overlapping circles: every combination is possible	Doctors, Men, Indians

Diagram	Relationship	Example
D5	two separate circles inside a large one	Tea, Coffee, Beverages
D6	one circle inside another, with a third separate	Mangoes, Fruits, Vegetables
D7	one circle inside another, with a third overlapping both	Males, Fathers, Doctors
D10	a chain: the middle circle overlaps both others, which do not meet	Boys, Students, Girls

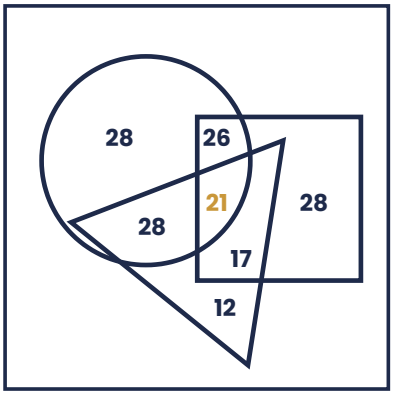
20.1.2 Subset, overlap and disjoint relationships

SHORTCUT TRICK
Look for a "general" word (animals, beverages, Asia): it is usually the big outer circle. Words that can never describe the same thing (boys and girls, tea and coffee) are separate circles.

20.2 Reading Venn diagrams

20.2.1 Identifying regions by conditions

Each region is described by which shapes it is inside and which it is outside. "Circle and triangle but not square" is the region inside the circle and the triangle but outside the square.



INSIDE CIRCLE AND TRIANGLE, OUTSIDE SQUARE: 28

20.2.2 Counting values in regions

Two groups: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. Three groups: $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

20.3 Mixed-shape diagrams

20.3.1 Triangles, squares and circles representing different groups

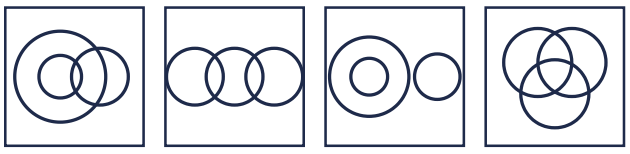
Many tests use a circle, a triangle and a square for three groups, with a number in each region. Translate the question into conditions on the shapes, then add the numbers of the regions that satisfy all of them.

WATCH OUT
"Who are A and B" usually includes those who are also C; "only A and B" or "A and B but not C" excludes them. Read the wording carefully.

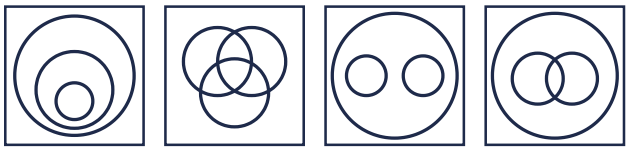
20.4 Practice questions

20.4.1 Basic level

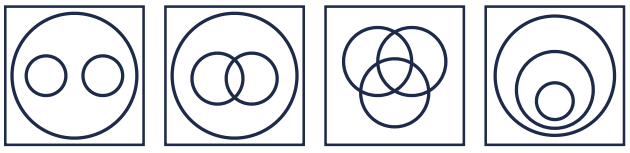
Q1 Which diagram best represents the relationship between **Doctors, Men, Indians**?


(a) (b) (c) (d)

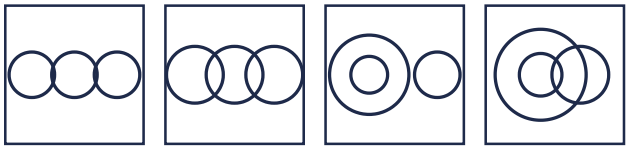
Q2 Which diagram best represents the relationship between **Delhi, India, Asia**?


(a) (b) (c) (d)

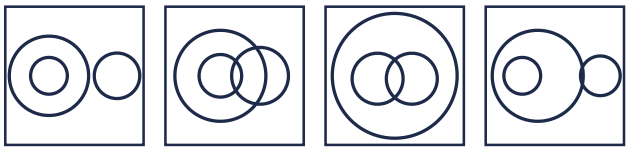
Q3 Which diagram best represents the relationship between **Tea, Coffee, Beverages**?


(a) (b) (c) (d)

Q4 Which diagram best represents the relationship between **Sun, Moon, Earth**?

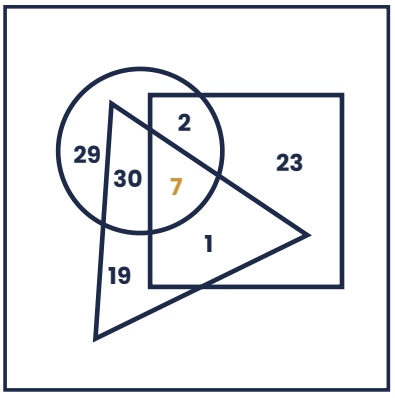

(a) (b) (c) (d)

Q5 Which diagram best represents the relationship between **Mangoes, Fruits, Vegetables**?


(a) (b) (c) (d)

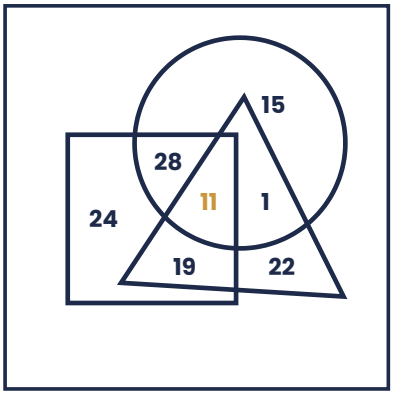
Q6 How many are inside the circle only?

(a) 58 **(b)** 32 **(c)** 82 **(d)** 29



Q7 What is the total of the numbers inside the triangle?

(a) 68 **(b)** 52 **(c)** 67 **(d)** 53

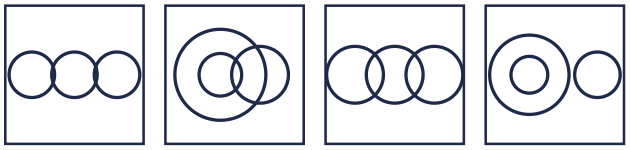


Q8 In a group of 75 students, 49 like tea, 46 like coffee and 21 like both. How many like neither?

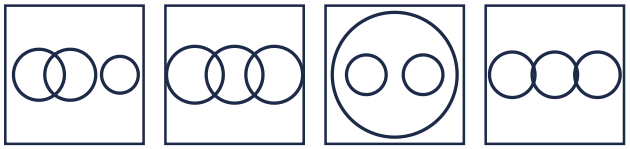
(a) 22 **(b)** 1 **(c)** 21 **(d)** 5

20.4.2 Intermediate level

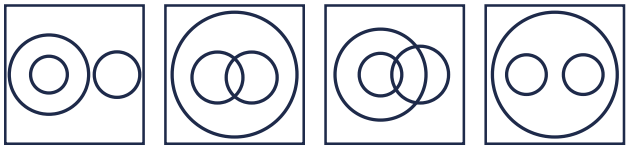
Q9 Which diagram best represents the relationship between **Males, Fathers, Doctors**?



Q10 Which diagram best represents the relationship between **Boys, Students, Girls**?

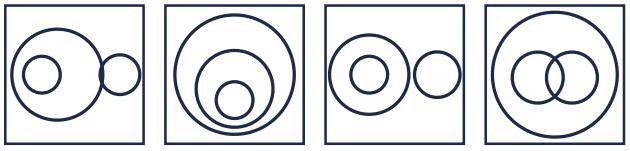


Q11 Which diagram best represents the relationship between **Whales, Mammals, Fish**?



(a) (b) (c) (d)

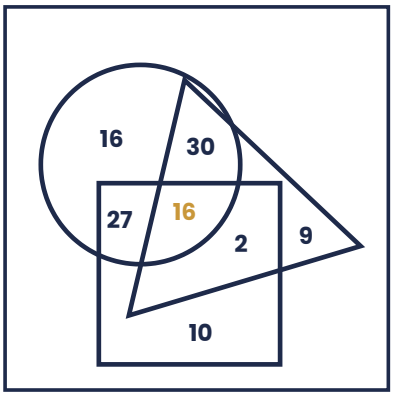
Q12 Which diagram best represents the relationship between **Cricketers, Footballers, Sportspersons**?



(a) (b) (c) (d)

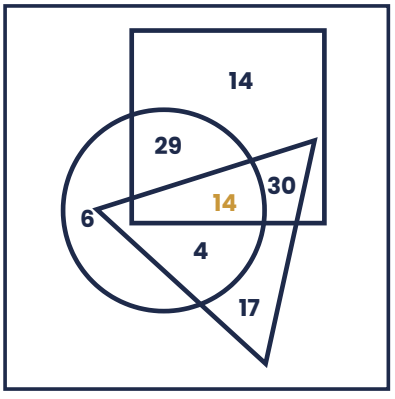
Q13 How many are inside the circle and the square but not the triangle?

(a) 43 **(b)** 30 **(c)** 27 **(d)** 83



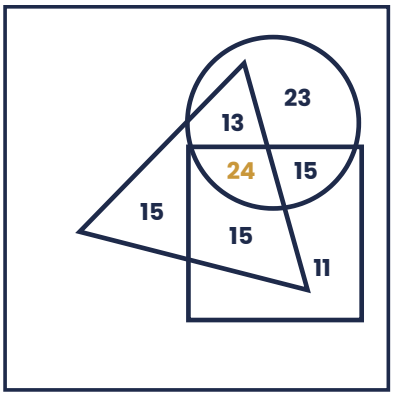
Q14 How many are inside exactly two shapes?

(a) 63 **(b)** 69 **(c)** 59 **(d)** 51



Q15 How many are inside the square but outside the circle?

(a) 49 **(b)** 15 **(c)** 90 **(d)** 26

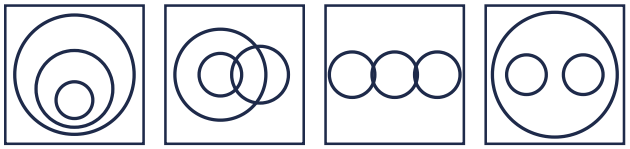


Q16 34 people read P. Of these, 19 also read Q, 15 also read R, and 5 read all three. How many read P only?

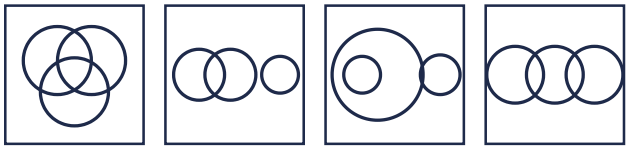
(a) 0 **(b)** 5 **(c)** 15 **(d)** 10

20.4.3 Advanced level

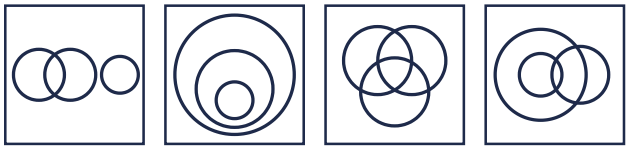
Q17 Which diagram best represents the relationship between **Parents, Mothers, Teachers**?



Q18 Which diagram best represents the relationship between **Odd numbers, Prime numbers, Even numbers**?

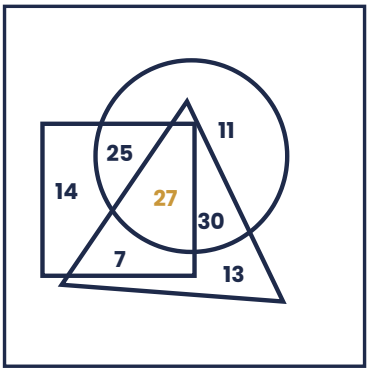


Q19 Which diagram best represents the relationship between **Teachers, Women, Graduates**?



Q20 How many Indian women are not doctors?

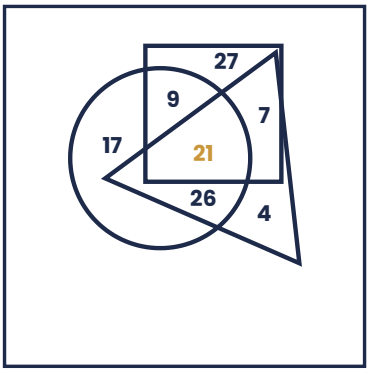
(a) 18 **(b)** 10 **(c)** 120 **(d)** 7



Circle = Doctors
Triangle = Women
Square = Indians

Q21 How many doctors are either women or Indian, but not both?

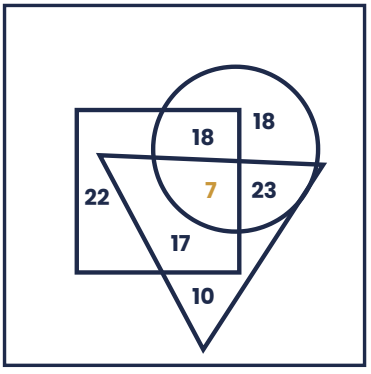
(a) 52 **(b)** 26 **(c)** 76 **(d)** 35



Circle = Doctors
Triangle = Women
Square = Indians

Q22 How many people belong to exactly one group?

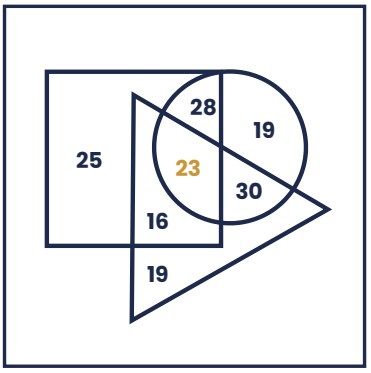
(a) 50 **(b)** 68 **(c)** 40 **(d)** 65



Circle = Doctors
Triangle = Women
Square = Indians

Q23 How many women are doctors (Indian or not)?

(a) 53 **(b)** 72 **(c)** 30 **(d)** 107



Circle = Doctors
Triangle = Women
Square = Indians

Q24 In a survey, 26 people read newspaper P, 37 read Q and 36 read R; 16 read P and Q, 26 read Q and R, 21 read R and P, and 14 read all three. How many read at least one newspaper?

(a) 50 **(b)** 36 **(c)** 99 **(d)** 64

Q25 In a group of 85 students, 54 like tea, 33 like coffee and 2 like both. How many like neither?

(a) 2 **(b)** 4 **(c)** 0 **(d)** 1

20.5 Solutions

Q1 **Answer: (d)**

Some doctors are men, some are Indian, and some men are Indian; all combinations occur. This is three mutually overlapping circles: every combination is possible: answer **(d)**.

Q2 **Answer: (a)**

Delhi is part of India, which is part of Asia. This is one circle inside another inside a third: each group is wholly contained in the next: answer **(a)**.

Q3 **Answer: (a)**

Tea and coffee are different drinks, but both are beverages. This is two separate circles inside a large one: answer **(a)**.

Q4 **Answer: (a)**

The three are entirely separate objects. This is three separate circles: no two groups share members: answer **(a)**.

Q5 **Answer: (a)**

Mangoes are fruits; vegetables are a separate group. This is one circle inside another, with a third separate: answer **(a)**.

Q6 **Answer: (d) 29**

Pick out every region that meets all the conditions and add the numbers in them: 29 = **29**.

Q7 **Answer: (d) 53**

Pick out every region that meets all the conditions and add the numbers in them: 22 + 1 + 19 + 11 = **53**.

Q8 **Answer: (b) 1**

Liking at least one = 49 + 46 − 21 = 74. Neither = 75 − 74 = **1**.

Q9 **Answer: (b)**

All fathers are males; some doctors are fathers, some are males but not fathers, and some are not male. This is one circle inside another, with a third overlapping both: answer **(b)**.

Q10 **Answer: (b)**

Some boys are students and some girls are students, but no one is both a boy and a girl. This is a chain: the middle circle overlaps both others, which do not meet: answer **(b)**.

Q11 **Answer: (a)**

Whales are mammals; fish are a separate group. This is one circle inside another, with a third separate: answer **(a)**.

Q12 **Answer: (d)**

Some people play both cricket and football, and all of them are sportspersons. This is two overlapping circles inside a large one: answer **(d)**.

Q13 **Answer: (c) 27**

Pick out every region that meets all the conditions and add the numbers in them: 27 = **27**.

Q14 **Answer: (a) 63**

Pick out every region that meets all the conditions and add the numbers in them: 4 + 29 + 30 = **63**.

Q15 **Answer: (d) 26**

Pick out every region that meets all the conditions and add the numbers in them: 11 + 15 = **26**.

Q16 **Answer: (b) 5**

P only = 34 − 19 − 15 + 5 = **5** (the 5 who read all three were subtracted twice, so add them back once).

Q17 **Answer: (b)**

All mothers are parents; some teachers are mothers, some are parents but not mothers, and some are neither. This is one circle inside another, with a third overlapping both: answer **(b)**.

Q18 **Answer: (d)**

Some primes are odd and one (2) is even, but no number is both odd and even. This is a chain: the middle circle overlaps both others, which do not meet: answer **(d)**.

Q19 **Answer: (c)**

All combinations occur: women teachers who are graduates, women who are neither, and so on. This is three mutually overlapping circles: every combination is possible: answer **(c)**.

Q20 **Answer: (d) 7**

Pick out every region that meets all the conditions and add the numbers in them: 7 = **7**.

Q21 **Answer: (d) 35**

Pick out every region that meets all the conditions and add the numbers in them: 26 + 9 = **35**.

Q22 **Answer: (a) 50**

Pick out every region that meets all the conditions and add the numbers in them: 18 + 10 + 22 = **50**.

Q23 **Answer: (a) 53**

Pick out every region that meets all the conditions and add the numbers in them: 30 + 23 = **53**.

Q24 **Answer: (a) 50**

$n(P \cup Q \cup R) = 26 + 37 + 36 - 16 - 26 - 21 + 14 =$ **50**.

Q25 **Answer: (c) 0**

Liking at least one = 54 + 33 - 2 = 85. Neither = 85 - 85 = **0**.

For each pair of groups decide subset, overlap or disjoint; the three answers fix the diagram. To read a diagram, turn the question into inside/outside conditions and add the matching regions. For totals use inclusion–exclusion: add the groups, subtract the pairs, add back the triple.

MODULE 21 · ADVANCED

Diagrammatic and flowchart reasoning

Minimum duration **2 days** | Level **Advanced**

Diagrammatic reasoning tests (Capgemini, SHL and similar) show operator symbols that transform a figure or a string of items; you work out what each operator does from examples and apply it. This module covers operators that reorder, reverse or change items, finding operators from examples, flowcharts with conditional branches, input-to-output, output-to-input and missing-operator questions, and switch-challenge games, followed by 25 solved questions in which a program checked that the examples fix every operator uniquely.

IN THIS MODULE

21.1 Operators and transformations
21.2 Flowchart reasoning
21.3 Sequences of operations
21.4 Switch and switch-challenge games
21.5 Practice questions
21.6 Solutions

21.1 Operators and transformations

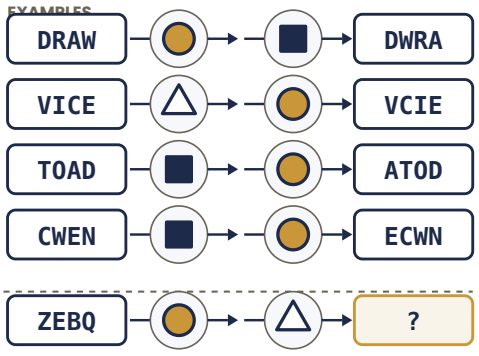
21.1.1 Symbols that rotate, reflect, recolour, add or remove elements

In a diagrammatic test, each symbol stands for one operation on the item passing through it. In figure versions the item is a shape (turn it, reflect it, change its colour); in string versions it is a row of letters (reverse them, swap two, move one, change each letter).

Operator type	Example effect on BKTM
reverses the order	MTKB
swaps the first two items	KBTM
swaps the first and last items	MKTB
moves the first item to the end	KTMB
moves the last item to the front	MBKT
replaces each letter by the next letter of the alphabet	CLUN
replaces each letter by the previous letter	AJSL
swaps the third and fourth items	BKMT

21.1.2 Finding what each operator does from examples

Start with a row that uses a single operator: its effect can be read directly. Then use it to peel that operator off longer rows and discover the rest. Check every conclusion against every example row.



EXAMPLES, THEN A QUESTION ROW

SHORTCUT TRICK

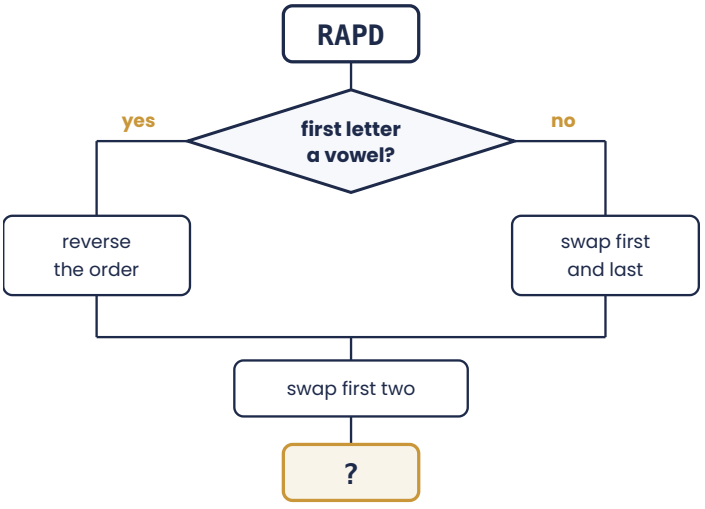
Compare letters and positions separately. If the output uses the same letters, the operator only moves them; if the letters change but stay in place, it changes them. Many operators do just one of these.

21.2 Flowchart reasoning

21.2.1 Following figures through a series of operators

Work strictly in order, writing down the intermediate result after each box. Most errors come from trying to do two operators in your head at once.

21.2.2 Conditional branches (if a shape is shaded, go left)



A FLOWCHART WITH A DECISION

21.3 Sequences of operations

21.3.1 Predicting the output from a given input

Apply the operators from left to right, writing each intermediate string.

21.3.2 Finding the input from a given output

Undo the operators from right to left using their inverses (reverse undoes reverse; "move first to end" is undone by "move last to front"; "next letter" is undone by "previous letter"), or test each option forwards.

21.3.3 Identifying the missing operator

Compute forwards from the input up to the gap and backwards from the output to the gap; the missing operator is whatever changes the first string into the second.

21.4 Switch and switch-challenge games

21.4.1 Matching symbol reorderings in game-based assessments

In switch challenges (used by Capgemini and others), a switch reorders a row of symbols and you choose the code that describes it, against the clock. Digit i of the code says which input position ends up in output position i.



A SWITCH REORDERS FOUR SYMBOLS

WATCH OUT
The most common trap is the inverse code, which describes where each input goes instead of where each output came from. Always read output position by output position.

21.5 Practice questions

21.5.1 Basic level

Q1 The switch rearranges the symbols on the left into the order on the right. Which code describes the switch? (Code digit i tells which input position goes to output position i.)

(a) 2143 **(b)** 2431 **(c)** 4231 **(d)** 4321



Q2 The switch rearranges the symbols on the left into the order on the right. Which code describes the switch? (Code digit i tells which input position goes to output position i.)

(a) 2431 **(b)** 4132 **(c)** 2143 **(d)** 4231



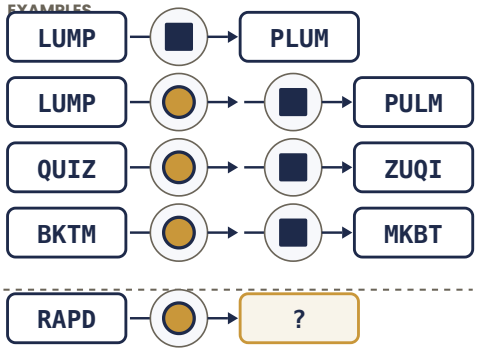
Q3 The switch rearranges the symbols on the left into the order on the right. Which code describes the switch? (Code digit i tells which input position goes to output position i.)

(a) 2314 **(b)** 2431 **(c)** 1243 **(d)** 3241



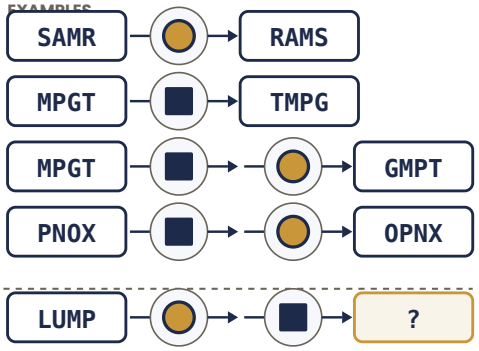
Q4 Work out what each symbol does from the examples. What is the output of the last row?

(a) ARPD **(b)** DRAP **(c)** SBQE **(d)** RADP



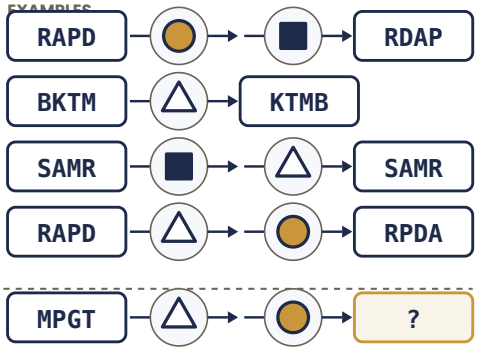
Q5 Work out what each symbol does from the examples. What is the output of the last row?

(a) LPMU **(b)** LPUM **(c)** LMUP **(d)** PMLU



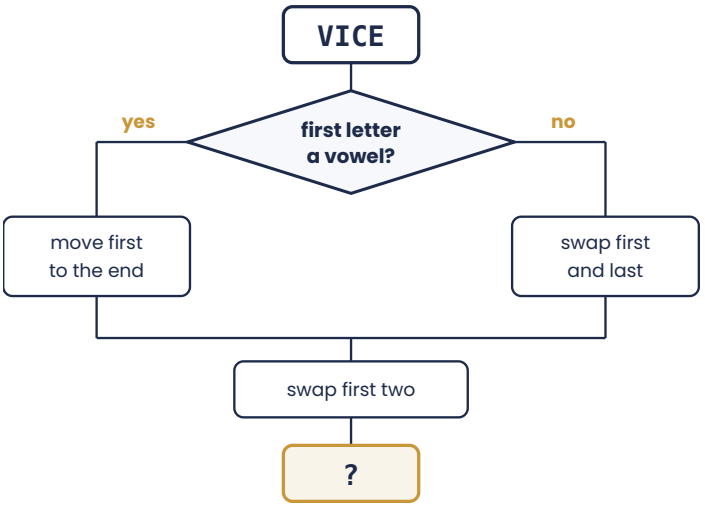
Q6 Work out what each symbol does from the examples. What is the output of the last row?

(a) PMTG **(b)** MGTP **(c)** UNQH **(d)** QHUN



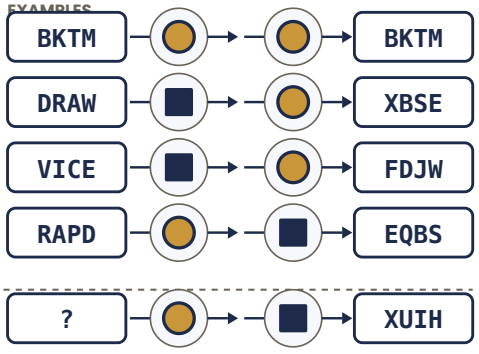
Q7 Follow the flowchart with the input shown. What is the output?

(a) CEIV **(b)** HUBD **(c)** VICE **(d)** IECV



Q8 Work out what each symbol does from the examples. Which input produces the output shown in the last row?

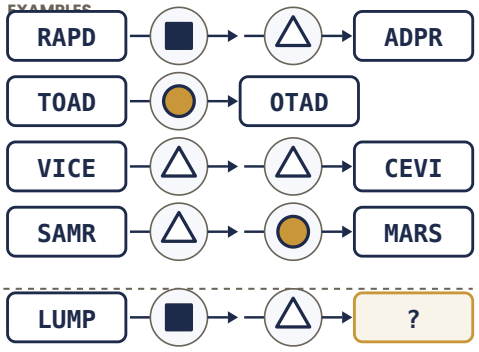
(a) GHTW **(b)** HIUX **(c)** HGTW **(d)** WTHG



21.5.2 Intermediate level

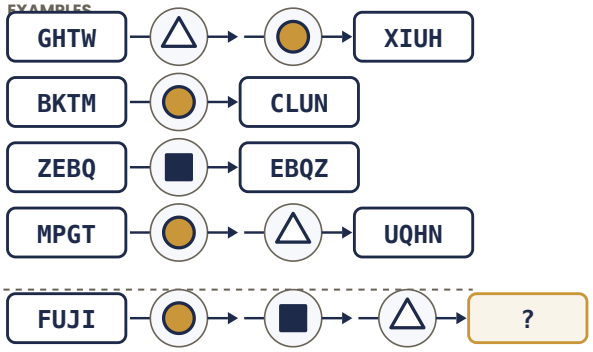
Q9 Work out what each symbol does from the examples. What is the output of the last row?

(a) UPML **(b)** PLMU **(c)** VMNQ **(d)** LUMP



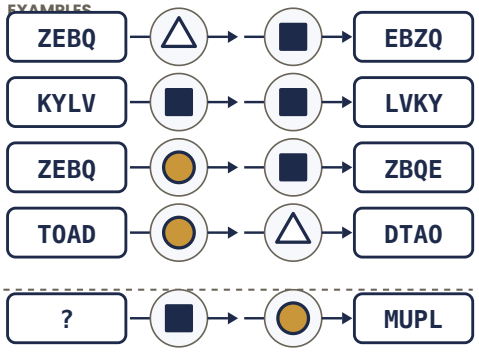
Q10 Work out what each symbol does from the examples. What is the output of the last row?

(a) JFIU **(b)** IJFU **(c)** GKJV **(d)** UJIF



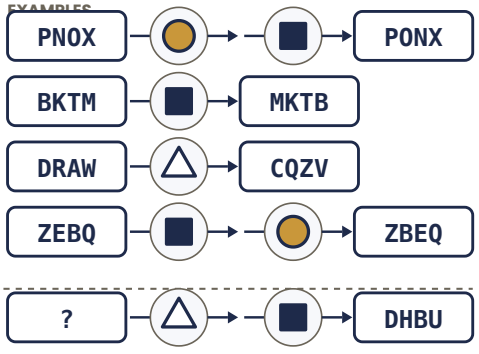
Q11 Work out what each symbol does from the examples. Which input produces the output shown in the last row?

(a) LUMP **(b)** MVNQ **(c)** PMUL **(d)** ULMP



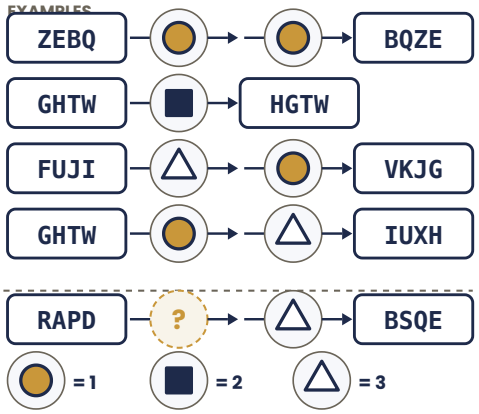
Q12 Work out what each symbol does from the examples. Which input produces the output shown in the last row?

(a) VICE **(b)** IVCE **(c)** WJDF **(d)** ICEV



Q13 Work out what each symbol does from the examples. Which symbol is missing from the last row? (Symbols are numbered in the key.)

(a) symbol 1 **(b)** symbol 2 **(c)** symbol 3 **(d)** none of these



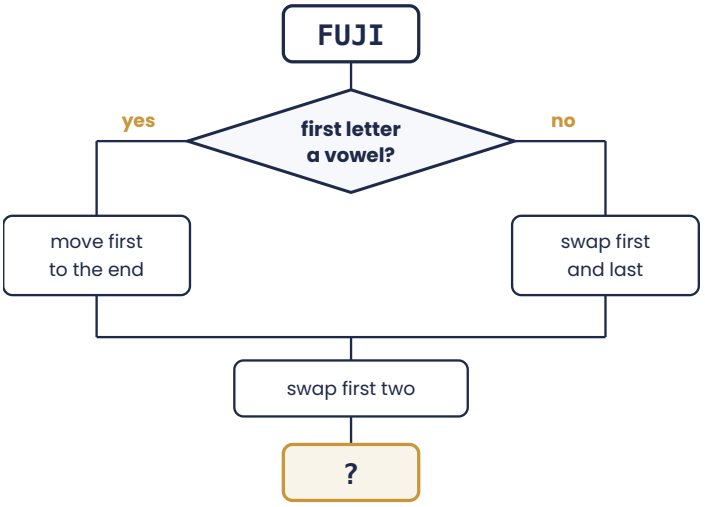
Q14 The switch rearranges the symbols on the left into the order on the right. Which code describes the switch? (Code digit i tells which input position goes to output position i.)

(a) 25413 **(b)** 24531 **(c)** 41532 **(d)** 42513



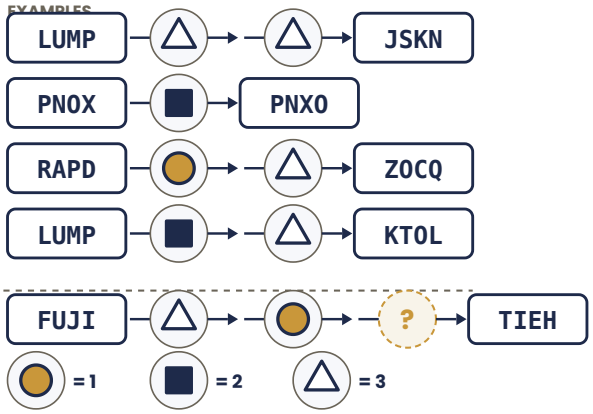
Q15 Follow the flowchart with the input shown. What is the output?

(a) IUJF **(b)** UIJF **(c)** UFIJ **(d)** JIUF



Q16 Work out what each symbol does from the examples. Which symbol is missing from the last row? (Symbols are numbered in the key.)

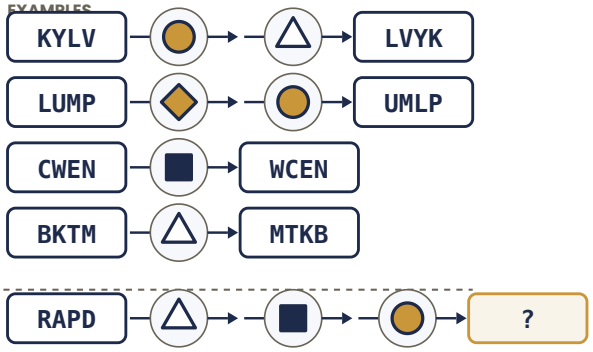
(a) symbol 1 **(b)** symbol 2 **(c)** symbol 3 **(d)** none of these



21.5.3 Advanced level

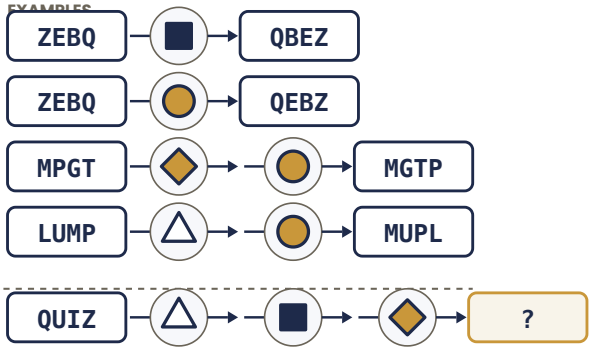
Q17 Work out what each symbol does from the examples. What is the output of the last row?

(a) RADP **(b)** OZCQ **(c)** QCOZ **(d)** PDRA



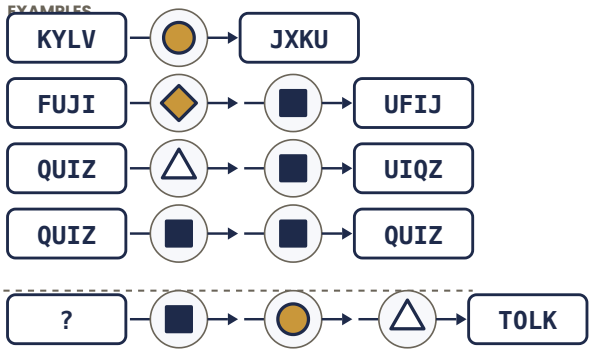
Q18 Work out what each symbol does from the examples. What is the output of the last row?

(a) HYTP **(b)** ZUQI **(c)** YPHT **(d)** JAVR



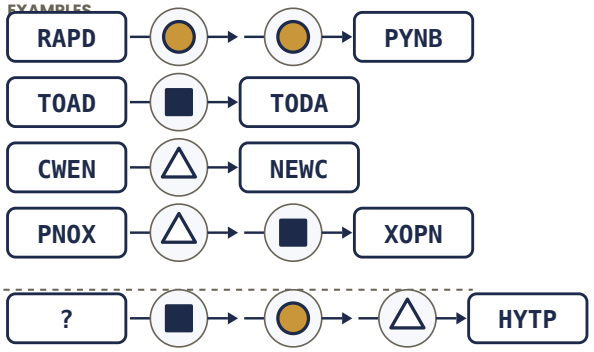
Q19 Work out what each symbol does from the examples. Which input produces the output shown in the last row?

(a) UMPL **(b)** PLUM **(c)** MVNQ **(d)** LUMP



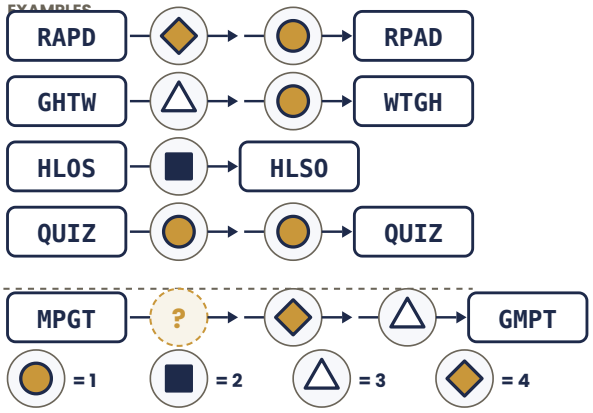
Q20 Work out what each symbol does from the examples. Which input produces the output shown in the last row?

(a) RVJA **(b)** ZQUI **(c)** UIZQ **(d)** QUIZ



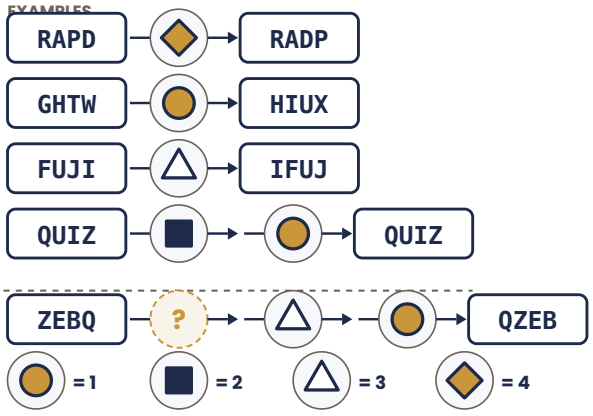
Q21 Work out what each symbol does from the examples. Which symbol is missing from the last row? (Symbols are numbered in the key.)

(a) symbol 1 **(b)** symbol 2 **(c)** symbol 3 **(d)** symbol 4



Q22 Work out what each symbol does from the examples. Which symbol is missing from the last row? (Symbols are numbered in the key.)

(a) symbol 1 **(b)** symbol 2 **(c)** symbol 3 **(d)** symbol 4



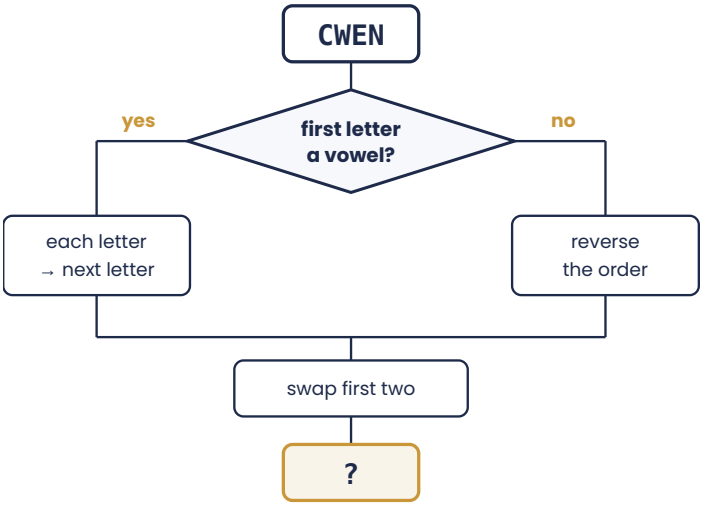
Q23 The switch rearranges the symbols on the left into the order on the right. Which code describes the switch? (Code digit i tells which input position goes to output position i.)

(a) 31542 **(b)** 25143 **(c)** 13425 **(d)** 34152



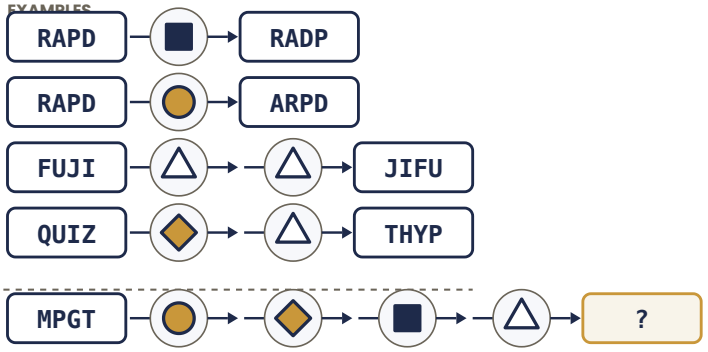
Q24 Follow the flowchart with the input shown. What is the output?

(a) ENWC **(b)** XDFO **(c)** CWEN **(d)** EWNC



Q25 Work out what each symbol does from the examples. What is the output of the last row?

(a) LSFO **(b)** MGPT **(c)** FOLS **(d)** SOLF



21.6 Solutions

Q1 **Answer: (c) 4231**

Read the output from left to right and find where each symbol was in the input: output 1 came from input 4, output 2 came from input 2, output 3 came from input 3, output 4 came from input 1. Code = **4231**. (The trap option is the inverse code, which describes where each input goes.)

Q2 **Answer: (c) 2143**

Read the output from left to right and find where each symbol was in the input: output 1 came from input 2, output 2 came from input 1, output 3 came from input 4, output 4 came from input 3. Code = **2143**. (The trap option is the inverse code, which describes where each input goes.)

Q3 **Answer: (c) 1243**

Read the output from left to right and find where each symbol was in the input: output 1 came from input 1, output 2 came from input 2, output 3 came from input 4, output 4 came from input 3. Code = **1243**. (The trap option is the inverse code, which describes where each input goes.)

Q4 **Answer: (a) ARPD**

From the examples: symbol 1 (circle) swaps the first two items; symbol 2 (square) moves the last item to the front. Applying the operators in order: RAPD → ARPD. Answer **(a) ARPD**.

Q5 **Answer: (b) LPUM**

From the examples: symbol 1 (circle) swaps the first and last items; symbol 2 (square) moves the last item to the front. Applying the operators in order: LUMP → PUML → LPUM. Answer **(b) LPUM**.

Q6 **Answer: (b) MGTP**

From the examples: symbol 1 (circle) swaps the first and last items; symbol 2 (square) moves the last item to the front; symbol 3 (triangle) moves the first item to the end. Applying the operators in order: MPGT → PGTM → MGTP. Answer **(b) MGTP**.

Q7 **Answer: (d) IECV**

VICE begins with a consonant, so take the no branch: it swaps the first and last items, giving EICV. Then swap the first two: IECV. Answer **(d) IECV**.

Q8 **Answer: (a) GHTW**

From the examples: symbol 1 (circle) reverses the order; symbol 2 (square) replaces each letter by the next letter of the alphabet. Undo the operators from right to left, or test each option forwards: only GHTW gives XUIH. Answer **(a) GHTW**.

Q9 **Answer: (a) UPML**

From the examples: symbol 1 (circle) swaps the first two items; symbol 2 (square) swaps the third and fourth items; symbol 3 (triangle) moves the first item to the end. Applying the operators in order: LUMP → LUPM → UPML. Answer **(a) UPML**.

Q10 **Answer: (c) GKJV**

From the examples: symbol 1 (circle) replaces each letter by the next letter of the alphabet; symbol 2 (square) moves the first item to the end; symbol 3 (triangle) swaps the first and last items. Applying the operators in order: FUJI → GVKJ → VKJG → GKJV. Answer **(c) GKJV**.

Q11 **Answer: (a) LUMP**

From the examples: symbol 1 (circle) swaps the first two items; symbol 2 (square) moves the first item to the end; symbol 3 (triangle) swaps the first and last items. Undo the operators from right to left, or test each option forwards: only LUMP gives MUPL. Answer **(a) LUMP**.

Q12 **Answer: (a) VICE**

From the examples: symbol 1 (circle) reverses the order; symbol 2 (square) swaps the first and last items; symbol 3 (triangle) replaces each letter by the previous letter. Undo the operators from right to left, or test each option forwards: only VICE gives DHBU. Answer **(a) VICE**.

Q13 **Answer: (b) symbol 2**

From the examples: symbol 1 (circle) moves the first item to the end; symbol 2 (square) swaps the first two items; symbol 3 (triangle) replaces each letter by the next letter of the alphabet. Trying each symbol in the gap, only symbol 2 (square) turns RAPD into BSQE. Answer **(b)**.

Q14 **Answer: (c) 41532**

Read the output from left to right and find where each symbol was in the input: output 1 came from input 4, output 2 came from input 1, output 3 came from input 5, output 4 came from input 3, output 5 came from input 2. Code = **41532**. (The trap option is the inverse code, which describes where each input goes.)

Q15 **Answer: (b) UIJF**

FUJI begins with a consonant, so take the no branch: it swaps the first and last items, giving IUJF. Then swap the first two: UIJF. Answer **(b) UIJF**.

Q16 **Answer: (b) symbol 2**

From the examples: symbol 1 (circle) moves the first item to the end; symbol 2 (square) swaps the third and fourth items; symbol 3 (triangle) replaces each letter by the previous letter. Trying each symbol in the gap, only symbol 2 (square) turns FUJI into TIEH. Answer **(b)**.

Q17 **Answer: (d) PDRA**

From the examples: symbol 1 (circle) swaps the third and fourth items; symbol 2 (square) swaps the first two items; symbol 3 (triangle) reverses the order; symbol 4 (diamond) moves the first item to the end. Applying the operators in order: RAPD → DPAR → PDAR → PDRA. Answer **(d) PDRA**.

Q18 **Answer: (b) ZUQI**

From the examples: symbol 1 (circle) swaps the first and last items; symbol 2 (square) reverses the order; symbol 3 (triangle) swaps the third and fourth items; symbol 4 (diamond) moves the first item to the end. Applying the operators in order: QUIZ → QUZI → IZUQ → ZUQI. Answer **(b) ZUQI**.

Q19 **Answer: (d) LUMP**

From the examples: symbol 1 (circle) replaces each letter by the previous letter; symbol 2 (square) swaps the third and fourth items; symbol 3 (triangle) moves the first item to the end; symbol 4 (diamond) swaps the first two items. Undo the operators from right to left, or test each option forwards: only LUMP gives TOLK. Answer **(d) LUMP**.

Q20 **Answer: (d) QUIZ**

From the examples: symbol 1 (circle) replaces each letter by the previous letter; symbol 2 (square) swaps the third and fourth items; symbol 3 (triangle) reverses the order. Undo the operators from right to left, or test each option forwards: only QUIZ gives HYTP. Answer **(d) QUIZ**.

Q21 **Answer: (a) symbol 1**

From the examples: symbol 1 (circle) reverses the order; symbol 2 (square) swaps the third and fourth items; symbol 3 (triangle) swaps the first two items; symbol 4 (diamond) swaps the first and last items. Trying each symbol in the gap, only symbol 1 (circle) turns MPGT into GMPT. Answer **(a)**.

Q22 **Answer: (b) symbol 2**

From the examples: symbol 1 (circle) replaces each letter by the next letter of the alphabet; symbol 2 (square) replaces each letter by the previous letter; symbol 3 (triangle) moves the last item to the front; symbol 4 (diamond) swaps the third and fourth items. Trying each symbol in the gap, only symbol 2 (square) turns ZEBQ into QZEB. Answer **(b)**.

Q23 **Answer: (b) 25143**

Read the output from left to right and find where each symbol was in the input: output 1 came from input 2, output 2 came from input 5, output 3 came from input 1, output 4 came from input 4, output 5 came from input 3. Code = **25143**. (The trap option is the inverse code, which describes where each input goes.)

Q24 **Answer: (a) ENWC**

CWEN begins with a consonant, so take the no branch: it reverses the order, giving NEWC. Then swap the first two: ENWC. Answer **(a) ENWC**.

Q25 **Answer: (a) LSFO**

From the examples: symbol 1 (circle) swaps the first two items; symbol 2 (square) swaps the third and fourth items; symbol 3 (triangle) moves the first item to the end; symbol 4 (diamond) replaces each letter by the previous letter. Applying the operators in order: MPGT → PMGT → OLFS → OLSF → LSFO. Answer **(a) LSFO**.

Read single-operator rows first, then peel known operators off longer rows; check every example. Apply operators strictly in order, writing each intermediate result. To find an input, undo operators right to left. For a missing operator, work forwards and backwards to the gap. In switch codes, digit i names the input position that lands in output position i.

MODULE 22 · ADVANCED

Exam strategy, mock tests and company-specific preparation

Minimum duration **3 days** | Level **Advanced**

The final module turns visual reasoning skills into consistent test performance. It covers the element checklist under time pressure, eliminating options by one rule, when to skip complex figures, managing eye fatigue, company-specific patterns (Capgemini, Accenture, Cognizant, TCS, Infosys, Wipro, AMCAT, SHL, Aon cut-e and game-based assessments), mock analysis with an error log, and a quick-revision sheet, followed by a 25-question mixed non-verbal mock test drawn from every module, fully solved.

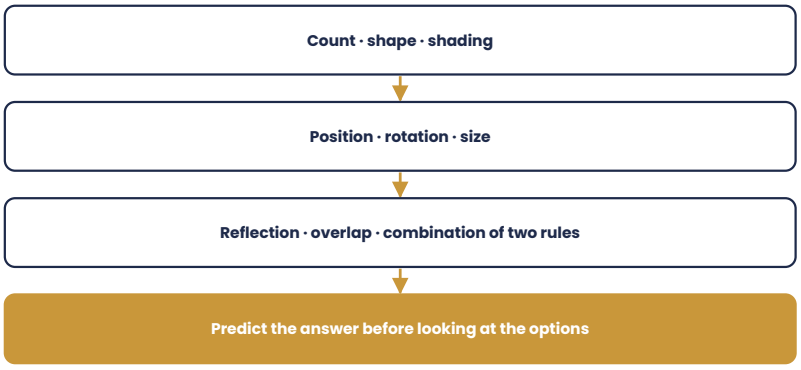
IN THIS MODULE

22.1 Strategy
22.2 Company-specific patterns
22.3 Mock tests and analysis
22.4 Revision plan
22.5 Mixed mock test
22.6 Solutions

22.1 Strategy

22.1.1 Using the element checklist under time pressure

Run the same checklist on every figure question, in the same order, until it is automatic: **count** (elements, sides, lines), **shape**, **shading**, **position**, **rotation** (angle and direction), **size**, **reflection**. The first attribute that changes in a regular way is usually the rule.



THE ELEMENT CHECKLIST

22.1.2 Eliminating options quickly by one rule

As soon as you have found one rule, apply it to all four options and cross out those that break it. Often only one or two options remain, and a second rule settles it in seconds. This is faster than checking every rule on every option.

22.1.3 When to skip complex figures

Give each question a budget (usually 45 to 75 seconds in non-verbal sections). If after twice the budget you have no rule, mark the question, guess if there is no negative marking, and move on. Complex questions are worth the same marks as easy ones.

22.1.4 Managing eye fatigue during long tests

Long visual tests tire the eyes and cause careless mistakes. Every 10 to 15 minutes look away from the screen at something distant for a few seconds; blink deliberately; keep the screen at arm's length and its brightness close to the room's. Use your finger or the cursor to track elements instead of holding them all in your head.

22.2 Company-specific patterns

Patterns change from year to year; always read the latest official notice. The table describes the broad shape of each test and what to prioritise.

Test	Non-verbal content	Priority modules
Capgemini	diagrammatic reasoning, switch challenge and other game-based rounds	19, 21
Accenture	abstract reasoning: series, odd one out, Set A/Set B, matrices	3, 5, 13, 19
Cognizant (GenC)	figure series, analogy, classification, embedded figures	3, 4, 5, 8
TCS NQT	visual reasoning within reasoning ability: series, mirror and water images, paper folding	3, 6, 7, 10, 11
Infosys	harder visual puzzles, cubes and dice, patterns	9, 13, 16, 17
Wipro	figure series, analogy, odd one out	3, 4, 5
AMCAT / SHL / Aon cut-e	inductive and deductive figure tests, spatial rotation, cube nets	13, 17, 18, 19, 21
Game-based assessments	grid challenge (spatial memory), motion challenge (moving blocks), inductive and deductive games	18, 19, 21

22.2.1 Capgemini diagrammatic and game-based tests

Expect operator puzzles like those in Module 21 and fast switch-challenge rounds. Practise writing intermediate strings, and in switch rounds always read output positions one by one.

22.2.2 Accenture abstract reasoning

Mostly series and odd-one-out questions under tight time. Use the element checklist and elimination by one rule.

22.2.3 Cognizant, TCS, Infosys and Wipro non-verbal questions

Classic figure questions: series, analogy, classification, mirror and water images, paper folding and cutting, embedded figures and counting. Modules 3 to 12 cover them fully.

22.2.4 AMCAT, SHL and Aon (cut-e) abstract and spatial tests

Inductive (find the rule) and deductive (apply the given rules) formats, plus spatial items on rotation, nets and views. Speed comes from recognising the format in the first few seconds.

22.2.5 Game-based assessments

Grid challenge: remember the positions of dots while doing another task; rehearse positions as a pattern ("top-left, centre, bottom-right diagonal"). Motion challenge: slide blocks to free a ball in the fewest moves; plan backwards from the exit. Inductive and deductive games: hypothesise, test, revise, exactly as in Module 19.

22.3 Mock tests and analysis

22.3.1 Weekly timed mocks

Take one full-length non-verbal mock each week under exam conditions. Record your score, the time used, and the number of questions skipped.

22.3.2 Error log by question type

Date	Question type	What went wrong	Rule to remember
—	Figure series	Missed the second element moving anticlockwise	Track each element separately
—	Dice	Assumed a standard die	Only assume opposite faces sum to 7 if told
—	Mirror image	Flipped top and bottom	Vertical mirror swaps left and right only

22.4 Revision plan

22.4.1 Quick-revision sheet

Topic	Key fact
Rotation	90° clockwise = 270° anticlockwise; 180° is the same either way; 45° steps take 8 moves to return
Mirror (vertical)	swaps left and right; top and bottom stay
Water image	swaps top and bottom; left and right stay
Mirror time	11:60 - time (subtract from 23:60 for 24-hour clocks)
Water time	18:30 - time
Paper folding	each fold doubles the layers; unfold in reverse, reflecting holes across each fold
Painted n-cube	3 faces 8; 2 faces $12(n-2)$; 1 face $6(n-2)^2$; none $(n-2)^3$

Topic	Key fact
Dice	standard: opposite faces sum to 7; two common faces → third faces opposite
Counting	triangle with n parts from a vertex: $n(n+1)/2$; $n \times n$ grid: Σk^2 squares; $m \times n$ grid: $C(m+1, 2) \times C(n+1, 2)$ rectangles
Cube nets	eleven nets; no 2×2 block; no row of five
Venn	$n(A \cup B \cup C) = \Sigma A - \Sigma(A \cap B) + A \cap B \cap C$

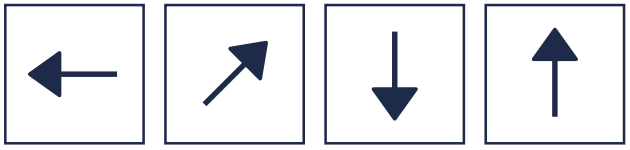
22.5 Mixed mock test

22.5.1 Section A level

Q1 Which figure comes next?



Problem figures



Answer figures

Q2 Which figure comes next?

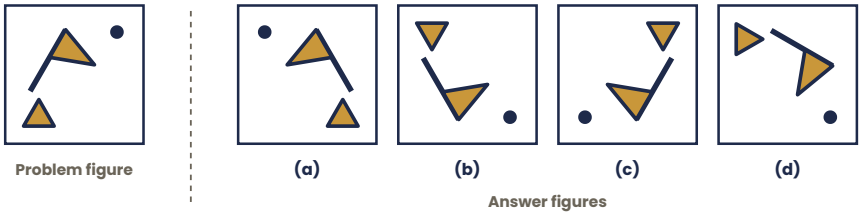


Problem figures

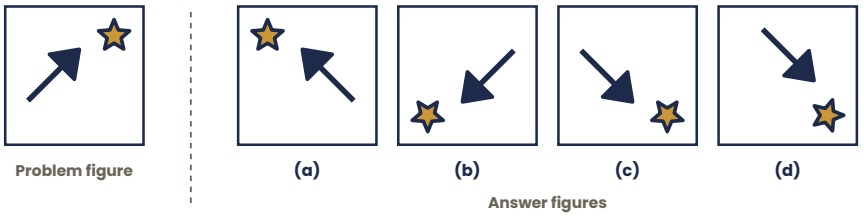


Answer figures

Q3 Which answer figure is the mirror image of the problem figure (mirror placed vertically on the right)?

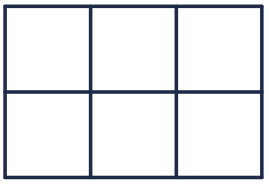


Q4 Which answer figure is the water image of the problem figure?

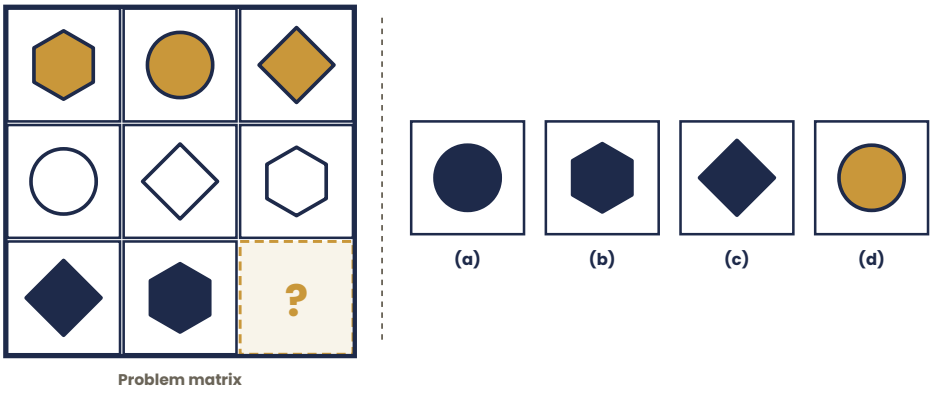


Q5 How many rectangles are there in the figure?

(a) 18 **(b)** 16 **(c)** 20 **(d)** 22



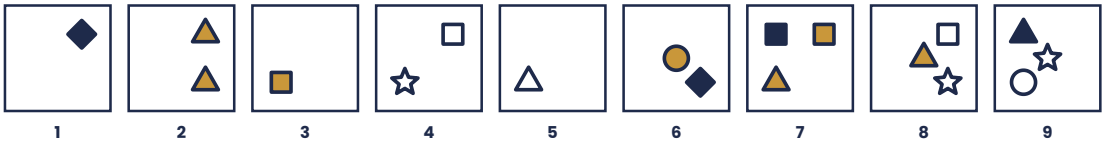
Q6 Which answer figure completes the matrix?



Q7 Divide the nine figures into three groups of three, each group sharing a common property.

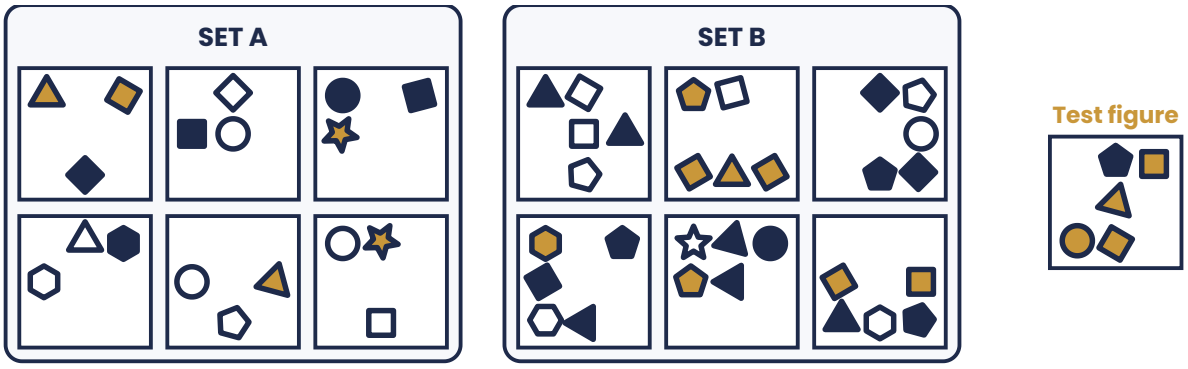
(a) 1, 2, 4; 3, 5, 8; 6, 7, 9 **(b)** 1, 7, 8; 2, 4, 6; 3, 5, 9

(c) 1, 2, 4; 3, 5, 6; 7, 8, 9 **(d)** 1, 3, 5; 2, 4, 6; 7, 8, 9



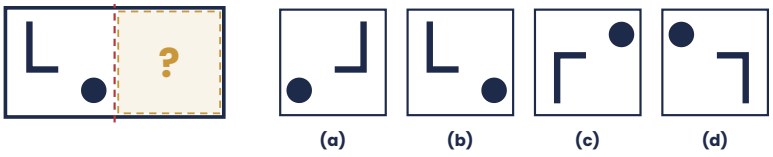
Q8 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither

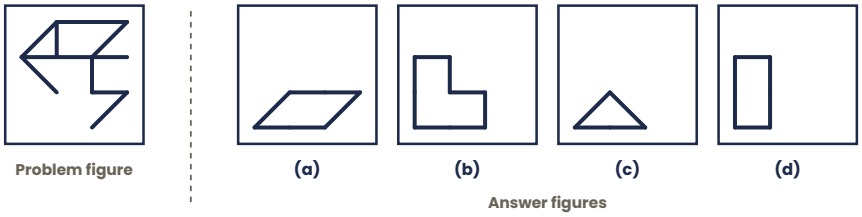


22.5.2 Section B level

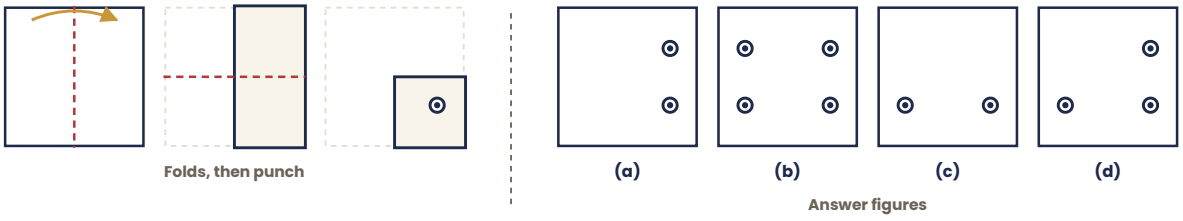
Q9 The dashed red line is a line of symmetry. Which answer figure completes the figure?



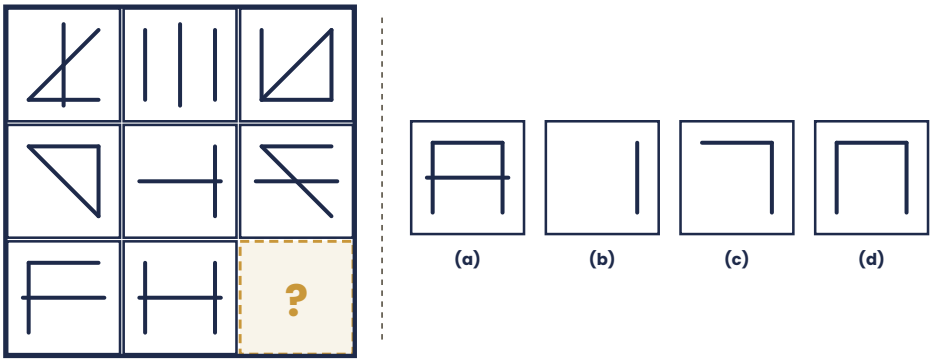
Q10 Which answer figure is hidden in the problem figure (same size and orientation)?



Q11 A square sheet is folded as shown and then punched. How does it look when unfolded?

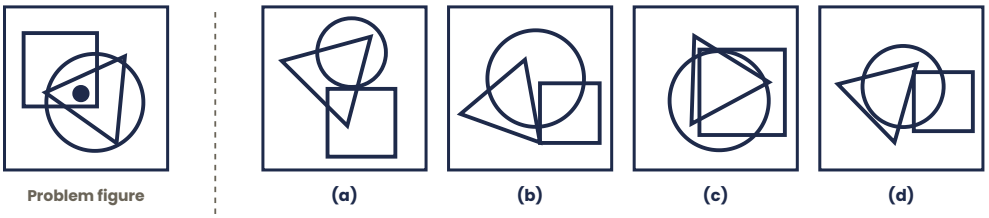


Q12 Which answer figure completes the matrix?



Problem matrix

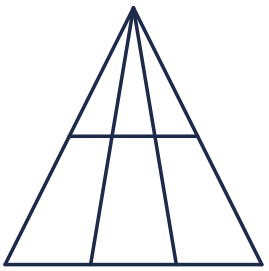
Q13 In which answer figure can a dot be placed so that it satisfies the same conditions as the dot in the problem figure?



Answer figures

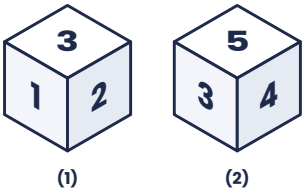
Q14 How many triangles are there in the figure?

(a) 10 **(b)** 12 **(c)** 14 **(d)** 16

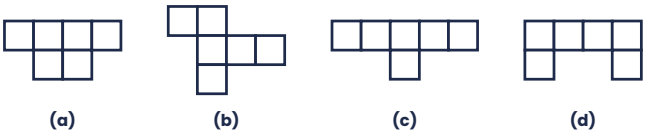


Q15 2 different positions of the same die are shown. Which number is opposite **5**?

(a) 1 **(b)** 3 **(c)** 2 **(d)** 4



Q16 Which of the following figures can be folded to make a cube?

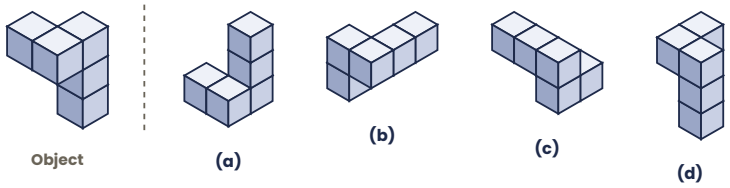


22.5.3 Section C level

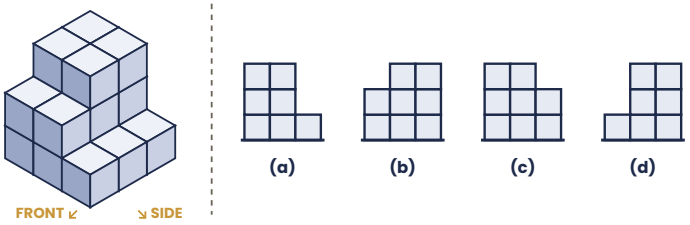
Q17 A cube painted on all faces is cut into 216 identical cubes. How many have exactly one painted face?

(a) 64 **(b)** 48 **(c)** 24 **(d)** 96

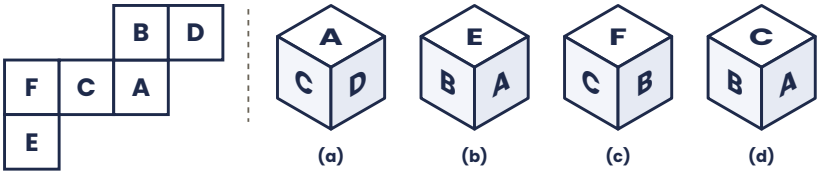
Q18 Which answer figure shows the same object as the one on the left, only turned to a different position?



Q19 Which answer figure is the side view (looking at the face marked SIDE) of the block structure?

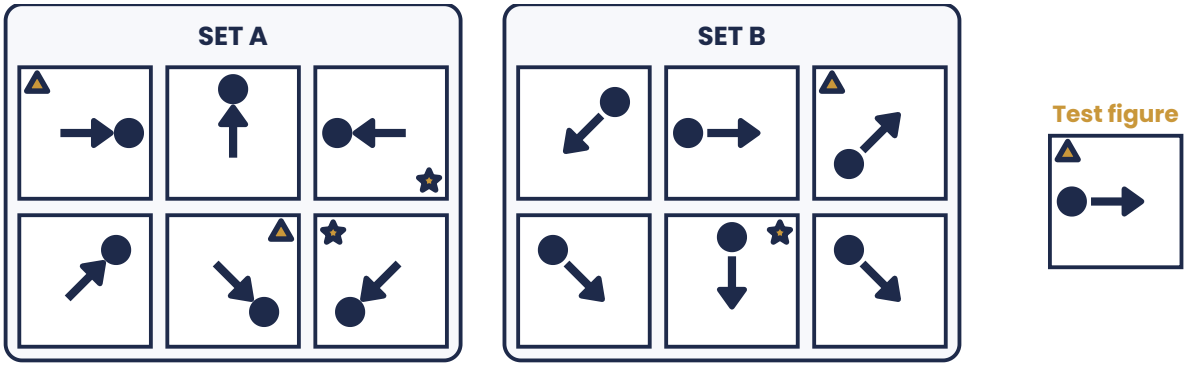


Q20 The net is folded into a cube. Which of the answer figures shows that cube?

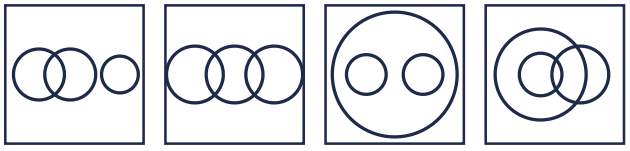


Q21 Does the test figure belong to Set A, Set B or neither?

(a) Set A **(b)** Set B **(c)** Neither

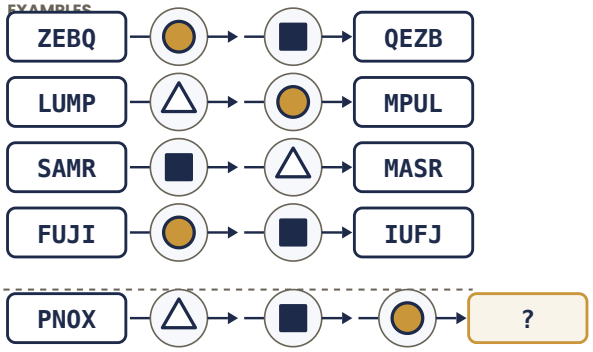


Q22 Which diagram best represents the relationship between **Odd numbers, Prime numbers, Even numbers**?



Q23 Work out what each symbol does from the examples. What is the output of the last row?

(a) XPON **(b)** QOPY **(c)** PONX **(d)** ONPX

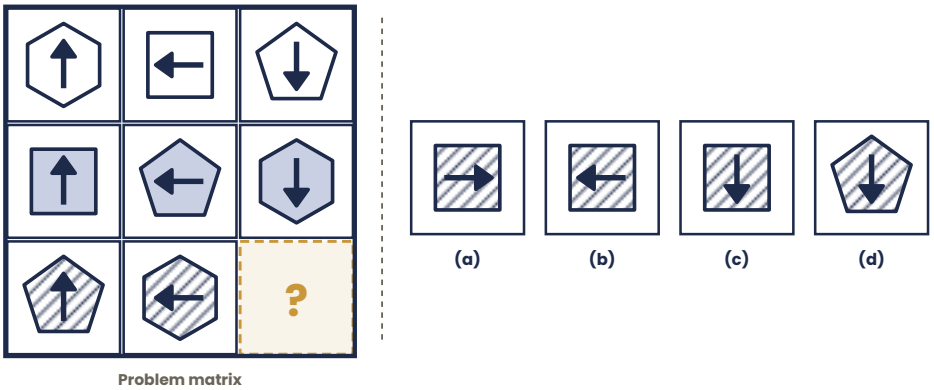


Q24 The switch rearranges the symbols on the left into the order on the right. Which code describes the switch? (Code digit i tells which input position goes to output position i.)

(a) 4312 **(b)** 3142 **(c)** 3421 **(d)** 3412



Q25 Which answer figure completes the matrix?



Problem matrix

22.6 Solutions

Q1 **Answer: (c)**

The arrow turns 135° clockwise at every step, so the next figure is turned another 135° clockwise: answer **(c)**.

Q2 **Answer: (b)**

The star moves 3 places clockwise around the frame each time; continuing the movement gives **(b)**.

Q3 **Answer: (a)**

A vertical mirror swaps left and right but keeps top and bottom: answer **(a)**.

Q4 **Answer: (c)**

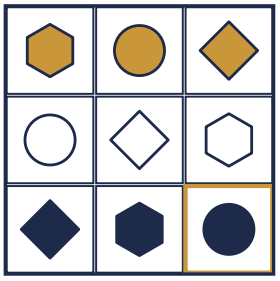
A water image swaps top and bottom but keeps left and right: answer **(c)**.

Q5 **Answer: (a) 18**

Labelling every point and testing every combination gives **18** rectangles.

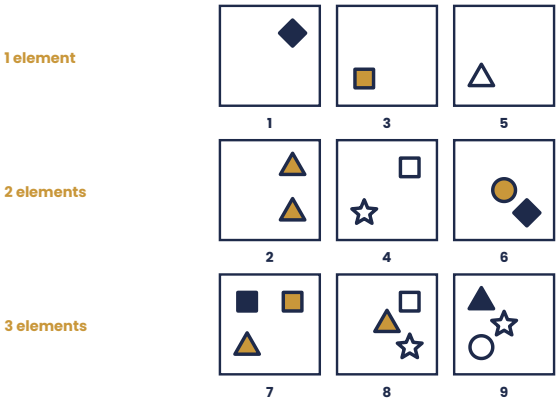
Q6 **Answer: (a)**

Each row contains a hexagon, a circle and a diamond exactly once, and all the figures in a row have the same shading. Row 3 is missing the circle, shaded black: **(a)**.



Q7 **Answer: (d) 1, 3, 5; 2, 4, 6; 7, 8, 9**

The figures group by the number of elements in each figure, whatever their shape or shading. 1 element: 1, 3, 5; 2 elements: 2, 4, 6; 3 elements: 7, 8, 9. Answer **(d)**.



Q8 **Answer: (b) Set B**

Every Set A figure has exactly 3 shapes; every Set B figure has exactly 5. Shape and colour vary at random. The test figure belongs to **Set B**: it has 5 shapes.

Q9 **Answer: (a)**

The right half must be the mirror image of the left half across the red line: **(a)**.

Q10 **Answer: (a)**

The parallelogram can be traced inside the problem figure (shown in gold); none of the other shapes appears with the same size and orientation. Answer **(a)**.



traced

Q11 **Answer: (b)**

Unfold in reverse order, reflecting every hole across each fold line. 2 folds give up to 4 layers, so the 1 punch produce 4 holes: answer **(b)**.

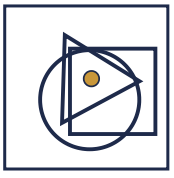
Q12 **Answer: (c)**

In each row the first two figures are overlaid, and lines that appear in **both** cancel out; lines in only one figure remain. Doing this for row 3 gives **(c)**.



Q13 **Answer: (c)**

The dot lies inside the circle and the square and the triangle. Only figure (c) has a region that is inside the circle and the square and the triangle; the dot can go there (shown in gold). In every other figure that region does not exist. Answer **(c)**.



(c)

Q14 **Answer: (b) 12**

Labelling every point and testing every combination gives **12** triangles.

Q15 **Answer: (c) 2**

The face 3 appears in both views. Reading the other two faces clockwise from 3 in each view and matching positions pairs each face with the one opposite it. The checked result: 5 is opposite 2. Answer **(c) 2**.

Q16 **Answer: (b)**

Only (b) folds into a closed cube: every face lands on a different side. In each of the others two squares fold onto the same side of the cube, leaving one side open. Answer **(b)**.

Q17 **Answer: (d) 96**

n = 6. One face: $6(n - 2)^2 = 6 \times 16 =$ **96**.

Q18 **Answer: (d)**

Figure (d) is the object turned in space. The other three are its **mirror image** in various positions: however they are turned, they can never be made to match the original, just as a left hand cannot be turned into a right hand.

Q19 **Answer: (b)**

In a side view each column shows the tallest stack in that line of sight. Reading the columns from left to right gives heights 2, 3, 3: answer **(b)**.

Q20 **Answer: (c)**

Check opposite pairs first: two faces that are opposite in the net can never be seen together. Then check the clockwise order around a corner. Only (c) passes both tests.

Q21 **Answer: (b) Set B**

In Set A the arrow points at the black circle; in Set B it points directly away from it. The gold corner shapes are distractors. The test figure belongs to **Set B**: the arrow points away from the circle.

Q22 **Answer: (b)**

Some primes are odd and one (2) is even, but no number is both odd and even. This is a chain: the middle circle overlaps both others, which do not meet: answer **(b)**.

Q23 **Answer: (a) XPON**

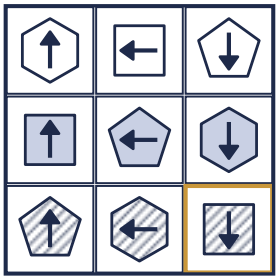
From the examples: symbol 1 (circle) swaps the first two items; symbol 2 (square) moves the last item to the front; symbol 3 (triangle) reverses the order. Applying the operators in order: PNOX → XONP → PXON → XPON. Answer **(a) XPON**.

Q24 **Answer: (c) 3421**

Read the output from left to right and find where each symbol was in the input: output 1 came from input 3, output 2 came from input 4, output 3 came from input 2, output 4 came from input 1. Code = **3421**. (The trap option is the inverse code, which describes where each input goes.)

Q25 **Answer: (c)**

Three rules work together: (1) each outer shape appears once per row; (2) the shading of the outer shape is fixed for each row; (3) the arrow turns 90° anticlockwise from column to column (pointing up in column 1). Row 3 needs the square, shaded like the rest of row 3, with the arrow turned 180°: **(c)**.



Run the element checklist on every figure, eliminate options by the first rule you find, budget 45 to 75 seconds per question and skip what does not yield, rest your eyes regularly, and keep an error log. Revise the quick-revision sheet the day before every test.